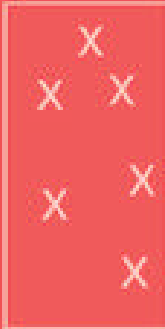
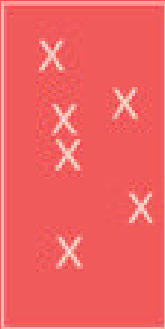
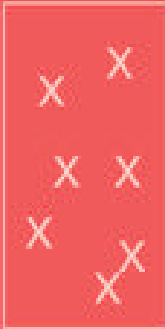
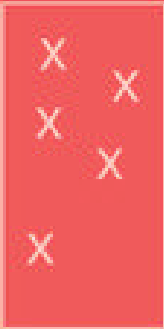


Applied Longitudinal Data Analysis for Epidemiology

Second Edition



Jos W. R. Twisk

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Applied Longitudinal Data Analysis for Epidemiology

A Practical Guide

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Second Edition

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An eye for an eye
A tooth for a tooth
And anyway I told the truth
And I'm not afraid to die

NICK CAVE

To Marjon, Mike, and Nick

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Preface

The most important feature of this book is the word “applied” in the title. This implies that the emphasis of this book lies more on the application of statistical techniques for longitudinal data analysis and not so much on the mathematical background. In most other books on the topic of longitudinal data analysis, the mathematical background is the major issue, which may not be surprising since (nearly) all the books on this topic have been written by statisticians. Although statisticians fully understand the difficult mathematical material underlying longitudinal data analysis, they often have difficulty in explaining this complex material in a way that is understandable for the researchers who have to use the technique or interpret the results. Therefore, this book is not written by a statistician, but by an epidemiologist. In fact, an epidemiologist is not primarily interested in the basic (difficult) mathematical background of the statistical methods, but in finding the answer to a specific research question; the epidemiologist wants to know how to apply a statistical technique and how to interpret the results. Owing to their different basic interests and different level of thinking, communication problems between statisticians and epidemiologists are quite common. This, in addition to the growing interest in longitudinal studies, initiated the writing of this book: a book on longitudinal data analysis, which is especially suitable for the “non-statistical” researcher (e.g. the epidemiologist). The aim of this book is to provide a practical guide on how to handle epidemiological data from longitudinal studies. The purpose of this book is to build a bridge over the communication gap that exists between statisticians and epidemiologists when addressing the complicated topic of longitudinal data analysis.

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I am very grateful to all my colleagues and students who came to me with (mostly) practical questions on longitudinal data analysis. This book is based on all those questions. Furthermore, I would like to thank Dick Bezemer, Maarten Boers, Bernard Uitdehaag, Wieke de Vente, Michiel de Boer, and Martijn Heymans who critically read preliminary drafts of some chapters and provided very helpful comments.

Introduction

1.1 Introduction

Longitudinal studies are defined as studies in which the outcome variable is repeatedly measured; i.e. the outcome variable is measured in the same subject on several occasions. In longitudinal studies the observations of one subject over time are not independent of each other, and therefore it is necessary to apply special statistical techniques, which take into account the fact that the repeated observations of each subject are correlated. The definition of longitudinal studies (used in this book) implicates that statistical techniques like survival analyses are beyond the scope of this book. Those techniques basically are not longitudinal data analysing techniques because (in general) the outcome variable is an irreversible endpoint and therefore strictly speaking is only measured at one occasion. After the occurrence of an event no more observations are carried out on that particular subject.

Why are longitudinal studies so popular these days? One of the reasons for this popularity is that there is a general belief that with longitudinal studies the problem of causality can be solved. This is, however, a typical misunderstanding and is only partly true. [Table 1.1](#) shows the most important criteria for causality, which can be found in every epidemiological textbook (e.g. Rothman and Greenland, 1998). Only one of them is specific for a longitudinal study: the rule of temporality. There has to be a time-lag between outcome variable Y (effect) and covariate X (cause); in time the cause has to precede the effect. The question of whether or not causality exists can only be (partly) answered in specific longitudinal studies (i.e. experimental studies) and certainly not in all longitudinal studies. What then is the advantage of performing a longitudinal study? A longitudinal study is expensive, time consuming, and difficult to analyze. If there are no advantages over cross-sectional studies why bother? The main advantage of a longitudinal study compared to a cross-sectional study is that the individual development of a certain outcome variable over time can be studied. In addition to this, the individual development

Table 1.1 Criteria for causality

Strength of the relationship
Consistency in different populations and under different circumstances
Specificity (cause leads to a single effect)
Temporality (cause precedes effect in time)
Biological gradient (dose–response relationship)
Biological plausibility
Experimental evidence

of a certain outcome variable can be related to the individual development of other variables.

1.2 General approach

The general approach to explain the statistical techniques covered in this book will be “the research question as basis for analysis.” Although it may seem quite obvious, it is important to realize that a statistical analysis has to be carried out in order to obtain an answer to a particular research question. The starting point of each chapter in this book will be a research question, and throughout the book many research questions will be addressed. The book is further divided into chapters regarding the characteristics of the outcome variable. Each chapter contains extensive examples, accompanied by computer output, in which special attention will be paid to interpretation of the results of the statistical analyses.

1.3 Prior knowledge

Although an attempt has been made to keep the complicated statistical techniques as understandable as possible, and although the basis of the explanations will be the underlying epidemiological research question, it will be assumed that the reader has some prior knowledge about (simple) cross-sectional statistical techniques such as linear regression analysis, logistic regression analysis, and analysis of variance.

1.4 Example

In general, the examples used throughout this book will use the same longitudinal dataset. This dataset consists of an outcome variable (Y) that is continuous and is measured six times on the same subjects. Furthermore there are four covariates, which differ in distribution (continuous or dichotomous) and in whether they are time-dependent or time-independent. X_1 is a continuous time-independent

Table 1.2 Descriptive information^a for an outcome variable Y and covariates X_1 to X_4 ^b measured at six occasions

Time-point	Y	X_1	X_2	X_3	X_4
1	4.43 (0.67)	1.98 (0.22)	3.26 (1.24)	143/4	69/78
2	4.32 (0.67)	1.98 (0.22)	3.36 (1.34)	136/11	69/78
3	4.27 (0.71)	1.98 (0.22)	3.57 (1.46)	124/23	69/78
4	4.17 (0.70)	1.98 (0.22)	3.76 (1.50)	119/28	69/78
5	4.67 (0.78)	1.98 (0.22)	4.35 (1.68)	99/48	69/78
6	5.12 (0.92)	1.98 (0.22)	4.16 (1.61)	107/40	69/78

^aFor outcome variable Y and the continuous covariates (X_1 and X_2) mean and standard deviation are given, for the dichotomous covariates (X_3 and X_4) the numbers of subjects in the different categories are given.

^b Y is serum cholesterol in mmol/l; X_1 is maximal oxygen uptake (in (dl/min)/kg^{2/3}); X_2 is the sum of four skinfolds (in cm); X_3 is smoking (non-smokers vs. smokers); X_4 is gender (males vs. females).

covariate, X_2 is a continuous time-dependent covariate, X_3 is a dichotomous time-dependent covariate, and X_4 is a dichotomous time-independent covariate. All time-dependent independent variables are measured at the same six occasions as the outcome variable Y .

In the chapter dealing with dichotomous outcome variables (i.e. [Chapter 7](#)), the continuous outcome variable Y is dichotomized (i.e. the highest tertile versus the other two tertiles) and in the chapter dealing with categorical outcome variables (i.e. [Chapter 8](#)), the continuous outcome variable Y is divided into three equal groups (i.e. tertiles).

The dataset used in the examples is taken from the Amsterdam Growth and Health Longitudinal Study, an observational longitudinal study investigating the longitudinal relationship between lifestyle and health in adolescence and young adulthood (Kemper, 1995). The abstract notation of the different variables (Y , X_1 to X_4) is used since it is basically unimportant what these variables actually are. The continuous outcome variable Y could be anything, a certain psychosocial variable (e.g. a score on a depression questionnaire, an indicator of quality of life, etc.) or a biological parameter (e.g. blood pressure, albumin concentration in blood, etc.). In this particular dataset the outcome variable Y was total serum cholesterol expressed in mmol/l, X_1 was fitness level at baseline (measured as maximal oxygen uptake on a treadmill), X_2 was body fatness (estimated by the sum of the thickness of four skinfolds), X_3 was smoking behavior (dichotomized as smoking versus non-smoking), and X_4 was gender. [Table 1.2](#) shows descriptive information for the variables used in the example.

All the example datasets used throughout the book are available from the following website: <http://www.jostwisk.nl>.

1.5 Software

The relatively simple analyses of the example dataset were performed with SPSS (version 18; SPSS, 1997, 1998). For sophisticated longitudinal data analysis, other software packages were used. Generalized estimating equations (GEE) analysis and mixed model analysis were performed with Stata (version 11; Stata, 2001). Stata is chosen as the main software package for sophisticated longitudinal analysis, because of the simplicity of its output. In Chapter 12, an overview (and comparison) will be given of other software packages such as SAS (version 8; Littell et al., 1991, 1996), R (version 2.13), and MLwiN (version 2.25; Goldstein et al., 1998; Rasbash et al., 1999). In all these packages algorithms to perform sophisticated longitudinal data analysis are implemented in the main software. Both syntax and output will accompany the overview of the different packages. For detailed information about the different software packages, reference is made to the software manuals.

1.6 Data structure

It is important to realize that different statistical software packages need different data structures in order to perform longitudinal analyses. In this respect a distinction must be made between a “long” data structure and a “broad” data structure. In the “long” data structure each subject has as many data records as there are measurements over time, while in a “broad” data structure each subject has one data record, irrespective of the number of measurements over time (Figure 1.1).

1.7 Statistical notation

The statistical notation will be very simple and straightforward. Difficult matrix notation will be avoided as much as possible. Throughout the book the number of subjects will be denoted as $i = 1$ to N , the number of times a particular subject is measured will be denoted as $t = 1$ to T , and the number of covariates will be noted as $j = 1$ to J . Furthermore, the outcome variable will be called Y , and the covariates will be called X . All other notations will be explained below the equations where they are used.

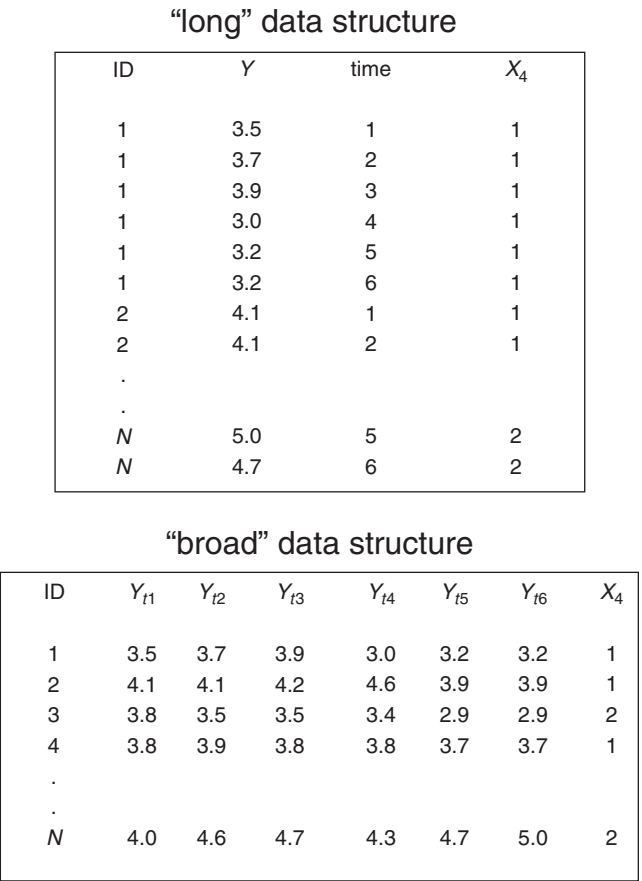


Figure 1.1 Illustration of two different data structures.

1.8 What’s new in the second edition?

Throughout the book changes are made to make some of the explanations clearer, and several chapters are totally rewritten. This holds for [Chapter 9](#) (Analysis of experimental studies) and [Chapter 10](#) (Missing data in longitudinal studies). Furthermore, two new chapters are added to the book: in [Chapter 5](#), the role of the time variable in longitudinal data analysis will be discussed, while in [Chapter 13](#) some new features of longitudinal data analysis will be briefly introduced.

Study design

2.1 Introduction

Epidemiological studies can be roughly divided into observational and experimental studies ([Figure 2.1](#)). Observational studies can be further divided into case-control studies and cohort studies. Case-control studies are never longitudinal, in the way that longitudinal studies were defined in [Chapter 1](#). The outcome variable Y (a dichotomous outcome variable distinguishing “case” from “control”) is measured only once. Furthermore, case-control studies are always retrospective in design. The outcome variable Y is observed at a certain time-point, and the covariates are measured retrospectively.

In general, observational cohort studies can be divided into prospective, retrospective, and cross-sectional cohort studies. A prospective cohort study is the only cohort study that can be characterized as a longitudinal study. Prospective cohort studies are usually designed to analyze the longitudinal development of a certain characteristic over time. It is argued that this longitudinal development concerns growth processes. However, in studies investigating the elderly, the process of deterioration is the focus of the study, whereas in other developmental processes growth and deterioration can alternately follow each other. Moreover, in many epidemiological studies one is interested not only in the actual growth or deterioration over time, but also in the longitudinal relationship between several characteristics over time. Another important aspect of epidemiological observational prospective studies is that sometimes one is not really interested in growth or deterioration, but rather in the “stability” of a certain characteristic over time. In epidemiology this phenomenon is known as tracking ([Twisk et al., 1994, 1997, 1998a, 1998b, 2000](#)).

Experimental studies, which in epidemiology are often referred to as (clinical) trials, are by definition prospective, i.e. longitudinal. The outcome variable Y is measured at least twice (the classical “pre-test,” “post-test” design), and other intermediate measures are usually also added to the research design (e.g. to evaluate short-term and long-term effects). The aim of an experimental (longitudinal)

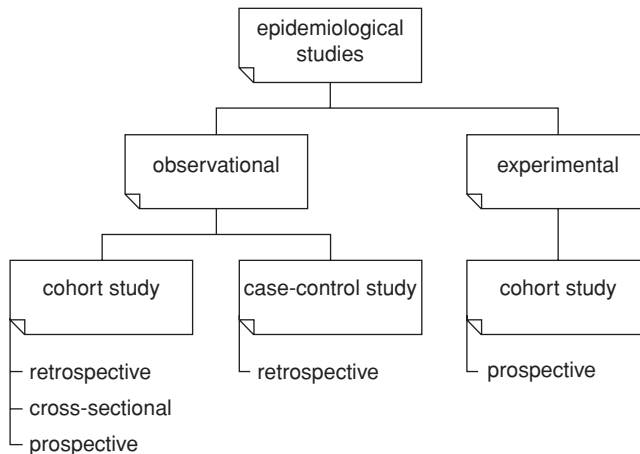


Figure 2.1 Schematic illustration of different epidemiological study designs.

study is to analyze the effect of one or more interventions on a certain outcome variable Y .

In [Chapter 1](#), it was mentioned that some misunderstanding exists with regard to causality in longitudinal studies. However, an experimental study is basically the only epidemiological study design in which the issue of causality can be covered. With observational longitudinal studies, on the other hand, the question of probable causality remains unanswered.

Most of the statistical techniques in the examples covered in this book will be illustrated with data from an observational longitudinal study. In a separate chapter ([Chapter 9](#)), examples from experimental longitudinal studies will be discussed extensively. Although the distinction between experimental and observational longitudinal studies is obvious, in most situations the statistical techniques discussed for observational longitudinal studies are also suitable for experimental longitudinal studies.

2.2 Observational longitudinal studies

In observational longitudinal studies investigating individual development, each measurement taken on a subject at a particular time-point is influenced by three factors: (1) age (time from date of birth to date of measurement); (2) period (time or moment at which the measurement is taken); and (3) birth cohort (group of subjects born in the same year). When studying individual development, one is mainly interested in the age effect. One of the problems of most of the designs used in studies of development is that the main age effect cannot be distinguished from the two other “confounding” effects (i.e. period and cohort effects).

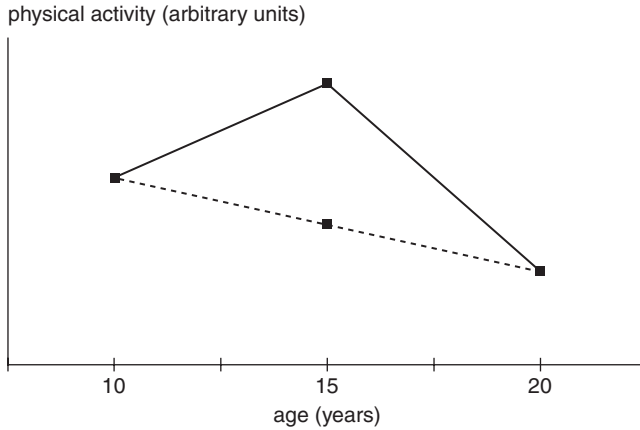


Figure 2.2 Illustration of a possible time of measurement effect (– – – “real” age trend, — observed age trend).

2.2.1 Period and cohort effects

There is an extensive amount of literature describing age, period and cohort effects (e.g. Lebowitz, 1996; Robertson et al., 1999; Holford et al., 2005). However, most of the literature deals with classical age–period–cohort models, which are used to describe and analyze trends in (disease-specific) morbidity and mortality (e.g. Kupper et al., 1985; Mayer and Huinink, 1990; Holford, 1992; McNally et al., 1997; Robertson and Boyle, 1998; Rosenberg and Anderson, 2010). In this book, the main interests are the individual development over time, and the longitudinal relationship between different variables. In this respect, period effects or time of measurement effects are often related to a change in measurement method over time, or to specific environmental conditions at a particular time of measurement. A hypothetical example is given in Figure 2.2. This figure shows the longitudinal development of physical activity with age. Physical activity patterns were measured with a five-year interval, and were measured during the summer in order to minimize seasonal influences. The first measurement was taken during a summer with normal weather conditions. During the summer when the second measurement was taken, the weather conditions were extremely good, resulting in activity levels that were very high. At the time of the third measurement the weather conditions were comparable to the weather conditions at the first measurement, and therefore the physical activity levels were much lower than those recorded at the second measurement. When all the results are presented in a graph, it is obvious that the observed age trend is highly biased by the “period” effect at the second measurement.



Figure 2.3 Illustration of a possible cohort effect (– – – cohort specific, — observed).

One of the most striking examples of a cohort effect is the development of body height with age. There is an increase in body height with age, but this increase is highly influenced by the increase in height of the birth cohort. This phenomenon is illustrated in Figure 2.3. In this hypothetical study, two repeated measurements were carried out in two different cohorts. The purpose of the study was to detect the age trend in body height. The first cohort had an initial age of 5 years; the second cohort had an initial age of 10 years. At the age of 5, only the first cohort was measured, at the age of 10, both cohorts were measured, and at the age of 15 only the second cohort was measured. The body height obtained at the age of 10 is the average value of the two cohorts. Combining all measurements in order to detect an age trend will lead to a much flatter age trend than the age trends observed in both cohorts separately.

Both cohort and period effects can have a dramatic influence on interpretation of the results of longitudinal studies. An additional problem is that it is very difficult to disentangle the two types of effects. They can easily occur together. Logical considerations regarding the type of variable of interest can give some insight into the plausibility of either a cohort or a period effect. When there are (confounding) cohort or period effects in a longitudinal study, one should be very careful with the interpretation of age-related results.

It is sometimes argued that the design that is most suitable for studying individual growth/deterioration processes is a so-called “multiple longitudinal design.” In such a design the repeated measurements are taken in more than one cohort with overlapping ages (Figure 2.4). With a “multiple longitudinal design” the main age effect can be distinguished from cohort and period effects. Because subjects of the same age are measured at different time-points, the difference in outcome variable

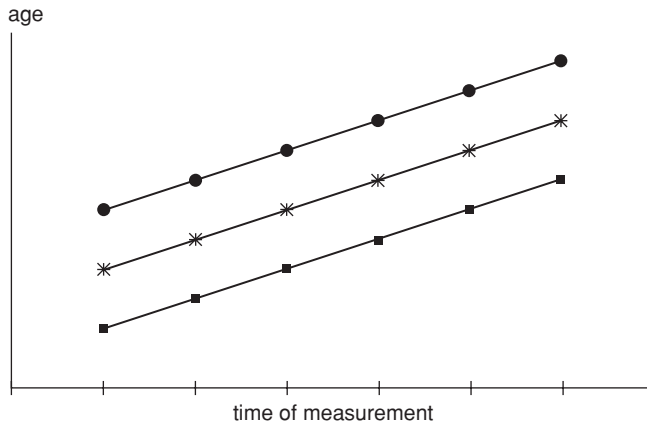


Figure 2.4 Principle of a multiple longitudinal design; repeated measurements of different cohorts with overlapping ages (■ cohort 1, * cohort 2, ● cohort 3).

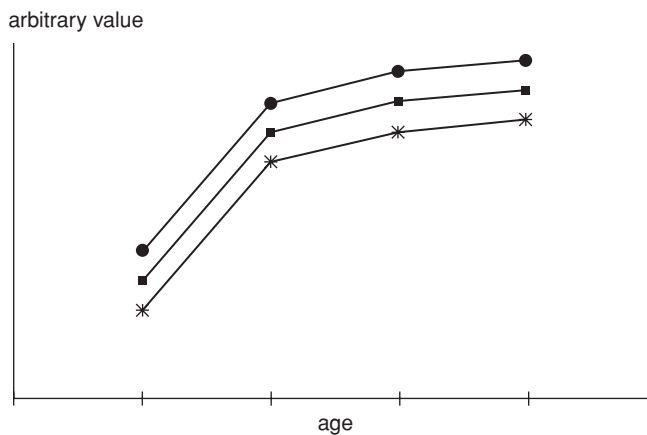


Figure 2.5 Possibility of detecting cohort effects in a "multiple longitudinal design" (* cohort 1, ■ cohort 2, ● cohort 3).

Y between subjects of the same age, but measured at different time-points, can be investigated in order to detect cohort effects. Figure 2.5 illustrates this possibility: different cohorts have different values at the same age.

Because the different cohorts are measured at the same time-points, it is also possible to detect possible time of measurement effects in a "multiple longitudinal design." Figure 2.6 illustrates this phenomenon. All three cohorts show an increase in the outcome variable at the second measurement, which indicates a possible time of measurement effect.

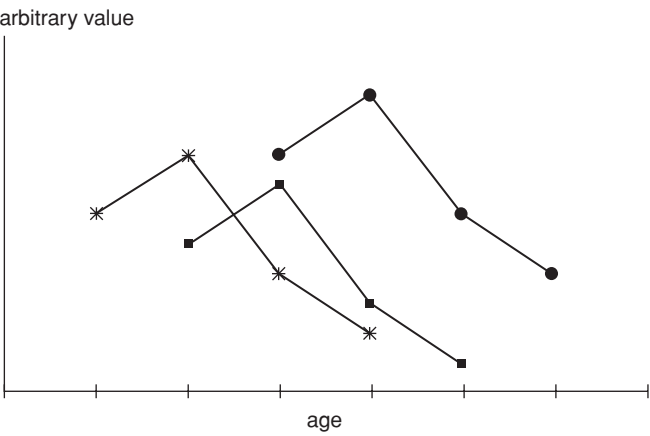


Figure 2.6 Possibility of detecting time of measurement effects in a “multiple longitudinal design” (* cohort 1, ■ cohort 2, ● cohort 3).

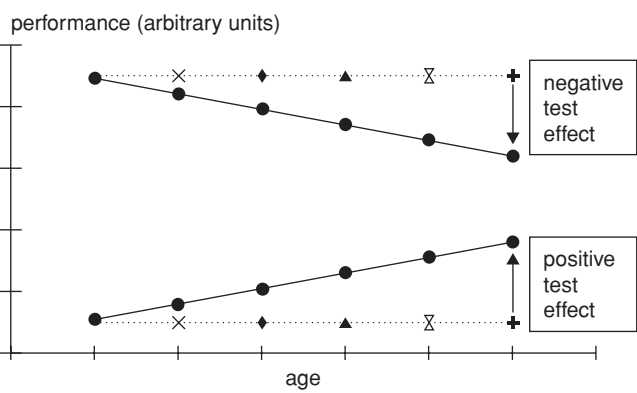


Figure 2.7 Test or learning effects; comparison of repeated measurements of the same subjects with non-repeated measurements in comparable subjects (different symbols indicate different subjects, cross-sectional, — longitudinal).

2.2.2 Other confounding effects

In studies investigating development, in which repeated measurements of the same subjects are performed, cohort and period effects are not the only possible confounding effects. The individual measurements can also be influenced by a changing attitude towards the measurement itself, a so-called test or learning effect. This test or learning effect, which is illustrated in [Figure 2.7](#), can be either positive or negative.

One of the most striking examples of a positive test effect is the measurement of memory in older subjects. It is assumed that with increasing age, memory decreases.

Table 2.1 The IPCs for outcome variable Y

	Y_{t1}	Y_{t2}	Y_{t3}	Y_{t4}	Y_{t5}	Y_{t6}
Y_{t1}	–	0.76	0.70	0.67	0.64	0.59
Y_{t2}		–	0.77	0.78	0.67	0.59
Y_{t3}			–	0.85	0.71	0.63
Y_{t4}				–	0.74	0.65
Y_{t5}					–	0.69
Y_{t6}						–

However, even when the time interval between subsequent measurements is as long as three years, an increase in memory performance with increasing age can be observed: an increase which is totally due to a learning effect (Dik et al., 2001).

Furthermore, missing data or drop-outs during follow-up can have important implications for the interpretation of the results of longitudinal data analysis. This important issue will be discussed in detail in [Chapter 10](#).

Analysis based on repeated measurements of the same subject can also be biased by a low degree of reproducibility of the measurement itself. This is quite important because the changes over time within one subject can be “overruled” by a low reproducibility of the measurements. An indication of reproducibility can be provided by analysing the inter-period correlation coefficients (IPCs) (van ‘t Hof and Kowalski, 1979). It is assumed that the IPCs can be approximated by a linear function of the time interval. The IPC will decrease as the time interval between the two measurements under consideration increases. The intercept of the linear regression line between the IPC and the time interval can be interpreted as the instantaneous measurement–remeasurement reproducibility (i.e. the correlation coefficient with a time interval of zero). Unfortunately, there are a few shortcomings in this approach. For instance, a linear relationship between the IPC and the time interval is assumed, and it is questionable whether that is the case in every situation. When the number of repeated measurements is low, the regression line between the IPC and the time interval is based on only a few data points, which makes the estimation of this line rather unreliable. Furthermore, there are no objective rules for the interpretation of this reproducibility coefficient. However, it must be taken into account that low reproducibility of measurements can seriously influence the results of longitudinal analysis.

2.2.3 Example

[Table 2.1](#) shows the IPCs for outcome variable Y in the example dataset. To obtain a value for the measurement–remeasurement reproducibility, a linear regression

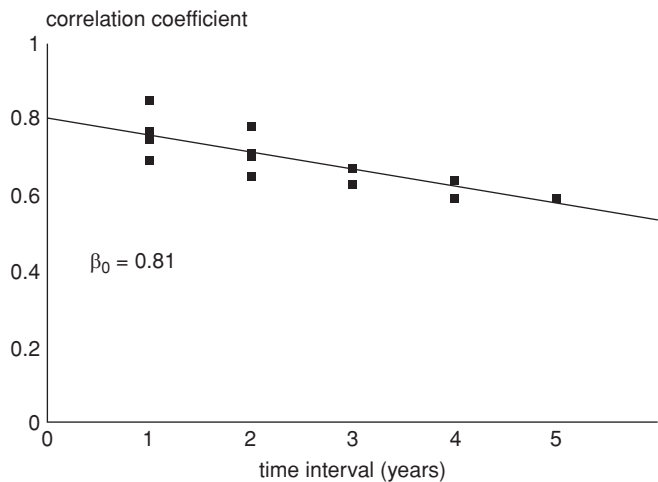


Figure 2.8 Linear regression line between the inter-period correlation coefficients and the length of the time interval.

analysis between the length of the time interval and the IPCs was carried out. The value of the intercept of that particular regression line can be seen as the IPC for a time interval with a length of zero, and can therefore be interpreted as a reproducibility coefficient (Figure 2.8).

The result of the regression analysis shows an intercept of 0.81, i.e. the reproducibility coefficient of outcome variable *Y* is 0.81. It has already been mentioned that it is difficult to provide an objective interpretation of this coefficient. Another important issue is that the interpretation of the coefficient highly depends on the explained variance (R^2) of the regression line (which is 0.67 in this example). In general, the lower the explained variance of the regression line, the more variation in IPCs with the same time interval, and the less reliable the estimation of the reproducibility coefficient.

2.3 Experimental (longitudinal) studies

Experimental (longitudinal) studies are by definition prospective cohort studies and in a classical experimental longitudinal study the experimental group is compared to a control group. A distinction can be made between randomized and non-randomized experimental studies. In randomized experimental studies the subjects are randomly assigned to either the experimental group or the control group. The main reason for this randomization is to make the groups to be compared as equal as possible at the start of the intervention.

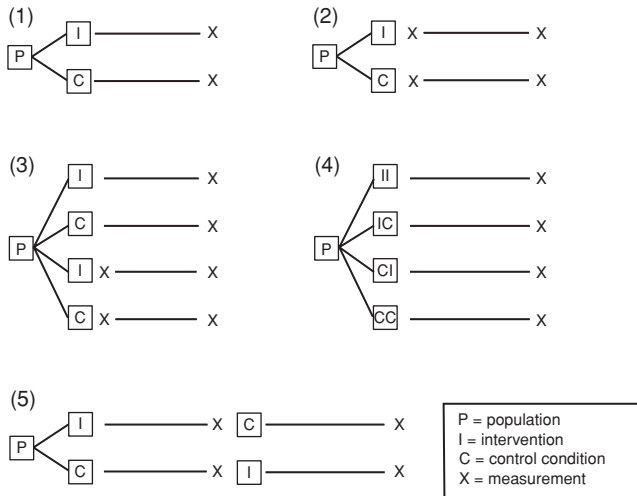


Figure 2.9 An illustration of a few experimental longitudinal designs: (1) “classic” experimental design; (2) “classic” experimental design with baseline measurement; (3) “Solomon four group” design; (4) factorial design; and (5) “cross-over” design.

It is not the purpose of this book to give a detailed description of all possible experimental designs. [Figure 2.9](#) summarizes a few commonly used experimental designs. For an extensive overview of this topic, reference is made to other books (e.g. Pocock, 1983; Judd et al., 1991; Rothman and Greenland, 1998).

In epidemiology a randomized experimental study is often referred to as a randomized controlled trial (RCT). In an RCT, the population under study is randomly divided into an intervention group and a non-intervention group which is referred to as the control group (e.g. a placebo group or a group with “usual” care). The groups are then measured after a certain period of time to investigate the differences between the groups in the outcome variable. Usually, however, a baseline measurement is performed before the start of the intervention. The so-called “Solomon four group” design is a combination of the design with and without a baseline measurement. The idea behind a “Solomon four group” design is that when a baseline measurement is performed there is a possibility of test or learning effects, and with a “Solomon four group” design these test or learning effects can be detected. In a factorial design, two or more interventions are combined into one experimental study.

In the experimental designs discussed before, the subjects are randomly assigned to two or more groups. In studies of this type, basically all subjects have missing data for all other conditions, except the intervention to which they have been assigned. In contrast, it is also possible that all of the subjects are assigned to all possible

interventions, but that the sequence of the different interventions is randomly assigned to the subjects. Experimental studies of this type are known as “cross-over trials.” They are very efficient and very powerful, but they can only be performed for short-lasting outcome measures.

Basically, all the “confounding” effects described for observational longitudinal studies (Section 2.2) can also occur in experimental studies. In particular, missing data or drop-outs are a major problem in experimental studies (see [Chapter 10](#)). Test or learning effects can be present, but cohort and time of measurement effects are less likely to occur.

It has already been mentioned that for the analysis of data from experimental studies, all techniques that will be discussed in the following chapters, with examples from an observational longitudinal study that can also be used. However, [Chapter 9](#), especially, will provide useful information regarding the analysis of data from experimental studies.

Continuous outcome variables

3.1 Two measurements

The simplest form of longitudinal study is that in which a continuous outcome variable Y is measured twice in time (Figure 3.1). With this simple longitudinal design the following question can be answered: “Does the outcome variable Y change over time?” Or, in other words: “Is there a difference in the outcome variable Y between $t = 1$ and $t = 2$?”

To obtain an answer to this question, a paired t -test can be used. Consider the hypothetical dataset presented in Table 3.1. The paired t -test is used to test the hypothesis that the mean difference between Y_{t1} and Y_{t2} equals zero. Because the individual differences are used in this statistical test, it takes into account the fact that the observations within one individual are dependent on each other. The test statistic of the paired t -test is the average of the differences divided by the standard deviation of the differences divided by the square root of the number of subjects (Equation 3.1).

$$t = \frac{\bar{d}}{\left(\frac{s_d}{\sqrt{N}} \right)} \quad (3.1)$$

where t is the test statistic, $\bar{d}$ is the average of the differences, s_d is the standard deviation of the differences, and N is the number of subjects.

This test statistic follows a t -distribution with $(N - 1)$ degrees of freedom. The assumptions for using the paired t -test are twofold, namely (1) that the observations of different subjects are independent and (2) that the differences between the two measurements are approximately normally distributed. In research situations in which the number of subjects is quite large (say above 25), the paired t -test can be used without any problems. With smaller datasets, however, the assumption of normality becomes important. When the assumption is violated, the non-parametric

Table 3.1 Hypothetical dataset for a longitudinal study with two measurements

i	Y_{t1}	Y_{t2}	Difference (d)
1	3.5	3.7	−0.2
2	4.1	4.1	0.0
3	3.8	3.5	0.3
4	3.8	3.9	−0.1
N	4.0	4.6	−0.6

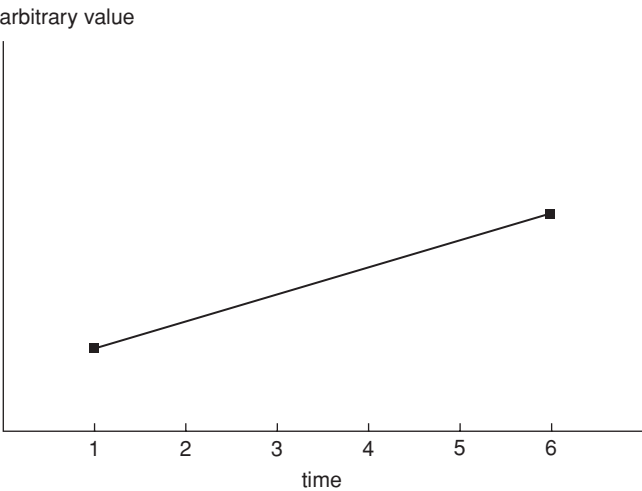


Figure 3.1 Longitudinal study with two measurements.

equivalent of the paired t -test can be used (see Section 3.2). In contrast to its non-parametric equivalent, the paired t -test is not only a testing procedure. With this statistical technique the average of the paired differences with the corresponding 95% confidence interval can also be estimated.

It should be noted that when the differences are not normally distributed and the sample size is rather large, the paired t -test provides valid results, but interpretation of the average differences can be complicated, because the average is not a good indicator of the mid-point of the distribution.

3.1.1 Example

One of the limitations of the paired t -test is that the technique is only suitable for two measurements over time. It has already been mentioned that the example dataset used throughout this book consists of six repeated measurements. To illustrate the

Output 3.1 Results of a paired *t*-test performed on the example dataset

Paired Samples Statistics							
	Mean	N	Std. Deviation	Std. Error	Mean		
continuous outcome variable Y at t=1	4.435	147	0.6737	0.0556			
continuous outcome variable Y at t=6	5.1216	147	0.92353	0.07617			

Paired Samples Test							
Paired Differences							
95% Confidence							
	Std.	Std.	Interval of the				Sig.
Mean	Deviation	Error	Difference		T	df	(2-
		Mean	Lower	Upper			tailed)
-0.68687	0.75977	0.06266	-0.81072	-0.56302	-10.961	146	0.000

paired *t*-test in the example dataset, only the first and last measurements of this dataset are used. The question to be answered is: “Is there a difference in outcome variable *Y* between $t = 1$ and $t = 6$?” [Output 3.1](#) shows the results of the paired *t*-test.

The first lines of the output give descriptive information (i.e. mean values, standard deviation (SD), etc.), which is not really important in the light of the postulated question. The second part of the output provides the more important information. First of all, the mean of the paired differences is given (i.e. -0.68687), and also the 95% confidence interval around this mean (-0.81072 to -0.56302). A negative value indicates that there is an increase in outcome variable *Y* between $t = 1$ and $t = 6$. Furthermore, the results of the actual paired *t*-test are given: the value of the test statistic ($t = -10.961$), with $(N - 1)$ degrees of freedom (146), and the corresponding *p*-value (0.000). The results indicate that the increase in outcome variable *Y* is statistically significant ($p < 0.001$). The fact that the increase over time is statistically significant was already clear in the 95% confidence interval of the mean difference, which did not include zero.

3.2 Non-parametric equivalent of the paired *t*-test

When the assumptions of the paired *t*-test are violated, it is possible to perform the non-parametric equivalent of the paired *t*-test, the (Wilcoxon) signed rank sum

Table 3.2 Hypothetical dataset for a longitudinal study with two measurements

<i>i</i>	<i>Y</i> _{<i>t</i>1}	<i>Y</i> _{<i>t</i>2}	Difference (<i>d</i>)	Rank number
1	3.5	3.7	−0.2	3
2	4.1	4.0	0.1	1.5 ^a
3	3.8	3.5	0.3	4
4	3.8	3.9	−0.1	1.5 ^a
5	4.0	4.4	−0.4	5
6	4.1	4.9	−0.8	7
7	4.0	3.4	0.6	6
8	5.1	6.8	−1.7	9
9	3.7	6.3	−2.6	10
10	4.1	5.2	−1.1	8

^aThe average rank is used for tied values.

test. This signed rank sum test is based on the ranking of the individual difference scores, and does not make any assumptions about the distribution of the outcome variable. Consider the hypothetical dataset presented in Table 3.2. The dataset consists of 10 subjects, who were measured on two occasions.

The signed rank sum test evaluates whether the sum of the rank numbers with a positive difference is equal to the sum of the rank numbers with a negative difference. When those two are equal, it suggests that there is no change over time. In the hypothetical dataset the sum of the rank numbers with a positive difference is 11.5 (i.e. 1.5 + 4 + 6), while the sum of the rank numbers with a negative difference is 43.5. The exact calculation of the level of significance is (very) complicated, and goes beyond the scope of this book. All statistical handbooks contain tables in which the level of significance can be found (see for instance Altman, 1991), and with all statistical software packages the levels of significance can be calculated. For the hypothetical example, the *p*-value is between 0.2 and 0.1, indicating no significant change over time.

The (Wilcoxon) signed rank sum test can be used in all longitudinal studies with two measurements. It is a testing technique which only provides *p*-values, without effect estimation. In “real life” situations, it will only be used when the sample size is very small (i.e. less than 25).

3.2.1 Example

Although the sample size in the example dataset is large enough to perform a paired *t*-test, in order to illustrate the technique the (Wilcoxon) signed rank sum test will

be used to test whether or not the difference between Y at $t = 1$ and Y at $t = 6$ is significant. [Output 3.2](#) shows the results of this analysis.

Output 3.2 Output of the (Wilcoxon) matched pairs signed rank sum test

Wilcoxon Matched-pairs Signed-ranks Test				
YT1	OUTCOME VARIABLE Y AT T1			
with YT6	OUTCOME VARIABLE Y AT T6			
Mean Rank	Cases			
34.84	29	- Ranks	(YT6 Lt YT1)	
83.62	118	+ Ranks	(YT6 Gt YT1)	
	0	Ties	(YT6 Eq YT1)	
	147	Total		
Z = -8.5637	2-tailed P = 0.0000			

The first part of the output provides the mean rank of the rank numbers with a negative difference and the mean rank of the rank numbers with a positive difference. It also gives the number of cases with a negative and a positive difference. A negative difference corresponds with the situation that Y at $t = 6$ is less than Y at $t = 1$. This corresponds with a decrease in outcome variable Y over time. A positive difference corresponds with the situation that Y at $t = 6$ is greater than Y at $t = 1$, i.e. corresponds with an increase in Y over time. The last line of the output shows the z -value. Although the (Wilcoxon) signed rank sum test is a non-parametric equivalent of the paired t -test, in many software packages a normal approximation is used to calculate the p -value. This z -value corresponds with a highly significant p -value (0.0000), which indicates that there is a significant change (increase) over time in outcome variable Y . Because there is a highly significant change over time, the p -value obtained from the paired t -test is the same as the p -value obtained from the signed rank sum test. In general, however, the non-parametric tests are less powerful than the parametric equivalents and will therefore give slightly higher p -values.

3.3 More than two measurements

In a longitudinal study with more than two measurements performed on the same subjects ([Figure 3.2](#)), the situation becomes somewhat more complex. A design

Table 3.3 Hypothetical dataset for a longitudinal study with more than two measurements

i	Y_{t1}	Y_{t2}	d_1	Y_{t3}	d_2	Y_{t6}	d_5
1	3.5	3.7	−0.2	3.9	−0.2	3.0	0.2
2	4.1	4.1	0.0	4.2	−0.1	4.6	0.0
3	3.8	3.5	0.3	3.5	0.0	3.4	−0.4
4	3.8	3.9	−0.1	3.8	0.1	3.8	0.3
N	4.0	4.6	−0.6	4.7	−0.1	4.3	0.1

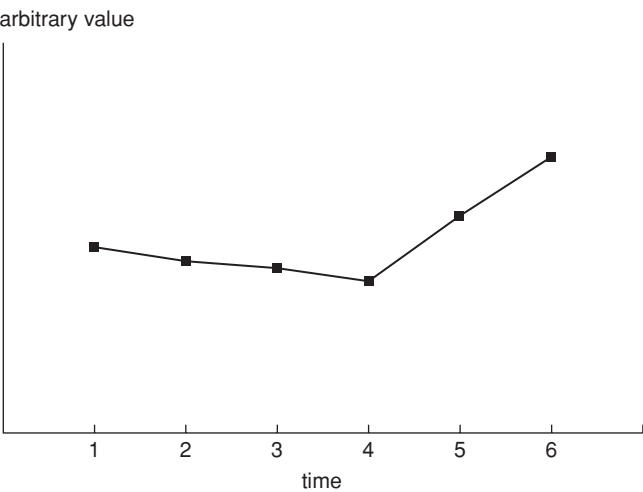


Figure 3.2 Longitudinal study with six measurements.

with only one outcome variable, which is measured several times on the same subjects, is known as a “one-within” design. This refers to the fact that there is only one factor of interest (i.e. time) and that this factor varies only within subjects. In a situation with more than two repeated measurements, a paired t -test cannot be carried out. Consider the hypothetical dataset, which is presented in [Table 3.3](#).

The question: “Does the outcome variable Y change over time?” can be answered with multivariate analysis of variance (MANOVA) for repeated measurements. The basic idea behind this statistical technique, which is also known as “generalized linear model (GLM) for repeated measures” is the same as for the paired t -test. The statistical test is carried out for the $T - 1$ differences between subsequent

measurements. In fact, MANOVA for repeated measurements is a multivariate analysis of these $T - 1$ differences between subsequent time-points. Multivariate refers to the fact that $T - 1$ differences are used simultaneously as outcome variable. The $T - 1$ differences and corresponding variances and covariances form the test statistic for the MANOVA for repeated measurements (Equation 3.2).

$$F = \left(\frac{N - T + 1}{(N - 1)(T - 1)} \right) H^2 \quad (3.2a)$$

$$H^2 = \frac{N y_d^t y_d}{S_d^2} \quad (3.2b)$$

where F is the test statistic, N is the number of subjects, T is the number of repeated measurements, y_d^t is the row vector of differences between subsequent measurements, y_d is the column vector of differences between subsequent measurements, and S_d^2 is the variance/covariance matrix of the differences between subsequent measurements.

The F -statistic follows an F -distribution with $(T - 1)$, $(N - T + 1)$ degrees of freedom. For a detailed description of how to calculate H^2 using Equation 3.2b, reference should be made to other textbooks (Crowder and Hand, 1990; Hand and Crowder, 1996; Stevens, 1996).¹ As with all statistical techniques, MANOVA for repeated measurements is based on several assumptions. These assumptions are more or less comparable with the assumptions of a paired t -test: (1) observations of different subjects at each of the repeated measurements need to be independent; and (2) the observations need to be multivariate normally distributed, which is comparable but slightly more restrictive than the requirement that the differences between subsequent measurements be normally distributed. The calculation procedure described above is called the “multivariate” approach because several differences are analyzed together. However, to answer the same research question, a “univariate” approach can also be followed. This “univariate” approach is comparable to the procedures carried out in simple analysis of variance (ANOVA) and is based on the “sum of squares,” i.e. squared differences between observed values and average values. The “univariate” approach is only valid when, in addition to the earlier mentioned assumptions, another assumption is met: the assumption of “sphericity.” This assumption is also known as the “compound symmetry” assumption. It implies, firstly, that all correlations in outcome variable Y between repeated measurements are equal, irrespective of the time interval between the

¹ H^2 is also known as Hotelling’s T^2 , and is often referred to as T^2 . Because throughout this book T is used to denote the number of repeated measurements, H^2 is the preferred notation for this statistic.

Table 3.4 Hypothetical longitudinal dataset with four measurements in six subjects

i	Y_{t1}	Y_{t2}	Y_{t3}	Y_{t4}	Mean
1	31	29	15	26	25.25
2	24	28	20	32	26.00
3	14	20	28	30	23.00
4	38	34	30	34	34.00
5	25	29	25	29	27.00
6	30	28	16	34	27.00
Mean	27.00	28.00	22.33	30.83	27.00

measurements. Secondly it implies that the variances of outcome variable Y are the same at each of the repeated measurements.

Whether or not the assumption of sphericity is met can be expressed by the sphericity coefficient epsilon (noted as ε). In an ideal situation the sphericity coefficient will equal one, and when the assumption is not entirely met, the coefficient will be less than one. When the assumption is not met, the degrees of freedom of the F -test used in the “univariate” approach can be changed: instead of $(T - 1)$, $(N - 1)(T - 1)$, the degrees of freedom will be $\varepsilon(T - 1)$, $\varepsilon(N - 1)(T - 1)$. It should be noted that the degrees of freedom for the “univariate” approach are different from the degrees of freedom for the “multivariate” approach. In many software packages, when MANOVA for repeated measurements is carried out, the sphericity coefficient is automatically estimated and the degrees of freedom are automatically adapted. The sphericity coefficient can also be tested for significance (with the null hypotheses tested: sphericity coefficient $\varepsilon = 1$). However, one must be very careful with the use of this test. If the sample size is large, the test for sphericity will (almost) always give a significant result, whereas in a study with a small sample size the test for sphericity will (almost) never give a significant result. In the first situation, the test is over-powered, which means that even very small violations of the assumption of sphericity will be detected. In studies with small sample sizes, the test will be under-powered, i.e. the power to detect a violation of the assumption of sphericity is too low.

In the next section a numerical example will be given to explain the “univariate” approach within MANOVA for repeated measurements.

3.3.1 The “univariate” approach: a numerical example

Consider the simple longitudinal dataset presented in [Table 3.4](#).

When ignoring the fact that each subject is measured four times, the question of whether there is a difference between the various time-points can be answered by applying a simple ANOVA, considering the measurements at the four time-points as four independent groups. The ANOVA is then based on a comparison between the “between group” (in this case “between time”) sum of squares (SS_b) and the “within group” (i.e. “within time”) sum of squares (SS_w). The latter is also known as the “error” sum of squares. The sums of squares are calculated as follows:

$$SS_b = N \sum_{t=1}^T (\bar{y}_t - \bar{y})^2 \quad (3.3)$$

where N is the number of subjects, T is the number of repeated measurements, $\bar{y}_t$ is the average value of outcome variable Y at time-point t , and $\bar{y}$ is the overall average of outcome variable Y .

$$SS_w = \sum_{t=1}^T \sum_{n=1}^N (y_{it} - \bar{y}_t)^2 \quad (3.4)$$

where T is the number of repeated measurements, N is the number of subjects, y_{it} is the value of outcome variable Y for individual i at time-point t , and $\bar{y}_t$ is the average value of outcome variable Y at time-point t .

Applied to the dataset presented in [Table 3.4](#), $SS_b = 6[(27 - 27)^2 + (28 - 27)^2 + (22.33 - 27)^2 + (30.83 - 27)^2] = 224.79$, and $SS_w = (31 - 27)^2 + (24 - 27)^2 + \dots + (29 - 30.83)^2 + (34 - 30.83)^2 = 676.17$. These sums of squares are used in the ANOVA’s F -test. In this test it is not the total sums of squares that are used, but the mean squares. The mean square (MS) is defined as the total sum of squares divided by the degrees of freedom. For SS_b , the degrees of freedom are $(T - 1)$, and for SS_w , the degrees of freedom are $(T) \times (N - 1)$. In the numerical example, $MS_b = 224.79/3 = 74.93$ and $MS_w = 676.17/20 = 33.81$. The F -statistic is equal to MS_b/MS_w and follows an F -distribution with $((T - 1), (T(N - 1)))$ degrees of freedom. Applied to the example, the F -statistic is 2.216 with 3 and 20 degrees of freedom. The corresponding p -value (which can be found in a table of the F -distribution, available in all statistical textbooks) is 0.12, i.e. no significant difference between the four time-points. [Output 3.3](#) shows the results of the ANOVA, applied to this numerical example.

It has already been mentioned that in the above calculation the dependency of the observations was ignored. It was ignored that the same subject was measured four times. In a design with repeated measurements, the “individual” sum of squares

Output 3.3 Results of an ANOVA with a simple longitudinal dataset, ignoring the dependency of observations

Source	Sum of squares	df	Mean square	F	Sig
Between groups	224.792	3	74.931	2.216	0.118
Within groups	676.167	20	33.808		
Total	900.958	23			

(SS_i) can be calculated (Equation 3.5).

$$SS_i = T \sum_{i=1}^N (\bar{y}_i - \bar{y})^2 \quad (3.5)$$

where T is the number of repeated measurements, N is the number of subjects, $\bar{y}_i$ is the average value of outcome variable Y at all time-points for individual i , and $\bar{y}$ is the overall average of outcome variable Y .

Applied to the example dataset presented in Table 3.4, $SS_i = 4[(25.25 - 27)^2 + (26 - 27)^2 + \dots + (27 - 27)^2] = 276.21$. It can be seen that a certain proportion ($276.21/676.17$) of the error sum of squares (i.e. the within-time sum of squares) can be explained by individual differences. So, in this design with repeated measurements, the total error sum of squares of 676.17 is split into two components. The part which is due to individual differences (276.21) is now removed from the error sum of squares for the time effect. The latter is reduced to 399.96 (i.e. $676.17 - 276.21$). The SS_b is still the same, because this sum of squares reflects the differences between the four time-points. Output 3.4 shows the computer output of this example.

Output 3.4 Results of a MANOVA for repeated measurements with a simple longitudinal dataset

Within-subjects effects					
Source	Sum of squares	df	Mean square	F	Sig
TIME	224.792	3	74.931	2.810	0.075
Error (TIME)	399.958	15	26.664		
Between-subjects effects					
Source	Sum of squares	df	Mean square	F	Sig
Intercept	17550.042	1	17550.042	317.696	0.000
Error	276.208	5	55.242		

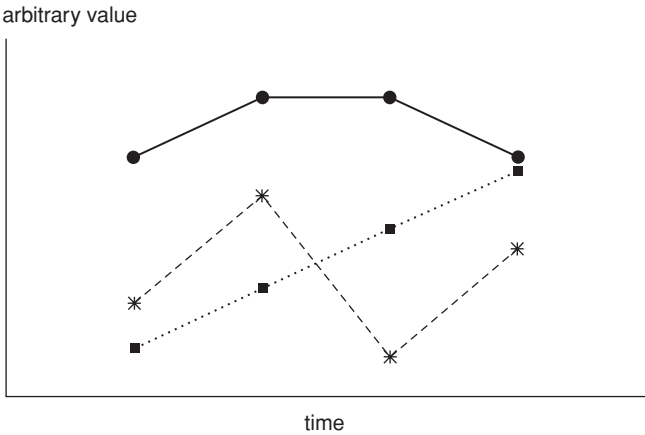


Figure 3.3 A few possible shapes of relationship between an outcome variable Y and time (■ linear, ● — quadratic, * - - - cubic).

As mentioned before for the ANOVA, to carry out the F -test, the total sum of squares is divided by the degrees of freedom to create the “mean square.” To obtain the appropriate F -statistic, the “mean square” of a certain effect is divided by the “mean square” of the error of that effect. The F -statistic is used in the testing procedure of that particular effect. As can be seen from [Output 3.4](#), the SS_b is divided by $(T - 1)$ degrees of freedom, while the corresponding error term is divided by $(T - 1) \times (N - 1)$ degrees of freedom. The p -value is 0.075, which indicates no significant change over time. Note, however, that this p -value is somewhat lower than the p -value obtained from the ANOVA, in which the dependency of the observations was ignored.

The intercept sum of squares, which is also provided in the output, is the sum of squares obtained when an overall average of zero is assumed. In this situation, the intercept sum of squares is useless, but it will be used in the analysis to investigate the shape of the relationship between the outcome variable Y and time.

3.3.2 The shape of the relationship between an outcome variable and time

In the foregoing sections of this chapter, the question of whether or not there is a change over time in outcome variable Y was answered. When such a change over time is found, this implies that there is some kind of relationship between the outcome variable Y and time. In this section the shape of the relationship between outcome variable Y and time will be investigated. In [Figure 3.3](#) a few possible shapes are illustrated.

It is obvious that this question is only of interest when there are more than two measurements. When there are only two measurements, the only possible

Table 3.5 Transformation “factors” used to test different shapes of the relationship between an outcome variable and time

	Linear	Quadratic	Cubic
Y_{t1}	−0.671	0.500	−0.224
Y_{t2}	−0.224	−0.500	0.671
Y_{t3}	0.224	−0.500	−0.671
Y_{t4}	0.671	0.500	0.224

relationship with time is a linear one. The question about the shape of the relationship can also be answered by applying MANOVA for repeated measurements. In MANOVA for repeated measurements, the relationship between the outcome variable Y and time is compared to a hypothetical linear relationship, a hypothetical quadratic relationship, and so on. When there are T repeated measurements, $T - 1$ possible functions with time can be tested. Although every possible relationship with time can be tested, it is important to have a certain idea or hypothesis of the shape of the relationship between the outcome variable Y and time. It is highly recommended only to test that particular hypothesis and not to test all possible relationships routinely.

For each possible relationship, an F -statistic is calculated which follows an F -distribution with $(1), (N - 1)$ degrees of freedom. The shape of the relationship between the outcome variable and time can only be analyzed with the “univariate” estimation approach. In the following section this will be illustrated with a numerical example.

3.3.3 A numerical example

Consider the same simple longitudinal dataset that was used in Section 3.3.1 and presented in Table 3.4. To answer the question: “What is the shape of the relationship between the outcome variable Y and time?”, the outcome variable Y must be transformed. When there are four repeated measurements, Y is transformed into a linear component, a quadratic component and a cubic component. This transformation is made according to the transformation “factors” presented in Table 3.5.

Each value of the original dataset is now multiplied by the corresponding transformation “factor” to create a transformed dataset. Table 3.6 presents the linear transformed dataset. The asterisk above the name of a variable indicates that the variable is transformed.

Table 3.6 Original dataset transformed by linear transformation “factors”

i	Y_{t1}^*	Y_{t2}^*	Y_{t3}^*	Y_{t4}^*	Mean
1	−20.8	−6.5	3.4	17.5	−1.62
2	−16.1	−6.3	4.5	21.5	0.89
3	−9.4	−4.5	6.3	20.1	3.13
4	−25.5	−7.6	6.7	22.8	−0.90
5	−16.8	−6.5	5.6	19.5	0.45
6	−20.1	−6.3	3.6	22.8	0.00
Mean					0.33

These transformed variables are now used to test the different relationships with time. Assume that one is interested in the possible linear relationship with time. Therefore, the individual sum of squares for the linear transformed variables is related to the individual sum of squares calculated when the overall mean value of the transformed variables is assumed to be zero (i.e. the intercept).

The first step is to calculate the individual sum of squares for the transformed variables according to Equation 3.5. For the transformed dataset $SS_i^* = 4[(-1.62 - 0.33)^2 + (0.89 - 0.33)^2 + \dots + (0.00 - 0.33)^2] = 54.43$. The next step is to calculate the individual sum of squares when the overall mean value is assumed to be zero. When this calculation is performed for the transformed dataset $SS_i^0 = 4[(-1.62 - 0.00)^2 + (0.89 - 0.00)^2 + \dots + (0.00 - 0.00)^2] = 56.96$.

The difference between these two individual sums of squares is called the “intercept” and is shown in the computer output (see [Output 3.5](#)). In the example, this intercept is equal to 2.546, and this value is used to test for the linear development over time. The closer this difference comes to zero, the less likely it is that there is a linear relationship with time. In the example the p -value of the intercept is 0.65, which is far from significance, i.e. there is no significant linear relationship between the outcome variable and time.

Output 3.5 Results of MANOVA for repeated measurements, applied to the linear transformed dataset

Between-subjects effects					
Source	Sum of squares	df	Mean square	F	Sig
Intercept	2.546	1	2.546	0.234	0.649
Error	54.425	5	10.885		

When MANOVA for repeated measurements is performed on the original dataset used in Section 3.3.1, these transformations are automatically carried out and the related test values are shown in the output. Because the estimation procedure is slightly different to that explained here, the sums of squares given in this output are the sums of squares given in the output (see [Output 3.6](#)) multiplied by T . Because it is basically the same approach, the levels of significance are exactly the same.

Output 3.6 Results of MANOVA for repeated measurements, applied to the original dataset, analysing the linear relationship between the outcome variable and time

Within-subjects contrasts					
Source	Sum of squares	df	Mean square	F	Sig.
Time (linear)	10.208	1	10.208	0.235	0.649
Error (linear)	217.442	5	43.488		

Exactly the same procedure can be carried out to test for a possible second-order (quadratic) relationship with time and for a possible third-order (cubic) relationship with time.

3.3.4 Example

[Output 3.7](#) shows the results of the MANOVA for repeated measurements performed on the example dataset with six repeated measurements on 147 subjects. The analysis was performed to answer the question of whether there is a change over time in outcome variable Y (using the information of all six repeated measurements)

Output 3.7 Results of MANOVA for repeated measurements; a “one-within” design

Multivariate Tests ^a						
Effect		Value	F	Hypothesis df	Error df	Sig.
time	Pillai's Trace	0.666	56.615 ^b	5.000	142.000	0.000
	Wilks' Lambda	0.334	56.615 ^b	5.000	142.000	0.000
	Hotelling's Trace	1.993	56.615 ^b	5.000	142.000	0.000
	Roy's Largest Root	1.993	56.615 ^b	5.000	142.000	0.000

^aDesign: Intercept.
Within Subjects Design: time.
^bExact statistic.

Mauchly's Test of Sphericity^a

Measure: MEASURE_1

Within Subjects		Approx.			Epsilon ^b		
Effect	Mauchly's W	Chi-Square	df	Sig.	Greenhouse-Geisser	Huynh-Feldt	Lower-bound
time	0.435	119.961	14	0.000	0.741	0.763	0.200

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

^aDesign: Intercept.

Within Subjects Design: time.

^bMay be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
time	Sphericity Assumed	89.987	5	17.997	99.987	0.000
	Greenhouse-Geisser	89.987	3.707	24.273	99.987	0.000
	Huynh-Feldt	89.987	3.816	23.582	99.987	0.000
	Lower-bound	89.987	1.000	89.987	99.987	0.000
Error(time)	Sphericity Assumed	131.398	730	0.180		
	Greenhouse-Geisser	131.398	541.272	0.243		
	Huynh-Feldt	131.398	557.126	0.236		
	Lower-bound	131.398	146.000	0.900		

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	time	Type III Sum of Squares	df	Mean Square	F	Sig.
time	Linear	40.332	1	40.332	126.240	0.000
	Quadratic	44.283	1	44.283	191.356	0.000
	Cubic	1.547	1	1.547	11.424	0.001
	Order 4	1.555	1	1.555	12.537	0.001
	Order 5	2.270	1	2.270	25.322	0.000

Error (time)	Linear	46.646	146	0.319
	Quadratic	33.787	146	0.231
	Cubic	19.770	146	0.135
	Order 4	18.108	146	0.124
	Order 5	13.088	146	0.090

Tests of Between-Subjects Effects

Measure: MEASURE_1
Transformed Variable: Average

Source	Type III		Mean		Sig.
	Sum of Squares	Df	Square	F	
Intercept	17845.743	1	17845.743	7273.162	0.000
Error	358.232	146	2.454		

The first part of the output (multivariate tests) shows directly the answer to the question of whether there is a change over time for outcome variable Y , somewhere between $t = 1$ and $t = 6$. The F -values and the significance levels are based on the multivariate test. In the output there are several multivariate tests available to test the overall time effect. The various tests are named after the statisticians who developed the tests, and they all use slightly different estimation procedures. However, the final conclusions of the various tests are almost always the same.

The second part of [Output 3.7](#) provides information on whether or not the assumption of sphericity is met. In this example, the sphericity coefficient (epsilon) calculated by the Greenhouse–Geisser method is 0.741. The output also gives other values for epsilon (Huynh–Feldt and lower-bound), but these values are seldom used. The value of epsilon can be tested for significance by Mauchly’s test of sphericity. The results of this test (p -value 0.000) indicate that epsilon is significantly different from the ideal value of one. This indicates that the degrees of freedom of the F -test should be adjusted. In the computer output presented, this adjustment is automatically carried out and is shown in the next part of the output (tests of within-subject effects), which shows the result of the “univariate” estimation approach. The output of the “univariate” approach gives four different estimates of the overall time effect. The first estimate is the one which assumes sphericity. The other three estimates (Greenhouse–Geisser, Huynh–Feldt, and lower-bound) adjust for violations of the assumption of sphericity, by changing the degrees of freedom. The three techniques are slightly different, but it

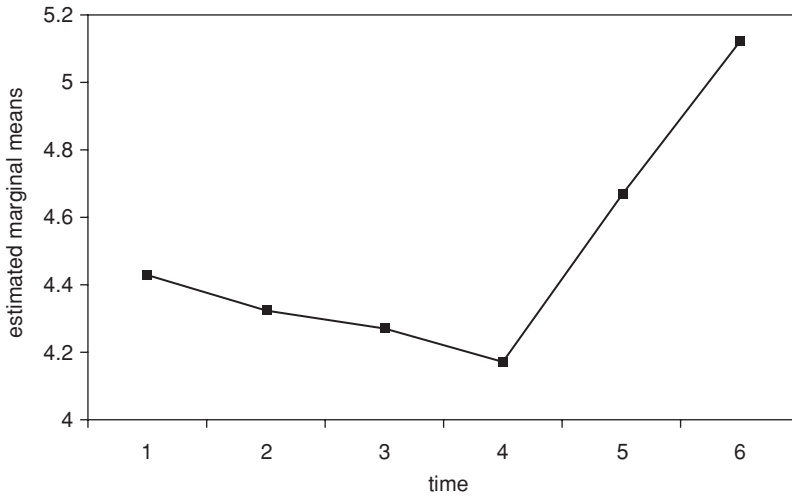


Figure 3.4 Results of the MANOVA for repeated measurements; graphical representation of a “one-within” design.

is recommended that the Greenhouse–Geisser adjustment is used, although this adjustment is slightly conservative. From the output it can be seen that the F -values and significance levels are equal for all estimation procedures. They are all highly significant, which indicates that there is a significant change over time in outcome variable Y . From the output, however, there is no indication of whether there is an increase, a decrease or whatever; it only shows a significant difference over time.

The last part of the output (tests of within-subjects contrasts) provides an answer to the second question (“What is the shape of the relationship with time?”). The first line (linear) indicates the test for a linear development. The F -value (obtained from the mean square (40.322) divided by the error mean square (0.319)) is very high (126.240), and is highly significant (0.000). This result indicates that there is a significant linear development over time. The following lines show the same values belonging to the other functions with time. The second line shows the second-order function (i.e. quadratic), the third line shows the third-order function (i.e. cubic), and so on. All F -values were significant, indicating that all other developments over time (second-order, third-order, etc.) are statistically significant. The magnitudes of the F -values indicate further that the best way to describe the development over time is a quadratic function, but the more simple linear function with time is also quite good. Again, from the results there is no indication of whether there is an increase or a decrease over time. In fact, the results of the MANOVA for repeated

measurements can only be interpreted correctly if a graphical representation of the change over time is made. [Figure 3.4](#) shows such a graphical representation. The figure shows that the significant development over time, which was found with MANOVA for repeated measurements, is first characterized by a small decrease, which is followed by an increase over time.

Within MANOVA for repeated measurements, there is also the possibility to obtain a magnitude of the strength of the effect (i.e. the within-subject time effect). This magnitude is reflected in a measure called “eta squared,” which can be seen as an indicator for the explained variance in the outcome variable *Y* due to a particular effect. Eta squared is calculated as the ratio between the sum of squares of the particular effect and the total sum of squares. [Output 3.8](#) shows part of the output of a MANOVA for repeated measurements including eta squared. From [Output 3.8](#) it can be seen that eta squared is 0.406 (i.e. $89.99/(131.40 + 89.99)$), which indicates that 41% of the variance in outcome variable *Y* is explained by the time effect.

Output 3.8 Results of MANOVA for repeated measurements; a “one-within” design including the explained variance

Tests of Within-Subjects Effects							
Measure: MEASURE_1							
Source		Type III		Mean		Partial Eta	
		Sum of Squares	df	Square	F	Sig.	Squared
time	Sphericity Assumed	89.987	5	17.997	99.987	0.000	0.406
	Greenhouse-Geisser	89.987	3.707	24.273	99.987	0.000	0.406
	Huynh-Feldt	89.987	3.816	23.582	99.987	0.000	0.406
	Lower-bound	89.987	1.000	89.987	99.987	0.000	0.406
	Sphericity Assumed	131.398	730	0.180			
Error(time)							
	Greenhouse-Geisser	131.398	541.272	0.243			
	Huynh-Feldt	131.398	557.126	0.236			
	Lower-bound	131.398	146.000	0.900			

To put the results of the MANOVA for repeated measurements in a somewhat broader perspective, the results of a “naïve” analysis are shown in [Output 3.9](#), naïve in the sense that the dependency of the repeated observations within one subject is ignored. Such a naïve analysis is an ANOVA, in which

the mean values of outcome variable Y are compared among all six measurements, i.e. six groups, each representing one time-point. For only two measurements, this comparison would be the same as the comparison between an independent sample t -test (the naïve approach) and a paired t -test (the adjusted approach).

Output 3.9 Results of a (naïve) ANOVA, ignoring the dependency of observations

Source	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	89.987	5	17.997	32.199	0.000
Within Groups	489.630	876	0.559		
Total	579.617	881			

From [Output 3.9](#) it can be seen that the F -statistic for the time effect (the effect in which we are interested) is 32.199, which is highly significant (0.000). This result indicates that at least one of the mean values of outcome variable Y at a certain time-point is significantly different from the mean value of outcome variable Y at one of the other time-points. However, as mentioned before, this approach ignores the fact that a longitudinal study is performed, i.e. that the same subjects are measured on several occasions. The most important difference between MANOVA for repeated measurements and the naïve ANOVA is that the “error sum of squares” in the ANOVA is much higher than the “error sum of squares” in the MANOVA for repeated measurements. In the ANOVA the residual mean square is 0.559 (see [Output 3.9](#)), while for the MANOVA for repeated measurements the residual mean square (indicated by Error (time) Sphericity Assumed) was more than three times lower, i.e. 0.180 (see [Output 3.7](#)). This has to do with the fact that in the MANOVA for repeated measurements the “individual” sum of squares is calculated to adjust for the dependency of the observations within a subject. This “individual” sum of squares is subtracted from the “error” sum of squares.

3.4 The “univariate” or the “multivariate” approach?

Within MANOVA for repeated measurements a distinction can be made between the “multivariate” approach (the multivariate extension of a paired t -test) and the “univariate” approach (an extension of ANOVA). The problem is that the two approaches do not produce the same results. So the question is: Which approach should be used?

One of the differences between the two approaches is the assumption of sphericity. For the “multivariate” approach this assumption is not necessary, while for the “univariate” approach it is an important assumption. The restriction of the assumption of sphericity (i.e. equal correlations and equal variances over time) leads to an increase in degrees of freedom, i.e. an increase in power for the “univariate” approach. This increase in power becomes more important when the sample size becomes smaller. Historically, the “multivariate” approach was developed later than the “univariate” approach, especially for situations when the assumption of sphericity does not hold. So, one could argue that when the assumption of sphericity is violated, the “multivariate” approach should be used. However, in the “univariate” approach, adjustments can be made when the assumption of sphericity is not met. So, in principle, both approaches can deal with a situation in which the assumption of sphericity does not hold. It is sometimes argued that when the number of subjects N is less than the number of (repeated) measurements plus 10, the “multivariate” approach should not be used. In every other situation, however, it is recommended that the results of both the “multivariate” and the “univariate” approach are used to obtain the most “valid” answer to the research question addressed. Only when both approaches lead to the same conclusion, it is fairly certain that there is either a significant or a non-significant change over time. When both approaches lead to different conclusions, the conclusions must be drawn with many restrictions and considerable caution. In such a situation, it is highly recommended not to use the approach with the lowest p -value!

3.5 Comparing groups

In the first sections of this chapter longitudinal studies were discussed in which one continuous outcome variable is repeatedly measured over time (i.e. the “one-within” design). In this section the research situation will be discussed in which the development of a certain continuous outcome variable Y is compared between different groups. This design is known as the “one-within, one-between” design. Time is the within-subject factor and the group variable is the between-subjects factor (Figure 3.5). This group indicator can be either dichotomous or categorical. The question to be addressed is: “Is there a difference in change over time for outcome variable Y between two or more groups?” This question can also be answered with MANOVA for repeated measurements. The same assumptions as have been mentioned earlier (Section 3.3) apply for this design, but it is also assumed that the covariance matrices of the different groups that are compared to each other are homogeneous. This assumption is comparable with the assumption

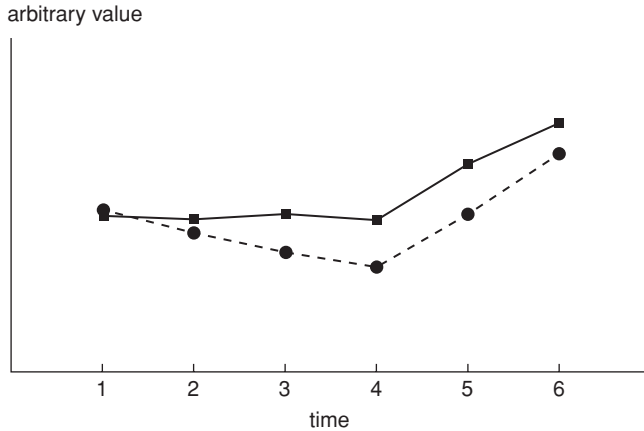


Figure 3.5 A longitudinal “one-within, one-between” design with six repeated measurements measured in two groups (■ — group 1, • - - - group 2).

of equal variances in two groups that are cross-sectionally compared with each other using the independent sample t -test. Although this is an important assumption, in reasonably large samples a violation of this assumption is generally not problematic.

From a “one-within, one-between” design the following “effects” can be obtained: (1) an overall time effect, i.e. “Is there a change over time in outcome variable Y for the total population?”, (2) a general group effect, i.e. “Is there on average over time a difference in outcome variable Y between the compared groups?”, (3) a group by time interaction effect, i.e. “Is the change over time in outcome variable Y different for the compared groups?” The within-subject effects can be calculated in two ways: the “multivariate” approach, which is based on the multivariate analysis of the differences between subsequent points of measurements, and the “univariate” approach, which is based on the comparison of several sums of squares (see Section 3.5.1). In epidemiological longitudinal studies the group by time interaction effect is probably the most interesting, because it gives an answer to the question of whether there is a difference in change over time between groups.

With respect to the shape of the relationship with time (linear, quadratic, etc.) specific questions can also be answered for the “one-within, one-between” design, such as “Is there a difference in the linear relationship with time between the groups?”, “Is there a difference in the quadratic relationship with time?”, etc. However, especially for interaction terms, the answers to those questions can be quite complicated, i.e. the results of the MANOVA for repeated measurements can be very difficult to interpret.

Table 3.7 Hypothetical longitudinal dataset with four measurements in six subjects divided into two groups

<i>I</i>	Group	Y_{t1}	Y_{t2}	Y_{t3}	Y_{t4}	Mean
1	1	31	29	15	26	25.25
2	1	24	28	20	32	26.00
3	1	14	20	28	30	23.00
Mean		23.00	25.67	21.00	29.33	24.75
4	2	38	34	30	34	34.00
5	2	25	29	25	29	27.00
6	2	30	28	16	34	27.00
Mean		31.00	30.33	23.67	32.33	29.33

It should be noted that an important limitation of MANOVA for repeated measurements is that the between-subjects factor can only be a time-independent dichotomous or categorical variable, such as treatment group, gender, etc.

3.5.1 The “univariate” approach: a numerical example

The simple longitudinal dataset used to illustrate the “univariate” approach in a “one-within” design will also be used to illustrate the “univariate” approach in a “one-within, one-between” design. Therefore, the dataset used in the earlier example, and presented in Table 3.4, is extended to include a group indicator. The “new” dataset is presented in Table 3.7.

To estimate the different “effects,” it should first be noted that part of the overall “error sum of squares” is related to the differences between the two groups. To calculate this part, the sum of squares for individuals (SS_i) must be calculated for each of the groups (see Equation 3.5). For group 1, $SS_i = 4[(25.25 - 24.75)^2 + (26 - 24.75)^2 + (23 - 24.75)^2] = 19.5$, and for group 2, $SS_i = 4[(34 - 29.33)^2 + (27 - 29.33)^2 + (27 - 29.33)^2] = 130.7$.

These two parts can be added together to give an overall “error sum of squares” of 150.2. If the group indication is ignored, the overall “error sum of squares” is 276.2 (see Section 3.3.1). This means that the between-subjects sum of squares caused by group differences is 126.0 (i.e. $276.2 - 150.2$). The next step is to calculate the SS and the SS_b for each group. This can be done in the same way as has been described for the whole population (see Equations 3.3 and 3.4). The results are summarized in Table 3.8.

The two within-subject “error sums of squares” can be added together to form the overall within-subject error sum of squares (adjusted for group). This total within-subject error sum of squares is 373.17. Without taking the group differentiation into

Table 3.8 Summary of the different sums of squares calculated for each group separately

	Group 1	Group 2
SS_b	116.9	134.7
SS	299.3	224.0
SS_i	19.5	130.7
Within-subject error sum of squares	$299.3 - 19.5 = 279.83$	$224.0 - 130.7 = 93.33$

account, a within-subject error sum of squares of 399.96 was found. The difference between the two is the sum of squares belonging to the interaction between the within-subject factor “time” and the between-subject factor “group.” This sum of squares is 26.79. [Output 3.10](#) shows the computerized results of the MANOVA for repeated measurements for this numerical example.

Output 3.10 Results of MANOVA for repeated measurements for a simple longitudinal dataset with a group indicator

Within-subjects effects					
Source	Sum of squares	df	Mean square	F	Sig
TIME	224.792	3	74.931	2.810	0.075
TIME × GROUP	26.792	3	8.931	0.287	0.834
Error (TIME)	373.167	12	31.097		
Between-subjects effects					
Source	Sum of squares	df	Mean square	F	Sig
Intercept	17550.042	1	17550.042	317.696	0.000
GROUP	126.042	1	126.042	3.357	0.141
Error	150.167	4	37.542		

3.5.2 Example

In the example dataset with six repeated measurements performed on 147 subjects, X_4 is a dichotomous time-independent covariate (i.e. gender), so this variable will be used as a between-subjects factor in this example. The results of the MANOVA for repeated measurements from a “one-within, one-between” design are shown in [Output 3.11](#).

Output 3.11 Results of MANOVA for repeated measurements; a “one-within, one-between” design

Tests of Between-Subjects Effects

Measure: MEASURE_1
Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Intercept	17715.454	1	17715.454	7486.233	0.000
x4	15.103	1	15.103	6.382	0.013
Error	343.129	145	2.366		

Multivariate Tests^a

Effect		Value	F	Hypothesis df	Error df	Sig.
time	Pillai's Trace	0.669	56.881 ^b	5.000	141.000	0.000
	Wilks' Lambda	0.331	56.881 ^b	5.000	141.000	0.000
	Hotelling's Trace	2.017	56.881 ^b	5.000	141.000	0.000
	Roy's Largest Root	2.017	56.881 ^b	5.000	141.000	0.000
time * x4	Pillai's Trace	0.242	8.980 ^b	5.000	141.000	0.000
	Wilks' Lambda	0.758	8.980 ^b	5.000	141.000	0.000
	Hotelling's Trace	0.318	8.980 ^b	5.000	141.000	0.000
	Roy's Largest Root	0.318	8.980 ^b	5.000	141.000	0.000

^aDesign: Intercept + x4.
Within Subjects Design: time.
^bExact statistic.

Mauchly's Test of Sphericity^a

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Epsilon ^b					
		Approx. Chi-Square	df	Sig.	Greenhouse-Geisser	Huynh-Feldt	Lower-bound
Time	0.433	119.736	14	0.000	0.722	0.748	0.200

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.
^aDesign: Intercept + x4
Within Subjects Design: time
^bMay be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square		F	Sig.
time	Sphericity Assumed	89.546	5	17.909	104.344	0.000	
	Greenhouse-Geisser	89.546	3.612	24.793	104.344	0.000	
	Huynh-Feldt	89.546	3.741	23.937	104.344	0.000	
	Lower-bound	89.546	1.000	89.546	104.344	0.000	
time * x4	Sphericity Assumed	6.962	5	1.392	8.113	0.000	
	Greenhouse-Geisser	6.962	3.612	1.928	8.113	0.000	
	Huynh-Feldt	6.962	3.741	1.861	8.113	0.000	
	Lower-bound	6.962	1.000	6.962	8.113	0.005	
Error(time)	Sphericity Assumed	124.436	725	0.172			
	Greenhouse-Geisser	124.436	523.707	0.238			
	Huynh-Feldt	124.436	542.443	0.229			
	Lower-bound	124.436	145.000	0.858			

Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square		F	Sig.
time	Linear	38.668	1	38.668	131.084	0.000	
	Quadratic	45.502	1	45.502	213.307	0.000	
	Cubic	1.602	1	1.602	11.838	0.001	
	Order 4	1.562	1	1.562	12.516	0.001	
	Order 5	2.212	1	2.212	24.645	0.000	
time * x4	Linear	3.872	1	3.872	13.127	0.000	
	Quadratic	2.856	1	2.856	13.388	0.000	
	Cubic	0.154	1	0.154	1.142	0.287	
	Order 4	0.008	1	0.008	0.060	0.806	
	Order 5	0.072	1	0.072	0.804	0.371	
Error(time)	Linear	42.773	145	0.295			
	Quadratic	30.931	145	0.213			
	Cubic	19.616	145	0.135			
	Order 4	18.100	145	0.125			
	Order 5	13.016	145	0.090			

Part of [Output 3.11](#) is comparable to the output of the “one-within” design, shown in [Output 3.7](#). The major difference is found in the first part of the output, in which the result of the “tests of between-subjects effects” is given. The F -value belonging to this test is 6.382 and the significance level is 0.013, which indicates that there is an overall (i.e. averaged over time) significant difference between the two groups indicated by X_4 . The other difference between the two outputs is the addition of a time by X_4 interaction term. This interaction is interesting, because it answers the question of whether there is a difference in development over time between the two groups indicated by X_4 (i.e. the difference in developments between males and females). The answer to that question can either be obtained with the “multivariate” approach (Pillai, Wilks, Hotelling, and Roy) or with the “univariate” approach. For the “multivariate” approach (multivariate tests), firstly the overall time effect is given and secondly the time by X_4 interaction. For the “univariate” approach, again the assumption of sphericity has to hold and from the output it can be seen that this is not the case (Greenhouse–Geisser epsilon = 0.722, and the significance of the sphericity test is 0.000). For this reason, in the univariate approach it is recommended that the Greenhouse–Geisser adjustment is used. From the output of the univariate analysis, firstly the overall time effect ($F = 104.344$, significance 0.000) and secondly the time by X_4 interaction effect ($F = 8.113$, significance 0.000) can be obtained. This result indicates that there is a significant difference in development over time between the two groups indicated by X_4 .

From the next part of [Output 3.11](#) (tests of within-subjects contrasts) it can be seen that this difference is significant for both the linear development over time and the quadratic development over time.

For all three effects, the explained variance (which is an indicator of the magnitude of the effect) can also be calculated (see [Output 3.12](#)). From [Output 3.12](#) it can be seen that 42% of the variance in outcome variable Y is explained by the “time effect,” that 5% is explained by the time by X_4 interaction, and that 4% of the variance in outcome variable Y is explained by the “overall group effect.” Care must be taken in the interpretation of these explained variances, because they cannot be interpreted together in a straightforward way. The explained variances for the time effect and the time–group interaction effect are only related to the within-subject “error sum of squares,” and not to the total “error sum of squares.”

As in the case for the “one-within” design, the results of the MANOVA for repeated measurements for a “one-within, one-between” design can only be interpreted correctly when a graphical representation is added to the results (see [Figure 3.6](#)).

Output 3.12 Results of MANOVA for repeated measurements; a “one-within, one-between” design including the explained variance

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	Df	Mean Square	F	Sig.	Partial Eta squared
time	Sphericity Assumed	89.546	5	17.909	104.344	0.000	0.418
	Greenhouse-Geisser	89.546	3.612	24.793	104.344	0.000	0.418
	Huynh-Feldt	89.546	3.741	23.937	104.344	0.000	0.418
	Lower-bound	89.546	1.000	89.546	104.344	0.000	0.418
time * x4	Sphericity Assumed	6.962	5	1.392	8.113	0.000	0.053
	Greenhouse-Geisser	6.962	3.612	1.928	8.113	0.000	0.053
	Huynh-Feldt	6.962	3.741	1.861	8.113	0.000	0.053
	Lower-bound	6.962	1.000	6.962	8.113	0.005	0.053
Error(time)	Sphericity Assumed	124.436	725	0.172			
	Greenhouse-Geisser	124.436	523.707	0.238			
	Huynh-Feldt	124.436	542.443	0.229			
	Lower-bound	124.436	145.000	0.858			

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	Df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	17715.454	1	17715.454	7486.233	0.000	0.981
x4	15.103	1	15.103	6.382	0.013	0.042
Error	343.129	145	2.366			

3.6 Comments

One of the problems with MANOVA for repeated measurements is that the time periods under consideration are weighted equally. A non-significant change over a short time period can be relatively greater than a significant change over a long time period. So, when the time periods are unequally spaced, the results of MANOVA

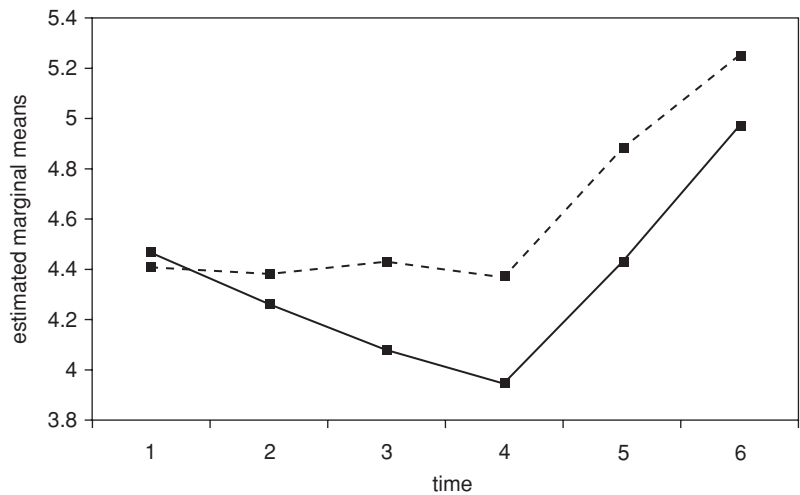


Figure 3.6 Results of MANOVA for repeated measurements; graphical representation of a “one-within, one-between” design (X_4 ; — males, - - - females).

for repeated measurements cannot be interpreted in a straightforward way. The length of the different time intervals must be taken into account.

Another major problem with MANOVA for repeated measurements is that it only takes into account the subjects with complete data, i.e. the subjects who are measured at all time-points. When a subject has no data available for a certain time-point, all other data for that subject are deleted from the analysis. In [Chapter 10](#), the problems and consequences of missing data in longitudinal studies and in the results obtained from a MANOVA for repeated measurements analysis will be discussed.

MANOVA for repeated measurements can also be used for more complex study designs, i.e. with more within-subject and/or more between-subjects factors. Because the ideas and the potential questions to be answered are the same as in the relatively simple designs discussed before, the more complex designs will not be discussed further. It should be kept in mind that the more groups that are compared to each other (given a certain number of subjects), or the more within-subject factors that are included in the design, the less power there will be to detect significant effects. This is important, because MANOVA for repeated measurements is basically a testing technique, so p -values are used to evaluate the development over time. In principle, no interesting effect estimations are provided by the procedure of the MANOVA for repeated measurements. As has been mentioned before, the explained variances can be calculated, but the importance of this indicator is rather limited.

3.7 Post-hoc procedures

With MANOVA for repeated measurements an “overall” time effect and an “overall” group effect can be obtained. As in cross-sectional ANOVA, post-hoc procedures can be performed to investigate further the observed “overall” relationships. In longitudinal analysis there are two types of these post-hoc procedures. (1) When there are more than two repeated measurements, it can be determined in which part of the longitudinal time period the observed “effects” occur. This can be done by performing MANOVA for repeated measurements for a specific (shorter) time period or by analyzing specific contrasts. (2) When there are more than two groups for which the longitudinal relationship is analyzed, a statistically significant between-subjects effect indicates that there is a difference between at least two of the compared groups. Further analysis can determine between which groups the differences occur. This can be carried out by applying the post-hoc procedures also used in the cross-sectional ANOVA (e.g. Tukey, Bonferroni, or Scheffe procedures). Each technique has its own particularities, but in essence multiple comparisons are made between all groups; each group is pairwise compared to the other groups and there is a certain “adjustment” for multiple testing.

3.7.1 Example

Output 3.13 shows a typical output of a post-hoc procedure following MANOVA for repeated measurements comparing three groups (the data are derived from a hypothetical dataset which will not be discussed any further).

The structure of the first part of the output is similar to the outputs discussed before. It shows the overall between-subjects group effect. The p -value belonging to this group effect is highly significant. The interpretation of this result is that there is at least one significant difference between two of the three groups, but this overall result gives no information about which groups actually differ from each other. To obtain an answer to that question, a post-hoc procedure can be carried out. In the output, the results of the three most commonly used post-hoc procedures are given. The first column of the output gives the name of the post-hoc procedure (Tukey, Scheffe, and Bonferroni). The second and third columns show the pairwise comparisons that are made, and the fourth and fifth columns give the overall mean difference between the compared groups and the standard error of that difference. The last column gives the p -value of the pairwise comparison. As can be seen from the output, there are only marginal differences between the three post-hoc procedures (in most research situations this will be the case). It can be seen that groups 1 and 2 do not differ significantly from each other, but that the average value of group 3 is totally different from that of the other two groups. One must realize that these post-hoc procedures deal with the overall between-subjects group

Output 3.13 Results of three post-hoc procedures in MANOVA for repeated measurements

Between-subjects effects					
Source	Sum of squares	Df	Mean square	F	Sig
Intercept	17845.743	1	17845.743	8091.311	0.000
GROUP	40.364	2	20.317	9.221	0.000
Error	317.598	144	2.206		
Post-hoc tests					
	Group1	Group2	Mean difference (1 - 2)	Std error	Sig
Tukey	1	2	-8.361 × 10 ⁻²	0.1225	0.774
		3	-0.4913	0.1225	0.000
	2	1	-8.361 × 10 ⁻²	0.1225	0.774
		3	-0.4077	0.1225	0.003
	3	1	0.4913	0.1225	0.000
		2	0.4077	0.1225	0.003
Scheffe	1	2	-8.361 × 10 ⁻²	0.1225	0.793
		3	-0.4913	0.1225	0.000
	2	1	-8.361 × 10 ⁻²	0.1225	0.793
		3	-0.4077	0.1225	0.005
	3	1	0.4913	0.1225	0.000
		2	0.4077	0.1225	0.005
Bonferroni	1	2	-8.361 × 10 ⁻²	0.1225	1.00
		3	-0.4913	0.1225	0.000
	2	1	-8.361 × 10 ⁻²	0.1225	1.00
		3	-0.4077	0.1225	0.003
	3	1	0.4913	0.1225	0.000
		2	0.4077	0.1225	0.003

effect, i.e. the difference between the average value over the different time-points. To obtain an answer to the question in which part of the longitudinal period the observed relationships occurred, a MANOVA can be performed for specific time periods or specific contrasts can be analyzed.

3.8 Different contrasts

In an earlier part of this chapter, attention was paid to answering the question: “What is the shape of the relationship between outcome variable Y and time?” In the example it was mentioned that the answer to that question can be found in the output section: test of within-subject contrasts. In the example a so-called

“polynomial” contrast was used in order to investigate whether one is dealing with a linear relationship with time, a quadratic relationship with time, and so on. In longitudinal research this is an important contrast, but there are many other possible contrasts (depending on the software package used). With a “simple” contrast, for instance, the value at each measurement is related to the first measurement. With a “difference” contrast, the value of each measurement is compared to the average of all previous measurements. A “Helmert” contrast is comparable to the “difference” contrast, however, the value at a particular measurement is compared to the average of all subsequent measurements. With the “repeated” contrast, the value of each measurement is compared to the value of the first subsequent measurement. In Section 3.3 it was mentioned that the testing of a “polynomial” contrast was based on transformed variables. In fact, the testing of all contrasts is based on transformed variables. However, for each contrast, different transformation coefficients are used.

3.8.1 Example

Outputs 3.14a to 3.14d show the results of MANOVA for repeated measurements with different contrasts performed on the example dataset with six repeated measurements on 147 subjects. The output obtained from the analysis with a polynomial contrast was already shown in Section 3.3.4 ([Output 3.7](#)).

With the “simple” contrast, each measurement is compared to the first measurement. From [Output 3.14a](#) it can be seen that all follow-up measurements differ significantly from the first measurement. From the output, however, it cannot be

Output 3.14a Results of MANOVA for repeated measurements with a “simple” contrast

Within-subject Contrasts					
Source	Sum of squares	df	Mean square	F	Sig
Level 2 vs. Level 1	1.830	1	1.830	8.345	0.004
Level 3 vs. Level 1	4.184	1	4.184	14.792	0.000
Level 4 vs. Level 1	10.031	1	10.031	32.096	0.000
Level 5 vs. Level 1	8.139	1	8.139	20.629	0.000
Level 6 vs. Level 1	69.353	1	69.353	120.144	0.000
Error					
Level 2 vs. Level 1	32.010	146	0.219		
Level 3 vs. Level 1	41.296	146	0.283		
Level 4 vs. Level 1	45.629	146	0.313		
Level 5 vs. Level 1	57.606	146	0.395		
Level 6 vs. Level 1	84.279	146	0.577		

Output 3.14b Results of MANOVA for repeated measurements with a “difference” contrast

Within-subject Contrasts

Source	Sum of squares	df	Mean square	F	Sig
Level 1 vs. Level 2	1.830	1	1.830	8.345	0.004
Level 2 vs. Previous	1.875	1	1.875	9.679	0.002
Level 3 vs. Previous	4.139	1	4.139	28.639	0.000
Level 4 vs. Previous	20.198	1	20.198	79.380	0.000
Level 5 vs. Previous	82.271	1	82.271	196.280	0.000
Error					
Level 1 vs. Level 2	32.010	146	0.219		
Level 2 vs. Previous	28.260	146	0.194		
Level 3 vs. Previous	21.101	146	0.145		
Level 4 vs. Previous	37.150	146	0.254		
Level 5 vs. Previous	61.196	146	0.419		

Output 3.14c Results of MANOVA for repeated measurements with a “Helmert” contrast

Within-subject Contrasts

Source	Sum of squares	df	Mean square	F	Sig
Level 1 vs. Later	0.852	1	0.852	4.005	0.047
Level 2 vs. Later	8.092	1	8.092	41.189	0.000
Level 3 vs. Later	22.247	1	22.247	113.533	0.000
Level 4 vs. Later	76.695	1	76.695	277.405	0.000
Level 5 vs. Level 6	29.975	1	29.975	63.983	0.000
Error					
Level 1 vs. Later	31.061	146	0.213		
Level 2 vs. Later	28.684	146	0.196		
Level 3 vs. Later	28.609	146	0.196		
Level 4 vs. Later	40.365	146	0.276		
Level 5 vs. Level 6	68.399	146	0.468		

Output 3.14d Results of MANOVA for repeated measurements with a “repeated” contrast

Within-subject Contrasts

Source	Sum of squares	df	Mean square	F	Sig
Level 1 vs. Level 2	1.830	1	1.830	8.345	0.004
Level 2 vs. Level 3	0.480	1	0.480	2.242	0.136
Level 3 vs. Level 4	1.258	1	1.258	8.282	0.005
Level 4 vs. Level 5	36.242	1	36.242	125.877	0.000
Level 5 vs. Level 6	29.975	1	29.975	63.983	0.000
Error					
Level 1 vs. Level 2	32.010	146	0.219		
Level 2 vs. Level 3	31.260	146	0.214		
Level 3 vs. Level 4	22.182	146	0.152		
Level 4 vs. Level 5	42.036	146	0.288		
Level 5 vs. Level 6	68.399	146	0.468		

seen whether the value at $t = 2$ is higher than the value at $t = 1$. It can only be concluded that there is a significant difference.

With the “difference” contrast, the value at each measurement is compared to the average value of all previous measurements. From [Output 3.14b](#) it can be seen that there is a significant difference between the value at each measurement and the average value of all previous measurements.

With the “Helmert” contrast ([Output 3.14c](#)), the same procedure is carried out as with the “difference” contrast, only the other way around. The value at each measurement is compared to the average value of all subsequent measurements. All these differences are also highly significant. Only if we compare the first measurement with the average value of the other five measurements, is the p -value of borderline significance (0.047).

With the “repeated” contrast, the value of each measurement is compared to the value of the first subsequent measurement. From [Output 3.14d](#) it can be seen that the value of outcome variable Y at $t = 2$ is not significantly different to the value of outcome variable Y at $t = 3$ ($p = 0.136$). All the other differences investigated were statistically significant. Again, it must be stressed that there is no information about whether the value at a particular time-point is higher or lower than the value at the first subsequent time-point. Like all other results obtained from MANOVA for repeated measurements, the results of the analysis with different contrasts can only be interpreted correctly if they are combined with a graphical representation of the development of outcome variable Y .

When there are more than two groups to be compared with MANOVA for repeated measurements, contrasts can also be used to perform post-hoc procedures for the “overall” group effect. With the traditional post-hoc procedures discussed in Section 3.7 all groups are pairwise compared, while with contrasts this is not the case. With a “simple” contrast for instance, the groups are compared to a certain reference category, and with a “repeated” contrast each group is compared to the next group (dependent on the coding of the group variable). The advantage of contrasts in performing post-hoc procedures is when an adjustment for certain covariates is applied. In that situation, the traditional post-hoc procedures cannot be performed, while with contrasts, the adjusted difference between groups can be obtained. Again, it is important to realize that the post-hoc procedures performed with different contrasts are only suitable (as are the traditional post-hoc procedures) for analysing the between-subjects effect.

3.9 Non-parametric equivalent of MANOVA for repeated measurements

When the assumptions of MANOVA for repeated measurements are violated, an alternative non-parametric approach can be applied. This non-parametric

Table 3.9 Absolute values and ranks (in parentheses) of the hypothetical dataset presented in [Table 3.4](#)

i	Y_{t1} (rank)	Y_{t2} (rank)	Y_{t3} (rank)	Y_{t4} (rank)
1	31 (4)	29 (3)	15 (1)	26 (2)
2	24 (2)	28 (3)	20 (1)	32 (4)
3	14 (1)	20 (2)	28 (3)	30 (4)
4	38 (4)	34 (2.5)	30 (1)	34 (2.5)
5	25 (1.5)	29 (3.5)	25 (1.5)	29 (3.5)
6	30 (3)	28 (2)	16 (1)	34 (4)
Total rank	15.5	16	8.5	20

equivalent of MANOVA for repeated measurements is called the Friedman test and can only be used in a “one-within” design. Like any other non-parametric test, the Friedman test does not make any assumptions about the distribution of the outcome variable under study. To perform the Friedman test, for each subject the outcome variable at T time-points is ranked from 1 to T . The Friedman test statistic is based on these rankings. In fact, the mean rankings (averaged over all subjects) at each time-point are compared to each other. The idea behind the Friedman test is that the observed rankings are compared to the expected rankings, assuming there is no change over time. The Friedman test statistic can be calculated according to Equation 3.6:

$$H = \frac{12 \sum_{t=1}^T R_t^2}{NT(T+1)} - 3N(T+1) \quad (3.6)$$

where H is the Friedman test statistic, R_t is the sum of the ranks at time-point t , N is the number of subjects, and T is the number of repeated measurements.

To illustrate this non-parametric test, consider again the hypothetical dataset presented earlier in [Table 3.4](#). In [Table 3.9](#) the ranks of this dataset are presented. Applied to the (simple) longitudinal dataset the Friedman test statistic (H) is equal to:

$$\frac{12(15.5^2 + 16^2 + 8.5^2 + 20^2)}{6 \times 4 \times 5} - 3 \times 6 \times 5 = 6.85$$

This value follows a chi-square distribution with $T - 1$ degrees of freedom. The corresponding p -value is 0.077. When this p -value is compared to the value obtained from a MANOVA for repeated measurements (see [Output 3.4](#)) it can be seen that they are almost the same. That the p -value from the non-parametric test is slightly

Output 3.15 Output of the non-parametric Friedman test

Friedman Two-way ANOVA			
Mean Rank	Variable		
3.49	YT1	OUTCOME VARIABLE Y AT T1	
2.93	YT2	OUTCME VARIABLE Y AT T2	
2.79	YT3	OUTCOME VARIABLE Y AT T3	
2.32	YT4	OUTCOME VARIABLE Y AT T4	
4.23	YT5	OUTCOME VARIABLE Y AT T5	
5.24	YT6	OUTCOME VARIABLE Y AT T6	
Cases	Chi-Square	DF	Significance
147	244.1535	5	0.0000

higher than the p -value from the parametric test has to do with the fact that non-parametric tests are in general less powerful than the parametric equivalents.

3.9.1 Example

Because the number of subjects in the example dataset is reasonably high (i.e. 147), in practice the Friedman test will not be used in this situation. However, for educational purposes the non-parametric Friedman test will be used to answer the question of whether there is a development over time in outcome variable Y . [Output 3.15](#) shows the results of this analysis.

From [Output 3.15](#) it can be seen that there is a significant difference between the measurements at different time-points. The Chi-square statistic is 244.1535, and with 5 degrees of freedom (the number of measurements minus one) this value is highly significant, i.e. a similar result to that found with the MANOVA for repeated measurements. The Friedman test statistic gives no direct information about the direction of the development, although from the mean rankings it can be seen that a decrease from the second to the fourth measurement is followed by an increase at the fifth and sixth measurements.

Continuous outcome variables – relationships with other variables

4.1 Introduction

With a paired t -test and multivariate analysis of variance (MANOVA) for repeated measurements it is possible to investigate changes in one continuous variable over time and to compare the development of a continuous variable over time between different groups. These methods, however, are not suitable for analysis of the longitudinal relationship between a continuous outcome variable and several covariates (which can be either continuous, dichotomous, or categorical). Before the development of “sophisticated” statistical techniques such as generalized estimating equations (GEE) analysis and mixed model analysis, “traditional” methods were used to analyze longitudinal data. The general idea of these “traditional” methods was to reduce the statistical longitudinal problem into a cross-sectional problem. Even nowadays these (limited) approaches are sometimes used in the analysis of longitudinal data.

4.2 “Traditional” methods

The greatest advantage of the “traditional” methods is that cross-sectional statistical techniques can be used to analyze the longitudinal data. The most commonly used technique for reducing the longitudinal problem to a cross-sectional problem is analysis of the relationships between changes in different parameters between two points in time ([Figure 4.1](#)). Because of its importance and its widespread use, a detailed discussion of the analysis of changes will be given in [Chapter 9](#).

Another traditional method with which to analyze the longitudinal relationship between several variables is the use of a single measurement at the end of the longitudinal period as outcome variable. This outcome variable is then related to a so-called “long-term exposure” to certain covariates, measured along the total longitudinal period ([Figure 4.2](#)).

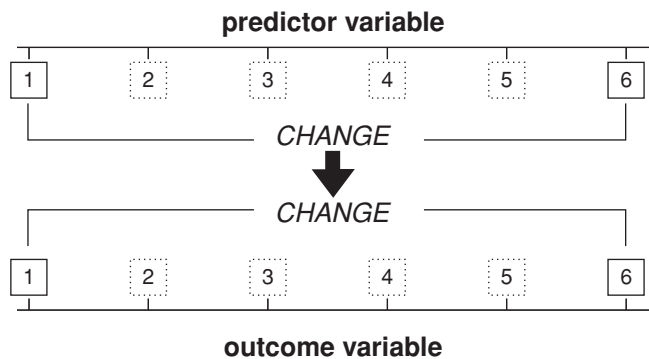


Figure 4.1 Changes in outcome variable *Y* between two subsequent measurements are related to changes in one or more covariate(s) *X* over the same time period.

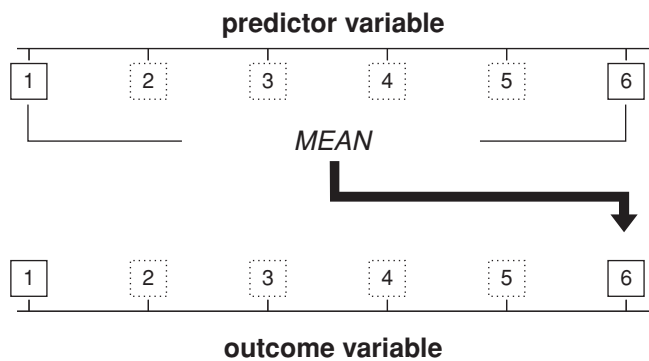


Figure 4.2 “Long-term exposure” to one or more covariate(s) *X* related to a single measurement of outcome variable *Y*.

It is obvious that a limitation of both methods is that if there are more than two repeated measurements, not all available longitudinal data are used in the analysis. Another cross-sectional possibility for analysing the longitudinal relationship between an outcome variable *Y* and (several) covariates *X*, using all the data, is to use individual regression lines with time. The first step in this procedure is to calculate the linear regression between the outcome variable *Y* and time for each subject. The regression coefficient with time (which is referred to as the slope) can be seen as an indicator for the change over the whole measurement period in the outcome variable *Y*. This regression coefficient with time is then used as the outcome variable in a cross-sectional regression analysis, in order to investigate the longitudinal relationship with other variables. The same procedure can be followed for the time-dependent covariates in order to analyze the relationship between the change in outcome variable *Y* and the change in the time-dependent covariates.

Table 4.1 Results of a linear regression analysis relating changes in covariates to changes in outcome variable Y between $t = 1$ and $t = 6$

	Regression coefficient	Standard error	p -Value
X_1	0.13	0.35	0.72
ΔX_2	0.05	0.04	0.29
ΔX_3	-0.05	0.14	0.71
X_4	0.36	0.15	0.02

However, the baseline value of the covariates can also be used in the final cross-sectional regression analysis. In the latter case it is obvious that a different research question is answered. The greatest disadvantage of this technique is the assumption of a linear relationship between the outcome variable Y and time, although it is possible to model a different individual regression function with time. Furthermore, it is questionable how well the individual regression line (or function), which is usually based on a few data points, fits the observed data.

4.3 Example

To illustrate the first cross-sectional technique that can be used to analyze the longitudinal relationship between a continuous outcome variable Y and several covariates X (Figure 4.1), the difference between Y at $t = 1$ and Y at $t = 6$ is first calculated. In the next step the differences in the time-dependent covariates X_2 and X_3 between $t = 1$ and $t = 6$ must be calculated. For X_3 , which is a dichotomous covariate, this is rather difficult, because the interpretation of the differences is not straightforward. In this example, the subjects were divided (according to X_3) into subjects who remained in the lowest category or “decreased” between $t = 1$ and $t = 6$ (i.e. the non-smokers and the subjects who quitted smoking), and subjects who remained in the highest category or “increased” between $t = 1$ and $t = 6$ (i.e. the ever-smokers and the subjects who started to smoke). Because the longitudinal problem is reduced into a cross-sectional problem, the relationships can be analyzed with cross-sectional linear regression analysis. The result of the analysis is shown in Table 4.1.

Because the longitudinal problem is reduced to a cross-sectional problem, and the data are analyzed with cross-sectional regression analysis, the regression coefficient can be interpreted in a straightforward way. For instance, the regression coefficient for X_4 indicates that the difference between Y at $t = 1$ and Y at $t = 6$ is

Table 4.2 Results of a linear regression analysis relating “long-term exposure” to covariates to the outcome variable Y at $t = 6$

	Regression coefficient	Standard error	p -value
X_1	0.72	0.41	0.08
$AverageX_2$	0.37	0.07	<0.001
$AverageX_3$	0.09	0.14	0.55
X_4	-0.07	0.18	0.69

0.36 higher for the group indicated by $X_4 = 2$ (i.e. females) compared to the group indicated by $X_4 = 1$ (i.e. males).

The second cross-sectional technique is slightly different. In this method “long-term exposure” to the covariates X is related to the outcome variable Y at $t = 6$ (Figure 4.2). For the time-dependent covariate X_2 , the average of the six measurements was used as indicator for “long-term exposure.” For the dichotomous covariate X_3 , the “long-term exposure” is coded as 0 when subjects report 0 (i.e. non-smoking) at all measurements, and coded 1 when subjects report 1 (i.e. smoking) at least at one of the measurements. The result of the linear regression analysis is shown in Table 4.2.

The results of the analysis show for instance a highly significant relationship between the average value of X_2 (calculated over all six repeated measurements) and the outcome variable Y at $t = 6$; a relationship which can be interpreted in such a way that a 1 point higher “long-term exposure” to X_2 is associated with a 0.37 point higher value for Y at $t = 6$.

The last mentioned cross-sectional method that can be used to analyze a longitudinal relationship is based on the individual linear regression lines between outcome variable Y and time. The individual regression coefficients with time (i.e. the slopes) are then used as outcome variable in a linear regression analysis relating the development of outcome variable Y to several covariates. It has already been mentioned that the covariates can be modeled in many different ways, depending on the research question at issue. In this example the relationship between the values of all covariates at $t = 1$ and the slopes of the individual regression lines of outcome variable Y was investigated, in order to obtain an answer to the question of whether or not the development in outcome variable Y can be predicted by covariates measured at the first measurement ($t = 1$). The result of this analysis is shown in Table 4.3.

In this analysis, both X_2 (measured at $t = 1$) and X_4 are significantly and positively related to the linear increase in the outcome variable Y between $t = 1$ and $t = 6$.

Table 4.3 Results of a linear regression analysis relating values of the covariates measured at $t = 1$ to the slopes of the individual regression lines between outcome variable Y and time

	Regression coefficient	Standard error	p -value
X_1	0.05	0.06	0.41
X_2	0.03	0.01	0.008
X_3	-0.02	0.07	0.74
X_4	0.06	0.03	0.02

Subjects with $X_4 = 2$ (i.e. females) have a 0.06 higher slope than the subjects with $X_4 = 1$ (i.e. males). The way the “slope” has to be interpreted depends on the way time is modeled. Because in the example dataset the outcome variable Y is measured at yearly intervals, the slope of the linear regression with time can be interpreted as the yearly increase in outcome variable Y .

So far, three relatively simple analyses have been performed to investigate the “longitudinal” relationship between the outcome variable Y and the four covariates X_1 to X_4 . It should be stressed that although all three analyses were based on the same dataset, and were performed to determine the longitudinal relationship between outcome variable Y and the four covariates, the different analyses produce different results. It should be realized that longitudinal relationships can be very complicated, and that different types of analysis could be performed to investigate different aspects of longitudinal relationships.

4.4 Longitudinal methods

With the development of (new) statistical techniques, such as GEE analysis and mixed model analysis, it has become possible to analyze longitudinal relationships using all available longitudinal data, without summarizing the longitudinal development of each subject into one value. The longitudinal relationship between a continuous outcome variable Y and one or more covariate(s) X can be described by Equation 4.1.

$$Y_{it} = \beta_0 + \sum_{j=1}^J \beta_{1j} X_{ijt} + \varepsilon_{it} \quad (4.1)$$

where Y_{it} are observations for subject i at time t , β_0 is the intercept, X_{ijt} is the covariate j for subject i at time t , β_{1j} is the regression coefficient for covariate j , J is the number of covariates, and ε_{it} is the “error” for subject i at time t .

This model is almost the same as a cross-sectional linear regression model, except for the subscripts t . These subscripts indicate that the outcome variable Y is repeatedly measured on the same subject (i.e. the definition of a longitudinal study), and that the covariate X can be repeatedly measured on the same subject. In this model the coefficients of interest are β_{1j} , because these regression coefficients show the magnitude of the longitudinal relationship between the outcome variable (Y_{it}) and the covariates (X_{ijt}).

Based on a long data structure (see [Figure 1.1](#)), the regression coefficients for each of the covariates can be estimated with a cross-sectional linear regression analysis. However, one of the assumptions of a cross-sectional linear regression analysis is that the observations are independent of each other. In a longitudinal dataset, the observations performed on the same subject are highly dependent on each other and therefore a cross-sectional linear regression analysis cannot be used to estimate the regression coefficients of Equation 4.1. Because of the dependency of the repeated observations within one subject, the relationship between X and Y must be adjusted for the subject (Equation 4.2).

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 id_number + \varepsilon_{it} \quad (4.2)$$

where, Y_{it} are observations for subject i at time t , β_0 is the intercept, X_{it} is the covariate for subject i at time t , β_1 is the regression coefficient for the covariate, β_2 is the regression coefficient for the variable representing subject i , id_number is the variable representing subject i , and ε_{it} is the “error” for subject i at time t .

When the id_number is added as a discrete or continuous variable to the regression model, the regression coefficient (β_2) has a very strange interpretation; i.e. when the id_number differs with one unit, the outcome variable Y differs with β_2 units. This assumes a linear relationship between the id_number and the outcome variable Y , which is rather strange. The problem is that the variable id_number is not a discrete or continuous variable, but it is a categorical one. When a categorical variable is added to a regression model, it should be represented by dummy variables. In the example dataset, there are 147 subjects, so 146 dummy variables are needed to adjust for the subject (Equation 4.3).

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 id_number2 + \beta_3 id_number3 + \dots \\ + \beta_{147} id_number147 + \varepsilon_{it} \quad (4.3)$$

where, Y_{it} are observations for subject i at time t , β_0 is the intercept, X_{it} is the covariate for subject i at time t , β_1 is the regression coefficient for the covariate, β_2 is the regression coefficient for the dummy variable representing subject two, β_3 is the regression coefficient for the dummy variable representing subject three, β_{147} is the regression coefficient for the dummy variable representing subject 147, and ε_{it} is the “error” for subject i at time t .

Using so many dummy variables in a cross-sectional linear regression model is a very inefficient way to adjust for the subject, especially because the magnitude of the differences in the outcome variable Y between the subjects (which is the interpretation for the regression coefficients belonging to the dummy variables representing the subjects) is neither interesting nor informative. In fact, because of this problem, longitudinal data analysing techniques are developed. The general idea behind a longitudinal data analysing technique is that the adjustment for the subject is performed in a very efficient way. The different sophisticated techniques that are available for the analyses of longitudinal data differ from each other in the way they perform this adjustment.

It should be noted that the regression models with the efficient adjustment for the subject are basically the same as cross-sectional regression models. This means that the covariates can be continuous, dichotomous, or categorical. For the latter, the same procedure as in cross-sectional linear regression analyses has to be followed, i.e. dummy variables must be created for each of the categories. In the following sections two sophisticated methods (GEE analysis and mixed model analysis) will be discussed. Both techniques are highly suitable for estimation of the regression coefficients of the general model given in Equation 4.1.

4.5 Generalized estimating equations

4.5.1 Introduction

With GEE analysis the relationships between the variables of the model at different time-points are analyzed simultaneously. So, the estimated β_1 reflects the longitudinal relationship between the outcome variable Y and the corresponding covariates X , using all available longitudinal data (Figure 4.3). GEE analysis is an iterative procedure, using quasi-likelihood to estimate the regression coefficients (Liang and Zeger, 1986; Zeger and Liang, 1986; Zeger et al., 1988; Zeger and Liang, 1992; Liang and Zeger, 1993; Lipsitz et al., 1994a). An extensive explanation of the details of quasi-likelihood goes beyond the scope of this book, but it can be found in several other publications (McCullagh, 1983; Nelder and Pregibon, 1987; Zeger and Qaqish, 1988; Nelder and Lee, 1992; Diggle et al., 1994).

4.5.2 Working correlation structures

It has already been mentioned that the general idea behind a longitudinal data analysis is an efficient adjustment for the subject. Within GEE analysis, this adjustment is carried out by assuming a priori a certain “working” correlation structure for the repeated measurements of the outcome variable Y . Depending on the software package used to estimate the regression coefficients, there is a choice between various correlation structures. The first possibility is an independent structure. With this structure the correlations between subsequent measurements are assumed to

be zero. In fact, this option is counterintuitive because a special technique is being used to adjust for the dependency of the observations and this correlation structure assumes independence of the observations:

	t_1	t_2	t_3	t_4	t_5	t_6
t_1	–	0	0	0	0	0
t_2	0	–	0	0	0	0
t_3	0	0	–	0	0	0
t_4	0	0	0	–	0	0
t_5	0	0	0	0	–	0
t_6	0	0	0	0	0	–

A second possible choice for a working correlation structure is an exchangeable structure. In this structure the correlations between subsequent measurements are assumed to be the same, irrespective of the length of the time interval:

	t_1	t_2	t_3	t_4	t_5	t_6
t_1	–	ρ	ρ	ρ	ρ	ρ
t_2	ρ	–	ρ	ρ	ρ	ρ
t_3	ρ	ρ	–	ρ	ρ	ρ
t_4	ρ	ρ	ρ	–	ρ	ρ
t_5	ρ	ρ	ρ	ρ	–	ρ
t_6	ρ	ρ	ρ	ρ	ρ	–

A third possible working correlation structure, the so-called (stationary) m -dependent structure assumes that the correlations t measurements apart are equal, the correlations $t + 1$ measurements apart are equal, and so on for $t = 1$ to $t = m$. Correlations more than m measurements apart are assumed to be zero. When, for instance, a “2-dependent correlation structure” is assumed, all correlations one measurement apart are assumed to be the same, all correlations two measurements apart are assumed to be the same, and the correlations more than two measurements apart are assumed to be zero:

	t_1	t_2	t_3	t_4	t_5	t_6
t_1	–	ρ_1	ρ_2	0	0	0
t_2	ρ_1	–	ρ_1	ρ_2	0	0
t_3	ρ_2	ρ_1	–	ρ_1	ρ_2	0
t_4	0	ρ_2	ρ_1	–	ρ_1	ρ_2
t_5	0	0	ρ_2	ρ_1	–	ρ_1
t_6	0	0	0	ρ_2	ρ_1	–

A fourth possibility is an autoregressive correlation structure, i.e. the correlations one measurement apart are assumed to be ρ ; correlations two measurements apart are assumed to be ρ^2 ; correlations t measurements apart are assumed to be ρ^t :

	t_1	t_2	t_3	t_4	t_5	t_6
t_1	—	ρ^1	ρ^2	ρ^3	ρ^4	ρ^5
t_2	ρ^1	—	ρ^1	ρ^2	ρ^3	ρ^4
t_3	ρ^2	ρ^1	—	ρ^1	ρ^2	ρ^3
t_4	ρ^3	ρ^2	ρ^1	—	ρ^1	ρ^2
t_5	ρ^4	ρ^3	ρ^2	ρ^1	—	ρ^1
t_6	ρ^5	ρ^4	ρ^3	ρ^2	ρ^1	—

The least restrictive correlation structure is the unstructured correlation structure. With this structure, all correlations are assumed to be different:

	t_1	t_2	t_3	t_4	t_5	t_6
t_1	—	ρ_1	ρ_2	ρ_3	ρ_4	ρ_5
t_2	ρ_1	—	ρ_6	ρ_7	ρ_8	ρ_9
t_3	ρ_2	ρ_6	—	ρ_{10}	ρ_{11}	ρ_{12}
t_4	ρ_3	ρ_7	ρ_{10}	—	ρ_{13}	ρ_{14}
t_5	ρ_4	ρ_8	ρ_{11}	ρ_{13}	—	ρ_{15}
t_6	ρ_5	ρ_9	ρ_{12}	ρ_{14}	ρ_{15}	—

In the literature it is assumed that GEE analysis is robust against a wrong choice of the correlation structure (i.e. it does not matter much which correlation structure is chosen, the results of the longitudinal analysis will be more or less the same). This robustness has to do with the fact that within a GEE analysis the Huber–White sandwich estimator is used for the estimation of the parameters of the model (Liang and Zeger, 1986; Zeger and Liang, 1986). However, this is only the case when there are no missing data and when the model correctly specifies the mean. So, when the results of analyses with different working correlation structures on real data are compared to each other, they can differ remarkably (Twisk et al., 1997; Twisk, 2004). It is therefore important to realize which correlation structure is most appropriate for the analysis. Unfortunately, within GEE analysis there is no straightforward way to determine which correlation structure should be used. One of the possibilities is to analyze the within-subject correlation structure of the observed data to find out which possible structure is the best approximation of the “real” correlation structure. Furthermore, the simplicity of the correlation structure has to be taken into account when choosing a certain working correlation structure. The number of parameters (in this case correlation coefficients) that need to be estimated differs for each of the various working correlation structures. For instance, for an exchangeable structure only one correlation coefficient has to be estimated, while for a stationary five-dependent structure, five correlation coefficients must be estimated. Assuming an unstructured correlation structure in a longitudinal study with six repeated measurements, 15 correlation coefficients must be estimated. As a result, the power and efficiency of the statistical analysis

is influenced by the choice of a certain structure. Basically, the best choice is the simplest correlation structure which fits the data well. It should be realized that, in fact, GEE analysis adjusts for correlated errors (ε_{it} in Equations 4.1–4.3). The correlated errors are caused by the correlated observations, but they are not exactly the same. Adding covariates to the longitudinal model, for instance, can lead to another correlation structure in the errors than the one approximated by the within-subject correlation structure of the observed data.

In order to enhance insight in GEE analysis, the estimation procedure can be seen as follows. First a “naïve” linear regression analysis is carried out, assuming the observations within subjects are independent. Then, based on the residuals of this analysis, the parameters of the working correlation structure are calculated. The last step is to re-estimate the regression coefficients, adjusting for the dependency of the observations. Although the whole procedure is slightly more complicated (i.e. the estimation process alternates between steps two and three, until the estimates of the regression coefficients and standard errors stabilize), it basically consists of the three above-mentioned steps (see Burton et al., 1998).

In GEE analysis, the within-subject correlation structure is treated as a “nuisance” variable (i.e. as a covariate). So, in principle, the way in which GEE analysis adjusts for the dependency of observations within one subject is the way that has been shown in Equation 4.4.

$$Y_{it} = \beta_0 + \sum_{j=1}^J \beta_{1j} X_{ijt} + \dots + CORR_{it} + \varepsilon_{it} \quad (4.4)$$

where Y_{it} are observations for subject i at time t , β_0 is the intercept, X_{ijt} is the covariate j for subject i at time t , β_{1j} is the regression coefficient for covariate j , J is the number of covariates, $CORR_{it}$ is the working correlation structure, and ε_{it} is the “error” for subject i at time t .

4.5.3 Interpretation of the regression coefficients derived from GEE analysis

Basically, the regression coefficient β_1 for a particular covariate relates the “vector” of outcomes over time to the “vector” of the covariate over time:

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \\ Y_5 \\ Y_6 \end{bmatrix} = \beta_0 + \beta_1 \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{bmatrix} + \dots$$

Unfortunately, there is no simple straightforward interpretation of the regression coefficient β_1 . In fact, GEE analysis based on the model presented here includes a “pooled” analysis of longitudinal and cross-sectional relationships; or in other words, it combines a within-subject relationship with a between-subjects relationship, resulting in one single regression coefficient. Although the interpretation of the regression coefficient seems to be different from the interpretation of a regression coefficient in a cross-sectional regression analysis, this is not the case. The regression coefficient of a cross-sectional regression analysis is interpreted as the difference in the outcome variable Y when the covariate X differs with one unit. This is further simplified by comparing two individuals who differ by one unit in the covariate X . In a longitudinal data analysis, only the latter is not the case, because the difference between two observations in a longitudinal dataset with a long data structure (see [Figure 1.2](#)) can be either within a subject or between subjects. This double interpretation has the following implications for the interpretation of the regression coefficient derived from a GEE analysis. Suppose that for a particular subject the value of an outcome variable Y is relatively high at each of the repeated measurements, and that this value does not change much over time. Suppose further that for that particular subject the value of a particular covariate X is also relatively high at each of the repeated measurements, and also does not change much over time. This indicates a longitudinal between-subjects relationship between outcome variable Y and covariate X . Suppose that for another subject the value of the outcome variable Y increases rapidly along the longitudinal period, and suppose that for the same subject this pattern is also found for covariate X . This indicates a within-subject relationship between outcome variable Y and covariate X . Both relationships are part of the overall longitudinal relationship between outcome variable Y and covariate X , so both should be taken into account in the analysis of the longitudinal relationship. The regression coefficient β_1 estimated with GEE analysis “combines” the two possible relationships into one regression coefficient. Both phenomena are illustrated in [Figure 4.3](#).

In [Chapter 6](#), alternative models will be discussed with which it is possible to obtain an estimation of only the within-subject relationships.

4.5.4 Example

4.5.4.1 Introduction

Before carrying out a GEE analysis, the within-subject correlation structure must be chosen. As mentioned before, a possible choice for this working correlation structure can be based on the correlation structure of the observed data. [Table 4.4](#) shows the observed correlation structure for outcome variable Y .

Table 4.4 Within-subject correlation structure for outcome variable Y

	Y_{t1}	Y_{t2}	Y_{t3}	Y_{t4}	Y_{t5}	Y_{t6}
Y_{t1}	–	0.76	0.70	0.67	0.64	0.59
Y_{t2}		–	0.77	0.78	0.67	0.59
Y_{t3}			–	0.85	0.71	0.63
Y_{t4}				–	0.74	0.65
Y_{t5}					–	0.69
Y_{t6}						–

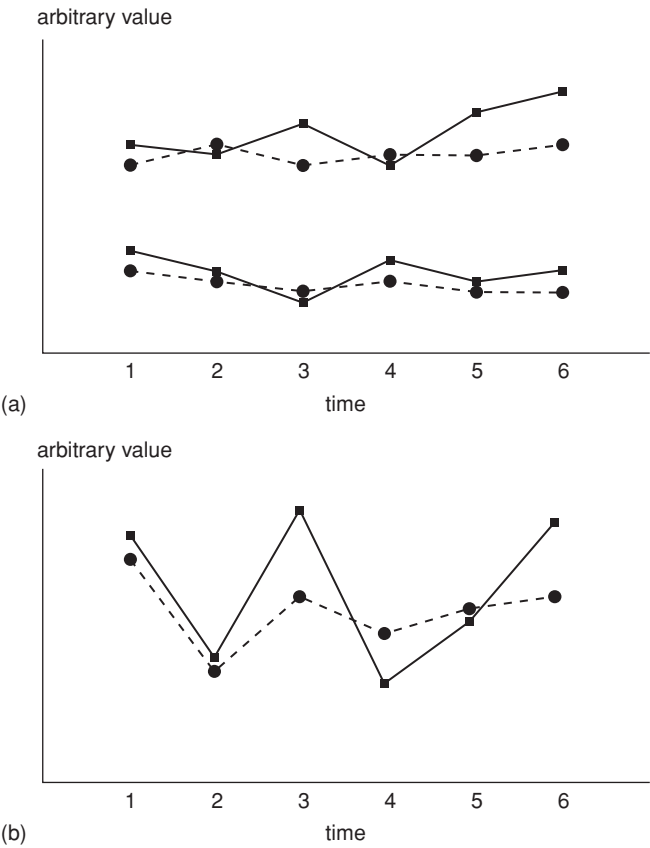


Figure 4.3 Illustration of the relationship between two continuous variables. The between-subjects relationship (a) and the within-subject relationship (b) (■ - - - outcome variable, ● - - - covariate).

The first correlation structure that should be considered is an independent structure, i.e. all correlations are assumed to be zero. From Table 4.4 it can be seen that the lowest correlation coefficient is 0.59, i.e. far from zero, so an independent correlation structure does not appear to fit the observed data. The second possibility is an exchangeable structure, i.e. all correlations are assumed to be the same. The correlation coefficients range from 0.59 to 0.85. They are not equal, but they are generally of the same magnitude. Another possible correlation structure to consider is an m -dependent structure. With six repeated measurements, the highest order for an m -dependent structure is a five-dependent structure (five time intervals). A lower-order-dependent structure does not appear to fit, because it implies that there are correlations close to zero, which is not the case in this particular situation. A five-dependent correlation structure indicates that all correlations one measurement apart are equal, all correlations two measurements apart are equal, etc. Looking at the observed correlation structure, the correlations one measurement apart range from 0.69 to 0.85; the correlations two measurements apart range between 0.65 and 0.78; the correlations three measurements apart range between 0.63 and 0.67; and the correlations four measurements apart range between 0.59 and 0.64. In other words, a five-dependent correlation structure fits the observed data quite well. From Table 4.4 it can be seen that an autoregressive correlation structure is less appropriate than a five-dependent correlation structure. An autoregressive correlation structure assumes a steep decrease in correlation coefficients when the time interval between measurements increases. Table 4.4 also shows that there is only a marginal decrease in the magnitude of the correlation coefficients with an increasing time interval. In every situation the unstructured correlation structure fits the data best, but it is questionable whether in this particular situation the loss of efficiency due to the estimation of 15 correlation coefficients is worthwhile – probably not.

So, neither an exchangeable structure nor a five-dependent structure are perfect, but both seem to fit the observed data well. In such a situation, the working correlation structure for which the least number of parameters need to be estimated is the best choice. Therefore, in this particular situation an exchangeable structure is chosen.

4.5.4.2 Results of a GEE analysis

Output 4.1 shows the results of a GEE analysis that was applied to investigate the longitudinal relationship between the outcome variable Y and the four covariates X_1 to X_4 .

Output 4.1 Results of a GEE analysis performed on the example dataset

GEE population-averaged model		Number of obs	=	882
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group: min	=	6
Family:	Gaussian	avg	=	6.0
Correlation:	exchangeable	max	=	6
		Wald chi2(4)	=	101.21
Scale parameter:	0.565295	Prob > chi2	=	0.0000

(Std. Err. adjusted for clustering on id)

ycont	Semirobust					
	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
x1	0.1481411	0.2661851	0.56	0.578	-0.3735721	0.6698543
x2	0.1848334	0.0225703	8.19	0.000	0.1405964	0.2290704
x3	0.0446735	0.0644485	0.69	0.488	-0.0816432	0.1709902
x4	0.0371133	0.1286247	0.29	0.773	-0.2149866	0.2892131
_cons	3.448975	0.6668432	5.17	0.000	2.141987	4.755964

The output is short, simple and straightforward. The left column of the first part of the output indicates that a linear GEE analysis was performed, i.e. a GEE analysis with a continuous outcome variable. This is indicated by the “link function” (i.e. identity) and the “family” (i.e. Gaussian). Furthermore it is indicated that an exchangeable correlation structure is chosen. In this part of the output also the scale parameter is given. The scale parameter is an indication of the unexplained variance of the model and can be used to acquire an indication of the explained variance of the model. To obtain this indication, Equation 4.5 must be applied.

$$R^2 = 1 - \left(\frac{S_{\text{model}}}{S_Y^2} \right) \quad (4.5)$$

where R^2 is percentage of explained variance, S_{model} is the variance of the model (given as Scale parameter in the GEE output), and S_Y^2 is the variance of the outcome variable Y , calculated over all available data.

The standard deviation of the outcome variable Y can be found in the descriptive information of the data, which is shown in Output 4.2. From Output 4.2 it can be seen that the standard deviation of outcome variable Y is 0.811. Applying Equation 4.5 to the data from the GEE analysis leads to an explained variance

Output 4.2 Descriptive information of data used in the GEE analysis

Variable	Obs	Mean	Std. Dev.	Min	Max
ycont	882	4.498141	0.811115	2.4	7.46
x1	882	1.975476	0.2188769	1.45625	2.525831
x2	882	3.743424	1.530071	1.57	12.21
x3	882	0.1746032	0.3798427	0	1
x4	882	1.530612	0.4993452	1	2

of $1 - (0.565)/(0.811)^2 = 14.1\%$. It should be stressed that this is only a (vague) indication of the explained variance of the model.

The right column of the first part of the output of the GEE analysis (Output 4.1) shows the number of observations (i.e. 882) and the number of subjects (i.e. 147). Furthermore, the average number of observation for each subject as well as the minimum and maximum number of observations is shown. Here, it can be seen that there are no missing data in the example dataset. The last part of the right column shows a Chi-square value and a corresponding p -value. The test performed here is an overall test of all covariates in the model. Because there are four covariates in the model, the number of degrees of freedom of the Chi-square test is equal to 4. It should be realized that the result of this statistical test is not very informative.

The second part of the output contains the most important part of the output. For each of the four covariates, the regression coefficient, the standard error, the z -statistic (which is obtained by dividing the regression coefficient by its standard error), the corresponding p -value and the 95% confidence interval around the regression coefficient are given. The latter can be obtained by taking the regression coefficient ± 1.96 times the standard error.

From Output 4.1 it can be seen that X_2 is the only covariate which is significantly related to the outcome variable Y , and that this association is positive. As has been mentioned before, the interpretation of the magnitude of the regression coefficient for a particular covariate is twofold: (1) The between-subjects interpretation indicates that a difference between two subjects of one unit in the covariate X_2 is associated with a difference of 0.185 units in the outcome variable Y . (2) The within-subject interpretation indicates that a change within one subject of one unit in the covariate X_2 is associated with a change of 0.185 units in the outcome variable Y . Again, the “real” interpretation of the regression coefficient is a combination of both relationships. However, from the analysis that has been performed it is not possible to determine the contribution of each part.

(Std. Err. adjusted for clustering on id)

GEE population-averaged model		Number of obs	=	882
Group and time vars:	id time	Number of groups	=	147
Link:	identity	Obs per group: min	=	6
Family:	Gaussian	avg	=	6.0
Correlation:	unstructured	max	=	6
		Wald chi2(4)	=	72.60
Scale parameter:	0.5829383	Prob > chi2	=	0.0000

(Std. Err. adjusted for clustering on id)

	Semirobust						
ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]		

x1	0.1364003	0.2905973	0.47	0.639	-0.4331599	0.7059605	
x2	0.1388265	0.0204921	6.77	0.000	0.0986627	0.1789902	
x3	0.0059453	0.0541737	0.11	0.913	-0.1002331	0.1121237	
x4	0.112248	0.1372446	0.82	0.413	-0.1567466	0.3812425	
_cons	3.635145	0.711379	5.11	0.000	2.240868	5.029422	

Table 4.5 Regression coefficients and standard errors estimated by GEE analysis with different correlation structures

	Correlation structure			
	Independent	Five-dependent	Exchangeable	Unstructured
X_1	0.19 (0.27)	0.12 (0.27)	0.15 (0.27)	0.14 (0.29)
X_2	0.20 (0.03)	0.17 (0.02)	0.18 (0.02)	0.14 (0.02)
X_3	0.08 (0.09)	0.03 (0.06)	0.04 (0.06)	0.01 (0.05)
X_4	0.02 (0.13)	0.05 (0.13)	0.04 (0.13)	0.11 (0.14)

Table 4.5 summarizes the results of the analysis with different working correlation structures. From Table 4.5 it can be seen that, although the conclusions based on p -values are the same, there are some differences in the magnitude of the regression coefficients. This is important, because it is far more interesting to estimate the magnitude of the association by means of the regression coefficients and the 95% confidence interval than just estimating p -values. Based on the results of Table 4.5, it is obvious that it is important to choose a suitable correlation structure before a GEE analysis is performed.

To put the importance of adjusting for the dependency of observations in a broader perspective the results of the GEE analysis can be compared to a “naïve” longitudinal analysis, ignoring the fact that repeated observations are carried out on the same subjects (i.e. a cross-sectional linear regression analysis carried out on a total longitudinal dataset). Output 4.4 shows the results of such a “naïve” linear regression analysis carried out on the example dataset.

A comparison between Output 4.3 and Output 4.4 indicates that the regression coefficients obtained from the “naïve” longitudinal analysis are exactly the same as the regression coefficients obtained from a GEE analysis with an independent correlation structure. The standard errors of the regression coefficients are however totally different. From both outputs it can be seen that ignoring the dependency of the observations leads to an under-estimation of the standard errors. This has to do with the fact that in the naïve analysis, it is assumed that each measurement within a particular subject provides 100% new information, while part of the information was already available in earlier measurements of that subject, reflected in the within-subject correlation coefficient. Depending on the magnitude of that coefficient, each repeated measurement within 1 subject provides less than 100% new information. This leads to larger standard errors in the adjusted analysis.

Output 4.4 Results of a “naïve” linear regression analysis performed on the example dataset

Source	SS	df	MS	Number of obs = 882			
-----+-----				F(4, 877) = 35.97			
Model	81.6909161	4	20.422729	Prob > F = 0.0000			
Residual	497.925634	877	0.56776013	R-squared = 0.1409			
-----+-----				Adj R-squared = 0.1370			
Total	579.616551	881	0.657907549	Root MSE = 0.7535			
-----+-----							
ycont	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]		
-----+-----							
x1	0.1884917	0.1461248	1.29	0.197	-0.0983034	0.4752869	
x2	0.2026222	0.0193829	10.45	0.000	0.16458	0.2406645	
x3	0.0750899	0.0682431	1.10	0.271	-0.0588491	0.2090288	
x4	0.021707	0.0651725	0.33	0.739	-0.1062053	0.1496193	
_cons	3.320943	0.3629074	9.15	0.000	2.608675	4.033211	
-----+-----							

4.6 Mixed model analysis

4.6.1 Introduction

The general idea behind mixed model analysis was initially developed in the social sciences, more specifically for educational research. Investigating the performance of pupils in schools, researchers realized that the performances of pupils within the same class are not independent, i.e. their performances are more or less correlated. Similarly, the performances of classes within the same school can be dependent on each other. This type of study design is characterized by a hierarchical structure. Students are nested within classes, and classes are nested within schools. Because various levels can be distinguished mixed model analysis is also known as multi-level analysis (Laird and Ware, 1982; Longford, 1993; Goldstein, 1995, Twisk 2006). Because the performances of pupils within one class are not independent of each other, an adjustment should be made for this dependency in the analysis of the performance of the pupils. Mixed model analysis is developed to adjust for this dependency, for instance by allowing for different regression coefficients for different classes. As this technique is suitable for correlated observations, it is obvious that it is also suitable for use in longitudinal studies. In longitudinal studies the

observations within one subject over time are correlated. The observations over time are nested within the subject.

4.6.2 Mixed models for longitudinal studies

As for all longitudinal data analyses, the general idea behind a mixed model analysis is that the adjustment for the subject is performed in a very efficient way. However, the way mixed model analysis adjusts for the subject is different from the way GEE analysis adjusts for the subject. To understand the general idea behind a mixed model analysis, it should be realized that within regression analysis an adjustment for a certain variable means that different intercepts are estimated for the different values of that particular variable. For instance, when in a cross-sectional regression analysis, an adjustment is made for gender, for males and females, different intercepts are calculated. In other words, when an adjustment is made for the subject (i.e. the *id_number*), for each subject different intercepts are calculated (see [Figure 4.4](#)). In the model in which the *id_number* is treated as a categorical variable and represented by dummy variables (see Equation 4.3), the different intercepts for each subject can be calculated by β_0 + the regression coefficient for the dummy variable representing that subject. However, it has already been mentioned that adding all the dummy variables to the model is not a very efficient way to adjust for the subject. Within a GEE analysis, the efficient adjustment for the subject was performed to estimate the within subject correlations. Within a mixed model analysis, the adjustment is performed the other way round; i.e. by estimating the differences between the subjects. The first step within a mixed model analysis is therefore to draw a normal distribution around the intercepts and in the second step the variance of that normal distribution is estimated. That variance is added to the longitudinal regression model in order to adjust for the subject in an efficient way. Because this variance is known as the random intercept, mixed model analysis is also known as random coefficient analysis. The corresponding statistical model is given in Equation 4.6.

$$Y_{it} = \beta_{0i} + \beta_1 X_{it} + \varepsilon_{it} \quad (4.6)$$

where Y_{it} are observations for subject i at time t , β_{0i} is the random intercept, X is the covariate, β_1 is the regression coefficient for X , and ε_{it} is the “error” for subject i at time t .

It should be noted that the general idea behind a mixed model analysis is the same as the general idea behind a MANOVA for repeated measurements. In Section 3.3.1 the individual sum of squares was introduced in order to take into account that the measurements over time were performed in the same subject. The individual sum of squares was in fact an estimation of the differences between the mean values over time between the subjects. The latter is actually the same as the differences between

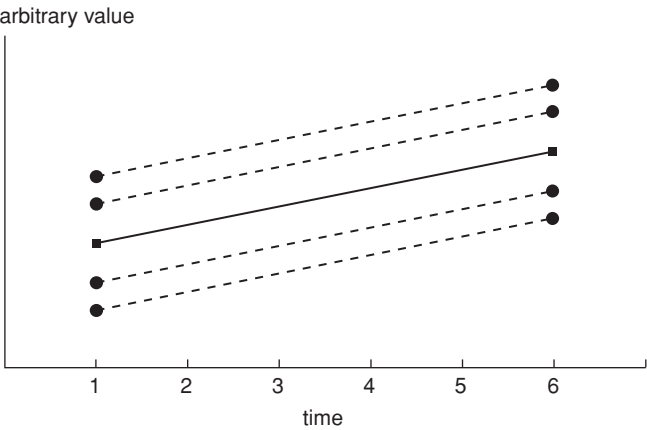


Figure 4.4 Development over time of a particular outcome variable Y ; different intercepts for different subjects (■ - - - population, ● - - - subjects 1 to n).

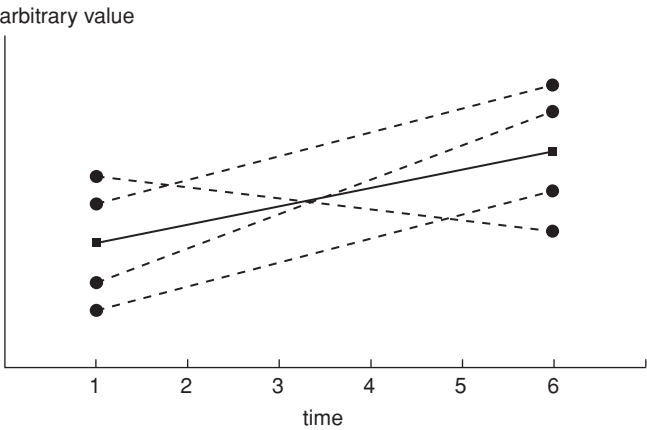


Figure 4.5 Development over time of a particular outcome variable Y ; different intercepts and different slopes for different subjects (■ - - - population, ● - - - subjects 1 to n).

the intercepts of each subject (Figure 4.4). In this way, a mixed model analysis can be seen as an extension of a MANOVA for repeated measurements.

As can be seen from Figure 4.4, a model with a random intercept allows the intercepts to differ between the subjects, but the regression coefficient for the covariate X is the same for all the subjects. In a longitudinal study it is not uncommon that besides the intercepts also the regression coefficients for X differ between the subjects (Figure 4.5). When regression coefficients for X differ between subjects, there is an interaction between the covariate X and the subject. As for the adjustment for the subject, also the interaction with the subject has to be added to a cross-sectional regression model with dummy variables; i.e. for each dummy variable,

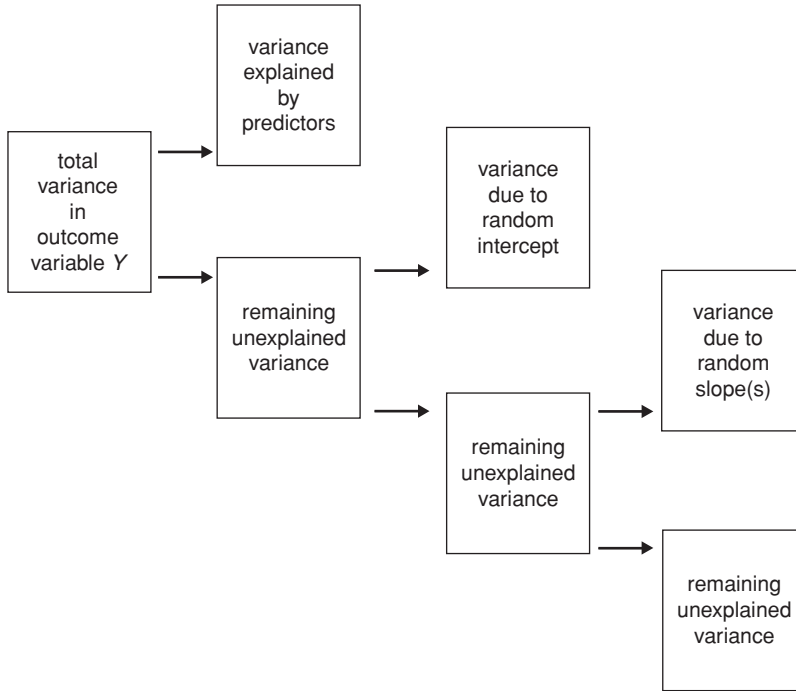


Figure 4.6 Schematic illustration of mixed model analysis; the unexplained variance in outcome variable Y is divided into different components.

an interaction term has to be created. In the example dataset with 147 subjects, for each of the 146 dummy variables representing the subjects, an interaction term has to be created. This means that in total 292 additional coefficients must be estimated, which is far from efficient. Within a mixed model analysis, the solution for this inefficiency is the same as the solution for the different intercepts. A normal distribution is drawn around the different regression coefficients and from that normal distribution the variance is estimated. This variance is then added to the regression model. Analogue to the random intercept, the variance around the different regression coefficients is known as a random slope. The corresponding statistical model is given in Equation 4.7.

$$Y_{it} = \beta_{0i} + \beta_{1i} X_{it} + \varepsilon_{it} \quad (4.7)$$

where Y_{it} are observations for subject i at time t , β_{0i} is the random intercept, X is the covariate, β_{1i} is the random regression coefficient for X , and ε_{it} is the “error” for subject i at time t .

The general idea behind a mixed model analysis is that the unexplained variance in outcome variable Y is divided into different components. One of the components is related to the random intercept and another component is related to random slopes (Figure 4.6).

Output 4.5 shows the results of a mixed model analysis in which the outcome variable Y is related to the four covariates X_1 to X_4 and in which a random intercept is modeled (Equation 4.6).

The first line of Output 4.5 refers to the fact that a mixed model analysis has been performed. Furthermore it is mentioned that it is a REML regression analysis. REML stands for Restricted Maximum Likelihood, which is the default estimation procedure for mixed model analysis within Stata. In Section 4.6.4 the difference between ordinary maximum likelihood and restricted maximum likelihood will be further discussed.

Output 4.5 Results of mixed model analysis with a random intercept

Mixed-effects REML regression	Number of obs	=	882				
Group variable: id	Number of groups	=	147				
	Obs per group: min	=	6				
	avg	=	6.0				
	max	=	6				
	Wald chi2(4)	=	103.70				
Log restricted-likelihood = -835.70823	Prob > chi2	=	0.0000				

ycont		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+							
x1		0.1479072	0.2720621	0.54	0.587	-0.3853247	0.6811391
x2		0.1847196	0.0195253	9.46	0.000	0.1464507	0.2229885
x3		0.0445538	0.059106	0.75	0.451	-0.0712918	0.1603993
x4		0.0372171	0.1199952	0.31	0.756	-0.1979691	0.2724033
_cons		3.449725	0.6648278	5.19	0.000	2.146687	4.752764

Random-effects Parameters			Estimate	Std. Err.	[95% Conf. Interval]		
-----+							
id: Identity							
var(_cons)			0.2993728	0.0407707	0.2292396 0.3909623		
-----+							
var(Residual)			0.2741017	0.0143153	0.2474324 0.3036455		

LR test vs. linear regression: chibar2(01) =				348.63	Prob >= chibar2 = 0.0000		

In the last line of the left column, the log (restricted) likelihood value of the model is given (−835.70823). The log (restricted) likelihood is an indication of the adequacy (or “fit”) of the model. The value by itself is useless, but can be used in the likelihood ratio test. This likelihood ratio test is exactly the same as known from logistic regression analysis (Hosmer and Lemeshow, 1989; Kleinbaum, 1994).

In the right column of the first part of the output, the same information is given as in the output of a GEE analysis, i.e. the number of observations (882), the number of subjects (147) and the average, minimum and maximum number of observations within a subject. Again, here can be seen that there are no missing data. The Chi-square test shown in the last line of the right column is a test to evaluate the significance of the whole model. In other words, it is a test to evaluate the importance of the four covariates in the model. In principle, this test is based on the difference in $-2 \log$ (restricted) likelihood between the model presented and the model with only an intercept (including the variance around the intercept, i.e. including the random intercept). Presumably this difference in $-2 \log$ (restricted) likelihood is 103.70. This difference follows a Chi-square distribution with 4 degrees of freedom, which gives a value $p < 0.001$.

The second part of the output shows the regression coefficients of the four covariates. The information provided is comparable to the information obtained from the earlier discussed GEE analysis: For each of the covariates, the output gives the regression coefficient, the standard error, the z -statistic (which is obtained by dividing the regression coefficient by its standard error), the corresponding p -value and the 95% confidence interval around the regression coefficient. The latter is calculated in the usual way (i.e. the regression coefficient ± 1.96 times the standard error).

The last part of the output shows the random part of the model. In general, the idea of a mixed model analysis is that the overall “error” variance is divided into different parts. In this example, the overall “error” variance is divided into two parts, one which is related to the random variation around the intercept (i.e. $\text{var}(_cons)$), and one which is the remaining “error” variance (i.e. $\text{var}(\text{Residual})$). Besides the variances (0.2993728 and 0.2741017 respectively), also the standard errors and 95% confidence intervals around the variances are given. Because the variance is skewed to the right, the 95% confidence intervals around the variances are not symmetric. However, both the standard errors and 95% confidence intervals around the variances are not very informative. Based on the variation around the intercept (i.e. the differences between the subjects), the so-called intraclass correlation coefficient (ICC) can be calculated. The ICC is an indication of the within-subject dependency (Twisk, 2006) and can be calculated as the variance around the intercept divided by the total variance. In this example the ICC is equal to $0.2993728 / (0.2993728 + 0.2741017) = 52\%$.

Output 4.6 Results of a “naïve” regression analysis without a random intercept

Mixed-effects REML regression				Number of obs	=	882	
				Wald chi2(4)	=	143.88	
Log restricted-likelihood = -1010.0256				Prob > chi2	=	0.0000	

ycont		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

x1		0.1884917	0.1461248	1.29	0.197	-0.097907	0.4748911
x2		0.2026222	0.0193829	10.45	0.000	0.164632	0.240612
x3		0.0750899	0.0682431	1.10	0.271	-0.0586643	0.208844
x4		0.021707	0.0651725	0.33	0.739	-0.1060288	0.1494428
_cons		3.320943	0.3629074	9.15	0.000	2.609658	4.032228

Random-effects Parameters			Estimate	Std. Err.		[95% Conf. Interval]	
var(Residual)			0.5677601	0.0270362		0.5171678	0.6233017

For illustrative purposes, Output 4.6 shows the results of an analysis in which no random intercept is considered (i.e. a naïve cross-sectional regression analysis). First of all, it can be seen that the total error variance (i.e. 0.5677601) is comparable to the sum of the two error variances shown in Output 4.5. Secondly, the log (restricted) likelihood of this model is -1010.0256 . Performing the likelihood ratio test between the model with and without a random intercept gives (as expected from Output (4.6)) a value of 348.63, which is highly significant.

It should be realized that the likelihood ratio test performed to evaluate whether or not a random intercept must be considered is not very useful in a longitudinal study. Not adding a random intercept to the model is theoretically wrong, because such a model ignores the dependency of repeated observations within one subject. Or in other words, a model without a random intercept ignores the longitudinal nature of the data.

In the two models considered, the regression coefficients for the four covariates are considered to be fixed (i.e. not assumed to vary between individuals). The next step in the modeling process is to add (a) random slope(s) to the model, i.e. to let the regression coefficients vary among subjects (Equation 4.7). Adding a random slope to the model is only possible for covariates that vary over time. In the present example, a random slope can, therefore, only be added for X_2 and X_3 . The result of a mixed model analysis with both a random intercept and a random slope for X_2 is shown in Output 4.7.

Output 4.7 Results of mixed model analysis with both a random intercept and a random slope for X_2

Mixed-effects REML regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
	Wald chi2(4)	=	83.26			
Log restricted-likelihood = -834.28274	Prob > chi2	=	0.0000			

ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
x1	0.187152	0.2772225	0.68	0.500	-0.3561942	0.7304981
x2	0.1916599	0.0223166	8.59	0.000	0.1479202	0.2353996
x3	0.0430607	0.0590371	0.73	0.466	-0.0726499	0.1587712

x4		0.0316736	0.1218081	0.26	0.795	-0.207066	0.2704131
_cons		3.364458	0.6779774	4.96	0.000	2.035646	4.693269

Random-effects Parameters			Estimate	Std. Err.	[95% Conf. Interval]		

id: Unstructured							
	var(x2)		0.0102836	0.0073646	0.0025267	0.0418539	
	var(_cons)		0.4929815	0.1619076	0.258985	0.9383974	
	cov(x2,_cons)		-0.044978	0.0324376	-0.1085546	0.0185986	

	var(Residual)		0.2636027	0.0150232	0.2357427	0.2947552	

LR test vs. linear regression:			chi2(3) =	351.49	Prob > chi2 = 0.0000		

Output 4.7 looks more or less the same as the output of a mixed model analysis with only a random intercept (Output 4.5). The important information in the first part of the output is the log (restricted) likelihood (i.e. -834.28274). This is the likelihood value related to the total model, including the regression coefficients (the fixed part) and the variance components (the random part). This value can be used to evaluate the importance of the inclusion of a random slope for X_2 in the model. Therefore, the -2 log (restricted) likelihood of this model must be compared to the -2 log (restricted) likelihood of the model without a random slope for X_2 . The difference between the -2 log (restricted) likelihoods is $1671.416 - 1668.565 = 2.85$. This value follows a Chi-square distribution with a number of degrees of freedom equal to the difference in the number of parameters estimated by the two models. Although only a random slope is added to the model it can be seen from the output that two additional parameters are estimated. Obviously, one of the estimated parameters is the variance of the slope ($\text{var}(x2)$), and the other (not so obviously) is the covariance between the random intercept and the random slope ($\text{cov}(x2, \text{_cons})$). The magnitude and direction of the covariance between the random intercept and the random slope give information about the interaction between random intercept and slope. When a negative covariance is found, subjects with a high intercept have lower slopes. When a positive covariance is found, subjects with a high intercept also have a high slope (Figure 4.7).

Because the covariance between the random intercept and random slope is also added to the random part of the model, the model with a random slope has two more parameters than the model with only a random intercept. So, the value calculated earlier with the likelihood ratio test (i.e. 2.85) follows a

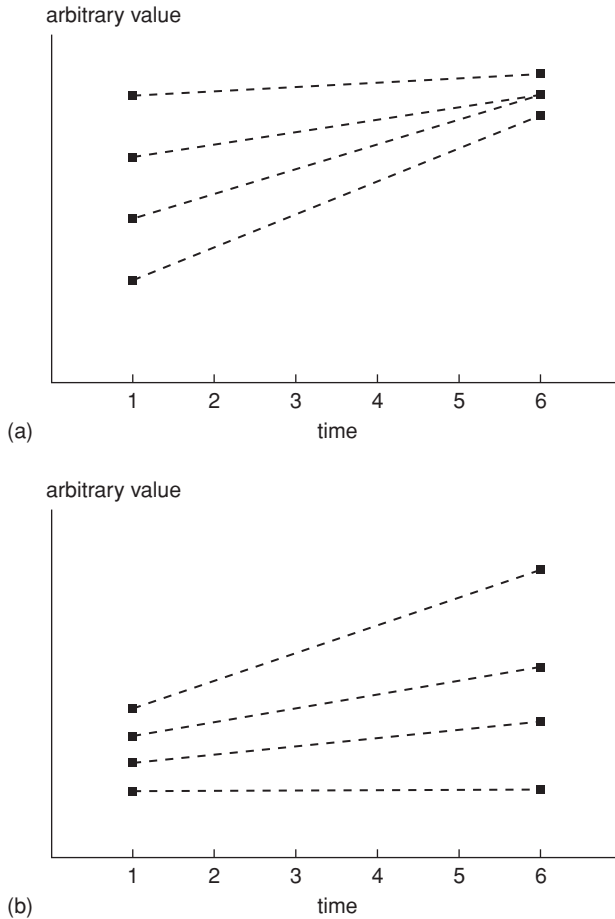


Figure 4.7 (a) Negative correlation between slope and intercept. (b) Positive correlation between slope and intercept.

Chi-square distribution with 2 degrees of freedom. This value is not statistically significant, so in this situation a random slope for X_2 does not seem to be important. As mentioned before, also for X_3 a random slope can be added to the model. Output 4.8 shows the results of a mixed model analysis with both a random intercept and a random slope for X_3 .

Based on the log (restricted) likelihood of the model with both a random intercept and a random slope for X_3 , it can be concluded that also for X_3 the slopes are not significantly different for each subject. The model with an additional random slope for X_3 is not significantly better than a model with only a random intercept. So, the “best” mixed model to analyze the relationship between the outcome variable Y and the four covariates is a model with only a random intercept.

Output 4.8 Results of mixed model analysis with both a random intercept and a random slope for X_3

Mixed-effects REML regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
	Wald chi2(4)	=	102.70			
Log restricted-likelihood = -835.63169	Prob > chi2	=	0.0000			

ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+						
x1	0.1467612	0.2721704	0.54	0.590	-0.386683	0.6802054
x2	0.1844302	0.019566	9.43	0.000	0.1460816	0.2227788
x3	0.0444509	0.0609785	0.73	0.466	-0.0750647	0.1639666
x4	0.0373861	0.1200335	0.31	0.755	-0.1978751	0.2726474
_cons	3.452987	0.665159	5.19	0.000	2.1493	4.756675

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+						
id: Unstructured						
var(x3)		0.0161873	0.0432701		0.0000859	3.051447
var(_cons)		0.3002905	0.0422681		0.2278919	0.3956893
cov(x3,_cons)		-0.0055446	0.0406274		-0.0851728	0.0740837
-----+						
var(Residual)		0.2724566	0.0148091		0.244924	0.3030843

LR test vs. linear regression:		chi2(3) =	348.79	Prob > chi2 = 0.0000		

The interpretation of the regression coefficients of the four covariates from a mixed model analysis is exactly the same as the interpretation of the regression coefficients estimated with GEE analysis, so the interpretation is twofold: (1) The between-subjects interpretation indicates that a difference between two subjects of 1 unit in, for instance, the covariate X_2 is associated with a difference of 0.185 units in the outcome variable Y . (2) The within-subject interpretation indicates that a change within one subject of 1 unit in the covariate X_2 is associated with a

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
id: Identity					
	var(_cons)	0.2918447	0.0394608	0.2239029	0.3804032
	var(Residual)	0.2734503	0.014266	0.2468707	0.3028917
LR test vs. linear regression: chibar2(01) = 345.07 Prob >= chibar2 = 0.0000					

From Output 4.9 it can be seen that all the estimated parameters (i.e. regression coefficients, standard errors, log likelihood, and random variances) of a mixed model analysis with only a random intercept estimated with maximum likelihood are slightly different from the parameters estimated with a restricted maximum likelihood estimation procedure (Output 4.5). However, the differences are very small.

4.7 Comparison between GEE analysis and mixed model analysis

In the foregoing paragraphs the general ideas behind GEE analysis and mixed model analysis were discussed. Both methods are highly suitable for the analysis of longitudinal data, because in both methods an adjustment is made for the dependency of the observations within one subject: in GEE analysis by assuming a certain working correlation structure, and in mixed model analysis by allowing the regression coefficients to vary between subjects. The question then arises: “Which of the two methods is better?” Also, which method is the most appropriate to answer the research question: “What is the longitudinal relationship between the outcome variable *Y* and several covariates *X*?” Unfortunately, no clear answer can be given. Theoretically, GEE analysis with an exchangeable correlation structure is the same as a mixed model analysis with only a random intercept. The adjustment for the dependency of observations with an exchangeable “working correlation” structure is the same as allowing subjects to have different intercepts. When an exchangeable correlation structure is appropriate, and there is no random variation in one of the estimated regression coefficients (except the intercept), GEE analysis, and mixed model analysis are equally appropriate. When an exchangeable correlation structure is not appropriate, GEE analysis with a different correlation structure can be used and when there is significant and relevant variation in one (or more) of the regression coefficients, mixed model analysis has the additional possibility of allowing other coefficients to vary between subjects. The latter makes mixed model analysis slightly more flexible compared to GEE analysis,

especially because the necessity of random slopes can be statistically evaluated with the likelihood ratio test.

It should be noted that in the example dataset a mixed model analysis with only a random intercept estimated with maximum likelihood estimation procedure give exactly the same results as a GEE analysis with an exchangeable correlation structure without a robust estimation of the standard errors (Output 4.10).

Output 4.10 Results of a GEE analysis without a robust estimation of the standard errors

GEE population-averaged model		Number of obs	=	882
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group: min	=	6
Family:	Gaussian	avg	=	6.0
Correlation:	exchangeable	max	=	6
		Wald chi2(4)	=	104.54
Scale parameter:	0.565295	Prob > chi2	=	0.0000

ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
x1	0.1481411	0.2690742	0.55	0.582	-0.3792346	0.6755168
x2	0.1848334	0.019469	9.49	0.000	0.1466748	0.222992
x3	0.0446735	0.0589844	0.76	0.449	-0.0709338	0.1602808
x4	0.0371133	0.1186888	0.31	0.755	-0.1955125	0.269739
_cons	3.448975	0.6575944	5.24	0.000	2.160114	4.737837

To put the difference between GEE analysis and mixed model analysis in a broader perspective, Table 4.6 summarizes the results of GEE analyses with and without a robust estimation of the standard errors and mixed model analyses performed with a maximum and restricted maximum likelihood estimation procedure.

It is very important to realize that the differences and equalities between GEE analysis and mixed model analysis described in this section only hold for continuous outcome variables and for datasets without missing data. For dichotomous and categorical outcome variables, the situation is different (see Chapters 7 and 8), as well as for datasets with missing data (see Chapter 10).

To facilitate the discussion, all the models have been restricted to simple linear models, i.e. no squared terms, no interactions between covariates, etc. This does not mean that it is not possible to use more complicated models. In fact, both GEE analysis and mixed model analysis with a continuous outcome variable are extensions of cross-sectional linear regression analysis. This means that all assumptions

Table 4.6 Summary of the results of different GEE analyses with an exchangeable correlation structure and different mixed model analyses with only a random intercept

	GEE analysis		Mixed model analysis	
	Not robust	Robust	ML ^a	REML ^b
X_1	0.148 (0.269)	0.148 (0.267)	0.148 (0.269)	0.148 (0.272)
X_2	0.185 (0.019)	0.185 (0.023)	0.185 (0.019)	0.185 (0.020)
X_3	0.045 (0.059)	0.045 (0.064)	0.045 (0.059)	0.045 (0.059)
X_4	0.037 (0.119)	0.037 (0.129)	0.037 (0.119)	0.037 (0.120)

^aML, maximum likelihood.
^bREML, restricted maximum likelihood.

known for cross-sectional linear regression analysis also hold for both GEE analysis and mixed model analysis. Besides that, the way for instance confounding and effect modification is investigated in cross-sectional linear regression analysis is exactly the same as for GEE analysis and mixed model analysis.

4.7.1 The “adjustment for covariance” approach

In some software packages (e.g. SAS and SPSS) the adjustment for the correlated observations within the subject can be performed with the “adjustment for covariance” approach. This approach is comparable to a GEE analysis, but instead of using a correlation matrix, a covariance matrix is used to adjust for the correlated observations within the subject. The “covariance” between two measurements is a combination of the correlation between the two measurements and the variances of the two measurements (Equation 4.8).

$$cov(Y_t, Y_{t+1}) = corr(Y_t, Y_{t+1}) \times sd(Y_t) \times sd(Y_{t+1})$$

(4.8)

where $cov(Y_t, Y_{t+1})$ is the covariance between Y at time t and Y at time $t + 1$, $corr(Y_t, Y_{t+1})$ is the correlation between Y at t and Y at $t + 1$, $sd(Y_t)$ is the standard deviation of Y at time t and $sd(Y_{t+1})$ is the standard deviation of Y at time $t + 1$.

Comparable to the adjustment for the correlation between the repeated observations used in GEE analysis, there are many different possibilities for the adjustment for “covariance” between repeated observations (see for instance Jennrich and Schluchter, 1986; Littel et al., 2000). Again, basically the adjustment is made for the “error covariance,” which is equal to the observed covariance of the repeated observations in an analysis without any covariates. In Section 12.7 the “adjustment for covariance” approach is further explained and illustrated.

4.7.2 Extensions of mixed model analysis

To summarize, in theory, a GEE analysis with an exchangeable correlation structure is the same as a mixed model analysis with only a random intercept. The assumption of mixed model analysis is that the intercepts of different individuals are normally distributed with mean zero and a certain variance (which is estimated by the statistical software package and given in the output). Although this assumption is quite sufficient in many situations, sometimes the individual intercepts are not normally distributed. This problem is mostly caused by skewness of the outcome variable of interest and can be solved by a proper transformation of the outcome variable. However, some software packages provide the possibility of modeling (to some extent) the distribution of the variation in the regression coefficients (see for instance Rabe-Hesketh et al., 2001a).

In GEE analysis there is some flexibility in modeling the correlation structure, which is not available in mixed model analysis. Therefore, in some software packages, the mixed model analysis can be extended by adding an additional correlation structure to the model. The possible correlation structures are basically the same as has been described for GEE analysis (see Section 4.5.2). In fact, this additional adjustment can be carried out when the random coefficients are not sufficient to adjust for the dependency of observations. In more technical terms, despite the adjustment made by the random coefficients, the “error” is still correlated within subjects, which indicates that an additional adjustment is necessary.

4.7.3 Comments

In the foregoing sections it was often mentioned that sophisticated analyses are needed to adjust for correlated observations. However, basically the adjustment in longitudinal data analysis is carried out for correlated “errors.” When there are no covariates in the model, the magnitude of the within-subject correlation in the observations is equivalent to the magnitude of the within-subject correlation in the “errors.” It is possible that by adding certain covariates to the model (part of) the within-subject correlation is “explained.” Because of this in the literature “correlated observations” is sometimes followed by “given the covariates in the statistical model.” This issue can be illustrated nicely by comparing the observed correlation matrix for the outcome variable Y (Table 4.4) with the estimated within-subject correlation matrix derived from a GEE analysis. Output 4.11 shows the estimated within-subject correlation matrix derived from a GEE analysis to analyze the longitudinal relationship between the outcome variable Y and the covariates X_1 to X_4 .

Output 4.11 Correlation matrix derived from a GEE analysis with an exchangeable correlation structure to analyze the longitudinal relationship between the outcome variable Y and the covariates X_1 to X_4

Estimated within-id correlation matrix R:

	c1	c2	c3	c4	c5	c6
r1	1.0000					
r2	0.5163	1.0000				
r3	0.5163	0.5163	1.0000			
r4	0.5163	0.5163	0.5163	1.0000		
r5	0.5163	0.5163	0.5163	0.5163	1.0000	
r6	0.5163	0.5163	0.5163	0.5163	0.5163	1.0000

From Output 4.11 it is obvious that the correlation of the estimated within-subject correlation matrix is much lower than the average observed within-subject correlation in the outcome variable Y . This indicates that part of the within-subject correlations are “explained” by the four covariates.

The modeling of time

5.1 The development over time

In [Chapter 3](#), multivariate analysis of variance (MANOVA) for repeated measurements was introduced as a way to analyze the development over time in a continuous outcome variable. With the sophisticated methods introduced in [Chapter 4](#), i.e. GEE analysis and mixed model analysis, it is also possible to analyze the development over time. To do so, the longitudinal regression analysis has to be performed with the time variable as covariate of interest. The simplest way of analyzing the development over time in a continuous outcome variable is to assume a linear development. [Output 5.1](#) shows the results of a GEE analysis in which the linear development over time is analyzed in the continuous outcome variable *Y*.

Output 5.1 Linear development over time in the continuous outcome variable *Y*. Results of a GEE analysis with an exchangeable correlation structure

GEE population-averaged model		Number of obs	=	882
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group: min	=	6
Family:	Gaussian	avg	=	6.0
Correlation:	exchangeable	max	=	6
		Wald chi2(1)	=	126.24
Scale parameter:	.6114334	Prob > chi2	=	0.0000
(Std. Err. adjusted for clustering on id)				

		Semirobust		
ycont		Coef.	Std. Err.	z P> z [95% Conf. Interval]

time		0.1252128	0.0111442	11.24 0.000 0.1033705 0.1470552
_cons		4.059896	0.0563758	72.01 0.000 3.949401 4.17039

From Output 5.1 it can be seen that there is a significant increase over time in the outcome variable *Y*. Because time is coded as yearly intervals (1, 2, 3, 4, 5, and 6) this increase is with 0.1252128 units per year. The same analysis can also be performed with a mixed model analysis. Output 5.2 shows the results of a mixed model analysis with only a random intercept.

Output 5.2 Linear development over time in the continuous outcome variable *Y*. Results of a mixed model analysis with only a random intercept

Mixed-effects REML regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
	Wald chi2(1)	=	163.51			
Log restricted-likelihood = -807.80797	Prob > chi2	=	0.0000			

ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+						
time	0.1252128	0.0097921	12.79	0.000	0.1060206	0.144405
_cons	4.059896	0.0629008	64.54	0.000	3.936612	4.183179

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]			
-----+						
id: Identity						
var(_cons)	0.3678297	0.047911			0.2849542	0.4748084
-----+						
var(Residual)	0.2466653	0.0128758			0.2226772	0.2732376

LR test vs. linear regression: chibar2(01) = 465.43 Prob >= chibar2 = 0.0000						

As expected, the magnitude of the regression coefficient for time is exactly the same as the one estimated with a GEE analysis with an exchangeable correlation structure. The difference between the two analyses is the standard error of the regression coefficient, which is a bit higher for the coefficient estimated with GEE analysis. This is caused by the robust estimation of the standard error within a GEE analysis (see Section 4.5.2). The next step in the mixed model analysis is to add a random slope for time to the model. Output 5.3 shows the result of that analysis.

Output 5.3 Linear development over time in the continuous outcome variable Y. Results of a mixed model analysis with both a random intercept and a random slope for time

Mixed-effects REML regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
	Wald chi2(1)	=	126.24			
Log restricted-likelihood = -798.63666	Prob > chi2	=	0.0000			

ycont		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----						
time		0.1252128	0.0111442	11.24	0.000	0.1033705 0.1470551
_cons		4.059896	0.0563758	72.01	0.000	3.949401 4.17039

Random-effects Parameters			Estimate	Std. Err.	[95% Conf. Interval]	
-----+-----						
id: Unstructured						
	var(time)		0.0051947	0.0022685	0.0022072	0.0122257
	var(_cons)		0.2690938	0.0558888	0.1791081	0.4042891
	cov(time,_cons)		0.005445	0.0087543	-0.0117131	0.0226031
-----+-----						
	var(Residual)		0.2285831	0.0133312	0.2038924	0.2562637

LR test vs. linear regression:			chi2(3)	=	483.78	Prob > chi2 = 0.0000

It was already mentioned in Section 4.6.3 that within mixed model analysis, the likelihood ratio test can be used to evaluate whether or not it is necessary to add a random slope to the model. Therefore, the $-2 \log$ likelihood of the model with only a random intercept (Output 5.2) can be compared to the $-2 \log$ likelihood of the model with both a random intercept and a random slope for time (Output 5.3). The difference between the two $-2 \log$ likelihoods follows a Chi-square distribution with (in this case) 2 degrees of freedom, because besides the variance of the slopes also the covariance between the random intercept and the random slope is estimated. In the example, the difference between the two

$-2 \log$ likelihoods is equal to 18.34 ($[807.80797 - 798.63666] * 2$). On a Chi-square distribution with two degrees of freedom, this is highly significant, i.e. the model with both a random intercept and a random slope for time is significantly better than the model with only a random intercept. It is very interesting to note, that the results (both the regression coefficient and the standard error) obtained from a mixed model analysis with both a random intercept and a random slope for time are exactly the same as the result obtained from a GEE analysis with an exchangeable correlation structure.

One of the disadvantages of the analyses performed to investigate the development over time in the outcome variable Y is that a linear development is assumed, while the data suggest a more quadratic development over time (see [Chapter 3](#)). Within GEE analysis or mixed model analysis, this can be done by adding a quadratic term to the models. In statistical terms, this means that a second order polynomial function with time is modeled. [Output 5.4](#) shows the results of this GEE analysis.

Output 5.4 Quadratic development over time in the continuous outcome variable Y . Results of a GEE analysis with an exchangeable correlation structure

GEE population-averaged model		Number of obs	=	882
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group: min	=	6
Family:	Gaussian	avg	=	6.0
Correlation:	exchangeable	max	=	6
		Wald chi2(2)	=	248.92
Scale parameter:	.5612264	Prob > chi2	=	0.0000
(Std. Err. adjusted for clustering on id)				

		Semirobust		
ycont		Coef.	Std. Err.	z P> z [95% Conf. Interval]

time		-0.5035797	0.0435872	-11.55 0.000 -0.5890091 -0.4181503
time_2		0.0898275	0.0064936	13.83 0.000 0.0771002 0.1025548
_cons		4.898286	0.0768462	63.74 0.000 4.74767 5.048901

The question that should be answered now is whether this model (assuming a quadratic development over time) is better than the model assuming a linear development over time. This can be done by evaluating the importance of the quadratic term in the model, which is normally done by looking at the significance level of the regression coefficient for the quadratic term. From [Output 5.4](#) it can be seen that the p -value belonging to the quadratic term is very low, i.e. highly

significant and therefore it can be concluded that the development over time can better be described with a quadratic function instead of a linear one. Of course, it is also possible to perform the same analysis with a mixed model analysis. Output 5.5 shows the results of a mixed model analysis assuming a quadratic development over time, with a random intercept, a random slope for time (the linear term), and a random slope for time squared (the quadratic term).

Output 5.5 Quadratic development over time in the continuous outcome variable Y. Results of a mixed model analysis with a random intercept and random slopes for time and time squared

Mixed-effects REML regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
	Wald chi2(2)	=	248.92			
Log restricted-likelihood = -668.58418	Prob > chi2	=	0.0000			

ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
time	-0.5035797	0.0435872	-11.55	0.000	-0.5890091	-0.4181503
time_2	0.0898275	0.0064936	13.83	0.000	0.0771002	0.1025548
_cons	4.898286	0.0768462	63.74	0.000	4.74767	5.048901

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]			
-----+-----						
id: Unstructured						
var(time)	0.1043057	0.0347465	0.0542943	0.2003833		
var(time_2)	0.0027767	0.0007612	0.0016225	0.0047521		
var(_cons)	0.4592842	0.1052672	0.2930809	0.7197398		
cov(time,time_2)	-0.0163864	0.0050602	-0.0263042	-0.0064686		
cov(time,_cons)	-0.115505	0.0534108	-0.2201882	-0.0108218		
cov(time_2,_cons)	0.0184654	0.0076737	0.0034253	0.0335055		
-----+-----						
var(Residual)	0.1277499	0.0086031	0.1119535	0.1457751		

LR test vs. linear regression:	chi2(6)	=	675.83	Prob > chi2	= 0.0000	

Table 5.1 Example dataset with time as a continuous variable and as a categorical variable with dummy variable coding

ID	Time (continuous)	Time (categorical)				
		Dummy 1	Dummy 2	Dummy 3	Dummy 4	Dummy 5
1	1	0	0	0	0	0
1	2	1	0	0	0	0
1	3	0	1	0	0	0
1	4	0	0	1	0	0
1	5	0	0	0	1	0
1	6	0	0	0	0	1
2	1	0	0	0	0	0
2	2	1	0	0	0	0
2	3	0	1	0	0	0
2	4	0	0	1	0	0
2	5	0	0	0	1	0
2	6	0	0	0	0	1

From Output 5.5 it can be seen that the results of the mixed model analysis with a random intercept, two random slopes, and all the covariances between the random intercept and slopes give exactly the same results as the GEE analysis with an exchangeable correlation structure. This holds for the regression coefficient as well as the standard errors.

In Chapter 3, it was already shown that the development over time for the outcome variable Y was quadratic, so it is not necessary to model the development over time with more complicated functions, such as a cubic function (i.e. an S-shaped curve). However, the procedure to evaluate a higher order function is exactly the same as has been described for a quadratic function.

The use of a mathematical function to model the development over time always assumes a particular shape of the development over time. A very elegant solution is to model time as a categorical variable instead of a continuous one. With time as a categorical variable, the development over time is modeled, without assuming a certain shape of the development. Table 5.1 illustrates part of the example dataset with time as a categorical variable.

A limitation of the use of time as a categorical variable is the fact that this is only possible when the time intervals between the repeated measurements are the same for each subject. It is obvious that with unequal time intervals between subjects, the dummy coding goes wrong. On the other hand, when the time intervals are the same for each subject, but are not equal within a subject, time can be treated as a categorical variable. The latter situation is not uncommon in longitudinal studies.

Output 5.6 Development over time in the continuous outcome variable *Y* with time treated as a categorical variable, represented by five dummy variables. Results of a GEE analysis with an exchangeable correlation structure

GEE population-averaged model		Number of obs	=	882		
Group variable:	id	Number of groups	=	147		
Link:	identity	Obs per group: min	=	6		
Family:	Gaussian	avg	=	6.0		
Correlation:	exchangeable	max	=	6		
		Wald chi2(5)	=	291.05		
Scale parameter:	.5551359	Prob > chi2	=	0.0000		
(Std. Err. adjusted for clustering on id)						

		Semirobust				
ycont		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----						
_Itime_2		-0.1115646	0.0386198	-2.89	0.004	-0.187258 -0.0358712
_Itime_3		-0.1687075	0.0438651	-3.85	0.000	-0.2546815 -0.0827335
_Itime_4		-0.2612245	0.046109	-5.67	0.000	-0.3515964 -0.1708526
_Itime_5		0.2353061	0.0518083	4.54	0.000	0.1337638 0.3368484
_Itime_6		0.6868707	0.0626649	10.96	0.000	0.5640498 0.8096917
_cons		4.434694	0.0555691	79.81	0.000	4.32578 4.543607

In the examples presented in this chapter, each subject was assumed to be measured at the same time-points. Time was simply coded as [1, 2, 3, 4, 5, 6]. However, with both GEE analysis and mixed model analysis it is possible to model the actual time of each measurement. For instance, the number of days or weeks after at the first measurement can be used as a time indicator (Table 5.2). This is sometimes more realistic, because subjects are almost never measured at exactly the same time. For each subject this indicates that a different time sequence of the measurements is modeled, which directly implies that time cannot be modeled as a categorical variable, represented by dummy variables.

Output 5.6 shows the results of a GEE analysis with an exchangeable correlation structure in order to analyze the development over time in the continuous outcome variable *Y*, with time treated as a categorical variable, represented by five dummy variables.

The regression coefficients of the five dummy variables (_Itime_2 to _Itime_6) can be interpreted as follows: compared to the first measurement (which is the reference “category”), there is a decrease in outcome variable *Y* at the second measurement ($\beta = -0.1115646$). At the third measurement the decrease continues (the

Table 5.2 Example of a dataset with four repeated measurements ($N = 3$) with time as a continuous variable with equal measurement points and time as the actual date of measurement

ID	Time (continuous)	Time (in days)
1	1	0
1	2	20
1	3	45
1	4	100
2	1	0
2	2	30
2	3	40
2	4	80
3	1	0
3	2	25
3	3	50
3	4	70

regression coefficient for the second dummy variable (-0.1687075) represents the difference between the third measurement and the first measurement), and at the fourth measurement the lowest point is reached. At the fifth and the sixth measurements the value of outcome variable Y is higher than at the first measurement, indicating a steep increase during the last two measurements. The same results can be obtained from a mixed model analysis with a random intercept, random slopes for all the dummy variables and all the covariances between the random intercept and the random slopes (Output 5.7). From Output 5.7 it can be seen, however, that although the regression coefficients and the standard errors are the same as from the GEE analysis, the random part of the model is estimated with a lot of error, i.e. the standard errors of the estimates are huge.

Output 5.7 Development over time in the continuous outcome variable Y with time treated as a categorical variable, represented by five dummy variables. Results of a mixed model analysis with a random intercept and random slopes for all five dummy variables

Mixed-effects REML regression	Number of obs	=	882
Group variable: id	Number of groups	=	147
	Obs per group: min	=	6
	avg	=	6.0
	max	=	6
	Wald chi2(5)	=	291.05
Log restricted-likelihood = -630.81988	Prob > chi2	=	0.0000

ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
+						
_Itime_2	-0.1115646	0.0386198	-2.89	0.004	-0.187258	-0.0358713
_Itime_3	-0.1687075	0.0438651	-3.85	0.000	-0.2546815	-0.0827335
_Itime_4	-0.2612245	0.046109	-5.67	0.000	-0.3515965	-0.1708525
_Itime_5	0.2353061	0.0518082	4.54	0.000	0.1337638	0.3368484
_Itime_6	0.6868707	0.0626649	10.96	0.000	0.5640498	0.8096917
_cons	4.434694	0.0555691	79.81	0.000	4.32578	4.543607

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		

id: Unstructured						
var(_Itime_2)		0.1242063	2.95098	7.43e-22	2.08e+19	
var(_Itime_3)		0.1878075	2.951029	7.92e-15	4.45e+12	
var(_Itime_4)		0.2174856	2.951076	6.13e-13	7.72e+10	
var(_Itime_5)		0.2995195	2.951239	1.23e-09	7.30e+07	
var(_Itime_6)		0.4822104	2.951651	2.97e-06	78255.36	
var(_cons)		0.4064036	1.476371	0.0003287	502.5489	
cov(_Itime_2,_Itime_3)		0.0964736	1.47563	-2.795708	2.988656	
cov(_Itime_2,_Itime_4)		0.1144373	1.475667	-2.777816	3.006691	
cov(_Itime_2,_Itime_5)		0.0794312	1.475691	-2.812871	2.971733	
cov(_Itime_2,_Itime_6)		0.0659768	1.475778	-2.826495	2.958449	
cov(_Itime_2,_cons)		-0.0674175	1.47568	-2.959698	2.824863	
cov(_Itime_3,_Itime_4)		0.1742019	1.475789	-2.718291	3.066695	
cov(_Itime_3,_Itime_5)		0.1299077	1.475794	-2.762595	3.02241	
cov(_Itime_3,_Itime_6)		0.1165638	1.475906	-2.776158	3.009286	
cov(_Itime_3,_cons)		-0.0713802	1.47575	-2.963796	2.821036	
cov(_Itime_4,_Itime_5)		0.1620663	1.475846	-2.730538	3.05467	
cov(_Itime_4,_Itime_6)		0.1477313	1.475967	-2.745111	3.040573	
cov(_Itime_4,_cons)		-0.0894499	1.475787	-2.98194	2.80304	
cov(_Itime_5,_Itime_6)		0.2041443	1.476134	-2.689025	3.097314	
cov(_Itime_5,_cons)		-0.0707122	1.475873	-2.96337	2.821946	
cov(_Itime_6,_cons)		-0.0416162	1.476054	-2.93463	2.851397	

var(Residual)		0.0475211	1.47542	1.77e-28	1.27e+25	

LR test vs. linear regression:		chi2(21) =	744.70	Prob > chi2 = 0.0000		

5.2 Comparing groups

In [Chapter 3](#) it was mentioned that with a MANOVA for repeated measurements, it is possible to compare the development over time in a continuous outcome variable between two (or more) groups. With GEE analysis and mixed model analysis, it is also possible to compare the development over time between two (or more) groups. Therefore, the interaction between the time variable and the group variable must be added to the model. In the example dataset, X_4 is a dichotomous time-independent covariate (i.e. comparing males and females) and this variable is used to illustrate this analysis. [Output 5.8](#) shows the results of the GEE analysis in order to investigate the difference in development over time between the two groups of the covariate variable X_4 .

Output 5.8 Difference in linear development over time between two groups. Results of a GEE analysis with an exchangeable correlation structure

GEE population-averaged model		Number of obs	=	882			
Group variable:	id	Number of groups	=	147			
Link:	identity	Obs per group: min	=	6			
Family:	Gaussian	avg	=	6.0			
Correlation:	exchangeable	max	=	6			
		Wald chi2(3)	=	153.54			
Scale parameter:	.5899192	Prob > chi2	=	0.0000			
(Std. Err. adjusted for clustering on id)							

		Semirobust					
ycont		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----							
time		0.0839627	0.0141201	5.95	0.000	0.0562878	0.1116377
x4		-0.0098837	0.1112563	-0.09	0.929	-0.227942	0.2081746
int_x4_time		0.0777406	0.0211624	3.67	0.000	0.036263	0.1192181
_cons		4.06514	0.0708058	57.41	0.000	3.926363	4.203917

From [Output 5.8](#) it can be seen that the interaction between X_4 and time is statistically significant (the z -value = 3.67 and the p -value is < 0.001). This indicates that the linear development over time is significantly different for males and females. For males (coded 0 in the example dataset) there is a yearly increase of 0.0839627 units per year, while for females (coded 1 in the example dataset) there is a yearly increase of $0.0839627 + 0.0777406 = 0.1617033$ units per year.

It should be noted that this analysis assumes a linear development over time, while in Section 5.1 it was already shown that the development over time in the continuous outcome variable Y could better be described with a quadratic development over time, or could be better analyzed with time as a categorical variables, represented by dummy variables. Especially the latter is very popular these days and especially in experimental studies in which the effect of an intervention is evaluated at different time-points (see [Chapter 9](#)). Output 5.9 shows the results of a GEE analysis investigating the difference in quadratic development over time between males and females (i.e. X_4 in the example dataset).

Output 5.9 Difference in quadratic development over time between two groups. Results of a GEE analysis with an exchangeable correlation structure

GEE population-averaged model		Number of obs	=	882		
Group variable:	id	Number of groups	=	147		
Link:	identity	Obs per group: min	=	6		
Family:	Gaussian	avg	=	6.0		
Correlation:	exchangeable	max	=	6		
		Wald chi2(5)	=	285.69		
Scale parameter:	.5364744	Prob > chi2	=	0.0000		
(Std. Err. adjusted for clustering on id)						

		Semirobust				
ycont		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----						
time		-0.7146061	0.0617876	-11.57	0.000	-0.8357077 -0.5935045
time_2		0.1140813	0.0093314	12.23	0.000	0.0957921 0.1323704
x4		-0.4365011	0.1493388	-2.92	0.003	-0.7291999 -0.1438024
int_x4_time		0.3977036	0.0813613	4.89	0.000	0.2382383 0.5571689
int_x4_tim~2		-0.045709	0.0124918	-3.66	0.000	-0.0701925 -0.0212255
_cons		5.129899	0.1068381	48.02	0.000	4.9205 5.339297

From Output 5.9 it can be seen that both the interaction between X_4 and time and the interaction between X_4 and time squared are statistically significant, which indicates that the difference between the quadratic development over time between the two groups of X_4 is statistically significant. However, the interpretation of the regression coefficients of this analysis is rather complicated. Therefore, if possible, time is better treated as a categorical variable, represented by dummy variables. Output 5.10 shows the results of a GEE analysis with time as a categorical variable

in order to investigate the difference in development over time between males and females (i.e. X_4 in the example dataset).

Output 5.10 Difference in development over time between two groups with time treated as a categorical variable, represented by five dummy variables. Results of a GEE analysis with an exchangeable correlation structure

GEE population-averaged model		Number of obs	=	882		
Group variable:	id	Number of groups	=	147		
Link:	identity	Obs per group: min	=	6		
Family:	Gaussian	avg	=	6.0		
Correlation:	exchangeable	max	=	6		
		Wald chi2(11)	=	343.73		
Scale parameter:	.5301184	Prob > chi2	=	0.0000		
(Std. Err. adjusted for clustering on id)						

	Semirobust					
ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
x4	-0.0547938	0.1101086	-0.50	0.619	-0.2706027	0.1610152
_Itime_2	-0.2043478	0.0604314	-3.38	0.001	-0.3227913	-0.0859044
_Itime_3	-0.3884058	0.0577875	-6.72	0.000	-0.5016673	-0.2751443
_Itime_4	-0.5130435	0.0621048	-8.26	0.000	-0.6347666	-0.3913204
_Itime_5	-0.032029	0.0654325	-0.49	0.624	-0.1602743	0.0962163
_Itime_6	0.5092754	0.0850456	5.99	0.000	0.3425891	0.6759616
_ItimXx4_2	0.1748606	0.0768525	2.28	0.023	0.0242325	0.3254888
_ItimXx4_3	0.4140468	0.0807563	5.13	0.000	0.2557673	0.5723263
_ItimXx4_4	0.4745819	0.0838169	5.66	0.000	0.3103039	0.63886
_ItimXx4_5	0.5038239	0.0944792	5.33	0.000	0.3186481	0.6889996
_ItimXx4_6	0.334699	0.1218257	2.75	0.006	0.0959251	0.5734729
_cons	4.463768	0.0734918	60.74	0.000	4.319727	4.607809

Although Output 5.10 looks a bit complicated, the regression coefficients can be interpreted in a straightforward way. The regression coefficients of the dummy variables have the same interpretation as in Output 5.6, although they now represent the development over time in the outcome variable Y for males (X_4 coded zero). For females (X_4 coded 1), the difference between the second and first measurement is equal to: $-0.2043478 + 0.1748606 = -0.0294872$, indicating a small decrease between the first and the second measurement. In the same way the difference

Table 5.3 Means and standard deviations (between brackets) of cholesterol and body weight at three repeated measurements

	32 years	36 years	42 years
Number of subjects	437	379	338
Cholesterol (mmol/l)	4.4 (0.9)	5.0 (0.9)	5.5 (0.8)
Body weight (kg)	72.7 (12.3)	75.4 (13.1)	77.6 (13.6)

From Output 5.11 it can be seen that the difference between the first and second measurement for females is equal to -0.0294872 . This number was already known from the analysis reported in Output 5.10. However, from Output 5.11, it can now be seen that the p -value which belongs to this difference is equal to 0.535.

For the sake of simplicity, in this section only the results of the GEE analyses were reported. However, of course, the same research questions can be answered with a mixed model analysis as well. The results will be exactly the same.

5.3 The adjustment for time

In Chapter 4, the longitudinal relationship between an outcome variable Y and four covariates (X_1 to X_4) was investigated. In all analyses the time variable was not included in the model. A major point of discussion in all longitudinal analyses is how to deal with the time variable in a situation when the analysis of the development over time is not the main purpose of the study. In some studies, an a priori adjustment for time is performed, while in other studies (as in the examples in Chapter 4) the time variable is ignored. In most studies, however, it is not clear whether or not a time variable is part of the statistical model. In this section, the influence of the time variable in a longitudinal data analysis will be discussed. For this purpose, a different example dataset will be used. The dataset is also taken from the Amsterdam Growth and Health Longitudinal Study (AGAHLS), but in this example a selection is made of three repeated measurements performed between 32 and 42 years of age. The purpose of the example was to investigate the relationship between cholesterol as the outcome variable and body weight as the time-dependent covariate. Table 5.3 shows descriptive information regarding the data used in this example.

In the first analysis, a GEE analysis with an exchangeable correlation structure was performed without an adjustment for time. In the second analysis, time (a continuous variable coded 1 to 3) was added to the model and the same GEE analysis was performed. Table 5.4 shows the results of the two analyses.

Table 5.4 Results of two GEE analyses regarding the longitudinal relationship between cholesterol and body weight

	Regression coefficient	Standard error	z-Value	p-Value
“Crude”	0.026	0.003	9.950	<0.001
Adjusted for time	0.016	0.002	6.735	<0.001

The results of the GEE analyses show that the regression coefficient for body weight decreases by about 40% when time is added to the model. So obviously, adding time to the model has a huge influence on the magnitude of the regression coefficient of interest. It is therefore important to address the question which of the two results gives the “real” longitudinal relationship between cholesterol and body weight?

In [Chapter 4](#) it was already mentioned that one of the most difficult parts of longitudinal data analysis is the interpretation of the regression coefficient; especially the interpretation of a regression coefficient of a time-dependent covariate, as in this example. It was already discussed that when both the outcome variable and the covariate are time-dependent the regression coefficient has to be interpreted as a combination of a between-subjects and a within-subject association (see [Section 4.5.3](#)). It is often suggested that when the time variable is added to the longitudinal model, the interpretation of the regression coefficient of a time-dependent covariate is limited to the within-subject interpretation (i.e. a change of one unit in the time-dependent covariate is associated with a change of regression coefficient units in the outcome variable). This is, however, not true. An adjustment for time only leads to a more “relative” interpretation of the regression coefficient. [Figure 5.1](#) shows an (extreme) example, which nicely illustrates the impact of an adjustment for time in a longitudinal data analysis.

From [Figure 5.1](#) it can be seen that both the outcome variable and the covariate increase over time. Furthermore, it can be seen that at each time-point both variables are hardly associated. A longitudinal analysis between the Y and X variable without adjusting for time would reveal a strong positive relationship. Adjustment for time will attenuate the relationship enormously. In fact, in an adjusted analysis, no relationship would be found between Y and X . Theoretically, by adding time to the model, the variance between the time-points is removed from the statistical model. In epidemiological terminology, time can be seen as a huge confounder in the relationship between the outcome variable Y and the covariate X . From the results reported in [Tables 5.3](#) and [5.4](#) it can be seen that in the cholesterol and body weight example more or less the same phenomenon occurs; both variables increase

Table 5.5 Results of a GEE analysis regarding the longitudinal relationship between cholesterol and body weight by using time-specific z-scores for both variables

	Regression coefficient	Standard error	z-Value	p-Value
z-Scores	0.240	0.035	6.786	<0.001

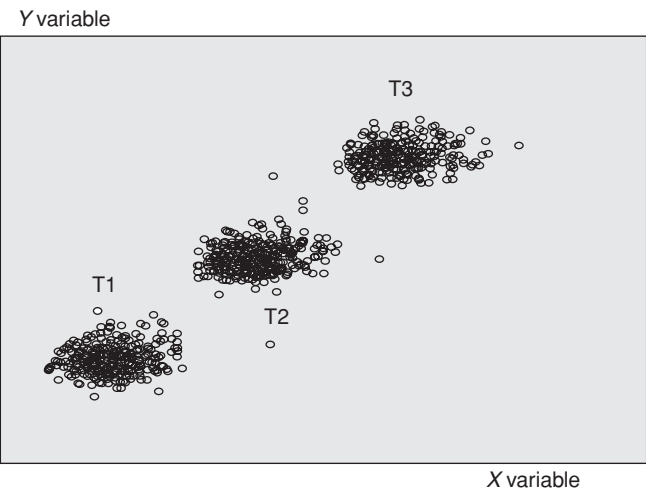


Figure 5.1 Example of the influence of an adjustment for time in a longitudinal data analysis

over time, which causes the “crude” analysis to reveal a much stronger effect than the time adjusted analysis.

The same idea can be achieved by performing an analysis with time specific z-scores (calculated at each time point separately by subtracting the mean value from the observed value and divided by the standard deviation) of the outcome variable and the covariate instead of the observed values. Basically, using time-specific z-scores also removes the variance between the time-points. In longitudinal studies, the use of time-specific z-scores is quite common, for instance, when there is a change in measurement equipment over time. To illustrate the effect of using time-specific z-scores, Table 5.5 shows the result of the GEE analysis performed on time-specific z-scores of both cholesterol and body weight instead of the observed values.

Although the regression coefficient and standard error are different from the earlier analyses reported in Table 5.4, which is due to the fact that the scale of

both variables have changed, the test statistic (z-value) is almost equal to the test statistic obtained from the time-adjusted analysis which shows that both analyses are similar.

It is important to realize that the potential confounding effect of time only holds for covariates that are time-dependent. When the longitudinal relationship is analyzed with covariates that are time-independent, an adjustment for time is not necessary. This has to do with the epidemiological definition of a confounder: a variable can only be a confounder in a particular relationship when the possible confounder is related to both the outcome and the covariate. Because time is not associated with a time-dependent covariate, time cannot be a confounder in such a relationship.

So, although it depends on the research question, time should be treated as a potential confounder; therefore, it is recommended to report both the results (i.e. with and without an adjustment for time) and interpret the possible differences between the two results. However, again this only holds for time-dependent covariates.

Other possibilities for modeling longitudinal data

6.1 Introduction

In [Chapter 4](#), GEE analysis and mixed model analysis were introduced as two (sophisticated) methods that can be used to analyze the longitudinal relationship between an outcome variable Y and several covariates X . In this chapter, the models described in [Chapter 4](#) (which are known as standard models) are slightly altered in order to answer specific research questions.

6.2 Alternative models

6.2.1 Time-lag model

It is assumed that the greatest advantage of a longitudinal study design in epidemiological research is that causal relationships can be detected. However, in fact this is only partly true for experimental designs (see [Chapter 9](#)). In observational longitudinal studies in general, no answer can be given to the question of whether a certain relationship is causal or not. With the standard models described in [Chapter 4](#), it is only possible to detect longitudinal associations between an outcome variable Y and one (or more) covariate(s) X . When there is some rationale about possible causation in observational longitudinal studies, these associations are called “quasi-causal relationships.” In every epidemiological textbook a list of arguments can be found which can give an indication as to whether or not an observed relationship is causal (see [Table 1.1](#)). One of these concerns the temporal sequence of the relationship. When the covariate X precedes the outcome variable Y , the observed relationship may be causal ([Figure 6.1](#)).

With a small change in the standard models described in [Chapter 4](#), this time sequence between covariates X and outcome variable Y can be modeled. In this so-called time-lag model the covariates X are modeled prior in time to the outcome

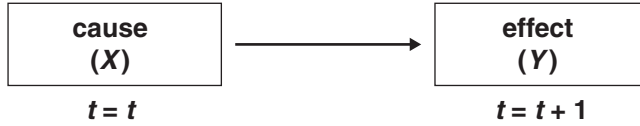


Figure 6.1 Temporal sequence between covariate (cause) and outcome variable (effect).

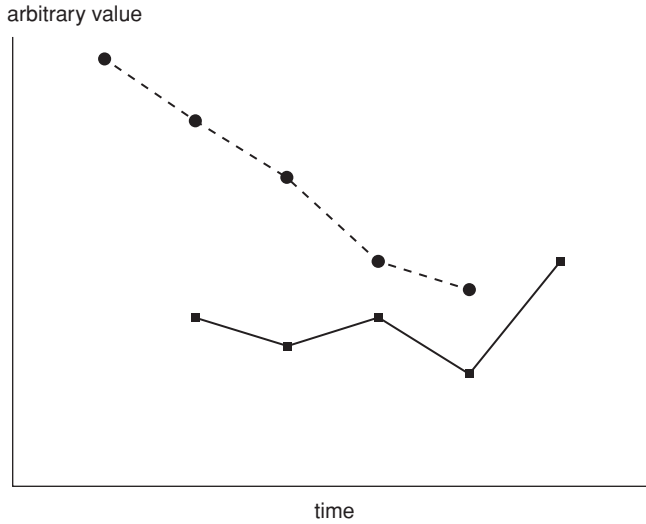


Figure 6.2 Illustration of the time-lag model; covariate X is modeled prior to the outcome variable Y (■ - - - outcome variable, ● — covariate).

variable Y (Figure 6.2). The corresponding equation is:

$$Y_{it} = \beta_0 + \sum_{j=1}^J \beta_{1j} X_{ijt-1} + \dots \quad (6.1)$$

where Y_{it} are observations for subject i at time t , β_0 is the intercept, X_{ijt-1} is the covariate j for subject i at time $t - 1$, β_{1j} is the regression coefficient for covariate j , and J is the number of covariates.

In a time-lag model the covariates X at time-point $t - 1$ are related to the outcome variable Y at time-point t .

Both the time-lag model and the standard model pool together within-subject and between-subjects relationships into one regression coefficient (see Section 4.5.3). The only difference between the time-lag model and the standard model is that the time-lag model takes into account the temporal sequence of a possible cause and effect. The question then arises: should a time-lag model be

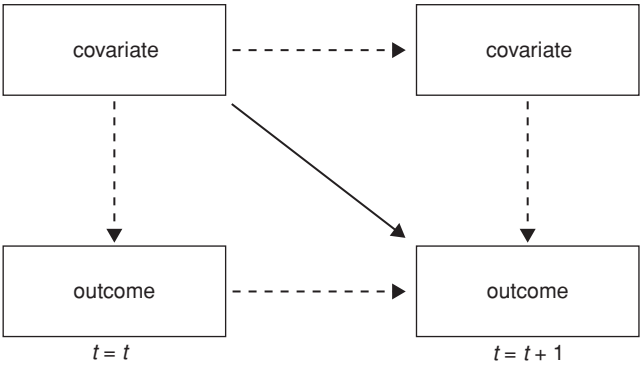


Figure 6.3 A time-lag relationship can be a reflection of the standard relationship when the relative stability of both the outcome variable and the covariate is high.

used in every situation in which a causal relationship is suspected? The answer is no! In fact, a time-lag model can only be useful when the time periods between subsequent measurements are short. When the time periods are long, the biological plausibility of a time lag between covariate X and outcome variable Y is not very clear. Furthermore, sometimes a time lag is already taken into account in the way a certain covariate is measured. For instance, when a lifestyle parameter such as dietary intake or physical inactivity is used as covariate in relation to some sort of disease outcome, both lifestyle parameters are often measured by some method of retrospective recall (e.g. measurement of the average amount of dietary intake of a certain nutrient over the previous three months). In other words, when a time lag is included in the method of measuring the covariate X , a statistical time-lag model is not very appropriate. In general, the usefulness of a time-lag model depends on the biological plausibility of a time lag in the relationship analyzed.

It is also possible that the results of a time-lag model are a reflection of the results that would have been found in a standard model. This occurs when the correlation between subsequent measurements of both the outcome variable and the covariate of interest is rather high. In fact, the standard relationships carry over to the time-lag relationship through the high correlation of the variables involved in the relationship investigated. [Figure 6.3](#) illustrates this phenomenon.

6.2.2 Model of changes

As mentioned before, both the standard model and the time-lag model pool together between-subjects and within-subject relationships. Although this is an important strength of the analysis, it also limits the interpretation of the results in such a way that no separation can be made between the two aspects of longitudinal

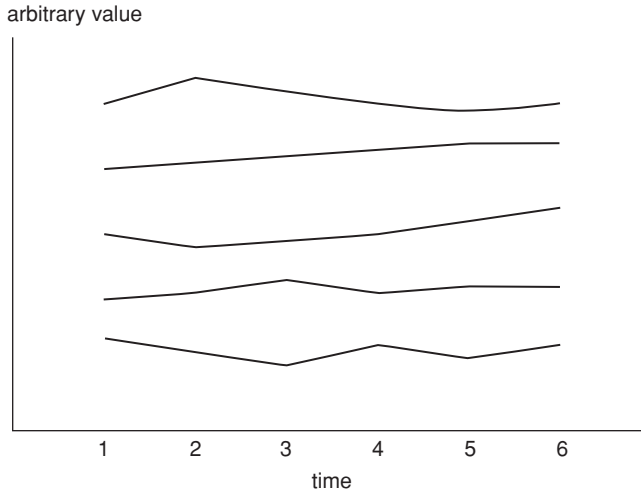


Figure 6.4 Illustration of a longitudinal study in which the changes over time within one subject are less than the differences between subjects; the cross-sectional relationships will “overrule” the longitudinal relationships.

relationships. This can be a problem especially when the variation in absolute values between subjects exceeds the changes over time within subjects. In this particular situation, in the pooled analysis the longitudinal within-subject relationships will be more or less overruled by the cross-sectional between-subjects relationships (Figure 6.4). This problem arises in particular when the time periods between subsequent measurements are relatively short, or when there is a strong (mostly non-observable) influence from the background variables (see Section 6.4).

Because of this limitation of both the standard model and the time-lag model, a model can be used in which the between-subjects part is more or less “removed” from the analysis. One possibility is not to model the values at each time-point, but to model differences between two subsequent measurements of both the outcome variable Y and the covariates X (Figure 6.5). The corresponding equation is:

$$(Y_{it} - Y_{it-1}) = \beta_0 + \sum_{j=1}^J \beta_{1j}(X_{ijt} - X_{ijt-1}) + \dots \quad (6.2)$$

where Y_{it} are observations for subject i at time t , Y_{it-1} is the observation for subject i at time $t-1$, β_0 is the intercept, X_{ijt} is the covariate j for subject i at time t , X_{ijt-1} is the covariate j for subject i at time $t-1$, β_{1j} is the regression coefficient for covariate j , and J is the number of covariates.

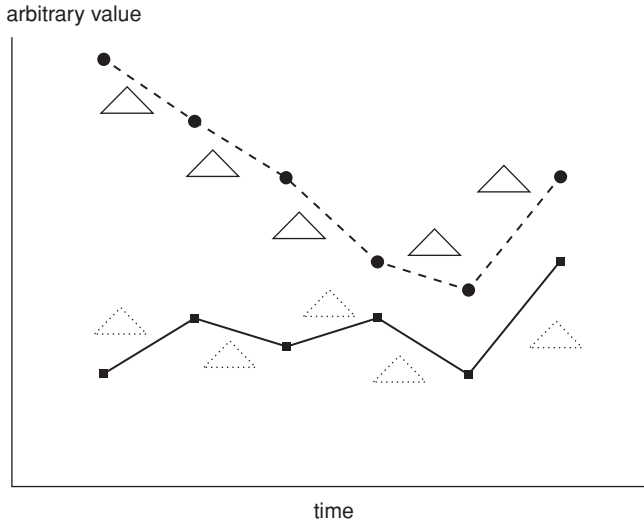


Figure 6.5 Illustration of a model of changes; changes in covariate X are related to changes in the outcome variable Y (■ - - - outcome variable, ● — covariate).

The model of changes looks quite simple, but it can be very complicated, owing to the difficulty of defining changes between subsequent measurements (see [Chapter 9](#)).

6.2.3 Autoregressive model

Another way in which to “remove” the between-subjects part of the relationship is to use an autoregressive model. Autoregressive models are also known as Markov models, conditional models, or transition models, and an extensive amount of literature has been devoted to these types of models (Rosner et al., 1985; Rosner and Munoz, 1988; Zeger and Qaqish, 1988; Stanek et al., 1989; Lindsey, 1993). The corresponding equation is:

$$Y_{it} = \beta_0 + \sum_{j=1}^J \beta_{1j} X_{ijt-1} + \beta_2 Y_{it-1} + \dots \quad (6.3)$$

where Y_{it} are observations for subject i at time t , β_0 is the intercept, X_{ijt-1} is the covariate j for subject i at time $t-1$, β_{1j} is the regression coefficient for covariate j , J is the number of covariate, Y_{it-1} is the observation for subject i at time $t-1$, and β_2 is the autoregression coefficient.

In an autoregressive model the value of the outcome variable Y at time-point t is related not only to the value of the covariate X at time-point $t-1$, but also to the

standard model: $\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \\ Y_5 \\ Y_6 \end{bmatrix} = \beta_0 + \beta_1 \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{bmatrix} + \dots$	time-lag model: $\begin{bmatrix} Y_2 \\ Y_3 \\ Y_4 \\ Y_5 \\ Y_6 \end{bmatrix} = \beta_0 + \beta_1 \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix} + \dots$
modeling of changes: $\begin{bmatrix} Y_2 - Y_1 \\ Y_3 - Y_2 \\ Y_4 - Y_3 \\ Y_5 - Y_4 \\ Y_6 - Y_5 \end{bmatrix} = \beta_0 + \beta_1 \begin{bmatrix} X_2 - X_1 \\ X_3 - X_2 \\ X_4 - X_3 \\ X_5 - X_4 \\ X_6 - X_5 \end{bmatrix} + \dots$	autoregressive model: $\begin{bmatrix} Y_2 \\ Y_3 \\ Y_4 \\ Y_5 \\ Y_6 \end{bmatrix} = \beta_0 + \beta_1 \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix} + \beta_2 \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \\ Y_5 \end{bmatrix} + \dots$

Figure 6.6 Overview of different models used to analyze the longitudinal relationship between an outcome variable Y and an independent variable X .

value of the outcome variable Y at $t - 1$. The model shown in Equation 6.3 is called a “first-order” autoregressive model, because the outcome variable Y at time-point t is only related to the value of the outcome variable Y at $t - 1$. In a “second-order” or “third-order” autoregressive model, the outcome variable Y at time-point t is also related to the value of the outcome variable Y at $t - 2$ or $t - 3$. The idea underlying the autoregressive model is that the value of an outcome variable at each time-point is primarily influenced by the value of this variable one measurement earlier. To estimate the “real” influence of the covariates on the outcome variable, the model should therefore adjust for the value of the outcome variable at time-point $t - 1$.

6.2.4 Overview

Figure 6.6 gives an overview of the way in which the regression coefficients of interest longitudinally relate the outcome variable Y to a particular covariate X .

6.2.5 Example

6.2.5.1 Introduction

In the following example the results of the three mentioned alternative models will be compared to the standard model (described in Chapter 4). For all three

models both GEE analysis and mixed model analysis will be used to estimate the regression coefficients. When GEE is used an adjustment is made for the within-subjects correlations by assuming a certain “working correlation structure.” It was argued that the choice of a particular structure can be based on the observed within-subject correlations of the outcome variable Y . When the longitudinal analysis is limited to the within-subject relationships (i.e. modeling of changes and the autoregressive model), the adjustment for the within-subject correlations is somewhat different than for the standard model. When changes are modeled, the within-subject correlations between the changes are in general much lower than the within-subject correlations of the observed values. The same is true for the autoregressive model. The latter is perhaps a bit difficult to understand, but it has to do with the fact that in GEE analysis an adjustment is made for the “correlated errors,” rather than for the correlated observations. Although in most situations the structure of the correlation between the observations is comparable to the structure of the correlations between the errors, for an autoregressive model this is not the case. In an autoregressive model part of the correlations in the observations is “explained” by the addition of the outcome variable Y at $t - 1$ to the model. In an autoregressive model, the within-subject correlations of the “errors” are therefore different from the within-subject correlations of the “observations.” In fact, in both the model of changes and the autoregressive model, correction for the within-subject correlations is part of the model. In those situations, an independent correlation structure will (often) be the most appropriate choice.

In the modeling of changes yet another problem arises, because changes between subsequent measurements can be defined in different ways (see [Chapter 9](#)). In the following examples the changes are simply defined as the difference between subsequent measurements.

6.2.5.2 Data structure for alternative models

There is no statistical software available which is capable of performing one of the alternative models automatically. For the alternative modeling of longitudinal data, the dataset has to be restructured in such a way that the standard software can be used for either GEE analysis or mixed model analysis. The way in which the data should be reconstructed is illustrated in [Table 6.1](#).

6.2.5.3 GEE analysis

The output of the GEE analysis, used to answer the question of whether there is a longitudinal relationship between outcome variable Y and the covariates X_1

Table 6.1 Data structures for various models used to analyze longitudinal relationships

Standard model				Time-lag model			
<i>I</i>	<i>Y</i>	<i>X</i>	Time	<i>i</i>	<i>Y</i>	X_{t-1}	Time
1	3.5	1.53	1	1	3.7	1.53	2
1	3.7	1.43	2	1	3.9	1.43	3
1	3.9	2.01	3	1	3.0	2.01	4
1	3.0	2.22	4	1	3.2	2.22	5
1	3.2	2.12	5	1	3.2	2.12	6
1	3.2	1.95	6	2	4.2	2.00	2
2	4.1	2.00	1	<i>N</i>	4.7	1.78	6
2	4.2	2.30	2				
<i>N</i>	5.0	1.78	5				
<i>N</i>	4.7	1.99	6				

Model of changes				Autoregressive model				
<i>i</i>	$Y_t - Y_{t-1}$	$X_t - X_{t-1}$	Time	<i>i</i>	<i>Y</i>	X_{t-1}	Y_{t-1}	Time
1	0.2	0.10	1		3.7	1.53	3.5	2
1	0.2	0.58	2	1	3.9	1.43	3.7	3
1	-0.9	0.20	3	1	3.0	2.01	3.9	4
1	0.2	-0.10	4	1	3.2	2.22	3.0	5
1	0.0	-0.17	5	1	3.2	2.12	3.2	6
2	0.1	0.30	1	2	4.2	2.00	4.1	2
<i>N</i>	-0.3	0.21	5	<i>N</i>	4.7	1.78	5.0	6

to X_4 , based on alternative longitudinal models is shown in Output 6.1 (time-lag model), Output 6.2 (model of changes), and Output 6.3 (autoregressive model).

In addition to the changes in outcome variable Y , in the model of changes, the changes in the covariate X_2 are added to the model. For all other covariates the observed values, either at the first measurement (for the time-independent covariate X_1 and X_4) or at the start of the time period over which the changes are calculated (for the time-dependent covariate X_3), are added to the model. For both the model of changes and the autoregressive model, an independent correlation structure was chosen.

In the output of an autoregressive model (Output 6.3) a new covariate is present (yt_1). This variable is the value of outcome variable Y measured one time-point earlier (see explanation of the autoregressive model in Section 6.2.3). From the output it can be seen that the value of Y at $t = t - 1$ is highly positively related to

Output 6.1 Results of a GEE analysis with a time-lag model

GEE population-averaged model		Number of obs	=	735
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group: min	=	5
Family:	Gaussian	avg	=	5.0
Correlation:	exchangeable	max	=	5
		Wald chi2(4)	=	103.80
Scale parameter:	0.5634565	Prob > chi2	=	0.0000

(Std. Err. adjusted for clustering on id)

	Semirobust					
ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
x1	0.3706314	0.2820497	1.31	0.189	-0.1821759	0.9234386
x2_t_1	0.2534181	0.0315411	8.03	0.000	0.1915987	0.3152376
x3_t_1	0.3038025	0.0691453	4.39	0.000	0.1682802	0.4393248
x4	0.0297545	0.1330179	0.22	0.823	-0.2309558	0.2904648
_cons	2.758252	0.7162779	3.85	0.000	1.354373	4.162131

Output 6.2 Results of a GEE analysis with a model of changes

GEE population-averaged model		Number of obs	=	735
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group: min	=	5
Family:	Gaussian	avg	=	5.0
Correlation:	independent	max	=	5
		Wald chi2(4)	=	50.38
Scale parameter:	0.3252727	Prob > chi2	=	0.0000
Pearson chi2(735):		239.08	Deviance	= 239.08
Dispersion (Pearson):		0.3252727	Dispersion	= 0.3252727

(Std. Err. adjusted for clustering on id)

	Semirobust					
dely	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
x1	0.1187375	0.0792937	1.50	0.134	-0.0366753	0.2741502
delx2	0.0819317	0.0279457	2.93	0.003	0.0271591	0.1367042
x3	0.2984187	0.0601012	4.97	0.000	0.1806224	0.4162149
x4	0.085376	0.0342716	2.49	0.013	0.018205	0.152547
_cons	-0.2888282	0.1962749	-1.47	0.141	-0.6735198	0.0958635

Table 6.2 Regression coefficients and standard errors regarding the longitudinal relationship (estimated by GEE analysis) between outcome variable Y and several covariates (X_1 to X_4); four different longitudinal models

	Standard model	Time-lag model	Model of changes	Autoregressive model
X_1	0.15 (0.27)	0.37 (0.28)	0.12 (0.08)	0.26 (0.11)
X_2	0.18 (0.02)	0.25 (0.03)	0.08 (0.03)	0.11 (0.02)
X_3	0.04 (0.06)	0.30 (0.07)	0.30 (0.06)	0.23 (0.06)
X_4	0.04 (0.13)	0.03 (0.13)	0.08 (0.03)	0.02 (0.05)

Output 6.3 Results of a GEE analysis with an autoregressive model

GEE population-averaged model		Number of obs	=	735
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group: min	=	5
Family:	Gaussian	avg	=	5.0
Correlation:	independent	max	=	5
		Wald chi2(5)	=	1121.20
Scale parameter:	0.297639	Prob > chi2	=	0.0000
Pearson chi2(735):	218.76	Deviance	=	218.76
Dispersion (Pearson):	0.297639	Dispersion	=	0.297639

(Std. Err. adjusted for clustering on id)

		Semirobust					
ycont		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

x1		0.2621347	0.1085393	2.42	0.016	0.0494015	0.4748678
x2		0.1113411	0.0178615	6.23	0.000	0.0763331	0.146349
x3		0.2335782	0.056743	4.12	0.000	0.122364	0.3447923
x4		0.0203888	0.0461873	0.44	0.659	-0.0701368	0.1109143
yt_1		0.7597378	0.0262596	28.93	0.000	0.7082699	0.8112057
_cons		0.195264	0.2833408	0.69	0.491	-0.3600739	0.7506018

the development of outcome variable Y , which is of course not really surprising. The regression coefficient of the independent variable yt_1 is also known as the autoregression coefficient. In [Table 6.2](#) the results of the different GEE analyses are summarized.

6.2.5.4 Mixed model analysis

The output of a mixed model analysis to answer the question of whether there is a longitudinal relationship between outcome variable Y and the covariates X_1 to X_4 , based on alternative longitudinal models, is shown in Output 6.4 (time-lag model), Output 6.5 (model of changes), and Output 6.6 (autoregressive model). The covariates in the mixed model analysis were modeled in the same way as has been described for the corresponding GEE analysis.

Output 6.4 Results of a mixed model analysis with a time-lag model

Mixed-effects REML regression	Number of obs	=	735			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	5			
	avg	=	5.0			
	max	=	5			
	Wald chi2(4)	=	175.05			
Log restricted-likelihood = -704.3169	Prob > chi2	=	0.0000			

ycont	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+						
x1	0.3709133	0.2782447	1.33	0.183	-0.1744362	0.9162628
x2_t_1	0.253466	0.0233232	10.87	0.000	0.2077533	0.2991787
x3_t_1	0.3043998	0.0671814	4.53	0.000	0.1727268	0.4360729
x4	0.0297325	0.1229799	0.24	0.809	-0.2113037	0.2707687
_cons	2.757461	0.6806106	4.05	0.000	1.423489	4.091434

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+						
id: Identity						
	var(_cons)	0.3044154	0.042381	0.231719	0.3999188	
-----+						
	var(Residual)	0.2677659	0.0156413	0.2387992	0.3002461	

LR test vs. linear regression: chibar2(01) =				273.53	Prob >= chibar2 = 0.0000	

Output 6.5 Results of a mixed model analysis with model of changes

Mixed-effects REML regression	Number of obs	=	735			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	5			
	avg	=	5.0			
	max	=	5			
	Wald chi2(4)	=	38.85			
Log restricted-likelihood = -641.39337	Prob > chi2	=	0.0000			

dely	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
x1	0.1187375	0.1191357	1.00	0.319	-0.1147641	0.3522391
delx2	0.0819317	0.022234	3.68	0.000	0.0383539	0.1255095
x3	0.2984187	0.0593362	5.03	0.000	0.1821219	0.4147155
x4	0.085376	0.0515989	1.65	0.098	-0.0157559	0.186508
_cons	-0.2888282	0.2900386	-1.00	0.319	-0.8572934	0.279637

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+-----						
id: Identity						
	var(_cons)	2.26e-25	9.00e-25		9.40e-29	5.45e-22
-----+-----						
	var(Residual)	0.3275006	0.0171422		0.2955684	0.3628826

LR test vs. linear regression: chibar2(01) =				0.00	Prob >= chibar2 = 1.0000	

From the output of both the model of changes and the autoregressive model, it can be seen that the variance of the random intercept is close to zero. So, adding a random intercept to the model of changes and the autoregressive model is not really necessary. In fact, this finding is comparable to the fact that in the GEE analysis for these two alternative models an independent correlation structure is considered to be the most appropriate choice for a “working correlation structure.” In [Table 6.3](#) the results of the different mixed model analyses are summarized.

6.3 Comments

Although the magnitude of the regression coefficients for the different models cannot be interpreted in the same way, a comparison between the regression coefficients and standard errors of the different models shows directly that the results obtained from the different models are quite different. Using an alternative model can lead to different conclusions than when using the standard model. On the one hand this is strange, because all analyses attempt to answer the question of whether there is a longitudinal relationship between outcome variable Y and the four covariates X_1 to X_4 . On the other hand, however, with the four models, different parts of the longitudinal relationships are analyzed, and the results of the models should be interpreted in different ways. To obtain the most general answer to the question of whether there is a longitudinal relationship between the outcome variable Y and the four independent variables, the results of several models could be combined (Twisk, 1997). In practice, however, this almost never happens: a priori the most appropriate model is chosen (usually the “standard” model), and only those results are reported.

In [Chapter 4](#) it has already been mentioned that GEE analyses do not give reliable information about the “fit” of the statistical model, whereas with mixed model analysis likelihood values can be obtained. However, when deciding which model should be used to obtain the best answer to a particular research question, comparing the “fit” of the models will not provide much interesting information. First of all, only the time-lag model and the autoregressive model can be directly compared to each other, because the autoregressive model can be seen as an extension of the time-lag model. The model of changes is totally different, while in the standard model more observations are used than in the alternative models. The problem is that the number of observations highly influences the likelihood of a particular analysis. Looking at the fit of the models, it is obvious for instance that an autoregressive model provides a much better fit than a time-lag model. This is due to the fact that a high percentage of variance of the outcome variable Y at time-point t is explained by the value of the outcome variable Y at time-point $t - 1$. This can be seen from the values of the scale parameter presented in the GEE output and the log likelihood presented in the output of the mixed model analysis. Both values are much lower in the autoregressive model than in the time-lag model. However, this does not mean that the autoregressive model should be used to obtain the best answer to the question of whether there is a longitudinal relationship between outcome variable Y and one (or more) covariate(s) X . In general, it should be realized that it is better to base the choice of a specific longitudinal model on logical considerations instead of statistical ones. If, for instance, it is expected that a covariate

measured at time-point $t - 1$ will influence the outcome variable at time-point t , then a time-lag model is suitable. If, however, it is expected that the independent and outcome variables are more directly related, a time-lag model is not suitable, and so forth.

It has already been mentioned that with the model of changes and with the autoregressive model the between-subject part of the analysis is more or less removed from the analysis. With both models only the within subject relationship is analyzed. It is therefore surprising that the results of the longitudinal analyses with the model of changes and the autoregressive model are quite different (see [Tables 6.1](#) and [6.2](#)). One reason for the difference in results is that both alternative models use a different model of change. This can be explained by assuming a longitudinal study with just two measurements. In the autoregressive model, $Y_2 = \beta_0 + \beta_1 Y_1$, while in the model of changes, $Y_2 - Y_1 = \beta_0$ (where β_0 is the difference between subsequent measurements), which is equal to $Y_2 = \beta_0 + Y_1$. The difference between the two equations is the coefficient β_1 . In the model of changes the “change” is a fixed parameter, while in the autoregressive model the “change” is a function of the value of Y_1 (for a detailed explanation of this phenomenon, see [Chapter 9](#)). Another reason for the differences in results between the model of changes and the autoregressive model is the different modeling of the covariates. It has already been mentioned that for the model of changes the changes in the time-dependent covariates were also modeled. In the autoregressive model, however, the covariates measured at $t - 1$ were used. It is obvious that different modeling of the covariates can lead to different results.

The most important message which emerges is that the modeling of the covariates can highly influence the results of the longitudinal analyses performed with alternative models. In other words, one should be very careful in the interpretation of the regression coefficients derived from such models.

6.4 Another example

One of the most striking examples to illustrate the necessity of using information from different models has been given in a study also based on data from the Amsterdam Growth and Health Longitudinal Study (AGAHLS) ([Twisk et al., 1998a](#)). The purpose of that study was to investigate the longitudinal relationship between smoking behavior and two lung function parameters: forced vital capacity (FVC) and forced expiratory volume in one second (FEV_1). Although the results of the standard model did not show any longitudinal relationship between smoking behavior and lung function parameters, the model of changes revealed a strong

Table 6.4 Standardized regression coefficients and 95% confidence intervals (calculated with GEE analysis) regarding the longitudinal relationship between lung function parameters (FVC and the FEV₁) and smoking behavior; a comparison between the standard model and the model of changes

	FVC ^a	FEV ₁ ^b
Standard model	−0.03 (−0.11 to 0.06)	−0.01 (−0.09 to 0.06)
Model of changes	−0.13 (−0.22 to −0.04)**	−0.14 (−0.25 to −0.04)**

** $p < 0.01$

^aFVC, forced vital capacity.

^bFEV₁, forced expiratory volume in one second.

inverse longitudinal relationship between smoking behavior and both lung function parameters (Table 6.4). So, although the values of the lung function parameters were not influenced by smoking behavior, the changes in lung function parameters over time were highly influenced by smoking behavior. This study is a nice example of the situation illustrated earlier in Figure 6.4.

Dichotomous outcome variables

7.1 Simple methods

7.1.1 Two measurements

When a dichotomous outcome variable is measured twice over time in the same subjects, a 2×2 table can be constructed as shown below (where n stands for the number of subjects and p stands for a proportion of the total number of subjects N).

		t_2		
		1	2	Total
t_1	1	$n_{11}(p_{11})$	$n_{12}(p_{12})$	$n_{1(t1)}(p_{1(t1)})$
	2	$n_{21}(p_{21})$	$n_{22}(p_{22})$	$n_{2(t1)}(p_{2(t1)})$
Total		$n_{1(t2)}(p_{1(t2)})$	$n_{2(t2)}(p_{2(t2)})$	$N(1)$

The simplest way to estimate the development over time is to compare the proportion of subjects in group 1 at $t_1(p_{1(t1)})$ with the proportion of subjects in group 1 at $t_2(p_{1(t2)})$. The difference in proportions is calculated as $(p_{1(t2)} - p_{1(t1)})$, and Equation 7.1 shows how to calculate the corresponding standard error:

$$SE(P_{1(t2)} - P_{1(t1)}) = \frac{\sqrt{n_{1(t2)} + n_{1(t1)}}}{N} \quad (7.1)$$

where SE is the standard error, $p_{1(t2)}$ is the proportion of subjects in group 1 at $t = 2$, $p_{1(t1)}$ is the proportion of subjects in group 1 at $t = 1$, $n_{1(t2)}$ is the number of subjects in group 1 at $t = 2$, $n_{1(t1)}$ is the number of subjects in group 1 at $t = 1$, and N is the total number of subjects.

The 95% confidence interval for the difference (difference ± 1.96 times the standard error) is used to answer the question of whether there is a significant

change over time. The problem with the difference in proportions is that it basically provides an indication of the difference between the changes in opposite directions. If all subjects from group 1 at $t = 1$ move to group 2 at $t = 2$, and all subjects from group 2 at $t = 1$ move to group 1 at $t = 2$, the difference in proportions reveals no changes over time.

A widely used method to determine whether there is a change over time in a dichotomous outcome variable is the McNemar test. This is an alternative Chi-square test, which takes into account the fact that the observed proportions in the 2×2 table are not independent. The McNemar test is, in principle, based on the difference between n_{12} and n_{21} , and the test statistic follows a Chi-square distribution with one degree of freedom (Equation 7.2).

$$\chi^2 = \frac{(n_{12} - n_{21} - 1)^2}{n_{12} + n_{21}} \quad (7.2)$$

where n_{12} is the number of subjects in group 1 at $t = 1$ and in group 2 at $t = 2$, and n_{21} is the number of subjects in group 2 at $t = 1$ and in group 1 at $t = 2$.

The McNemar test determines whether the change in one direction is equal to the change in another direction. So the McNemar test has the same disadvantages as have been mentioned above for the difference in proportions. It tests the difference between the changes in opposite directions.

A possible way in which to estimate the total change over time is to calculate the proportion of subjects who change from one group to another: i.e. $p_{12} + p_{21}$. This “proportion of change” can be tested for significance by means of the 95% confidence interval (± 1.96 times the standard error). The standard error of this proportion is calculated as:

$$SE(p_{change}) = \sqrt{\frac{p_{change}(1 - p_{change})}{N}} \quad (7.3)$$

where SE is the standard error, p_{change} is the “proportion of change” equal to $p_{12} + p_{21}$, and N is the total number of subjects.

If one is only interested in the proportion of subjects who change in a certain direction (i.e. only a “decrease” or “increase” over time) the same procedure can be followed for separate changes. In this respect, a “proportion of increase” equal to p_{12} or a “proportion of decrease” equal to p_{21} can be calculated and a 95% confidence interval can be constructed, based on the standard error calculated with Equation 7.3.

It should be noted that when all individuals belong to the same group at $t = 1$, the estimate of the change in opposite directions is equal to the estimate of the total change over time. In that situation, which often occurs in experimental studies, all methods discussed so far can be used to estimate the change over time in a dichotomous outcome variable.

7.1.2 More than two measurements

When more than two measurements are performed on the same subjects, the multivariate extension of the McNemar test can be used. This multivariate extension is known as Cochran's Q , and it has the same disadvantages as the McNemar test. It is a measure of the difference between changes in opposite directions, while in longitudinal studies one is generally interested in the total change over time. To analyze the total change over time, the "proportion of change" can be calculated in the same way as in the situation with two measurements. To do this, $(T - 1) 2 \times 2$ tables must first be constructed (for $t = 1$ and $t = 2$, for $t = 2$ and $t = 3$, etc.). The next step is to calculate the "proportion of change" for each 2×2 table. To calculate the total proportion of change, Equation 7.4 can be applied:

$$\bar{p} = \frac{1}{N(T - 1)} \sum_{i=1}^N c_i \quad (7.4)$$

where $\bar{p}$ is the total "proportion of change," N is the number of subjects, T is the number of measurements, and c_i is the number of changes for individual i over time.

7.1.3 Comparing groups

To compare the development over time between two groups, for a dichotomous outcome variable the "proportion of change" in the two groups can be compared. This can be done by applying the test for two independent proportions, i.e. $p_{g1} - p_{g2}$. The standard error of this difference (needed to create a 95% confidence interval and for testing whether there is a significant difference between the two groups) is calculated by Equation 7.5.

$$SE(p_{g1} - p_{g2}) = \sqrt{\left[\frac{p_{g1}(1 - p_{g1})}{N_{g1}} \right] + \left[\frac{p_{g2}(1 - p_{g2})}{N_{g2}} \right]} \quad (7.5)$$

where SE is the standard error, p_{g1} is the "proportion of change" in group 1, p_{g2} is the "proportion of change" in group 2, N_{g1} is the number of subjects in group 1, and N_{g2} is the number of subjects in group 2.

Of course, this procedure can also be carried out to determine the "proportion of change" in a certain direction (i.e. the "proportion of increase" or the "proportion of decrease"). It should be realized that the calculation of the "proportion of change" over a particular time period is primarily useful for the longitudinal analysis of datasets with only two repeated measurements. For more information on the analysis of proportions and differences in proportions, reference is made to the classical work of Fleiss (1981).

Table 7.1 A 2×2 table indicating the relationship between the outcome variable Y_{dich} at $t = 1$ and $t = 6$

Y_{dich} at $t = 1$	Y_{dich} at $t = 6$		Total
	0	1	
0	80	17	97
1	18	32	50
Total	98	49	147

7.1.4 Example

7.1.4.1 Introduction

The dataset used to illustrate longitudinal analysis with a dichotomous outcome variable is the same as that used to illustrate longitudinal analysis with continuous outcome variables. The only difference is that the outcome variable Y is dichotomized (Y_{dich}). This is done by means of the 66th percentile. At each of the repeated measurements the upper 33% are coded as “1,” and the lower 66% are coded as “0” (see Section 1.4).

7.1.4.2 Development over time

To analyze the development of a dichotomous outcome variable Y_{dich} over time, the situation with two measurements will first be illustrated. From the example dataset the first ($t = 1$) and the last ($t = 6$) measurements will be considered. Let us first investigate the 2×2 table, which is presented in [Table 7.1](#).

Because the dichotomization of outcome variable Y_{dich} was based on a fixed value (the 66th percentile) at each of the repeated measurements, by definition, there is no difference between the changes over time in opposite directions. The proportion of subjects in group 1 at $t = 1$ (33.3%) is almost equal to the proportion of subjects in group 1 at $t = 6$ (34.0%). Therefore, the McNemar test is useless in this particular situation. However, just as an example, the result of the McNemar test is presented in [Output 7.1](#).

As expected, the McNemar test statistic Chi-square = 0.0000 and the corresponding p -value is 1.0000, which indicates that there is no change over time for outcome variable Y_{dich} . The output of the McNemar test illustrates perfectly the limitation of the method, i.e. only the difference between the changes over time in opposite directions is taken into account.

From the 2×2 table, also the total “proportion of change” and the corresponding 95% confidence interval can be calculated. The “proportion of change” is $(18 + 17) / 147 = 0.24$. The standard error of this proportion, which is calculated according to [Equation 7.3](#), is 0.035. With these two components the 95% confidence

Output 7.1 Dichotomous variable Y at $t = 1$ *dichotomous variable Y at $t = 6$
crosstabulation

		dichotomous variable Y at $t = 6$		Total
		0.00	1.00	
dichotomous variable	0.00	80	17	97
Y at $t = 1$	1.00	18	32	50
Total		98	49	147

Chi-square tests

	Value	exact	Sig. (2-sided)
McNemar test			1.000 ^a
N of valid cases	147		

^a Binomial distribution used.

interval can be calculated, which leads to an interval that ranges from 0.17 to 0.31, indicating a highly significant change over time.

When the development over time of the outcome variable Y_{dich} is analyzed using all six measurements, the multivariate extension of the McNemar test (Cochran’s Q) can be used. However, Cochran’s Q has the same limitations as the McNemar test. So again it is useless in this particular situation, in which the groups are defined according to the same (fixed) percentile at each measurement. However, Output 7.2 shows the result of the Cochran’s Q test. As expected, the significance level of Cochran’s Q (0.9945) is close to one, indicating no difference between the changes over time in opposite directions.

Output 7.2 Output result of the Cochran’s Q test calculated for the longitudinal development of the dichotomized outcome variable Y_{dich} from $t = 1$ to $t = 6$, using data from all repeated measurements

Cochran Q Test				
Cases				
	=0.00	=1.00	Variable	
	97	50	YDICT1	OUTCOME VARIABLE Y AT T1 (2 GROUPS)
	99	48	YDICT2	OUTCOME VARIABLE Y AT T2 (2 GROUPS)
	96	51	YDICT3	OUTCOME VARIABLE Y AT T3 (2 GROUPS)
	98	49	YDICT4	OUTCOME VARIABLE Y AT T4 (2 GROUPS)
	99	48	YDICT5	OUTCOME VARIABLE Y AT T5 (2 GROUPS)
	98	49	YDICT6	OUTCOME VARIABLE Y AT T6 (2 GROUPS)
Cases	Cochran Q DF Significance			
147	0.4298	5	0.9945	

Table 7.2 Five 2×2 tables used to calculate the “proportion of change” when there are more than two measurements

Y_{dich} at $t = 1$	Y_{dich} at $t = 1$		Total	Y_{dich} at $t = 2$	Y_{dich} at $t = 3$		Total
	0	1			0	1	
0	83	14	97	0	83	16	99
1	16	34	50	1	13	35	48
Total	99	48	147	Total	96	51	147

Y_{dich} at $t = 3$	Y_{dich} at $t = 4$		Total	Y_{dich} at $t = 4$	Y_{dich} at $t = 5$		Total
	0	1			0	1	
0	84	12	96	0	86	12	98
1	14	37	51	1	13	36	49
Total	98	49	147	Total	99	48	147

Y_{dich} at $t = 5$	Y_{dich} at $t = 6$		Total
	0	1	
0	82	17	99
1	16	32	48
Total	98	49	147

To evaluate the total change over time, Equation 7.4 can be used. First of all, the $(T - 1) 2 \times 2$ tables must be constructed (Table 7.2). From these tables, the total “proportion of change” can be calculated.

The sum of the changes is 143, so the “proportion of change” is $143/(147 \times 5) = 0.19$. The corresponding 95% confidence interval (based on the standard error calculated with Equation 7.3) is [0.16 to 0.22], indicating a highly significant change over time.

7.1.4.3 Comparing groups

When the aim of the study is to investigate whether there is a difference in development over time between several groups, the “proportion of change” in the groups can be compared. In the example dataset the population can be divided into two groups, according to the time-independent covariate X_4 (i.e. males and females). For both groups a 2×2 table is constructed (Table 7.3), indicating the changes between $t = 1$ and $t = 6$ in Y_{dich} .

Table 7.3 Two 2×2 tables indicating the relationship between the outcome variable Y_{dich} at $t = 1$ and $t = 6$ for two groups divided by X_4 (i.e. gender)

X_4 equals 0				X_4 equals 1			
Y_{dich} at $t = 1$	Y_{dich} at $t = 6$		Total	Y_{dich} at $t = 1$	Y_{dich} at $t = 6$		Total
	0	1			0	1	
0	40	5	45	0	40	12	52
1	8	16	24	1	10	16	26
Total	48	21	69	Total	50	28	78

The next step is to calculate the “proportion of change” for both groups. For the group $X_4 = 0$, $p_{\text{change}} = 13/69 = 0.19$; while for the group $X_4 = 1$, $p_{\text{change}} = 0.28$. From these two proportions the difference and the 95% confidence interval can be calculated. The latter is based on the standard error calculated with Equation 7.5. The difference in “proportion of change” between the two groups is 0.09, with a 95% confidence interval ranging between -0.05 to 0.23 . So, there is a difference between the two groups (i.e. females have a 9% greater change over time), but this difference is not statistically significant.

When there are more than two measurements, Equation 7.4 can be used to calculate the “proportion of change” in both groups. After creating $(T - 1)$ separate 2×2 tables, for group $X_4 = 0$ this proportion equals 0.18, and for group $X_4 = 1$, this proportion equals 0.21. The difference in “proportion of change” between the two groups (i.e. 0.03) can be tested for significance by means of the 95% confidence interval. Based on the standard error, which is calculated with Equation 7.5, this interval ranges between -0.03 and 0.09 , so the (small) difference observed between the two groups is not statistically significant different from zero.

7.2 Relationships with other variables

7.2.1 “Traditional” methods

With the (simple) methods described in Section 7.1 it was possible to answer the question of whether there is a change over time in a certain dichotomous outcome variable, and whether there is a difference in change over time between two or more groups. Both questions can also be answered by using more complicated methods, which must be applied in any other situation than described above, for instance to answer the question of whether there is a longitudinal relationship between the dichotomous outcome variable Y_{dich} and one or more covariate(s) X . In

Table 7.4 Results of a logistic regression analysis relating “long-term exposures” to covariates X_1 to X_4 between $t = 1$ and $t = 6$ (using all available data) to the dichotomous outcome variable Y_{dich} at $t = 6$

	Regression coefficient	Standard error	p -value
X_1	1.44	1.12	0.20
Average X_2	0.86	0.21	<0.001
Average X_3	-0.12	0.39	0.75
X_4	-0.61	0.50	0.23

Section 4.2, it was discussed that for continuous outcome variables “traditional,” i.e. cross-sectional, methods are sometimes used to analyze these longitudinal relationships. For dichotomous outcome variables, comparable procedures are available. The most popular choice is the method illustrated in Figure 4.2, i.e. “long-term exposure” to certain independent variables is related to the dichotomous outcome variable at the end of the follow-up period. It is obvious that this analysis can be performed with cross-sectional logistic regression analysis.

7.2.2 Example

Table 7.4 presents the results of a logistic regression analysis, in which the “long-term exposures” to the covariates X_1 to X_4 between $t = 1$ and $t = 6$ (using all available data) are related to the outcome variable Y_{dich} at $t = 6$.

From the significance levels it can be seen that “long-term exposure” to X_2 is significantly associated with Y_{dich} at $t = 6$. The p -value is based on the Wald statistic, which is defined as the regression coefficient divided by its standard error, squared. The Wald statistic follows a Chi-square distribution with (in this case) one degree of freedom. The corresponding odds ratio can be calculated as $\text{EXP}(\text{regression coefficient})$, which is equal to 2.36, and the 95% confidence interval can be calculated as $\text{EXP}(\text{regression coefficient} \pm 1.96 \text{ times the standard error of the regression coefficient})$, which is [1.58 to 3.53]. The interpretation of the odds ratio is straightforward: a one point difference in the “long-term exposure” to X_2 between two subjects is associated with a 2.37 times higher odds of being in the upper tertile of the outcome variable Y_{dich} at $t = 6$. It should be noted that a 2.37 times higher odds is usually (loosely) interpreted as a 2.37 times greater “risk,” which is comparable but not the same.

7.2.3 Sophisticated methods

In general, when a dichotomous outcome variable is used in a longitudinal study, and the objective of the study is to analyze the longitudinal relationship between

such a variable and one or more covariates, it is possible to use the sophisticated methods mentioned before (i.e. generalized estimating equations (GEE) analysis and mixed model analysis). In [Chapter 4](#), it was extensively explained that for continuous outcome variables in longitudinal studies the sophisticated techniques can be considered as longitudinal linear regression analysis. Analogous to this, GEE analysis and mixed model analysis with a dichotomous outcome variable in longitudinal studies can be considered as longitudinal logistic regression analysis. So, comparable to Equation 4.3, the longitudinal logistic model can be formulated as in Equation 7.6.

$$\ln \left(\frac{\Pr(Y_{it} = 1)}{1 - \Pr(Y_{it} = 1)} \right) = \beta_0 + \sum_{j=1}^J \beta_{1j} X_{ijt} \quad (7.6a)$$

In a different notation:

$$\Pr(Y_{it} = 1) = \frac{1}{1 + \text{EXP} \left[- \left(\beta_0 + \sum_{j=1}^J \beta_{1j} X_{ijt} \right) \right]} \quad (7.6b)$$

where $\Pr(Y_{it} = 1)$ is the probability that the observations at t_1 to t_T of subject i equal 1 (where T is the the number of measurements and 1 means that subject i belongs to the group of interest), β_0 is the intercept, X_{ijt} is the covariate j of subject i at time t , β_{1j} is the regression coefficient of i covariate j , and J is the number of covariates.

Although the model looks quite complicated, it is in fact nothing more than an extension of a cross-sectional logistic regression model. The extension is presented in the subscript t , which indicates that the same individuals are repeatedly measured over time. In this model the probability of belonging to a group (coded 1) from t_1 to t_T (Y_{it}) is related to several covariates (X_{ijt}). Like in cross-sectional multiple logistic regression analysis, all covariates can be continuous, dichotomous or categorical, although in the latter situation dummy coding can or must be used. The coefficient of interest is β_1 , because this coefficient reflects the longitudinal relationship between a certain covariate (X_{it}) and belonging to the group of interest from t_1 to t_T . Like in cross-sectional logistic regression, this coefficient (β_1) can be transformed into an odds ratio ($\text{EXP}(\beta_1)$). The interpretation of the regression coefficients (i.e. odds ratios) is equivalent to the “combined” interpretation of the regression coefficients for continuous outcome variables (see the example in Section 7.2.4 for a detailed explanation).

Analogous to the situation with continuous outcome variables, with GEE analysis an adjustment is made for the within-subject correlations between the repeated measurements by assuming a “working correlation structure,” while with mixed

model analysis this adjustment is made by allowing different regression coefficients to vary between subjects, i.e. random intercept and random slopes.

7.2.4 Example

7.2.4.1 Generalized estimating equations

For dichotomous outcome variables, the GEE approach also requires the choice of a “working correlation structure.” Although there are the same possibilities as have been discussed for continuous outcome variables (see Section 4.5.2), it is not possible to use the correlation structure of the observed data as a guide for the choice of “working correlation structure.” In this example, an exchangeable correlation structure (which is the default option in many software packages) will be used.

Output 7.3 presents the result of the logistic GEE analysis in which Y_{dich} is related to the four covariates (X_1 to X_4).

Output 7.3 Results of the logistic GEE analysis performed on the example dataset

GEE population-averaged model		Number of obs	=	882			
Group variable:	id	Number of groups	=	147			
Link:	logit	Obs per group: min	=	6			
Family:	binomial	avg	=	6.0			
Correlation:	exchangeable	max	=	6			
		Wald chi2(4)	=	28.51			
Scale parameter:	1	Prob > chi2	=	0.0000			
(Std. Err. adjusted for clustering on id)							

		Semirobust					
ydich		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----							
x1		0.0497193	0.7540079	0.07	0.947	-1.428109	1.527548
x2		0.2881188	0.0583648	4.94	0.000	0.1737259	0.4025118
x3		-0.2585005	0.1916057	-1.35	0.177	-0.6340408	0.1170398
x4		0.1306231	0.3759594	0.35	0.728	-0.6062438	0.86749
_cons		-2.062825	1.91856	-1.08	0.282	-5.823134	1.697483

The output of the logistic GEE analysis is comparable to the output of a linear GEE analysis, which was discussed in Section 4.5.4.2. The outcome variable is Y_{dich} , which is the dichotomized version of the outcome variable Y , and the correlation structure used is “exchangeable.” The difference between the outputs is found

in the Link function and the Family. In a logistic regression analysis, the link function is the logit and the family is binomial. The second part of the output consists of the parameter estimates. For each of the covariates the magnitude of the regression coefficient, the standard error, the z -value (obtained from dividing the regression coefficient by its standard error), the corresponding p -value, and the 95% confidence interval around the regression coefficient are presented. The latter is calculated in the regular way, i.e. by the regression coefficient ± 1.96 times the standard error.

From the four covariates, only X_2 is significantly related to the dichotomous outcome variable Y_{dich} . The regression coefficient is 0.2881188, and the odds ratio is therefore $\text{EXP}[0.2881188] = 1.33$. The 95% confidence interval around the odds ratio ranges from $\text{EXP}[0.1737259] = 1.19$ to $\text{EXP}[0.4025118] = 1.50$. The interpretation of this odds ratio is somewhat complicated. As for the regression coefficients calculated for a continuous outcome variable, the odds ratios can be interpreted in two ways. (1) The between-subjects interpretation: a subject with a one-unit higher score for covariate X_2 , compared to another subject, has a 1.33 times higher odds of being in the highest group for the dichotomous outcome variable Y_{dich} , compared to the odds of being in the lowest group. (2) The within-subject interpretation: an increase of one unit in covariate X_2 within a subject over a certain time period is associated with a 1.33 times higher odds of moving to the highest group of the dichotomous outcome variable Y_{dich} compared to the odds of staying in the lowest group. The magnitude of the regression coefficient (i.e. the magnitude of the odds ratio) reflects both relationships, and it is not clear from the results of this analysis which is the most important component of the relationship.

As in the GEE analysis with a continuous outcome variable, the scale parameter (also known as dispersion parameter) is an indication of the variance of the model. The interpretation of this coefficient is, however, different to that in the situation with a continuous outcome variable. This has to do with the characteristics of the binomial distribution on which the logistic GEE analysis is based. In the binomial distribution the variance is directly linked to the mean value (Equation 7.7).

$$\text{var}(\bar{p}) = \bar{p}(1 - \bar{p}) \quad (7.7)$$

where var is the variance, and $\bar{p}$ is the average probability.

So, for the logistic GEE analysis, the scale parameter has to be one (i.e. a direct connection between the variance and the mean).

Comparable to the situation already described for continuous outcome variables, GEE analysis requires the choice of a particular “working correlation structure.”

Table 7.5 Results of the GEE analysis with different correlation structures

	Correlation structure			
	Independent	Exchangeable	5-Dependent	Unstructured
X_1	0.42 (0.78)	0.05 (0.75)	0.08 (0.76)	0.04 (0.76)
X_2	0.43 (0.08)	0.29 (0.06)	0.29 (0.06)	0.29 (0.06)
X_3	-0.23 (0.24)	-0.26 (0.19)	-0.22 (0.19)	-0.24 (0.19)
X_4	0.07 (0.40)	0.13 (0.38)	0.12 (0.38)	0.13 (0.38)

It has already been mentioned that for a dichotomous outcome variable it is not possible to base that choice on the correlation structure of the observed data. It is therefore interesting to investigate the difference in regression coefficients estimated when different correlation structures are chosen. Output 7.4 shows the results of several analyses with different correlation structures and Table 7.5 summarizes the results of the different GEE analyses.

Output 7.4 Results of the GEE analyses with different correlation structures

GEE population-averaged model		Number of obs	=	882			
Group variable:	id	Number of groups	=	147			
Link:	logit	Obs per group: min	=	6			
Family:	binomial	avg	=	6.0			
Correlation:	independent	max	=	6			
		Wald chi2(4)	=	32.26			
Scale parameter:	1	Prob > chi2	=	0.0000			
Pearson chi2(882):	874.03	Deviance	=	1049.79			
Dispersion (Pearson):	0.9909666	Dispersion	=	1.190236			
(Std. Err. adjusted for clustering on id)							

		Semirobust					
y dich		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

x1		0.4212592	0.7787442	0.54	0.589	-1.105051	1.94757
x2		0.4292022	0.0847053	5.07	0.000	0.2631829	0.5952216
x3		-0.2270176	0.237943	-0.95	0.340	-0.6933773	0.2393422
x4		0.0654474	0.3962265	0.17	0.869	-0.7111423	0.8420372
_cons		-3.233905	1.995321	-1.62	0.105	-7.144662	0.6768509

GEE population-averaged model		Number of obs		=	882		
Group and time vars:	id time	Number of groups		=	147		

Link:	logit	Obs per group: min =	6				
Family:	binomial	avg =	6.0				
Correlation:	stationary(5)	max =	6				
		Wald chi2(4)	= 30.43				
Scale parameter:	1	Prob > chi2	= 0.0000				
(Std. Err. adjusted for clustering on id)							

		Semirobust					
y dich		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

	+						
x1		0.0846049	0.7551044	0.11	0.911	-1.395373	1.564582
x2		0.2890992	0.0564381	5.12	0.000	0.1784826	0.3997158
x3		-0.2203473	0.1856134	-1.19	0.235	-0.5841429	0.1434483
x4		0.12252	0.3754818	0.33	0.744	-0.6134109	0.8584509
_cons		-2.124901	1.918413	-1.11	0.268	-5.884922	1.635119

GEE population-averaged model				Number of obs	=	882	
Group and time vars:	id time	Number of groups	=	147			
Link:	logit	Obs per group: min	=	6			
Family:	binomial	avg	=	6.0			
Correlation:	unstructured	max	=	6			
		Wald chi2(4)	=	31.39			
Scale parameter:	1	Prob > chi2	=	0.0000			
(Std. Err. adjusted for clustering on id)							

		Semirobust					
y dich		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

	+						
x1		0.0404175	0.757712	0.05	0.957	-1.444671	1.525506
x2		0.2853039	0.055181	5.17	0.000	0.177151	0.3934567
x3		-0.2351255	0.18657	-1.26	0.208	-0.6007961	0.130545
x4		0.1341775	0.3765546	0.36	0.722	-0.6038559	0.8722109
_cons		-2.055243	1.930124	-1.06	0.287	-5.838216	1.727729

The most important conclusion which can be drawn from [Table 7.5](#) is that the results of the GEE analysis with different (dependent) correlation structures are highly comparable. This finding is different from that observed in the analysis of a continuous outcome variable (see [Table 4.2](#)), for which a remarkable difference

was found between the results of the analysis with different correlation structures. So, probably, the statement in the literature that GEE analysis is robust against the wrong choice of a correlation structure is particularly true for dichotomous outcome variables (see for instance also Liang and Zeger, 1993).

Furthermore, from Table 7.5 it can be seen that there are remarkable differences between the results obtained from the analysis with an independent correlation structure and the results obtained from the analysis with the three dependent correlation structures. It should further be noted that comparable to the situation with a continuous outcome variable, the standard errors obtained from the analysis with an independent correlation structure are higher than those obtained from the analysis with a dependent correlation structure.

To put the results of the GEE analysis in a somewhat broader perspective, they can be compared with the results of a “naïve” logistic regression analysis, in which the dependency of observations is ignored. Output 7.5 presents the results of such a “naïve” logistic regression analysis.

Output 7.5 Results of a “naïve” logistic regression analysis performed on the example dataset, ignoring the dependency of the observations

Logistic regression	Number of obs	=	882
	LR chi2(4)	=	74.40
	Prob > chi2	=	0.0000
Log likelihood = -524.89426	Pseudo R2	=	0.0662

ydich	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
x1	0.421259	0.429491	0.98	0.327	-0.4205278	1.263046
x2	0.4292022	0.0583226	7.36	0.000	0.314892	0.5435123
x3	-0.2270175	0.2026592	-1.12	0.263	-0.6242223	0.1701874
x4	0.0654473	0.190579	0.34	0.731	-0.3080806	0.4389753
_cons	-3.233905	1.073166	-3.01	0.003	-5.337272	-1.130537

The comparison between the results of the “naïve” logistic regression and the results of the GEE analysis with an independent correlation structure is comparable to what has been observed for continuous outcome variables. The regression coefficients are exactly the same as the regression coefficients obtained from a GEE

analysis, while the standard errors obtained from the GEE analysis are higher than those calculated with the “naïve” logistic regression analysis.

7.2.4.2 Mixed model analysis

Comparable to the situation with continuous outcome variables, in the case of dichotomous outcome variables it is also possible to analyze the relationship with several covariates by means of a mixed model analysis. The first step is to perform an analysis with only a random intercept. Output 7.6 shows the result of this analysis.

Output 7.6 Results of a mixed model analysis with only a random intercept

Mixed-effects logistic regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
Integration points = 7	Wald chi2(4)	=	29.79			
Log likelihood = -402.29239	Prob > chi2	=	0.0000			

ydich	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+						
x1	0.1417343	1.442582	0.10	0.922	-2.685675	2.969144
x2	0.5716169	0.1137886	5.02	0.000	0.3485954	0.7946383
x3	-0.4710987	0.3383479	-1.39	0.164	-1.134248	0.192051
x4	0.3318951	0.6285065	0.53	0.597	-0.899955	1.563745
_cons	-4.367423	3.520139	-1.24	0.215	-11.26677	2.531922

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+						
id: Identity						
var(_cons)		6.816792	1.503748	4.423943	10.5039	

LR test vs. logistic regression: chibar2(01) =				245.20	Prob>=chibar2 = 0.0000	

The output of a mixed model analysis with a dichotomous outcome variable is comparable to the output observed for a continuous outcome variable.

The first part provides some general information about the model. It shows that a logistic mixed model analysis was performed, that there are 882 observations within 147 individuals (number of groups). Again there are no missing data; the minimum, maximum, and average number of observations within an individual is equal to six.

Furthermore, the log likelihood of the model (i.e. -402.29239) and the result of a Wald test ($\text{Wald } \chi^2(4) = 29.79$), and the corresponding p -value ($\text{prob} > \chi^2 = 0.000$) are presented. This Wald test is a generalized Wald test for all covariates. Because X_1 , X_2 , X_3 , X_4 are analyzed in the model, the generalized Wald statistic follows a Chi-square distribution with 4 degrees of freedom, which is highly significant. As for a linear mixed model analysis, the log likelihood can be used for the likelihood ratio test in order to evaluate whether random slopes should be added to the model.

The second part of the output shows the most important information obtained from the analysis, i.e. the (fixed) regression coefficients. This information is exactly the same as has been discussed for continuous outcome variables, although the regression coefficients can be transformed into odds ratios by taking $\text{EXP}[\text{regression coefficient}]$. Again the interpretation of the coefficients is the same as has been discussed for the logistic GEE analysis, i.e. a combined between-subjects and within-subject interpretation. For instance, for independent variable X_2 the between-subjects interpretation is that a subject with a one-unit higher score for the independent variable X_2 , compared to another subject, has an $\text{EXP}(0.5716169) = 1.77$ times higher odds of being in the highest group for the dichotomous outcome variable Y_{dich} . The within-subject interpretation is that an increase of one unit in independent variable X_2 within a subject (over a certain time period) is associated with a 1.77 times higher odds of moving to the highest group of the dichotomous outcome variable Y_{dich} .

The last part of the output shows information about the random part of the analysis. In this situation only the variance around the intercept is given, i.e. 6.816792. Also for the logistic mixed model analysis, the assumption is that the intercepts for all individuals are normally distributed. From this normal distribution, the variance is calculated. It should be noted that in the output of the logistic mixed model analysis no error variance is given. This has to do with the fact that in all logistic regression analyses the probability of belonging to a certain group is estimated without error. The error in the analysis is outside the model in the difference between the calculated probability and the observed dichotomous value (which is either 0 or 1).

As there is no error variance in the logistic mixed model analysis, the intraclass correlation coefficient (ICC) cannot be calculated in the same way as has been discussed for continuous outcome variables. For logistic mixed model analysis, the

ICC can be calculated by $\sigma_w^2 / (\sigma_w^2 + \pi^2 / 3)$ (Twisk, 2006). In the example, the ICC is therefore equal to $6.82 / (6.82 + (3.14)^2 / 3) = 67\%$.

The last line of the output gives the result of the likelihood ratio test comparing the model with a random intercept with a model without a random intercept, i.e. a naïve logistic regression analysis. Apparently, this difference is 245.20, which follows a Chi-square distribution with one degree of freedom (i.e. the random intercept), and which is highly significant. In other words, the results of the likelihood ratio test suggest that it is necessary to add a random intercept to the model. As already been mentioned in [Chapter 4](#), when a linear mixed model analysis was discussed, this likelihood ratio test is not very interesting, because theoretically, there must be an adjustment for the dependency of the observations within an individual, i.e. there has to be a random intercept.

The results of this likelihood ratio test can be verified by comparing the $-2 \log$ likelihood of the “naïve” logistic regression analysis presented in [Output 7.5](#) ($-2 * 524.89$) with the $-2 \log$ likelihood of the logistic mixed model analysis with only a random intercept presented in [Output 7.6](#) ($-2 * 402.29$). This difference is indeed equal to 245.20.

The next step in this logistic mixed model analysis is to evaluate the necessity of random slopes. As has been mentioned before, a random slope is only possible for time-dependent covariates. In the example a possible random slope can only be evaluated for the covariates X_2 and X_3 . When a random slope for the covariate X_2 is added to the model (depending on the software package used) it is possible that the model will not converge. This happens quite often when random slopes are added to a logistic mixed model analysis. The reason for this is that the mathematics behind logistic mixed model analysis is very complicated and therefore, it is sometimes not possible to add random slopes to the model (see also [Chapter 12](#)). For the covariate X_3 , however, it is possible to add a random slope to the model. [Output 7.7](#) shows the results of this analysis.

From the random part of [Output 7.7](#), it can be seen that a random slope for X_3 ($\text{var}(x3)$) is added to the model, as well as the covariance between the random intercept and the random slope for X_3 ($\text{cov}(x3_cons)$). To evaluate the necessity of a random slope for X_3 , the log likelihood of the model presented in [Output 7.7](#) (-399.08711) must be compared to the log likelihood of the model with only a random intercept (-402.29239 , [Output 7.6](#)). The difference between the two values is 2.42. Two times this difference follows a Chi-square distribution with 2 degrees of freedom (i.e. the random slope and the covariance between the random slope and the random intercept), which gives a p -value which is slightly higher than 5%. So following the basic rule of significance, allowing a random slope for X_3 is not really necessary. However, although the corresponding p -value is not significant, the difference in $-2 \log$ likelihood is substantial, so in this situation it is also possible

Output 7.7 Results of a mixed model analysis with a random intercept and a random slope for covariate X_3

Mixed-effects logistic regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
Integration points = 7	Wald chi2(4)	=	32.97			
Log likelihood = -399.08711	Prob > chi2	=	0.0000			

ydich	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
x1	0.1614655	1.326505	0.12	0.903	-2.438437	2.761368
x2	0.5762243	0.1121052	5.14	0.000	0.3565021	0.7959464
x3	0.0448755	0.3692629	0.12	0.903	-0.6788665	0.7686175
x4	0.3422037	0.5794259	0.59	0.555	-0.7934502	1.477858
_cons	-4.548733	3.23082	-1.41	0.159	-10.88102	1.783557

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+-----						
id: Unstructured						
	var(x3)	1.741933	1.269322		0.4176122	7.265909
	var(_cons)	8.411559	1.968913		5.316598	13.30819
	cov(x3,_cons)	-3.827841	1.696775		-7.153459	-0.5022236

LR test vs. logistic regression:			chi2(3) =	251.61	Prob > chi2 = 0.0000	

to present the results of a model with both a random intercept and a random slope for X_3 .

7.2.5 Comparison between GEE analysis and mixed model analysis

For continuous outcome variables it was seen that GEE analysis with an exchangeable correlation structure and a mixed model analysis with only a random intercept provided almost identical results in the analysis of a longitudinal dataset.

Table 7.6 Regression coefficients and standard errors (in parentheses) of longitudinal regression analyses with a dichotomous outcome variable; a comparison between logistic GEE analysis with an exchangeable correlation structure and a logistic mixed model analysis with only a random intercept

	GEE analysis ^a	Mixed model analysis ^b
X_1	0.05 (0.75)	0.14 (1.44)
X_2	0.29 (0.06)	0.57 (0.11)
X_3	−0.26 (0.19)	−0.47 (0.34)
X_4	0.13 (0.38)	0.33 (0.63)

^aGEE analysis with an exchangeable correlation structure.

^bMixed model analysis with only a random intercept.

For dichotomous outcome variables, however, the situation is more complex. In [Table 7.6](#) the results of the logistic GEE analysis with an exchangeable correlation structure and the logistic mixed model analysis with only a random intercept are summarized.

From [Table 7.6](#) it can be concluded that there are remarkable differences between the results of the GEE analysis and the results of the mixed model analysis. All regression coefficients and standard errors obtained from GEE analysis are much lower than those obtained from mixed model analysis.

In this respect, it is important to realize that the regression coefficients calculated with GEE analysis are “population averaged,” i.e. the average of the individual regression lines. The regression coefficients calculated with mixed model analysis can be seen as “subject specific.” In [Figure 7.1](#), this difference is illustrated for both the linear model (i.e. with a continuous outcome variable) and the logistic model (i.e. with a dichotomous outcome variable) with only a random intercept. For the linear longitudinal regression analysis, both GEE analysis and mixed model analysis produce exactly the same results, i.e. the “population-average” is equal to the “subject-specific” (see also Section 4.7). For the logistic longitudinal regression analysis, however, the two approaches produce different results. From [Figure 7.1](#) it can be seen that the regression coefficients calculated with a logistic GEE analysis will always be lower than the coefficients calculated with a logistic mixed model analysis (see also, for instance, Neuhaus et al., [1991](#); Hu et al., [1998](#)).

Because of the remarkable differences, the question then arises: “When a dichotomous outcome variable is analyzed in a longitudinal study, should GEE analysis or mixed model analysis be used?” If one is performing a population study and one is interested in the relationship between a dichotomous outcome variable and

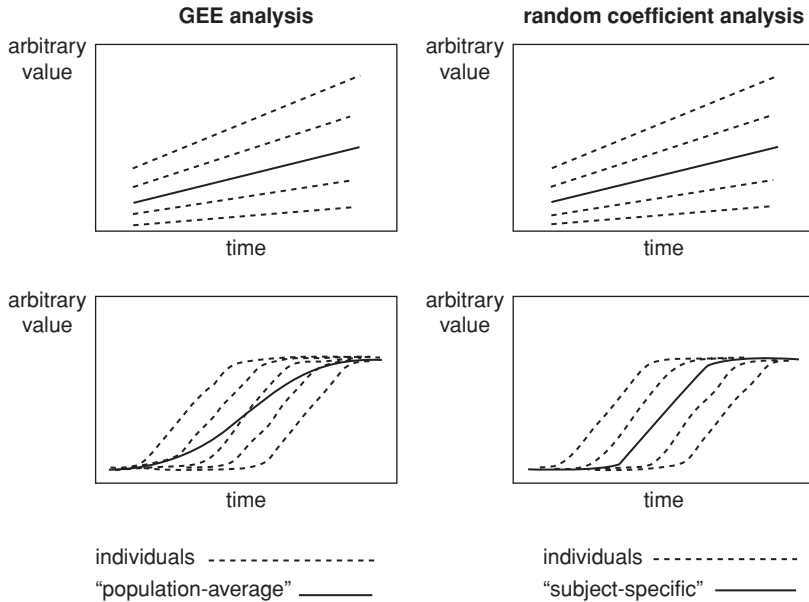


Figure 7.1 Illustration of the “population average” approach of GEE analysis and the “subject specific” approach of mixed model analysis, illustrating both the situation with a continuous outcome variable (upper graphs) and the situation with a dichotomous outcome variable (lower graphs).

several covariates, GEE analysis will probably provide the most “valid” results. However, looking at [Figure 7.1](#), most researchers will probably choose the logistic mixed model analysis as the most appropriate. It should, however, be noted that mixed model analyses with a dichotomous outcome variable are mathematically very complicated. Different software packages give different results, and within one software package there is (usually) more than one algorithm to estimate the coefficients. Unfortunately, these different estimation procedures often lead to different results (see also [Chapter 12](#)). In other words, although in theory mixed model analysis seems to be highly suitable in some situations, in practice one should be very careful in using this technique in the longitudinal analysis of a dichotomous outcome variable. Furthermore, when reviewing the epidemiological literature, it is obvious that logistic GEE analysis is far more used than logistic mixed model analysis.

7.2.6 Alternative models

In [Chapter 6](#) several alternative methods were introduced to analyze longitudinal relationships for continuous outcome variables (i.e. a time-lag model, a model of changes, and an autoregressive model). In principle, all the alternative models

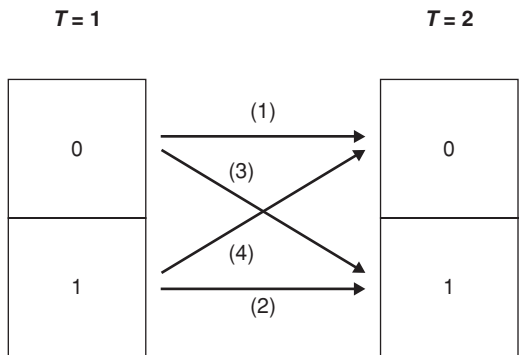


Figure 7.2 Changes in a dichotomous variable between two time-points lead to a categorical variable with four groups.

discussed for continuous outcome variables can also be used for the analysis of dichotomous outcome variables. The time-lag model can be used when one is interested in the analysis of possible causation, while an autoregressive model can be used when one is interested in the analysis of the within subject part of the relationship. However, a problem arises in the model of changes between subsequent measurements. This has to do with the fact that changes in a dichotomous outcome variable result in a categorical variable with four groups (i.e. subjects who stay in one group, subjects who stay in another group, and two groups in which subjects move from one group to another (Figure 7.2)), and with the fact that the longitudinal analysis of categorical outcome variables is rather complicated (see Chapter 8).

This chapter does not include a very detailed discussion of the results of alternative models to analyze dichotomous outcome variables, because this was already done in Chapter 6 with regard to continuous outcome variables. Basically, the same problems and advantages apply to dichotomous outcome variables. It is important to realize that the three models represent different aspects of the longitudinal relationships between a dichotomous outcome variable and several covariates, and that therefore the regression coefficients obtained from the different models should be interpreted differently.

7.2.7 Comments

In this chapter, the longitudinal analysis of a dichotomous outcome variable is explained in a rather simple way. It should be realized that the mathematical details of these analyses are very complicated. For these mathematical details reference is made to other publications (with regard to GEE analysis, see for instance Liang and Zeger, 1986; Prentice, 1988; Lipsitz et al., 1991; Carey et al., 1993; Diggle

et al., [1994](#); Lipsitz et al., [1994b](#); Wiliamson et al., 1995; Lipsitz and Fitzmaurice, [1996](#); and with regard to mixed model analysis, see for instance Conway, [1990](#); Goldstein, [1995](#); Rodriguez and Goldman, [1995](#); Goldstein and Rasbash, [1996](#); Gibbons and Hedeker, [1997](#); Barbosa and Goldstein, [2000](#); Yang, M. and Goldstein, [2000](#); Rodriguez and Goldman, [2001](#)).

Categorical and “count” outcome variables

8.1 Categorical outcome variables

8.1.1 Two measurements

Longitudinal analysis with a categorical outcome variable is more problematic than the longitudinal analysis of continuous or dichotomous outcome variables. Until recently, only simple methods were available to analyze such outcome variables. Therefore, categorical variables are sometimes treated as continuous, especially when they are ordinal and have a sufficient number (usually five or more) of categories. Another method is to reduce the categorical outcome variable into a dichotomous one by combining two or more categories. However, this results in a loss of information, and is only recommended when there are only a few subjects in one or more categories of the categorical variable.

The simplest form of longitudinal study with a categorical outcome variable is one where the categorical outcome variable is measured twice in time. This situation (when the categorical variable consists of three groups) is illustrated in the 3×3 table presented below (where n stands for number of subjects and p stands for proportion of the total number of subjects N).

		t_2			
		1	2	3	Total
t_1	1	$n_{11}(p_{11})$	$n_{12}(p_{12})$	$n_{13}(p_{13})$	$n_{1(t1)}(p_{1(t1)})$
	2	$n_{21}(p_{21})$	$n_{22}(p_{22})$	$n_{23}(p_{23})$	$n_{2(t1)}(p_{2(t1)})$
	3	$n_{31}(p_{31})$	$n_{32}(p_{32})$	$n_{33}(p_{33})$	$n_{3(t1)}(p_{3(t1)})$
	Total	$n_1(t_2)(p_{1(t2)})$	$n_2(t_2)(p_{2(t2)})$	$n_3(t_2)(p_{3(t2)})$	$N(1)$

To determine whether there is a change over time in the categorical outcome variable Y_{cat} , an extension of the McNemar test (which has been discussed for dichotomous outcome variables, see Section 7.1.1) can be used. This extension is

known as the Stuart–Maxwell test, and is only suitable for outcome variables with three categories. The Stuart–Maxwell test statistic follows a Chi-square distribution with one degree of freedom, and is defined as shown in Equation 8.1.

$$\chi^2 = \frac{\bar{n}_{23}d_1^2 + \bar{n}_{13}d_2^2 + \bar{n}_{12}d_3^2}{2(\bar{n}_{12}\bar{n}_{13} + \bar{n}_{12}\bar{n}_{23} + \bar{n}_{13}\bar{n}_{23})} \quad (8.1a)$$

$$n_{ij} = \frac{n_{ij} + n_{ji}}{2} \quad (8.1b)$$

$$d_i = n_{it1} - n_{it2} \quad (8.1c)$$

where n_{ij} is the number of subjects in group i at $t = 1$ and in group j at $t = 2$, and n_{ji} is the number of subjects in group j at $t = 1$ and in group i at $t = 2$.

Just as the McNemar test, the Stuart–Maxwell test gives an indication of the differences between the changes over time in opposite directions, while the main interest is usually the total change over time. Therefore, the “proportion of change” can be calculated. This “proportion of change” is a summation of all the off-diagonal proportions of the categorical 3×3 table, which is equal to $1 - (p_{11} + p_{22} + p_{33})$. Around this proportion a 95% confidence interval can be calculated in the usual way. For calculation of the standard error of the “proportion of change,” Equation 8.2 can be used:

$$SE(p_{\text{change}}) = \sqrt{\frac{p_{\text{change}}(1 - p_{\text{change}})}{N}} \quad (8.2)$$

where SE is the standard error, p_{change} is the “proportion of change” equal to $p_{12} + p_{21}$, and N is the total number of subjects.

This 95% confidence interval provides an answer to the question of whether there is a significant change over time. As for the dichotomous outcome variables, this procedure can be carried out for the proportion of subjects that “increases” over time or the proportion of subjects that “decreases” over time. It is obvious that the calculation of the “proportion of change” is not limited to categorical variables with only three categories.

8.1.2 More than two measurements

When there are more than two measurements in a longitudinal study, the same procedure can be followed as has been described for dichotomous outcome variables, i.e. the “proportion of change” can be used as a measure of total change over time. To do so, $(T - 1) r \times c$ tables¹ must be constructed (for $t = 1$ and $t = 2$, for $t = 2$ and $t = 3$, and so on); then for each table the “proportion of change”

¹ $r \times c$ stands for row $\times$ column, and indicates that all types of categorical variables can be analyzed in this way.

Table 8.1 A 3×3 table for the relationship between outcome variable Y_{cat} at $t = 1$ and Y_{cat} at $t = 6$

Y_{cat} at $t = 1$	Y_{cat} at $t = 6$			Total
	1	2	3	
1	30	15	3	48
2	16	19	14	49
3	3	15	32	50
Total	49	49	49	147

can be calculated. To obtain the total “proportion of change,” Equation 8.3 can be applied:

$$\bar{p} = \frac{1}{N(T - 1)} \sum_{i=1}^N c_i \tag{8.3}$$

where $\bar{p}$ is the overall “proportion of change,” N is the number of subjects, T is the number of measurements, and c_i is the number of changes for individual i .

8.1.3 Comparing groups

In research situations in which the longitudinal development over time between several groups must be compared, the simple methods discussed for dichotomous outcome variables can also be used for categorical outcome variables, i.e. comparing the “proportion of change” between different groups, or comparing the “proportion of change” in a certain direction between different groups. When there are only two groups to compare, a 95% confidence interval can be constructed around the difference in proportions, so that this difference can be tested for significance. This should be done in exactly the same way as has been described for dichotomous outcome variables (see Section 7.1.3).

8.1.4 Example

For the example, the original continuous outcome variable Y of the example dataset was divided into three equal groups, according to the 33rd and the 66th percentile, in order to create Y_{cat} . This was done at each of the six measurements (see also Section 1.4). Most of the statistical methods are suitable for situations in which there are only two measurements, and therefore the change between the first and the last repeated measurement (between $t = 1$ and $t = 6$) for the categorical outcome variable Y_{cat} will be considered first. In Table 8.1 the 3×3 table for Y_{cat} at $t = 1$ and Y_{cat} at $t = 6$ is presented. From Table 8.1 the Stuart–Maxwell test statistic and the “proportion of change” can be calculated. Output 8.1 shows the results of the Stuart–Maxwell test.

Table 8.2 Five 3×3 tables to analyze the change over time in the outcome variable Y_{cat} between $t=1$ and $t=6$

Y_{cat} at $t=1$	Y_{cat} at $t=2$				Y_{cat} at $t=2$	Y_{cat} at $t=3$			
	1	2	3	Total		1	2	3	Total
1	35	11	2	48	1	34	11	1	46
2	8	29	12	49	2	18	20	15	53
3	3	13	34	50	3	2	11	35	48
Total	46	53	48	147	Total	54	42	41	147

Y_{cat} at $t=3$	Y_{cat} at $t=4$				Y_{cat} at $t=4$	Y_{cat} at $t=5$			
	1	2	3	Total		1	2	3	Total
1	45	9	0	54	1	36	16	0	52
2	7	23	12	42	2	10	24	12	46
3	0	14	37	51	3	3	10	36	49
Total	52	46	49	147	Total	49	50	48	147

Y_{cat} at $t=5$	Y_{cat} at $t=6$			
	1	2	3	Total
1	35	13	1	49
2	12	22	16	50
3	2	14	32	48
Total	49	49	49	147

the two categories of time-independent independent variable X_4 (i.e. gender). [Table 8.3](#) shows the two 3×3 tables. For both groups the “proportion of change” is exactly the same, i.e. 0.45. Around this (no) difference a 95% confidence interval can be constructed: $[-0.16 \text{ to } 0.16]$. The width of the confidence interval provides information about the precision of the calculated difference between the two groups.

To obtain an estimation of the possible differences in change over time for the two groups by using all six measurements, the overall “proportion of change” must be calculated for both groups. When this is done (by creating $(T - 1), 3 \times 3$ tables for both groups), the overall “proportion of change” for the group X_4 equals 0 is 0.47, while for group X_4 equals 1, the overall proportion of change is 0.44. Around this difference of 3% a 95% confidence interval can be calculated. To obtain a standard error for this difference, Equation 8.4 can be applied to these data, which results in a confidence interval of $[-0.05 \text{ to } 0.11]$, i.e. no significant difference

Table 8.3 Two 3×3 tables for the relationship between outcome variable Y_{cat} at t_1 and Y_{cat} at t_6 for the group in which X_4 equals 0 and for the group in which X_4 equals 1

$X_4 = 0$					$X_4 = 1$				
Y_{cat} at $t = 1$	Y_{cat} at $t = 6$				Y_{cat} at $t = 1$	Y_{cat} at $t = 6$			
	1	2	3	Total		1	2	3	Total
1	14	7	0	21	1	16	8	3	27
2	11	8	5	24	2	5	11	9	25
3	3	5	16	24	3	0	10	16	26
Total	28	20	21	69	Total	21	29	28	78

between the two groups in the overall proportion of change.

$$SE(p_{g1} - p_{g2}) = \sqrt{\left[\frac{p_{g1}(1 - p_{g1})}{N_{g1}} \right] + \left[\frac{p_{g2}(1 - p_{g2})}{N_{g2}} \right]} \quad (8.4)$$

where SE is the standard error, p_{g1} is the “proportion of change” in group 1, p_{g2} is the “proportion of change” in group 2, N_{g1} is the number of subjects in group 1, and N_{g2} is the number of subjects in group 2.

8.1.5 Relationships with other variables

8.1.5.1 “Traditional” methods

With the (simple) methods described in the foregoing sections, it was possible to answer the question of whether there is a change over time in a certain categorical outcome variable and/or the question of whether there is a difference in change over time between two or more groups. Both questions can also be answered by using more complicated methods, which must be applied in any situation other than that described above: for instance, to answer the question of whether there is a longitudinal relationship between a categorical outcome variable Y_{cat} and one or more covariates X . For categorical outcome variables, a comparable “cross-sectional” procedure is available, as has already been described for continuous and dichotomous outcome variables, i.e. “long-term exposure” to certain independent variables is related to the categorical outcome variable at the end of the follow-up period (see Figure 4.2). This analysis can be performed with multinomial logistic regression analysis, which is also known as polytomous logistic regression analysis and which is the categorical extension of logistic regression analysis.

Table 8.4 Results of a multinomial logistic regression analysis relating “long-term exposures” to covariates X_1 to X_4 between $t = 1$ and $t = 6$ (using all available data) to the categorical outcome variable Y_{cat} at $t = 6$

	Regression coefficient	Standard error	p -value
Group 2			
X_1	−0.45	1.23	0.71
Average X_2	0.30	0.25	0.24
Average X_3	0.54	0.42	0.20
X_4	0.19	0.55	0.73
Group 3			
X_1	1.21	1.23	0.35
Average X_2	1.04	0.26	<0.001
Average X_3	0.18	0.46	0.70
X_4	−0.53	0.58	0.36

8.1.5.2 Example

Table 8.4 presents the results of the multinomial logistic regression analysis, in which “long-term exposures” to the covariates X_1 to X_4 between $t = 1$ and $t = 6$ (using all available data) were related to the categorical outcome variable Y_{cat} at $t = 6$.

With multinomial logistic regression analysis, basically two logistic regression analyses are combined into one analysis, although the procedure is slightly different from performing two separate independent logistic regression analyses. In the multinomial logistic regression analysis performed in this example, the lower tertile of the outcome variable Y_{cat} is used as a reference category. The interpretation of the regression coefficients is exactly the same as for (simple) logistic regression analysis. From Table 8.4 it can be seen that “long-term exposure” to X_2 is significantly associated with the third group of the categorical outcome variable Y_{cat} at $t = 6$. The corresponding odds ratio can be obtained from the regression coefficient, i.e. the odds ratio is $\text{EXP}(\text{regression coefficient})$ and the corresponding 95% confidence interval is $\text{EXP}(\text{regression coefficient} \pm 1.96 \text{ times the standard error of the regression coefficient})$. The interpretation of the odds ratio is straightforward: a one-point difference in the “long-term exposure” to X_2 between subjects is associated with a 2.82 (i.e. $\text{EXP}(1.036)$) “higher” odds of being in the highest tertile of the outcome variable Y_{cat} at $t = 6$, compared to the odds of being in the lowest tertile. The corresponding 95% confidence intervals range between 1.70 and 4.67.

8.1.5.3 Sophisticated methods

In Chapters 4 and 7, it was argued that longitudinal data analysis with a continuous outcome variable is a longitudinal extension of linear regression analysis, and that longitudinal data analysis with a dichotomous outcome variable is a longitudinal extension of logistic regression analysis, i.e. both take into account the fact that the repeated measurements within one subject are correlated. Analogous to this, it is obvious that longitudinal data analysis with a categorical outcome variable is a longitudinal extension of multinomial logistic regression analysis. Multinomial logistic regression analysis for longitudinal data analysis was first described for GEE analysis (see for instance Liang et al., 1992; Miller et al., 1993; Lipsitz et al., 1994b). Surprisingly this multinomial logistic GEE approach is still not yet available in standard software packages, and will therefore not be discussed in detail. The general idea of this GEE approach is the same as for all other GEE approaches, i.e. an adjustment for the dependency of observations is performed by assuming a certain “working correlation structure.”

In recent years, a multinomial logistic mixed model analysis has also been described (Agresti et al., 2000; Rabe-Hesketh and Skondral, 2001; Rabe-Hesketh et al., 2001a). As with all other mixed model analyses described earlier, with this newly developed method all questions can be answered that were answered by the earlier mentioned simple methods. Moreover, it can also be used to analyze the longitudinal relationship between a categorical outcome variable and one or more covariates. The underlying procedures and the interpretation of the regression coefficients are comparable to what has been described for logistic mixed model analysis.

8.1.5.4 Example

The first step in the analysis to answer the question whether there is a relationship between the categorical outcome variable Y_{cat} and the four covariates (X_1 to X_4) is to perform a mixed model analysis with only a random intercept. Output 8.2 shows the results of this analysis.

The output of a multinomial logistic mixed model analysis has a slightly different structure than the outputs of linear and logistic mixed model analyses. This has to do with the fact that a multinomial logistic mixed model analysis within Stata can only be estimated with the GLLAMM procedure. GLLAMM stands for Generalized Linear Latent and Mixed Models and is a very flexible procedure with which many complicated mixed model analyses can be performed (see also Section 12.6).

In the output, first the “number of level 1 units” and the “number of level 2 units” are given. Level 1 and level 2 refers to multilevel analysis, which is another name for mixed model analysis. Level 1 refers to the repeated observations which are clustered within the subjects (which are the level 2 units). So, there are 882 observations performed within 147 subjects. This was already known, because that is

Output 8.2 Result of a mixed model analysis with a categorical outcome variable and only a random intercept

```
number of level 1 units = 882
```

number of level 2 units = 147

```
log likelihood = -788.11876
```

	ycat	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
c2							
	x1	-1.46495	1.709081	-0.86	0.391	-4.814687	1.884786
	x2	0.4555807	0.1448768	3.14	0.002	0.1716274	0.7395339
	x3	-0.101516	0.3717667	-0.27	0.785	-0.8301653	0.6271332
	x4	-0.0484343	0.7555654	-0.06	0.949	-1.529315	1.432447
	_cons	2.120387	4.184942	0.51	0.612	-6.081949	10.32272
c3							
	x1	-0.6976371	1.712847	-0.41	0.684	-4.054756	2.659482
	x2	0.7614764	0.143648	5.30	0.000	0.4799314	1.043021
	x3	-0.4543081	0.3824287	-1.19	0.235	-1.203855	0.2952385
	x4	0.0795641	0.757095	0.11	0.916	-1.404315	1.563443
	_cons	-0.7204713	4.196338	-0.17	0.864	-8.945142	7.504199

Variances and covariances of random effects

```
***level 2 (id)
```

```
var(1): 7.8643317 (1.7948303)
```

the structure of the example dataset. Next, the log likelihood of the model analyzed is presented (-788.11876). As for all other mixed model analyses, this number is only interesting in comparison to the log likelihood value of another model, which must be an extension of the present model. In the next part of the output, the regression coefficients and standard errors are given as well as the z -values, the corresponding p -values and the 95% confidence intervals around the regression coefficients. In the example dataset, Y_{cat} is a categorical outcome variable with three categories (i.e. tertiles), so there are two “tables” with regression coefficients. In the first “table” the second tertile of Y_{cat} is compared to the lowest tertile of Y_{cat} (which

is the reference category), while in the second “table” the highest tertile of Y_{cat} is compared to the lowest tertile. The interpretation of the regression coefficients is rather complicated. From Output 8.2 it can be seen that only X_2 is significantly related to the outcome variable Y_{cat} . For the comparison between the second tertile and the reference category (i.e. the lowest tertile) the regression coefficient (0.4555807) can be transformed into an odds ratio (i.e. $\text{EXP}(0.4555807) = 1.58$). As for all other longitudinal regression coefficients this odds ratio has a “combined” interpretation. (1) The between-subjects interpretation: a subject with a one-unit higher score for independent variable X_2 , compared to another subject, has a 1.58 times higher odds of being in the second tertile compared to the odds of being in the lowest tertile. (2) The within-subject interpretation: an increase of one unit in independent variable X_2 within a subject (over a certain time period) is associated with a 1.58 times higher odds of moving from the lowest tertile to the second tertile of the categorical outcome variable Y_{cat} , compared to the odds of staying in the lowest tertile. The regression coefficient of X_2 belonging to the comparison between the highest tertile and the lowest tertile ($\text{EXP}(0.7614764) = 2.14$) can be interpreted in the same way. (1) A subject with a one-unit higher score for independent variable X_2 , compared to another subject, has a 2.14 times higher odds of being in the highest tertile for the categorical outcome variable Y_{cat} compared to the odds of being in the lowest tertile. (2) An increase of one unit in independent variable X_2 within a subject (over a certain time period) is associated with a 2.14 times higher odds of moving from the lowest tertile to the highest tertile of the categorical outcome variable Y_{cat} , compared to the odds of staying in the lowest tertile. The magnitude of the regression coefficient (i.e. the magnitude of the odds ratio) reflects both relationships, and it is not clear from the results of this analysis, which is the most important component of the relationship. However, the relative contribution of both parts highly depends on the proportion of subjects who move from one category to another. In the example dataset for instance, the proportion of subjects who move from the lowest to the highest category is rather low, so for the comparison between the lowest and the highest tertile, the estimated odds ratio of 2.14 mainly reflects the between-subjects relationship. As for all other longitudinal data analyses, alternative models are available (e.g. an autoregressive model, see Section 6.2.3) in which the between-subjects and within-subject relationships can be more or less separated.

In the last part of Output 8.2 the variance around the intercept ($\text{var}(1) = 7.8643317$) and the corresponding standard error (1.7948303) are provided. Again, this variance is estimated assuming a normal distribution of the intercepts. It has been mentioned before, that in a longitudinal study it is not necessary to evaluate whether or not a random intercept should be added to the model. A model without a random intercept (i.e. a “naïve” multinomial logistic regression analysis) is theoretically wrong, because it ignores the longitudinal nature of the data.

Output 8.3 Results of a multinomial logistic mixed model analysis with a random intercept and a random slope for X_2

```
var(2): 0.19682367 (0.24550007)
```

Output 8.4 Results of a multinomial logistic mixed model analysis with a random intercept and a random slope for X_3

number of level 1 units = 882						
number of level 2 units = 147						
log likelihood = -787.2403						

	ycat	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]

+-----						
c2						
	x1	-1.611381	1.683657	-0.96	0.339	-4.911288 1.688526
	x2	0.4582562	0.1964272	2.33	0.020	0.073266 0.8432465
	x3	-0.0879496	0.3758449	-0.23	0.815	-0.8245922 0.6486929
	x4	-0.0608442	0.7106066	-0.09	0.932	-1.453607 1.331919
	_cons	2.409571	4.179902	0.58	0.564	-5.782887 10.60203

+-----						
c3						
	x1	-0.8392308	1.685215	-0.50	0.618	-4.142192 2.46373
	x2	0.7652847	0.1962792	3.90	0.000	0.3805846 1.149985
	x3	-0.4380944	0.3862101	-1.13	0.257	-1.195052 0.3188634
	x4	0.0667845	0.7112674	0.09	0.925	-1.327274 1.460843
	_cons	-0.4452304	4.187841	-0.11	0.915	-8.653248 7.762787

Variances and covariances of random effects						

***level 2 (id)						
var(1): 14.195731 (7.2059801)						
cov(1,2): -1.1782975 (1.213095) cor(1,2): -0.68329186						
var(2): 0.20947859 (0.22900647)						

First of all, in Outputs 8.3 and 8.4 it can be seen that the random part of the model (Variances and covariances of random effects) is extended compared to the model with only a random intercept. In both outputs a random slope (var(2)) is provides as well as the covariance between the particular random slope and the random intercept (cov(1,2)). In the output the correlation between the random slope and the random intercept (cor(1,2)) is also provided, although the latter is

not very informative. In both situations, the log likelihood of the model with both a random intercept and a random slope is slightly better compared to the model with only a random intercept. For both models, however, the improvement of the model, by adding a random slope to the model, is not statistically significant. So, in this example, a model with only a random intercept is preferred.

8.2 “Count” outcome variables

A special type of categorical outcome variable is a so-called “count” outcome variable (e.g. the number of asthma attacks in one year, the number of falls in elderly people, etc.). Because of the discrete and non-negative nature of the “count” outcome variables, they are assumed to follow a Poisson distribution. Longitudinal analysis with “count” outcome variables is therefore comparable to a cross-sectional Poisson regression analysis, the difference being that the longitudinal technique takes into account the within-subject correlations. It should further be noted that the longitudinal Poisson regression analysis is sometimes referred to as longitudinal log-linear regression analysis.

As for the longitudinal linear regression analysis, the longitudinal logistic regression analysis, and the longitudinal multinomial logistic regression analysis, the longitudinal Poisson regression analysis is, in fact, nothing more than an extension of the cross-sectional Poisson regression analysis, i.e. an adjustment for the dependency of the observations within the subject. With this analysis the longitudinal relationship between the “count” outcome variable and several covariates can be analyzed. As in cross-sectional Poisson regression analysis, all covariates can be continuous, dichotomous, or categorical, although of course in the latter situation dummy coding can or must be used. As in cross-sectional Poisson regression analysis, the regression coefficient can be transformed into a rate ratio ($\text{EXP}(\text{regression coefficient})$). For estimation of the regression coefficients (i.e. rate ratios) the same sophisticated methods can be used as were discussed before, i.e. GEE analysis and mixed model analysis. With GEE analysis, a correction for the within-subject correlations is made by assuming a “working correlation structure,” while with mixed model analysis the different regression coefficients are allowed to vary between individuals, i.e. a random intercept and random slopes (for technical details see, for instance, Diggle et al., 1994; Goldstein, 1995).

8.2.1 Example

8.2.1.1 Introduction

The example chosen to illustrate the analysis of a “count” outcome variable is taken from the same longitudinal study which was used to illustrate most of the other techniques, i.e. the Amsterdam Growth and Health Longitudinal Study (Kemper,

Table 8.5 Number of subjects with a particular cluster score (i.e. the number of CHD^a risk factors) measured at six time-points

Time-point	Number of CHD ^a risk factors				
	0	1	2	3	4
1	65	49	25	4	4
2	60	44	33	9	1
3	47	64	26	9	1
4	54	53	29	9	2
5	56	53	26	11	1
6	55	46	33	13	0

^aCHD, coronary heart disease.

1995). One of the aims of this study was to investigate the possible clustering of risk factors for coronary heart disease (CHD) and the longitudinal relationship with several “lifestyle” covariates. To construct a measure of clustering, at each of the six measurements “high risk” quartiles were formed for each of the following biological risk factors: (1) the ratio between total serum cholesterol and high density lipoprotein cholesterol; (2) diastolic blood pressure; (3) the sum of skinfolds; and (4) cardiopulmonary fitness. At each of the repeated measurements, clustering was defined as the number of biological risk factors that occurred in a particular subject. So, if a subject belonged to the “high risk” quartile for all biological risk factors, the clustering score at that particular measurement was 4, if the subject belonged to three “high risk” groups, the clustering score was 3, etc. This cluster score is a “count” outcome variable, and this outcome variable Y_{count} is related to four covariates: (1) the baseline Keys score (a time-independent continuous variable), which is an indicator of the amount of unfavorable fatty acids and cholesterol in the diet; (2) the amount of physical activity (a time-dependent continuous variable); (3) smoking behavior (a time-dependent dichotomous variable); and (4) gender (a time-independent dichotomous variable) (Twisk et al., 2001). In Tables 8.5 and 8.6, descriptive information about the example dataset is shown.

Again, the aim of this study was to investigate the longitudinal relationship between the four covariates and the clustering of CHD risk factors. In the example, both GEE analysis and mixed model analysis will be used to investigate the longitudinal relationships.

8.2.1.2 GEE analysis

As for all GEE analyses, the GEE analysis with a “count” outcome variable requires the choice of a working correlation structure. In principle, there are

Output 8.5 presents the results of the GEE analysis relating outcome variable Y_{count} to the four covariates. The output looks exactly the same as the output from a linear or logistic GEE analysis. The only difference is the different link function, which is log, and the different family, which is Poisson. The other information provided in the output is exactly the same as for the linear and the logistic GEE analysis. So, the most interesting part is the last part of the output in which the regression coefficients are given. As for all the other GEE outputs that have already been discussed, this part of the output shows besides the regression coefficients, the standard errors, the z -values, the corresponding p -values and the 95% confidence intervals around the regression coefficients.

As in the linear and logistic GEE analyses, the scale parameter (also known as dispersion parameter) is an indication of the variance of the model. The interpretation of this coefficient is comparable to the interpretation of the scale parameter for a logistic GEE analysis. This has to do with the characteristics of the Poisson distribution on which the Poisson GEE analysis is based. Within the Poisson distribution the variance is exactly the same as the mean value. So, for the Poisson GEE analysis, the scale parameter has to be one (i.e. a direct relationship between the variance and the mean).

Looking at the estimated parameters, it can be seen that there is a highly significant inverse relationship between the CHD risk cluster score and the amount of physical activity. The regression coefficient of -0.0849986 can be transformed into a rate ratio by taking $\text{EXP}[\text{regression coefficient}]$. The rate ratio for physical activity is therefore $\text{EXP}[-0.0849986] = 0.92$ with a 95% confidence interval from $\text{EXP}[-0.1231015] = 0.88$ to $\text{EXP}[-0.0468957] = 0.96$, and a highly significant p -value ($p < 0.001$). As for all other longitudinal data analyses, this rate ratio can be interpreted in two ways: (1) the between-subjects interpretation; i.e. a difference of one unit in physical activity between subjects is associated with a 9% ($1/0.92 = 1.09$) difference (i.e. lower) in the number of CHD risk factors; and (2) the within-subject interpretation; a change of one unit in physical activity within a subject (over a certain time period) is associated with a decrease of 9% in the number of CHD risk factors.

To investigate the influence of using a different correlation structure, the data were reanalyzed with an independent, a five-dependent, and an unstructured correlation structure. The results are presented in Output 8.6, and in Table 8.7 the results of the Poisson GEE analyses with different correlation structures are summarized.

From Table 8.7, it can be seen that the differences between the results are only marginal. In fact, the results obtained from the GEE analysis with the three dependent (i.e. exchangeable, 5-dependent and unstructured) correlation structures are almost the same. This was also observed for the GEE analysis with a dichotomous outcome variable. In other words, for the longitudinal analysis of a “count”

Table 8.7 Results of Poisson GEE analyses with different correlation structures

	Correlation structure			
	Independent	Exchangeable	5-Dependent	Unstructured
Keys score	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)
Activity	−0.10 (0.03)	−0.08 (0.02)	−0.08 (0.02)	−0.08 (0.02)
Smoking	0.12 (0.10)	0.02 (0.08)	0.04 (0.07)	0.02 (0.07)
Gender	−0.01 (0.12)	−0.00 (0.12)	−0.00 (0.12)	−0.01 (0.12)

Output 8.6 Results of Poisson GEE analyses with different correlation structures

GEE population-averaged model				Number of obs	=	882
Group variable:	id	Number of groups	=	147		
Link:	log	Obs per group: min	=	6		
Family:	Poisson	avg	=	6.0		
Correlation:	independent	max	=	6		
		Wald chi2(4)	=	24.45		
Scale parameter:	1	Prob > chi2	=	0.0001		
Pearson chi2(882):	817.57	Deviance	=	966.04		
Dispersion (Pearson):	0.9269469	Dispersion	=	1.095284		
(Std. Err. adjusted for clustering on id)						

		Semirobust				
ycount		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]

activity		-0.0969721	0.023296	-4.16	0.000	-0.1426314 -0.0513128
gender		-0.0100579	0.1186392	-0.08	0.932	-0.2425863 0.2224706
smoking		0.1285283	0.0984785	1.31	0.192	-0.064486 0.3215426
keys		0.0022342	0.0078617	0.28	0.776	-0.0131744 0.0176429
_cons		0.1791249	0.4672023	0.38	0.701	-0.7365748 1.094825

GEE population-averaged model				Number of obs	=	882
Group and time vars:	id time	Number of groups	=	147		
Link:	log	Obs per group: min	=	6		
Family:	Poisson	avg	=	6.0		
Correlation:	stationary(5)	max	=	6		
		Wald chi2(4)	=	24.34		
Scale parameter:	1	Prob > chi2	=	0.0001		
(Std. Err. adjusted for clustering on id)						

	Semirobust					
ycount	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
activity	-0.0838824	0.0181746	-4.62	0.000	-0.1195041	-0.0482608
gender	-0.0016009	0.1167685	-0.01	0.989	-0.230463	0.2272612
smoking	0.0444	0.070806	0.63	0.531	-0.0943773	0.1831773
keys	0.0020896	0.007858	0.27	0.790	-0.0133117	0.0174909
_cons	0.1399715	0.4639466	0.30	0.763	-0.7693471	1.04929

GEE population-averaged model		Number of obs	=	882
Group and time vars:	id time	Number of groups	=	147
Link:	log	Obs per group: min	=	6
Family:	Poisson	avg	=	6.0
Correlation:	unstructured	max	=	6
		Wald chi2(4)	=	21.65
Scale parameter:	1	Prob > chi2	=	0.0002

(Std. Err. adjusted for clustering on id)

	Semirobust					
ycount	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
activity	-0.0790183	0.0176332	-4.48	0.000	-0.1135787	-0.0444579
gender	-0.0140126	0.1155339	-0.12	0.903	-0.2404548	0.2124297
smoking	0.0161429	0.0714043	0.23	0.821	-0.123807	0.1560928
keys	0.001344	0.0076687	0.18	0.861	-0.0136863	0.0163743
_cons	0.1905495	0.4530101	0.42	0.674	-0.697334	1.078433

outcome variable, GEE analysis also seems to be quite robust against a “wrong” choice of a “working correlation structure.” For the analysis with an independent correlation structure, the regression coefficients are slightly different than for the analysis with the three dependent correlation structures, while the standard errors are slightly higher for all covariates.

8.2.1.3 Mixed model analysis

The first mixed model analysis performed on this dataset is an analysis with only a random intercept. Output 8.7 shows the result of this mixed model analysis.

Output 8.7 Results of a Poisson mixed model analysis with only a random intercept

Mixed-effects Poisson regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
Integration points = 7	Wald chi2(4)	=	14.90			
Log likelihood = -1051.4669	Prob > chi2	=	0.0049			

ycount	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+						
activity	-0.0879271	0.0235087	-3.74	0.000	-0.1340033	-0.041851
gender	-0.0298156	0.1290088	-0.23	0.817	-0.2826682	0.2230371
smoking	0.0482049	0.1057465	0.46	0.648	-0.1590544	0.2554642
keys	0.0024457	0.0082114	0.30	0.766	-0.0136483	0.0185397
_cons	-0.0016474	0.5056195	-0.00	0.997	-0.9926434	0.9893485

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]			
-----+						
id: Identity						
	var(_cons)	0.3871748	0.0779149	0.2609827 0.5743841		

LR test vs. Poisson regression:		chibar2(01) =	124.61	Prob>=chibar2 = 0.0000		

The output of the Poisson mixed model analysis looks the same as the output that has been discussed earlier for the linear and logistic mixed model analysis. The left column of the first part contains general information about the model. It shows that a Poisson mixed model analysis was performed, and it gives the log likelihood of the model (−1051.4669). The right column of the first part of the output shows the information about the number of observations, the number of subjects and the number of observations within the subjects. It also shows the result of a likelihood ratio test which is a comparison between this model and a model with a random intercept but without covariates. Because four covariates are analyzed (Keys score, activity, smoking, and gender), the difference between the two −2 log likelihoods follows a Chi-square distribution with 4 degrees of freedom. The corresponding *p*-value is prob chi2 = 0.0049, which is highly significant.

The second part of the output shows information about the (fixed) regression coefficients. First of all, the regression coefficients can be transformed into rate ratios by taking $\text{EXP}[\text{regression coefficient}]$, and secondly the interpretation of the regression coefficients is a combination of a between-subjects interpretation and a within-subject interpretation.

The last part of the output shows information about the random part of the model. It contains only the variance around the intercept. Furthermore, the result of the likelihood ratio test is given, which compares the model with a random intercept to a model without a random intercept (a “naïve” Poisson regression analysis with the four covariates). This difference is 124.61, and it follows a Chi-square distribution with one degree of freedom (i.e. the random intercept). The corresponding p -value ($\text{prob} > \text{chi2}$) is very low (i.e. highly significant), so it is, also statistically, necessary to allow a random intercept. This likelihood ratio test gives an indication about the importance of the random part of the model, while the earlier mentioned likelihood ratio test provided information about the importance of the fixed part of the model.

The next step in the analysis is to investigate the necessity of allowing a random slope with the time dependent covariates as well. A model with a random slope for physical activity was unfortunately not possible. For smoking on the other hand, it was possible to add a random slope to the model. Output 8.8 shows the result of the Poisson mixed model analysis with both a random intercept and a random slope for smoking.

To investigate the need for a random slope for smoking (in addition to the random intercept), the log likelihood of the model with only a random intercept can be compared to the log likelihood of the model with both a random intercept and a random slope for smoking. The difference between these log likelihoods is 0.2456. Because the likelihood ratio test is based on the difference between the $-2 \log$ likelihoods, the calculated difference has to be multiplied by two. This value (i.e. 0.4912) follows a Chi-square distribution with 2 degrees of freedom (i.e. the random slope for smoking and the covariance between the random intercept and random slope for smoking). The corresponding p -value is far from significant, so a random slope for smoking is not necessary in this situation.

8.2.2 Comparison between GEE analysis and mixed model analysis

In Table 8.8, the results of the longitudinal analysis with a “count” outcome variable performed with GEE analysis and mixed model analysis are summarized. When the results of the GEE analysis and the mixed model analysis are compared, it can be concluded that the differences observed for dichotomous outcome variables (see

Output 8.8 Results of a Poisson mixed model analysis with a random intercept and a random slope for smoking

Mixed-effects Poisson regression	Number of obs	=	882			
Group variable: id	Number of groups	=	147			
	Obs per group: min	=	6			
	avg	=	6.0			
	max	=	6			
Integration points = 7	Wald chi2(4)	=	15.44			
Log likelihood = -1051.2213	Prob > chi2	=	0.0039			

ycount	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+						
activity	-0.0878839	0.0235125	-3.74	0.000	-0.1339675	-0.0418004
gender	-0.0274273	0.128724	-0.21	0.831	-0.2797217	0.224867
smoking	0.0938232	0.1208499	0.78	0.438	-0.1430382	0.3306846
keys	0.00226	0.0081874	0.28	0.783	-0.013787	0.018307
_cons	-0.0031672	0.5040611	-0.01	0.995	-0.9911088	0.9847744

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+						
id: Unstructured						
var(smoking)		0.0069489	0.0194797		0.0000286	1.690682
var(_cons)		0.4072316	0.0861062		0.2690674	0.6163421
cov(smoking,_cons)		-0.053196	0.0768829		-0.2038837	0.0974916

LR test vs. Poisson regression:		chi2(3) =	125.10	Prob > chi2 = 0.0000		

Section 7.2.5) are not observed for a “count” outcome variable. In fact, the observed differences between the two sophisticated techniques are only marginal, although both the regression coefficients and the standard errors obtained from a Poisson mixed model analysis are, in general, slightly higher than those obtained from a Poisson GEE analysis. The fact that the “subject-specific” regression coefficients and standard errors are slightly higher than the “population-averaged” regression coefficients has to do with the characteristics of the Poisson model compared to the linear model. However, the differences are far less pronounced than has been discussed for the logistic model.

Table 8.8 Regression coefficients and standard errors (in parentheses) of longitudinal regression analyses with a “count” outcome variable; a comparison between GEE analysis and mixed model analysis

	GEE analysis ^a	Mixed model analysis ^b
Keys score	0.00 (0.01)	0.00 (0.01)
Activity	−0.08 (0.02)	−0.09 (0.02)
Smoking	0.02 (0.08)	0.05 (0.11)
Gender	−0.00 (0.12)	−0.03 (0.13)

^aGEE analysis with an exchangeable correlation structure.

^bMixed model analysis with only a random intercept.

8.3 Comments

As has been mentioned for the other longitudinal data analyses, an alternative model (i.e. an autoregressive model) can be used to obtain an estimate for the within-subject relationship. The procedures are the same as those described in [Chapter 6](#), so this will not be further discussed.

One of the characteristics of a Poisson distribution is the fact that the average value is equal to the variance. In our example with CHD risk factor clustering the average value of the count outcome variable was 0.97 and the variance 0.92. In other words, the count outcome has a nice Poisson distribution and therefore a longitudinal Poisson regression analysis was appropriate. In many situations, however, the variance of the outcome variable will be higher than the average value. This is known as overdispersion and when this occurs, a longitudinal negative binomial regression analysis is preferred above a longitudinal Poisson regression. For a detailed technical description of this phenomenon, reference is made to, for instance, Lindsey (1993), Diggle et al. (1994), and Agresti et al. (2000). It is also possible that the overdispersion is caused by an excess of zeros. In Section 13.2 this issue will be further discussed.

Analysis of experimental studies

9.1 Introduction

Experimental (longitudinal) studies differ from observational longitudinal studies in that experimental studies (in epidemiology often described as trials) include one or more interventions. In general, before the intervention (i.e. at baseline) the population is (randomly) divided into two or more groups. In the case of two groups, one of the groups receives the intervention of interest and the other group receives a placebo intervention, no intervention at all, or the “usual” treatment. The latter is known as the control group. Both groups are monitored over a certain period of time, in order to find out whether the groups differ with regard to a particular outcome variable. The outcome variable can be continuous, dichotomous, or categorical.

In epidemiology, the simplest form of an experimental longitudinal study is one in which a baseline measurement and only one follow-up measurement are performed ([Figure 9.1](#)). If the subjects are randomly assigned to the different groups (interventions), a comparison of the follow-up values between the groups will give an answer to the question of which intervention is more effective with regard to the particular outcome variable. The assumption is that random allocation at baseline will ensure that there is no difference between the groups at baseline (in fact, in this situation a baseline measure is not even necessary).

Another possibility is to analyze the changes between the values of the baseline and the follow-up measurement, and to compare these changes among the different groups. However, the definition of change can be rather complicated and, although this technique is widely used to analyze experimental studies, the interpretation of the results is more difficult than many researchers think (see [Section 9.2.1](#)).

In the past decade, experimental studies with only one follow-up measurement have become rare. At least one short-term follow-up measurement and one long-term follow-up measurement “must” be performed. However, more than two follow-up measurements are usually performed in order to investigate the

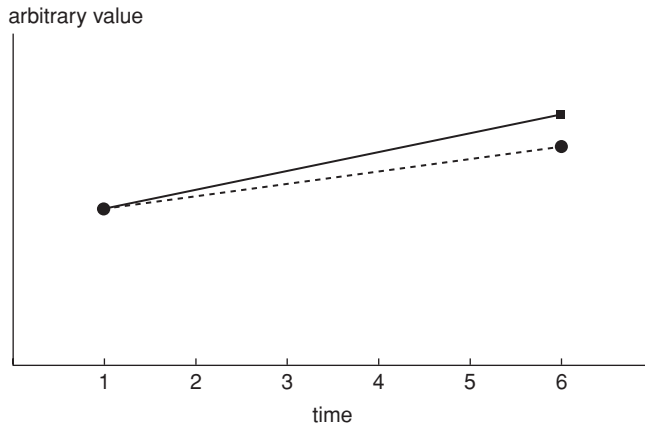


Figure 9.1 An experimental longitudinal study in which one intervention and one control group are compared with regard to a continuous outcome variable at one follow-up measurement (■ — intervention, ● - - - control).

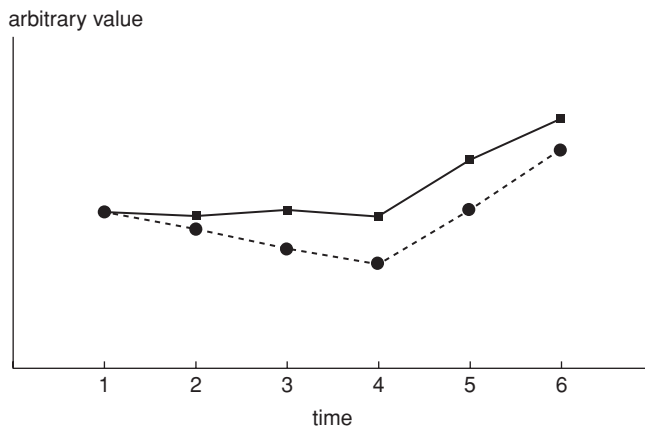


Figure 9.2 An experimental longitudinal study in which one intervention and one control group are compared with regard to a continuous outcome variable at more than one follow-up measurement (■ — intervention, ● - - - control).

“development over time” of the outcome variable, and to compare the “developments over time” among the groups (Figure 9.2). These more complicated experimental designs are often analyzed with simple cross-sectional methods, mostly by analysing the outcome at each follow-up measurement separately, or sometimes even by ignoring the information gathered from the in-between measurements, i.e. only using the last measurement as outcome variable to evaluate the effect of the intervention. This is even more surprising, in view of the fact that there are statistical methods available which can be used to analyze the difference in “development over time” of the outcome variable in two or more groups.

It is obvious that the methods that can be used for the statistical analysis of experimental (longitudinal) studies are exactly the same as have been discussed for observational longitudinal studies. The remainder of this chapter is devoted to extensive examples covering all aspects of the analysis of experimental studies. For educational purposes, various possible ways to analyze the data of an experimental study will be discussed. It should be realized that this will also include methods that are not really suitable in the situation of the example datasets. Like in most other chapters, separate sections will deal with continuous outcome variables and dichotomous outcome variables. Furthermore, in the examples a distinction is made between experimental studies with only one follow-up measurement and experimental studies with more than one follow-up measurement. Although experimental studies with only one follow-up measurement become rare, they are theoretically very important to discuss in detail.

9.2 Continuous outcome variables

9.2.1 Experimental studies with only one follow-up measurement

When the effect of an intervention is evaluated in an experimental study with only one follow-up measurement, mostly the change between the baseline measurement and the follow-up measurement in the continuous outcome variable is compared between the intervention group and the control group. The effect of the intervention can then be analyzed by a cross-sectional linear regression analysis (or even by an independent t -test). This is a very popular method, which greatly reduces the complexity of the statistical analysis. However, how to define the change between two repeated measurements is often more complicated than is usually realized.

In the literature, several methods with which to define changes in continuous outcome variables have been reported. The simplest method is to calculate the difference between two measurements over time (Equation 9.1), a method that is also used in the paired t -test and in the multivariate analysis of variance (MANOVA) for repeated measurements (see Chapter 3).

$$\Delta Y = Y_{it2} - Y_{it1} \quad (9.1)$$

where Y_{it2} are observations for subject i at time t_2 , and Y_{it1} are observations for subject i at time t_1 .

In some situations the relative difference between two subsequent measurements is used as an estimate of the change over time (Equation 9.2).

$$\Delta Y = \frac{(Y_{it2} - Y_{it1})}{Y_{it1}} \times 100\% \quad (9.2)$$

where Y_{it2} are observations for subject i at time t_2 and Y_{it1} are observations for subject i at time t_1 .

Both techniques are suitable in situations in which the continuous outcome variable theoretically ranges from 0 to $+\infty$, or from $-\infty$ to 0, or from $-\infty$ to $+\infty$. Some variables (e.g. scores on questionnaires) have maximal possible values (“ceilings”) and/or minimal possible values (“floors”). To take these “ceilings” and/or “floors” into account, the definition of change can be as shown in Equation 9.3:

$$\text{when } Y_{it2} > Y_{it1} \Delta Y = \frac{(Y_{it2} - Y_{it1})}{(Y_{\max} - Y_{it1})} \times 100\% \quad (9.3a)$$

$$\text{when } Y_{it2} < Y_{it1} \Delta Y = \frac{(Y_{it2} - Y_{it1})}{(Y_{it1} - Y_{\min})} \times 100\% \quad (9.3b)$$

$$\text{when } Y_{it2} = Y_{it1} \Delta Y = 0 \quad (9.3c)$$

where Y_{it2} are observations for subject i at time t_2 , Y_{it1} are observations for subject i at time t_1 , $Y_{\max}$ is the maximal possible value of Y (“ceiling”), and $Y_{\min}$ is the minimal possible value of Y (“floor”).

It is sometimes suggested to apply Equation 9.4 to take into account possible “floor” or “ceiling” effects.

$$\Delta Y = \frac{(Y_{it2} - Y_{it1})}{(Y_{\max} - Y_{\min})} \times 100\% \quad (9.4)$$

where Y_{it2} are observations for subject i at time t_2 , Y_{it1} are observations for subject i at time t_1 , $Y_{\max}$ is the maximal possible value of Y (“ceiling”), and $Y_{\min}$ is the minimal possible value of Y (“floor”).

However, Equation 9.4 is nothing more than a linear transformation of the difference (i.e. divided by the range and multiplied by 100). So, basically, the definition of change based on Equation 9.4 does not take into account possible “floors” or “ceilings.” In Section 13.2, the problem of “floor” and “ceiling” effects will be further discussed.

One of the typical problems related to the use of the change scores discussed so far is the phenomenon of regression to the mean. If the outcome variable at $t = 1$ is a sample of random numbers, and the outcome variable at $t = 2$ is also a sample of random numbers, then the subjects in the upper part of the distribution at $t = 1$ are less likely to be in the upper part of the distribution at $t = 2$, compared to the other subjects. In the same way, the subjects in the lower part of the distribution at $t = 1$ are less likely than the other subjects to be in the lower part of the distribution at $t = 2$. When the change over time in a whole population is analyzed, regression to the mean is not really a big problem. However, in an experimental study, when two groups are compared to each other, it is possible that the average baseline values of the outcome variable differ between the two groups.

As mentioned before, the general idea of random allocation at baseline is that the average values of the two groups are the same. However, that is theoretically only the case when the (source) population is extremely large. In real life practice, the two groups are of limited size and therefore it is highly possible that the average baseline values of the outcome variable differ between the two groups. When the two groups are derived from one (source) population, this difference is totally caused by chance. When the average baseline values of the outcome variable differ between the intervention and the control group, regression to the mean becomes a problem. Suppose that the aim of a particular intervention is to decrease the outcome variable and suppose further that the intervention group has a higher average baseline value compared to the control group. When the intervention has no effect at all, due to regression to the mean, the average value of the intervention group will go down, while the average value of the control group will go up. A comparison between the intervention and the control group will then reveal a favorable intervention effect. This is not a “real” effect, but an effect caused by regression to the mean.

Because of this regression to mean problem, there are methods available to define changes between subsequent measurements, more or less “adjusting” for regression to the mean. One of these methods is known as “analysis of covariance” (Equation 9.5). With this technique the value of the outcome variable Y at the second measurement is used as outcome variable in a linear regression analysis, with the baseline value of the outcome variable Y as one of the covariates,

$$Y_{it2} = \beta_0 + \beta_1 Y_{it1} + \beta_2 \text{Intervention} + \varepsilon_i \quad (9.5)$$

where Y_{it2} are observations for subject i at time t_2 , β_1 is the regression coefficient for the baseline value, Y_{it1} are observations for subject i at baseline (time t_1), β_2 is the regression coefficient for the intervention variable, and ε_i is the “error” for subject i .

In this model, β_2 reflects the effect of the intervention adjusted for possible differences at baseline, i.e. adjusted for regression to the mean. Analysis of covariance is almost, but not quite, the same as the calculation of the difference (see Equation 9.1). This can be seen when Equation 9.5 is written in a different way (Equation 9.6).

$$Y_{it2} - \beta_1 Y_{it1} = \beta_0 + \beta_2 \text{Intervention} + \varepsilon_i \quad (9.6)$$

where Y_{it2} are observations for subject i at time t_2 , β_1 is the regression coefficient for the baseline value, Y_{it1} are observations for subject i at baseline (time t_1), β_2 is the regression coefficient for the intervention variable, and ε_i is the “error” for subject i .

In the analysis of covariance, the change is defined relative to the value of Y at $t = 1$. This relativity is expressed in the regression coefficient β_1 , which is known as the autoregression coefficient (see also Section 6.2.3), and therefore it is assumed that this method adjusts for regression to the mean. This analysis is comparable to the analysis of “residual change,” which was first described by Blomquist (1977). The first step in this method is to perform a linear regression analysis between Y_{t2} and Y_{t1} . The second step is to calculate the difference between the observed value of Y_{t2} and the predicted value of Y_{t2} (predicted by the regression model with Y_{t1}). This difference is called the “residual change,” which is then used as outcome variable in a linear regression analysis with the intervention variable. The regression coefficient of the intervention variable is an estimate of the effect of the intervention adjusting for regression to the mean. Although the general idea behind “residual change” analysis is the same as for analysis of covariance, the results of both methods are not exactly the same. From the literature it is not only known that “residual change” analysis is not equivalent to analysis of covariance, but also that “residual change” analysis is not as good as analysis of covariance (Forbes and Carlin, 2005).

Some researchers argue that the best way to define changes, adjusting for regression to the mean, is a combination of Equations 9.1 and 9.5. They suggest calculating the change between Y_{t2} and Y_{t1} , adjusting for the value of Y_{t1} (Equation 9.7).

$$Y_{it2} - Y_{it1} = \beta_0 + \beta_1 Y_{it1} + \beta_2 \text{Intervention} + \varepsilon_i \quad (9.7)$$

where Y_{it2} are observations for subject i at time t_2 , Y_{it1} are observations for subject i at baseline (time t_1), β_1 is the regression coefficient for the baseline value, Y_{it1} , β_2 is the regression coefficient for the intervention variable, and ε_i is the “error” for subject i .

However, analyzing the change, adjusting for the initial value at $t = 1$, is exactly the same as the analysis of covariance described in Equation 9.5. This can be seen when Equation 9.7 is written in another way (Equation 9.8). The only difference between the models is that the regression coefficient for the initial value is different, i.e. the difference between the regression coefficients for the initial value is equal to one.

$$Y_{it2} = \beta_0 + (\beta_1 + 1)Y_{it1} + \beta_2 \text{Intervention} + \varepsilon_i \quad (9.8)$$

where Y_{it2} are observations for subject i at time t_2 , β_1 is the regression coefficient for the baseline value, Y_{it1} are observations for subject i at baseline (time t_1), β_2 is the regression coefficient for the intervention variable, and ε_i is the “error” for subject i .

Some researchers believe that using the relative change (Equation 9.5) also adjusts for regression to the mean. However, that is not the case. In Figure 9.3, this is nicely illustrated.

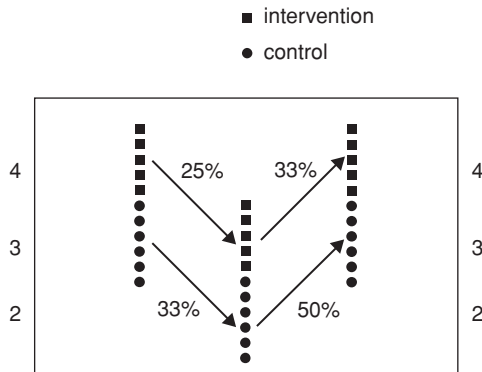


Figure 9.3 Illustration to show that the use of a relative change (Equation 9.5) does not adjust for regression to the mean.

In [Figure 9.3](#), two situations are illustrated. In the first situation, the first column represents the baseline value. It can be seen that the baseline value for the intervention group is higher compared to the control group. The next column reflects the situation after the intervention period. The difference in outcome variable in both groups is equal to one. However, because the intervention group has a higher baseline value compared to the control group, due to regression to the mean, the average value of the intervention group is expected to decrease, while the average value of the control group is expected to increase. In other words, the decrease of one point in the intervention group is easier to achieve than the one point decrease in the control group. When the relative change is calculated, the intervention group decreases by 25%, while the control group decreases by 33%. So, in this situation (when there is a decrease in the outcome variable), the use of the relative change works well.

However, the second part of the figure shows totally the opposite. Now the second column represents the baseline value. Again, there is a difference in baseline values between the two groups and again, the average baseline value of the intervention group is higher compared to the control group. In the second part the figure (where the last column reflects the values after the intervention), however, the outcome variable increases over time. Both groups increase with one point, but because the control group has a lower value at baseline, due to regression to the mean, the one point change is easier to achieve for the control group compared to the one point change in the intervention group. When in this situation the relative change is calculated, the intervention group increases by 33%, while the control group increases by 50%. So, based on the difference between the two relative changes, the control group performs “better” than the intervention group.

This is not true, because in fact, it is just the opposite; the intervention group performs “better” than the control group. In other words, when the outcome variable decreases over time, the use of the relative change more or less adjusts for regression to the mean, but when the outcome variable increases, it goes totally wrong.

9.2.1.1 Example

The example dataset used in this chapter is (of course) different from the one used in the foregoing chapters. Because this chapter deals with the analysis of data from experimental studies, the example dataset is derived from a randomized controlled trial (RCT), which was performed by Proper et al. (2003). The example nicely shows the influence of ignoring the regression to the mean phenomenon on the results and conclusions of an experimental study (and in particular an RCT). In brief, 299 civil servants working within municipal services in the Netherlands were randomized either into an intervention or a control group. All subjects randomized into the intervention group were offered seven consultations, each 20 minutes in duration. The intervention period was nine months, and counseling was focused primarily on the enhancement of the individual’s level of physical activity. Subjects in the control group received no individual counseling. After baseline measurements, they received only written information about several lifestyle factors. This information was the same as that handed out to the intervention group. Outcome variables were assessed before the intervention (i.e. at baseline) and directly after the completion of the last consultation.

In principle, the trial had three primary outcome variables (physical activity, cardio-respiratory fitness, and prevalence of musculoskeletal disorders [e.g., upper extremity complaints]) and three secondary outcomes (body composition [i.e., the percentage of body fat and the body mass index], blood pressure, and total serum cholesterol), but to make the example not too extensive, we selected two continuous outcome variables (physical activity and total serum cholesterol), which were selected because of differences between the groups at baseline (Table 9.1). Physical activity was assessed by the Baecke questionnaire (Baecke et al., 1982). With this questionnaire, physical activity during sport and leisure time was measured. Both were combined into one physical activity index.

From Table 9.1 it can be seen that for both the physical activity index and total serum cholesterol, the baseline value for the intervention group is higher than for the control group.

To analyze the effect of the intervention, several methods were used. Output 9.1 shows the results for total serum cholesterol of three methods described in Section 9.2.1, i.e. the comparison of changes, the comparison of changes, adjusted for baseline values and the analysis of covariance.

Table 9.1 Mean and standard deviation (in parentheses) for the variables used in the RCT example, i.e. physical activity index and total serum cholesterol

	Baseline	Follow-up
Physical activity index		
Intervention group	5.80 (1.08)	5.95 (0.95)
Control group	5.47 (1.07)	5.39 (1.04)
Total serum cholesterol (mmol/l)		
Intervention group	5.51 (1.04)	5.33 (0.99)
Control group	5.35 (0.94)	5.39 (0.95)

Output 9.1 Results from three methods to analyze the effect of the intervention on total serum cholesterol. The comparison of changes (a), the comparison of changes, adjusted for baseline values (b), and the analysis of covariance (c)

(a)						
del_chol	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
intervention	-0.220306	0.0888516	-2.48	0.014	-0.3955283	-0.0450836
_cons	0.0364762	0.0610664	0.60	0.551	-0.0839516	0.156904
(b)						
del_chol	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
intervention	-0.184309	0.0837584	-2.20	0.029	-0.3494925	-0.0191256
chol_base	-0.2212941	0.0424295	-5.22	0.000	-0.304971	-0.1376172
_cons	1.22021	0.2341	5.21	0.000	0.7585317	1.681888
(c)						
chol_t1	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
intervention	-0.184309	0.0837584	-2.20	0.029	-0.3494925	-0.0191256
chol_base	0.7787059	0.0424295	18.35	0.000	0.695029	0.8623828
_cons	1.22021	0.2341	5.21	0.000	0.7585317	1.681888

From Output 9.1 it can be seen that the effect of the intervention is overestimated when changes between the baseline and follow-up measurement are compared between the intervention and control group. This overestimation is caused by the differences between the groups at baseline. Because the intervention group starts at a higher level, and the intervention is intending to decrease total serum cholesterol values, regression to the mean is “helping” the intervention group to decrease. Analysis of covariance adjusts for the differences at baseline and therefore this analysis revealed a less strong intervention effect (i.e. -0.184309 versus -0.220306). As has been explained in Section 9.2.1, the analysis of covariance and the analysis of changes adjusted for the baseline are basically the same and therefore, they reveal the same intervention effect.

Although for both total serum cholesterol and the physical activity index the intervention group has higher values at baseline, for the physical activity index, the results show a different picture than for total serum cholesterol (Output 9.2).

Output 9.2 Results from three methods to analyze the effect of the intervention on the physical activity index. The comparison of changes (a), the comparison of changes, adjusted for baseline values (b), and the analysis of covariance (c).

(a)

del_act	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
intervention	0.2473911	0.096712	2.56	0.011	0.0566673	0.4381149
_cons	-0.0904762	0.0664688	-1.36	0.175	-0.221558	0.0406056

(b)

del_act	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
intervention	0.3328238	0.0887946	3.75	0.000	0.1577083	0.5079392
act_base	-0.2675616	0.0408708	-6.55	0.000	-0.3481645	-0.1869587
_cons	1.375379	0.2319074	5.93	0.000	0.9180252	1.832734

(c)

act_t1	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
intervention	0.3328238	0.0887946	3.75	0.000	0.1577083	0.5079392
act_base	0.7324384	0.0408708	17.92	0.000	0.6518355	0.8130413
_cons	1.375379	0.2319074	5.93	0.000	0.9180252	1.832734

From Output 9.2 it can be seen that the analysis of changes underestimates the effect of the intervention. Again this is due to the differences at baseline between the intervention and the control group. The difference between the analysis of the intervention effect for total serum cholesterol and the physical activity index has to do with the fact that the intervention aimed to increase physical activity and to decrease total serum cholesterol. So for the physical activity index, regression to the mean is “helping” the control group and, therefore, the analysis of changes results in an underestimation of the intervention effect.

In this example with only one follow-up measurement, it is also possible to use the sophisticated methods (i.e. generalized estimating equations (GEE) analysis or mixed model analysis) to analyze the intervention effect. The sophisticated methods can only be used when there are correlated observations in the outcome variable, so when a sophisticated method is used for the analysis of an RCT with only one follow-up measurement it directly implies that the baseline value must be part of the outcome variable. One of the possibilities often used in practice is given in Equation 9.9.

$$Y_{it} = \beta_0 + \beta_1 \text{Intervention} + \beta_2 \text{time} + \beta_3 \text{Intervention} \times \text{time} + \varepsilon_{it} \quad (9.9)$$

where Y_{it} are the observations for subject i at time t , β_1 is the regression coefficient for the intervention variable, β_2 is the regression coefficient for time, time is time of measurement (0 for the baseline measurement and 1 for the follow-up measurement), β_3 is the regression coefficient for the interaction between the intervention variable and time, and ε_{it} is the “error” of individual i at time t .

The coefficient of interest in this analysis is the regression coefficient for the interaction between the intervention variable and time (i.e. β_3). The regression coefficient for the intervention variable (β_1) reflects the differences between the two groups at baseline, while the summation of the regression coefficient for the intervention variable and the regression coefficient for the interaction term ($\beta_1 + \beta_3$) reflects the difference between the intervention and the control group at the follow-up measurement. So, the regression coefficient for the interaction term reflects the difference in the difference between the intervention and control group between the baseline and the follow-up measurement, which is an estimation of the intervention effect. Output 9.3 shows the results of the GEE analysis for both total serum cholesterol and the physical activity index, using Equation 9.9. For the sake of simplicity only the results of a GEE analysis are shown; the results of a comparable mixed model analysis are exactly the same.

As has been mentioned before, the coefficients of interest from the results of the GEE analysis shown in Output 9.3 are the regression coefficients belonging to the interaction terms. Those coefficients reflect the effect of the intervention for the two outcome variables. For cholesterol, the effect of the intervention was -0.220306 , while for physical activity, the effect of the intervention was 0.2473911 . When these

(a)

GEE population-averaged model		Number of obs	=	398
Group variable:	id	Number of groups	=	199
Link:	identity	Obs per group: min	=	2
Family:	Gaussian	avg	=	2.0
Correlation:	exchangeable	max	=	2
		Wald chi2(3)	=	9.85
Scale parameter:	0.9449958	Prob > chi2	=	0.0199

(Std. Err. adjusted for clustering on id)

	Semirobust					
chol	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
intervention	0.1626657	0.1406085	1.16	0.247	-0.1129219	0.4382532
time	0.0364762	0.0646368	0.56	0.573	-0.0902095	0.1631619
int_time	-0.220306	0.0879709	-2.50	0.012	-0.3927258	-0.0478862
_cons	5.349143	0.0912884	58.60	0.000	5.170221	5.528065

(b)

GEE population-averaged model		Number of obs	=	398
Group variable:	id	Number of groups	=	199
Link:	identity	Obs per group: min	=	2
Family:	Gaussian	avg	=	2.0
Correlation:	exchangeable	max	=	2
		Wald chi2(3)	=	19.25
Scale parameter:	1.072655	Prob > chi2	=	0.0002

(Std. Err. adjusted for clustering on id)

	Semirobust					
act	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
intervention	0.3193009	0.1528573	2.09	0.037	0.0197062	0.6188957
time	-0.0904762	0.062347	-1.45	0.147	-0.2126741	0.0317217
int_time	0.2473911	0.0971184	2.55	0.011	0.0570424	0.4377397
_cons	5.478571	0.1041608	52.60	0.000	5.27442	5.682723

(b)

GEE population-averaged model		Number of obs	=	398
Group variable:	id	Number of groups	=	199
Link:	identity	Obs per group: min	=	2
Family:	Gaussian	avg	=	2.0
Correlation:	exchangeable	max	=	2
		Wald chi2(2)	=	12.80
Scale parameter:	1.093279	Prob > chi2	=	0.0017

(Std. Err. adjusted for clustering on id)

	Semirobust					
act	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

time	-0.1222315	0.0607048	-2.01	0.044	-0.2412107	-0.0032523
int_time	0.3146177	0.0881056	3.57	0.000	0.141934	0.4873015
_cons	5.629397	0.0770809	73.03	0.000	5.478321	5.780473

From Output 9.4 it can be seen that the intervention effects estimated with this method are slightly different from the intervention effects estimated with the analysis of covariance. For total serum cholesterol the effect estimates were respectively -0.1871102 and -0.184309 while for the physical activity index the effect estimates were 0.3146177 and 0.3328238 . This was not really expected because both methods adjust for the differences at baseline. However, when a mixed model analysis is used with both a random intercept and a random slope for *time* the effect estimates were exactly the same as the ones estimated with the analysis of covariance (Output 9.5), despite the fact that the estimation of the random part of the model is estimated with a huge error.

Table 9.2 summarizes the results of the different analyses performed to evaluate the effect of the intervention on total serum cholesterol and the physical activity index.

From the results reported in Table 9.2 it can be seen that the estimation of the intervention effect highly depends on the method used. And although there is some debate whether or not an adjustment for baseline differences must be performed, it is more or less accepted that the adjustment has to be done (Steyerberg, 2000; Vickers and Altman, 2001; Twisk and Proper, 2004; Lingsma et al., 2010). In other words, it is strongly advised to use analysis of covariance to estimate the effect of an intervention in an RCT with only one follow-up measurement.

(a)			
Mixed-effects REML regression	Number of obs	=	398
Group variable: id	Number of groups	=	199
	Obs per group: min	=	2
	avg	=	2.0
	max	=	2
	Wald chi2(2)	=	7.22
Log restricted-likelihood = -460.17958	Prob > chi2	=	0.0270

	chol	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
	time	0.0194728	0.0592863	0.33	0.743	-0.0967262 0.1356717
	int_time	-0.1843094	0.0832616	-2.21	0.027	-0.3474992 -0.0211196
	_cons	5.42598	0.0700378	77.47	0.000	5.288708 5.563251

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured					
	var(time)	0.2027585	14.68195	4.68e-63	8.78e+60
	var(_cons)	0.8817129	7.34179	7.20e-08	1.08e+07
	cov(time,_cons)	-0.1215739	7.341217	-14.5101	14.26695

```
(b)
Mixed-effects REML regression
Group variable: id

Number of obs      =      398
Number of groups   =      199

Obs per group: min =         2
                avg  =        2.0
                max  =         2

Wald chi2(2)      =        14.73
Prob > chi2       =        0.0006

Log restricted-likelihood = -489.17425
```

act	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
time	-0.130831	0.0636453	-2.06	0.040	-0.2555736	-0.0060884
int_time	0.3328231	0.0876073	3.80	0.000	0.161116	0.5045302
_cons	5.629397	0.0770808	73.03	0.000	5.478321	5.780473

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured				
var(time)	0.2415794	26.03633	4.41e-93	1.32e+91
var(_cons)	1.070483	13.01894	4.76e-11	2.41e+10
cov(time,_cons)	-0.204484	13.01844	-25.72017	25.3112
var(Residual)	0.1118645	13.01816	9.8e-101	1.28e+98

LR test vs. linear regression:	chi2(3) =	192.45	Prob > chi2 =	0.0000
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Table 9.2 Regression coefficients and standard errors (in parentheses) derived from different analyses to estimate the intervention effect on both total serum cholesterol and the physical activity index

	Intervention effect (SE)
Total serum cholesterol	
Changes	-0.22 (0.09)
Changes adjusting for baseline	-0.18 (0.08)
Analysis of covariance	-0.18 (0.08)
GEE analysis without adjusting for baseline	-0.22 (0.09)
Mixed model analysis without adjusting for baseline	-0.22 (0.09)
GEE analysis adjusting for baseline	-0.19 (0.08)
Mixed model analysis adjusting for baseline	-0.18 (0.08)
Physical activity index	
Changes	0.25 (0.10)
Changes adjusting for baseline	0.33 (0.09)
Analysis of covariance	0.33 (0.09)
GEE analysis without adjusting for baseline	0.25 (0.10)
Mixed model analysis without adjusting for baseline	0.25 (0.10)
GEE analysis adjusting for baseline	0.31 (0.09)
Mixed model analysis adjusting for baseline	0.33 (0.09)

Table 9.3 Mean and standard deviation (in parentheses) in the treatment groups at three time-points

	Placebo group			Therapy group		
	t_1	t_2	t_3	t_1	t_2	t_3
N	71	74	66	68	69	65
Systolic blood pressure	130.7 (17.6)	129.1 (16.9)	126.3 (14.2)	126.5 (12.5)	122.5 (11.2)	121.6 (12.1)

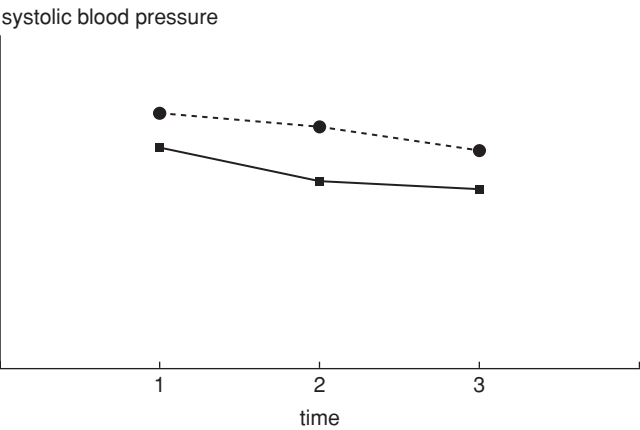


Figure 9.4 Development of systolic blood pressure over time in the two groups (■ — therapy, ● — — — placebo).

9.2.2 Experimental studies with more than one follow-up measurement

The example dataset used in this section is taken from an RCT in which a therapy intervention is compared to a placebo intervention with regard to the development of systolic blood pressure (Vermeulen et al., 2000). In this RCT, three measurements were carried out: one baseline measurement and two follow-up measurements with equally spaced time intervals. A total of 152 patients were included in the study, equally divided between the therapy and the placebo group.

Table 9.3 gives descriptive information about the variables used in the study, while Figure 9.4 illustrates the development of systolic blood pressure over time for both groups. It should be noted that the main purpose of the therapy intervention under study was not to lower the systolic blood pressure; this was investigated as a side effect. That is one of the reasons why the number of subjects at baseline was lower than the number of subjects at the first follow-up measurement. In the

Table 9.4 Results of independent sample t -tests to obtain a short-term and long-term effect of the intervention by comparing the mean systolic blood pressure at $t = 2$ and $t = 3$ between the therapy and placebo group

	Effect	95% confidence interval	p -value
Short-term effect (at $t = 2$)	-6.57	-11.34 to -1.80	0.007
Long-term effect (at $t = 3$)	-4.69	-9.26 to -0.11	0.044

example, the results of many different analyses will be shown. It should be realized that some of the analyses are not really appropriate in this particular situation. However, because these (wrong) analyses are often used in real life practice, it is important to show the impact of the use of such inappropriate analyses.

9.2.2.1 Simple analysis

The simplest way to answer the question of whether the therapy intervention is more effective than the placebo intervention is to compare systolic blood pressure values at the two follow-up measurements between the two groups. The hypothesis is that the systolic blood pressure will be lower in the group that received the therapy intervention than in the group that received the placebo intervention. In this example, a distinction can be made between the short-term effect and the long-term effect. To analyze the short-term effect, the mean systolic blood pressure measured at $t = 2$ can be compared between the therapy group and placebo group. For the long-term effect, the systolic blood pressure measured at $t = 3$ can be compared. The differences between the two groups can be analyzed with an independent sample t -test. In [Table 9.4](#) the results of these analyses are shown.

From the results in [Table 9.4](#) it can be seen that there is both a significant short-term and a significant long-term effect in favor of the therapy group, but that the long-term differences between the two groups are smaller than the short-term differences. This indicates that the short-term effect is stronger than the long-term effect. A slightly different approach is not to analyze the systolic blood pressure values at $t = 2$ and $t = 3$, but to analyze the short-term and long-term changes in systolic blood pressure. For this purpose the changes in systolic blood pressure between $t = 1$ and $t = 2$ and between $t = 1$ and $t = 3$ were calculated. Obviously, the change scores for the therapy and the placebo groups can be compared to each other with an independent sample t -test or with a cross-sectional linear regression analysis. [Table 9.5](#) shows the results of these analyses.

The results presented in [Table 9.5](#) show a different picture than the results in [Table 9.4](#); i.e. the analysis of the changes between baseline and follow-up

Table 9.5 Results of independent sample *t*-tests to obtain a short-term and long-term effect of the intervention by comparing the changes in systolic blood pressure between the therapy and placebo group

	Therapy	Placebo	Effect	95% confidence interval	<i>p</i> -value
Short-term difference ^a	3.38	0.64	−2.74	−6.84 to 1.37	0.189
Long-term difference ^b	4.23	3.13	−1.10	−5.46 to 3.25	0.616

^a(Systolic blood pressure at *t* = 1) − (systolic blood pressure at *t* = 2).

^b(Systolic blood pressure at *t* = 1) − (systolic blood pressure at *t* = 3).

Table 9.6 Results of analyses of covariance to obtain a short-term and long-term effect of the therapy intervention

	Effect	95% confidence interval	<i>p</i> -value
Short-term effect	−4.38	−8.11 to −0.65	0.022
Long-term effect	−2.96	−6.77 to 0.85	0.127

measurements show a non-significant beneficial effect of the therapy intervention compared with the placebo intervention. Although the changes for the therapy group were slightly greater than the changes for the placebo group in all comparisons, the independent sample *t*-test did not produce any significant difference. In conclusion, most of the assumed effect of the therapy was already present at baseline. So this “effect” could not be attributed to the therapy intervention.

In Section 9.2.1 it was shown that both analyses performed (i.e. the comparison of the follow-up measurements and the comparison of the changes between the baseline measurement and the follow-up measurements) are not appropriate in an RCT when the baseline values of the groups are different. From the descriptive information (see [Table 9.3](#)) it was shown that the baseline blood pressure of the two groups differed from each other (130.7 mmHg for the placebo group versus 126.5 mmHg for the intervention group). To adjust for the baseline differences, analyses of covariance should be performed to obtain a short-term and long-term effect of the intervention. [Table 9.6](#) shows the result of these analyses.

As expected, from [Table 9.6](#), it can be seen that the long-term effect is less strong than the short-term effect. Furthermore, both effects are a bit stronger than the effects estimated with the comparison between the changes between the baseline measurement and the follow-up measurements. This has to do with regression to the mean; because the baseline value of the therapy group is lower than the baseline

Table 9.7 Examples of summary statistics which are frequently used in experimental studies

The mean of all follow-up measurements
The highest (or lowest) value during follow-up
The time needed to reach the highest value or a certain predefined level
The area under the curve

value of the control group, it is more “difficult” for the therapy group to decrease its systolic blood pressure (see also Section 9.2.1).

9.2.2.2 Summary statistics

There are many summary statistics available with which to estimate the effect of an intervention in an experimental longitudinal study. Depending on the research question to be addressed and the characteristics of the outcome variable, different summary statistics can be used. The general idea of a summary statistic is to express the longitudinal development of a particular outcome variable as one quantity. Therefore, the (complicated) longitudinal problem is reduced to a cross-sectional problem. To evaluate the effect of the intervention, the summary statistics of the groups under study are compared to each other. Table 9.7 gives a few examples of summary statistics.

One of the most frequently used summary statistics is the area under the curve (AUC). The AUC is calculated as shown in Equation 9.11.

$$AUC = \frac{1}{2} \sum_{t=1}^T (t_{t+1} - t_t) (Y_t + Y_{t+1}) \tag{9.11}$$

where AUC is the area under the curve, T is the number of measurements, and Y_t is the observation of the outcome variable at time t .

The unit of the AUC is the multiplication of the unit used for the outcome variable Y and the unit used for time. This is often rather difficult, and therefore the AUC is often divided by the total time period under consideration in order to obtain a “weighted” average level over the time period. When the AUC is used as a summary statistic, the AUC must first be calculated for each subject; this is then used as an outcome variable to evaluate the effect of the therapy under study. Again, this comparison is simple to carry out with an independent t -test or a cross-sectional linear regression analysis. The result of the analysis is shown in Table 9.8.

Table 9.8 Area under the curve for systolic blood pressure between $t = 1$ and $t = 3$; a comparison between therapy and placebo group and p -value derived from an independent sample t -test

	Therapy	Placebo	Effect	95% confidence interval	p -value
Area under the curve	246.51	259.23	−12.72	−21.98 to −3.47	0.007

From [Table 9.8](#) it can be seen that a highly significant difference was found between the AUC values of the two groups. This will not directly indicate that the therapy intervention has an effect on the outcome variable. In the calculation, the difference in baseline value between the two groups is not taken into account. So, again, a difference in baseline value between groups can cause a difference in AUC.

When the time intervals are equally spaced (like in the example dataset), the AUC is comparable to the overall mean. The AUC becomes interesting when the time intervals in the longitudinal study are unequally spaced, because then the AUC reflects the “weighted” average in a certain outcome variable over the total follow-up period.

9.2.2.3 MANOVA for repeated measurements

With the simple methods described in Section 9.2.2.1, separate analyses for short-term and long-term effects were performed. The purpose of summary statistics, such as the AUC, is to summarize the total development of the outcome variable, in order to make a cross-sectional analysis possible. Another way to analyze the total development of the outcome variable and to answer the question of whether the therapy has an effect on a certain outcome variable, is to use MANOVA for repeated measurements (see [Chapter 3](#)). [Output 9.6](#) shows the result of the MANOVA for repeated measurements.

The output of the MANOVA for repeated measurements reveals that for systolic blood pressure there is an overall group effect ($F = 6.593$, $p = 0.011$) and an overall time effect ($F = 6.430$, $p = 0.002$), but no significant interaction between group and time ($F = 1.268$, $p = 0.283$). In particular, the information regarding the interaction is important, because this indicates that the observed overall group effect does not change over time. This means that from the results of the MANOVA for repeated measurements it can be concluded that the two groups on average differ over time (a significant group effect), but that this difference is present along the whole longitudinal period, including the baseline measurement (no significant interaction effect between group and time). So there is no “real” therapy effect. From [Figure 9.5](#) it can be seen that there is a decrease in systolic blood pressure over

Output 9.6 Result of MANOVA for repeated measurements for systolic blood pressure (only the “univariate” estimation procedure is presented)

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	Df	Mean Square	F	Sig.
time	Sphericity Assumed	816.415	2	408.207	6.430	0.002
	Greenhouse-Geisser	816.415	1.957	417.209	6.430	0.002
	Huynh-Feldt	816.415	2.000	408.207	6.430	0.002
	Lower-bound	816.415	1.000	816.415	6.430	0.013
time * therapy	Sphericity Assumed	160.953	2	80.476	1.268	0.283
	Greenhouse-Geisser	160.953	1.957	82.251	1.268	0.283
	Huynh-Feldt	160.953	2.000	80.476	1.268	0.283
	Lower-bound	160.953	1.000	160.953	1.268	0.263
Error(time)	Sphericity Assumed	14854.683	234	63.482		
	Greenhouse-Geisser	14854.683	228.951	64.881		
	Huynh-Feldt	14854.683	234.000	63.482		
	Lower-bound	14854.683	117.000	126.963		

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Intercept	5701685.315	1	5701685.315	12039.101	0.000
therapy	3122.558	1	3122.558	6.593	0.011
Error	55410.881	117	473.597		

time. Because a decrease in systolic blood pressure is considered to be beneficial, a beneficial development is observed in both groups, which seems to be independent of the therapy intervention.

9.2.2.4 MANOVA for repeated measurements adjusted for the baseline value

When the baseline values are different in the groups to be compared, it is often suggested that a MANOVA for repeated measurements should be performed, adjusting for the baseline value of the outcome variable. With this procedure the changes

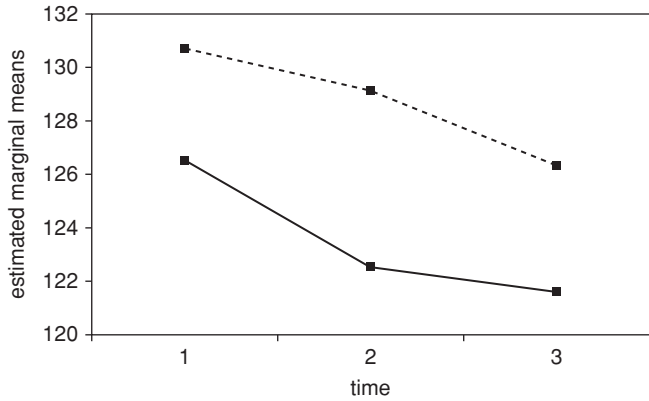


Figure 9.5 Graphical representation of the results of the MANOVA for repeated measurements (— placebo, - - - therapy).

between the baseline measurement and the first follow-up measurement as well as the changes between the first and the second follow-up measurements are adjusted for the baseline value.

It should be noted carefully that when this procedure (which is also known as multivariate analysis of covariance, i.e. MANCOVA for repeated measurements) is performed, the baseline value is both an outcome variable (i.e. to create the difference between the baseline value and the first follow-up measurement) and a covariate. In some software packages (such as SPSS) this is not possible, and therefore an exact copy of the baseline value must be added to the model. Output 9.7 shows the results of the MANCOVA for repeated measurements.

It should be noted that in a MANCOVA for repeated measurements the overall group effect is an indication of the effect of the therapy intervention. This has to do with the adjustment for the baseline value. From the results of Output 9.7 it can be seen that there is a significant therapy effect ($p = 0.018$). In addition, the interaction between time and therapy (obtained from the MANCOVA for repeated measurements) does not provide information about the “direct” therapy effect. The therapy by time interaction provides information about whether the observed therapy effect is stronger at the beginning or at the end of the follow-up period. From the results it can be seen that the time by therapy interaction is almost significant ($p = 0.062$), but it is not clear during which part of the follow-up period the effect is the strongest. Therefore, a graphical representation of the MANCOVA results is needed (Figure 9.6). From Figure 9.6 it can be seen that in the first part of the follow-up period, the therapy effect is the strongest. It should further be noted that the correction for baseline leads to equal starting points for both groups.

Output 9.7 Result of MANCOVA for repeated measurements for systolic blood pressure (only the “univariate” estimation procedure is presented)

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
Time	Sphericity Assumed	1959.230	2	979.615	18.057	0.000
	Greenhouse-Geisser	1959.230	1.953	1003.326	18.057	0.000
	Huynh-Feldt	1959.230	2.000	979.615	18.057	0.000
	Lower-bound	1959.230	1.000	1959.230	18.057	0.000
time * therapy	Sphericity Assumed	304.692	2	152.346	2.808	0.062
	Greenhouse-Geisser	304.692	1.953	156.033	2.808	0.064
	Huynh-Feldt	304.692	2.000	152.346	2.808	0.062
	Lower-bound	304.692	1.000	304.692	2.808	0.096
Error(time)	Sphericity Assumed	12586.272	232	54.251		
	Greenhouse-Geisser	12586.272	226.517	55.564		
	Huynh-Feldt	12586.272	232.000	54.251		
	Lower-bound	12586.272	116.000	108.502		

Tests of Between-Subjects Effects

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Intercept	3599.882	1	3599.882	38.534	0.000
therapy	539.478	1	539.478	5.775	0.018
Error	10836.937	116	93.422		

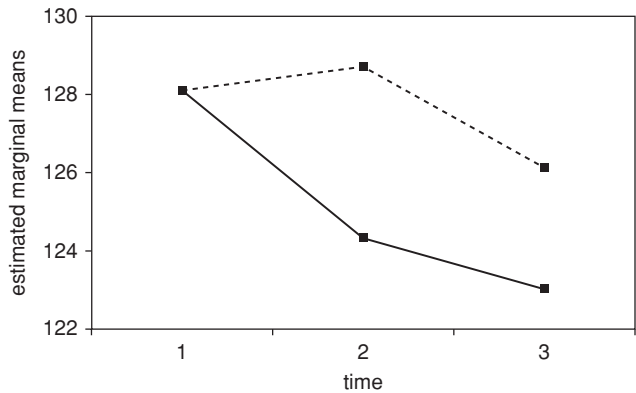


Figure 9.6 Graphical representation of the results of the MANCOVA for repeated measurements (— placebo, - - - therapy).

9.2.2.5 Sophisticated analysis

In the discussion regarding the modeling of time (Chapter 5) it has already been mentioned that the questions answered by MANOVA for repeated measurements could also be answered by sophisticated methods (GEE analysis and mixed model analysis). The advantage of the sophisticated methods is that all available data are included in the analysis, while with MANOVA for repeated measurements (and therefore also with MANCOVA) only those subjects with a complete dataset are included. In the present example the MANOVA for repeated measurements (and the MANCOVA) was carried out for 118 patients, whereas with GEE analysis and mixed model analysis all available data relating to all 152 patients can be used.

There are many possibilities to use the sophisticated methods to evaluate the effect of an intervention in an experimental study with more than one follow-up measurement. The different possibilities will be discussed step by step. Again it should be noted that not all possibilities are equally appropriate.

To analyze the effects of the therapy on systolic blood pressure, the following statistical model was first used (Equation 9.12).

$$Y_{it} = \beta_0 + \beta_1 therapy + \varepsilon_{it}$$
 (9.12)

where Y_{it} are observations for subject i at follow-up time t , β_0 is the intercept, β_1 is the regression coefficient for therapy versus placebo, and ε_{it} is the “error” for subject i at time t .

First a GEE analysis was performed. It should be noted that in the longitudinal data file used for this analysis, only the outcome variable Y at $t = 2$ and $t = 3$ are used. Output 9.8 shows the result of this GEE analysis.

Output 9.8 Results of the GEE analysis with an exchangeable correlation structure to analyze the effect of the therapy intervention on systolic blood pressure using Equation 9.12

GEE population-averaged model		Number of obs	=	274		
Group variable:	id	Number of groups	=	143		
Link:	identity	Obs per group: min	=	1		
Family:	Gaussian	avg	=	1.9		
Correlation:	exchangeable	max	=	2		
		Wald chi2(1)	=	6.93		
Scale parameter:	190.6738	Prob > chi2	=	0.0085		
(Std. Err. adjusted for clustering on id)						

		Semirobust				
sysbp		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]

therapy		-5.594428	2.124422	-2.63	0.008	-9.758219 -1.430638
_cons		127.6229	1.707014	74.76	0.000	124.2772 130.9686

From Output 9.8 it can be seen that the therapy variable is significantly inversely associated with systolic blood pressure. This indicates that on average over time, the therapy group has a 5.594428 mmHg lower systolic blood pressure compared to the control group. This does not mean that the therapy has an effect on systolic blood pressure, because it is possible that the difference between the groups was already present at baseline. In fact, the analysis performed is only suitable in a situation in which the baseline values of the outcome variable are equal for the therapy group and the placebo group. To analyze the “real” therapy effect, a second longitudinal analysis was carried out in which an adjustment was made for the baseline value (Equation 9.13).

$$Y_{it} = \beta_0 + \beta_1 \text{therapy} + \beta_2 Y_{i0} + \varepsilon_{it} \quad (9.13)$$

where Y_{it} are observations for subject i at follow-up time t , β_0 is the intercept, β_1 is the regression coefficient for therapy versus placebo, β_2 is the regression coefficient for the baseline value, Y_{i0} is the baseline value of the outcome variable Y and ε_{it} is the “error” for subject i at time t .

This model, which is known as a longitudinal analysis of covariance is an extension of Equation 9.5; The model looks similar to the autoregressive model which was described in Chapter 6 (Section 6.2.3), but it is slightly different. In an autoregressive analysis, an adjustment is made for the value of the outcome variable at $t = t - 1$ (Equation 9.14), while in the analysis of covariance, an adjustment is made for the baseline value of the outcome variable.

$$Y_{it} = \beta_0 + \beta_1 \text{therapy} + \beta_2 Y_{it-1} + \varepsilon_{it} \quad (9.14)$$

where Y_{it} are observations for subject i at follow-up time t , β_0 is the intercept, β_1 is the regression coefficient for therapy versus placebo, β_2 is the regression coefficient for the value of the outcome variable Y at $t - 1$, and ε_{it} is the “error” for subject i at time t .

Output 9.9 shows the results of the GEE analysis based on a longitudinal analysis of covariance, while Output 9.10 shows the results of an autoregressive analysis.

If the results obtained from a longitudinal analysis of covariance are compared to the results obtained from an autoregressive analysis, it can be seen that the therapy effect obtained from a longitudinal analysis of covariance is higher than the one obtained from an autoregressive analysis (-3.709455 versus -2.276566).

The difference between a longitudinal analysis of covariance and the autoregressive analysis is that in the latter basically the change between Y at $t = 1$ and Y at $t = 2$ and the change between Y at $t = 2$ and Y at $t = 3$ are combined in one analysis, while in the longitudinal analysis of covariance the change between Y at $t = 1$ and

Output 9.9 Results of a GEE analysis with an exchangeable correlation structure to analyze the effect of the therapy intervention on systolic blood pressure using Equation 9.13 (i.e. longitudinal analysis of covariance)

GEE population-averaged model		Number of obs	=	249
Group variable:	id	Number of groups	=	130
Link:	identity	Obs per group: min	=	1
Family:	Gaussian	avg	=	1.9
Correlation:	exchangeable	max	=	2
		Wald chi2(2)	=	70.70
Scale parameter:	108.8449	Prob > chi2	=	0.0000
(Std. Err. adjusted for clustering on id)				

		Semirobust		z	P> z	[95% Conf. Interval]	
sysbp		Coef.	Std. Err.				
therapy		-3.709455	1.539928	-2.41	0.016	-6.727659	-0.6912508
baseline		0.6157178	0.079142	7.78	0.000	0.4606024	0.7708332
_cons		48.2724	9.961027	4.85	0.000	28.74915	97.79566

Output 9.10 Results of a GEE analysis with an independent correlation structure to analyze the effect of the therapy intervention on systolic blood pressure based on Equation 9.14 (i.e. autoregressive analysis)

GEE population-averaged model		Number of obs	=	261
Group variable:	id	Number of groups	=	142
Link:	identity	Obs per group: min	=	1
Family:	Gaussian	avg	=	1.8
Correlation:	independent	max	=	2
		Wald chi2(2)	=	132.53
Scale parameter:	103.3972	Prob > chi2	=	0.0000
Pearson chi2(261):		26986.66	Deviance	= 26986.66
Dispersion (Pearson):		103.3972	Dispersion	= 103.3972
(Std. Err. adjusted for clustering on id)				

		Semirobust		z	P> z	[95% Conf. Interval]	
sysbp		Coef.	Std. Err.				
therapy		-2.276566	1.175734	-1.94	0.053	-4.580962	0.027829
sysbp_1		0.6458443	0.0614827	10.50	0.000	0.5253405	0.7663481
_cons		44.06264	7.809794	5.64	0.000	28.75572	59.36955

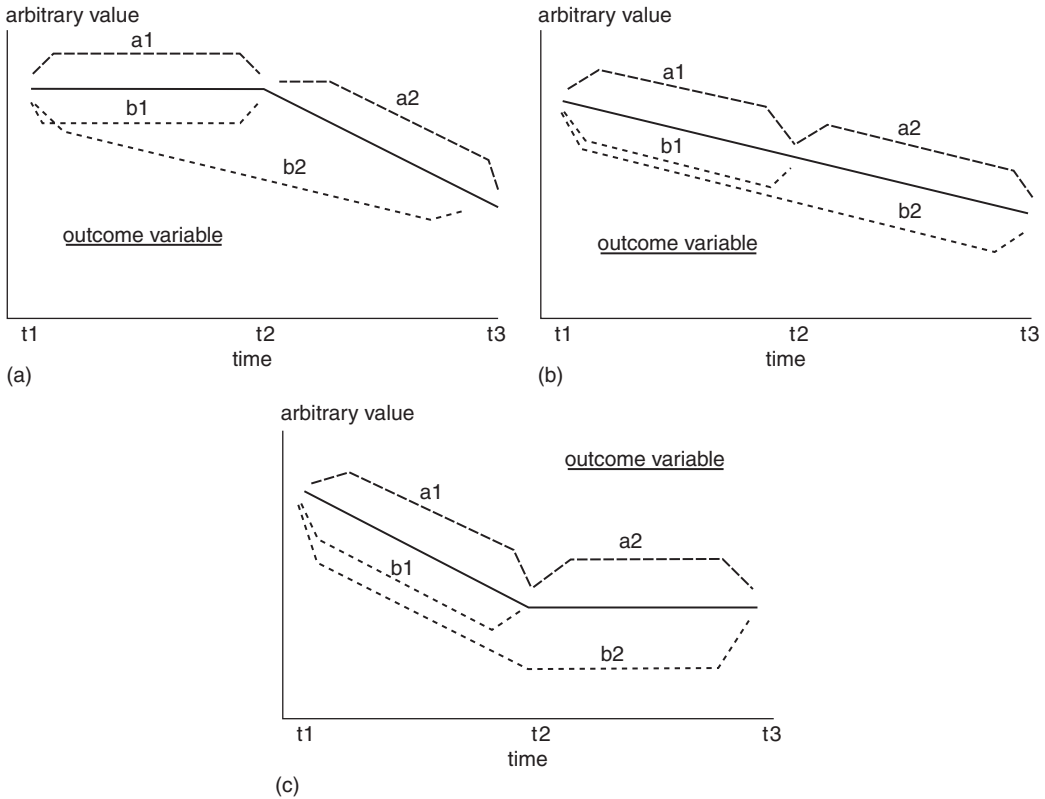


Figure 9.7 Illustration of the difference between two approaches that can be used in the analysis of an experimental longitudinal study. The effects a_1 and a_2 are detected by an autoregressive analysis (Equation 9.14), while the effects b_1 and b_2 are detected by the longitudinal analysis of covariance (Equation 9.13). For the situation in (a), the two methods will show comparable results ($a_1 = b_1$ and $a_2 = b_2$). For the situation shown in (b), longitudinal analysis of covariance will detect a stronger decline than the autoregressive analysis ($a_1 = b_1$ and $a_2 < b_2$). The situation in (c) will produce the same result as (b) (i.e. $a_1 = b_1$ and $a_2 < b_2$).

Y at $t = 2$ and the change between Y at $t = 1$ and Y at $t = 3$ are combined in one analysis. This means that in the longitudinal analysis of covariance the changes between Y at $t = 1$ and Y at $t = 2$ is analyzed twice. In fact, the longitudinal analysis of covariance seems to overestimate the therapy effect, because the short-term effect is doubled in the estimation of the overall therapy effect. This situation is illustrated in [Figure 9.7](#).

For illustrative purposes, Outputs [9.11](#) and [9.12](#) show the results of both the longitudinal analysis of covariance and the autoregressive analysis performed with a mixed model analysis.

Output 9.11 Results of a mixed model analysis to analyze the effect of the therapy intervention on systolic blood pressure using Equation 9.13 (i.e. longitudinal analysis of covariance)

Mixed-effects REML regression	Number of obs	=	249			
Group variable: id	Number of groups	=	130			
	Obs per group: min	=	1			
	avg	=	1.9			
	max	=	2			
	Wald chi2(2)	=	155.72			
Log restricted-likelihood = -929.04134	Prob > chi2	=	0.0000			

sysbp	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
therapy	-3.708664	1.58812	-2.34	0.020	-6.821322	-0.5960061
baseline	0.6142927	0.0522639	11.75	0.000	0.5118573	0.7167281
_cons	48.44556	6.913254	7.01	0.000	34.89583	61.99528

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+-----						
id: Identity						
	var(_cons)	44.03601	11.61538		26.2595	73.84642
-----+-----						
	var(Residual)	67.7939	9.049373		52.18782	88.06677

LR test vs. linear regression: chibar2(01) =				17.27	Prob >= chibar2 = 0.0000	

As expected, the results obtained from the mixed model analyses are the same as the results obtained from the GEE analyses. It should be noted that for the autoregressive model, in the GEE analysis, an independent correlation structure was chosen. This has to do with the fact that with the adjustment for the outcome variable at $t = t - 1$ all the correlation between the repeated measures is explained (see also Chapter 6). This can also be seen in the output of the autoregressive mixed model analysis, i.e. the variance of the random intercept is almost zero – see Output 9.12). For the longitudinal analysis of covariance the situation is slightly different. The adjustment for the baseline value explains only part of the correlation between the repeated measurements and therefore in the GEE analysis an exchangeable correlation structure was used. In the mixed model longitudinal

analysis of covariance, the variance of the random intercept is still quite high (see Output 9.11).

Output 9.12 Results of a mixed model analysis to analyze the effect of the therapy intervention on systolic blood pressure based on Equation 9.14 (i.e. autoregressive analysis)

Mixed-effects REML regression	Number of obs	=	261			
Group variable: id	Number of groups	=	142			
	Obs per group: min	=	1			
	avg	=	1.8			
	max	=	2			
	Wald chi2(2)	=	245.58			
Log restricted-likelihood = -976.28947	Prob > chi2	=	0.0000			

sysbp	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
therapy	-2.276566	1.292195	-1.76	0.078	-4.809222	0.2560891
sysbp_1	0.6458443	0.0433175	14.91	0.000	0.5609435	0.7307451
_cons	44.06264	5.710656	7.72	0.000	32.86996	55.25532

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+-----						
id: Identity						
	var(_cons)	2.27e-20	6.31e-20	9.71e-23	5.30e-18	
-----+-----						
	var(Residual)	104.5995	9.209683	88.02047	124.3012	

LR test vs. linear regression: chibar2(01) =				0.00	Prob >= chibar2 = 1.0000	

Although the use of an autoregressive analysis seems to be an adequate solution to estimate a “valid” effect of the therapy intervention, a new problem arises. The adjustment for baseline is performed because it is assumed that the groups to be compared are equal at baseline; i.e. all the differences at baseline are due to chance. At the first follow-up measurement the situation is different, though, because differences between the groups can now be caused by the fact that one group receives the intervention and the other group did not (Twisk and Proper, 2004; Boshuizen, 2005). So, except for the first follow-up measurement, adjustment for previous measurements is probably not necessary, i.e. not correct. Therefore, an alternative approach (i.e. the “combination” approach) is developed in which this problem is implemented.

In this approach the first follow-up measurement is adjusted for the baseline value while the second follow-up measurement is adjusted for neither the baseline value nor the value of the outcome variable at $t = t - 1$. The difficulty lies in the operationalization of the latter in a longitudinal dataset. It is not possible to leave the adjusted variable empty when no adjustment is needed, because when the value is left empty, that corresponding record will not be used in the analysis. So, instead of nothing, a constant value must be filled in to mimic the fact that no adjustment has to be made. To illustrate how this can be done, Equations 9.13 and 9.14 have to be written in a slightly different way (Equations 9.15 and 9.16).

$$(Y_{it} - Y_{it0}) = \beta_0 + \beta_1 therapy + \beta_2 Y_{it0} + \varepsilon_{it} \quad (9.15)$$

where $Y_{it} - Y_{it0}$ are the differences in observations for subject i between follow-up time t and t_0 , β_0 is the intercept, β_1 is the regression coefficient for therapy versus placebo, β_2 is the regression coefficient for the value of the outcome variable Y at t_0 , and ε_{it} is the “error” for subject i at time t .

$$(Y_{it} - Y_{it-1}) = \beta_0 + \beta_1 therapy + \beta_2 Y_{it-1} + \varepsilon_{it} \quad (9.16)$$

where $Y_{it} - Y_{it-1}$ are the differences in observations for subject i between follow-up time t and $t - 1$, β_0 is the intercept, β_1 is the regression coefficient for therapy versus placebo, β_2 is the regression coefficient for the value of the outcome variable Y at $t - 1$, and ε_{it} is the “error” for subject i at time t .

So, instead of modeling the observed values, it is also possible to model the changes between subsequent measurements (see also Section 9.2.1). It is important to realize that the estimated therapy effect using Equation 9.15 is exactly the same as the estimated therapy effect using Equation 9.13, while the estimated therapy effect using Equation 9.16 is exactly the same as the estimated therapy effect using Equation 9.14.

Equation 9.17 shows the equation for the “combination” approach.

$$(Y_{it} - Y_{it-1}) = \beta_0 + \beta_1 therapy + \beta_2 comb + \varepsilon_{it} \quad (9.17)$$

where $Y_{it} - Y_{it-1}$ are the differences in observations for subject i between follow-up time t and $t - 1$, β_0 is the intercept, β_1 is the regression coefficient for therapy versus placebo, β_2 is the regression coefficient for the “combination,” $comb$ is the baseline value for the first follow-up measurement and a constant for the other follow-up measurements for subject i , and ε_{it} is the “error” for subject i at time t .

By using Equation 9.17, the question arises which constant value should be used to obtain a “valid” estimation of the therapy effect. It was found that the average value of the outcome variable at $t = t - 1$ for the whole population is the best choice for the constant value (Twisk and de Vente, 2008). Table 9.9 shows part of the longitudinal dataset used to estimate the therapy effect with longitudinal analysis of covariance, autoregressive analysis and the “combination” approach.

Output 9.14 Results of a mixed model analysis to analyze the effect of the therapy intervention on systolic blood pressure based on Equation 9.17 (i.e. the “combination” approach)

Mixed-effects REML regression	Number of obs	=	261			
Group variable: id	Number of groups	=	142			
	Obs per group: min	=	1			
	avg	=	1.8			
	max	=	2			
	Wald chi2(2)	=	26.72			
Log restricted-likelihood = -993.00502	Prob > chi2	=	0.0000			

del_sysbp	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
therapy	-0.9450136	1.360874	-0.69	0.487	-3.612277	1.722249
sysbp_com	-0.3221572	0.06234	-5.17	0.000	-0.4443414	-0.199973
_cons	39.28461	8.056428	4.88	0.000	23.4943	55.07492

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
-----+-----						
id: Identity						
	var(_cons)	1.63e-21	4.35e-21	8.62e-24	3.07e-19	
-----+-----						
	var(Residual)	119.3461	10.50804	100.4298	141.8253	

LR test vs. linear regression: chibar2(01) =				0.00	Prob >= chibar2 = 1.0000	

Table 9.10 summarizes the results of the three different GEE analyses in order to estimate the therapy effect in the systolic blood pressure example.

From Table 9.10 it can be seen that there is a huge difference in therapy effects estimated with the different models. It is obvious that the conclusion regarding whether or not the therapy effect is statistically significant highly depends on the method used. In the present example a significant therapy effect is found with a longitudinal analysis of covariance, while with the autoregressive analysis a less strong and borderline significant therapy effect was found. With the “combination” approach, the estimated therapy effect was even weaker and far from significant.

Table 9.10 Summary of the results of three GEE analyses in order to estimate the effect of a therapy regarding the reduction of systolic blood pressure

	Effect	SE	<i>p</i> -value
Longitudinal analysis of covariance	−3.71	1.54	0.02
Autoregressive analysis	−2.28	1.18	0.05
“Combination” approach	−0.95	0.97	0.33

Surprisingly, all three methods are accepted and are used in practice. In most situations, a longitudinal analysis of covariance is used. However, it has been mentioned before that in this particular situation, this approach leads to an overestimation of the therapy effect, because the differences between baseline and the first follow-up measurement are more or less doubled. Regarding the autoregressive analysis and the “combination” approach, it is questionable whether the differences between the groups at the first follow-up measurement are totally due to the effect of the intervention. In the example dataset, probably most of the difference between the groups at the first follow-up measurement is a reflection of the differences observed at baseline. So, in this particular example, maybe, autoregressive analysis is the most appropriate way to estimate the effect of the therapy intervention.

The question whether or not the differences at a follow-up measurement are due to the intervention or a reflection of the differences at baseline is very difficult to answer. In most situations the differences at a follow-up measurement are partly due to the intervention and partly a reflection of the differences at baseline. When this is the case, the “real” intervention effect is somewhat between the effect estimated with autoregressive analysis and the effect estimated with the “combination” approach.

Up to now, the sophisticated analyses performed were aimed to estimate an overall intervention effect. Sometimes, however, one is more interested in the estimation of intervention effects at the different follow-up measurements. To obtain those effects, time and the interaction between the intervention variable and time should be added to the model (see Section 5.2). Output 9.15 shows the results of the three longitudinal analyses in order to obtain the effect of the therapy intervention at different time-points.

Table 9.11 summarizes the results of Output 9.15

An alternative approach to evaluate the effect of an intervention at different time-points is provided by Fitzmaurice et al. (2004). In this approach a GEE or mixed model analysis is performed in which all measurements are used as outcome (including the baseline measurement). The following model is then used (Equation 9.18).

Output 9.15 Results of three different GEE analyses to analyze the effect of the therapy intervention on systolic blood pressure at different time-points: (a) longitudinal analysis of covariance, (b) autoregressive analysis, (c) the longitudinal “combination” approach

(a)

GEE population-averaged model		Number of obs	=	249
Group variable:	id	Number of groups	=	130
Link:	identity	Obs per group: min	=	1
Family:	Gaussian	avg	=	1.9
Correlation:	independent	max	=	2
		Wald chi2(4)	=	30.84
Scale parameter:	107.8573	Prob > chi2	=	0.0000
Pearson chi2(249):	26856.48	Deviance	=	26856.48
Dispersion (Pearson):	107.8573	Dispersion	=	107.8573

(Std. Err. adjusted for clustering on id)

		Semirobust				
delsbp		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
therapy		-4.529225	1.805081	-2.51	0.012	-8.067119 -0.9913304
sysbpt0		-0.3769259	0.0761921	-4.95	0.000	-0.5262597 -0.2275922
time		-2.624029	1.673336	-1.57	0.117	-5.903708 0.6556502
ther_time		1.740922	2.137501	0.81	0.415	-2.448503 5.930347
_cons		48.62299	9.618919	5.05	0.000	29.77026 67.47573

(b)

GEE population-averaged model		Number of obs	=	261
Group variable:	id	Number of groups	=	142
Link:	identity	Obs per group: min	=	1
Family:	Gaussian	avg	=	1.8
Correlation:	independent	max	=	2
		Wald chi2(4)	=	40.11
Scale parameter:	101.9496	Prob > chi2	=	0.0000
Pearson chi2(261):	26608.85	Deviance	=	26608.85
Dispersion (Pearson):	101.9496	Dispersion	=	101.9496

(Std. Err. adjusted for clustering on id)

		Semirobust				
delsysbp		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
therapy		-4.41981	1.806539	-2.45	0.014	-7.960561 -0.8790593
sysbpt_1		-0.353887	0.0624284	-5.67	0.000	-0.4762443 -0.2315297

time	-3.192249	2.163966	-1.48	0.140	-7.433544	1.049046
ther_time	4.296427	2.723534	1.58	0.115	-1.041601	9.634454
_cons	45.61177	7.836752	5.82	0.000	30.25201	60.97152

(c)

GEE population-averaged model		Number of obs	=	261
Group variable:	id	Number of groups	=	142
Link:	identity	Obs per group: min	=	1
Family:	Gaussian	avg	=	1.8
Correlation:	independent	max	=	2
		Wald chi2(4)	=	22.35
Scale parameter:	114.8319	Prob > chi2	=	0.0002
Pearson chi2(261):	29971.12	Deviance	=	29971.12
Dispersion (Pearson):	114.8319	Dispersion	=	114.8319
	(Std. Err. adjusted for clustering on id)			

		Semirobust				
delsysbp	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
therapy	-4.379276	1.753098	-2.50	0.012	-7.815286	-0.943267
sysbp_comb	-0.345352	0.0903015	-3.82	0.000	-0.5223396	-0.1683643
time	-4.503601	2.484773	-1.81	0.070	-9.373666	0.3664634
ther_time	6.756199	3.1258	2.16	0.031	0.6297448	12.88265
_cons	44.49622	11.24251	3.96	0.000	22.46132	66.53113

$$Y_{it} = \beta_0 + \beta_1 therapy + \beta_2 time_1 + \beta_3 time_2 + \beta_4 therapy \times time_1 + \beta_5 therapy \times time_2 + \varepsilon_{it} \quad (9.18)$$

where Y_{it} are observations for subject i at follow-up time t , β_0 is the intercept, β_1 is the regression coefficient for therapy versus placebo, β_2 is the regression coefficient for the first dummy variable for time, β_3 is the regression coefficient for the second dummy variable for time, β_4 is the regression coefficient for the interaction between *therapy* and the first dummy variable for time, β_5 is the regression coefficient for the interaction between *therapy* and the second dummy variable for time, and ε_{it} is the “error” for subject i at time t .

In this model, the β_1 coefficient reflects the differences between the two groups at baseline, $\beta_1 + \beta_4$ reflects the differences at the first follow-up measurement, while $\beta_1 + \beta_5$ reflects the differences between the two groups at the second follow-up measurement. Although, this is a nice way of analysing the therapy effect at the different time-points, it does not adjust for the differences between

Table 9.11 Summary of the results of three different GEE analyses to analyze the effect of the therapy intervention on systolic blood pressure at different time-points

	First follow-up	Second follow-up
Longitudinal analysis of covariance	−4.53 (1.81)	−2.79 (1.93)
Autoregressive analysis	−4.42 (1.81)	−0.12 (1.78)
“Combination” approach	−4.38 (1.75)	2.38 (1.91)

the groups observed at baseline, or in other words, it does not adjust for possible regression to the mean. Output 9.16 shows the output of a GEE analysis performed on the example dataset.

Output 9.16 Results of a GEE analysis with an exchangeable correlation structure to analyze the effect of the therapy intervention on systolic blood pressure at different time-points based on Equation 9.18

GEE population-averaged model		Number of obs	=	413		
Group variable:	id	Number of groups	=	152		
Link:	identity	Obs per group: min	=	1		
Family:	Gaussian	avg	=	2.7		
Correlation:	exchangeable	max	=	3		
		Wald chi2(5)	=	21.74		
Scale parameter:	203.4487	Prob > chi2	=	0.0006		
(Std. Err. adjusted for clustering on id)						

	Semirobust					
sysbp	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

_Itime_2	-0.7897315	1.610396	-0.49	0.624	-3.94605	2.366587
_Itime_3	-3.839491	1.496127	-2.57	0.010	-6.771847	-0.9071357
therapy	-3.442182	2.498941	-1.38	0.168	-8.340016	1.455652
_ItimXther~2	-2.793464	2.00552	-1.39	0.164	-6.724212	1.137284
_ItimXther~3	-0.7465874	2.15294	-0.35	0.729	-4.966272	3.473097
_cons	129.9099	2.022707	64.23	0.000	125.9455	133.8743

From Output 9.16 it can be seen that at baseline the therapy group had a lower systolic blood pressure compared to the control group (−3.442182 mmHg). At the first follow-up measurement there is a bigger difference between the groups

($-3.442182 - 2.793464 = -6.235646$). At the second follow-up measurement the difference between the groups is somewhat less than observed at the first follow-up measurement (i.e. $-3.442182 - 0.7465874 = -4.1887694$). Due to missing values in the dataset, the differences between the therapy group and the control group at the different time-points are not exactly the same as the observed differences between the average values at the different time-points (see [Table 9.3](#)), but they are close. However, again, this approach does not adjust for the differences observed at baseline and, therefore, it is not the best way of analysing data from an RCT when there are differences in baseline values between the groups to be compared.

It is of course also possible to use an extension of Equation 9.10 to analyze the therapy effect (Equation 9.19).

$$Y_{it} = \beta_0 + \beta_1 time_1 + \beta_2 time_2 + \beta_3 therapy \times time_1 + \beta_4 therapy \times time_2 + \varepsilon_{it} \quad (9.19)$$

where Y_{it} are observations for subject i at follow-up time t , β_0 is the intercept, β_1 is the regression coefficient for the first dummy variable for time, β_2 is the regression coefficient for the second dummy variable for time, β_3 is the regression coefficient for the interaction between *therapy* and the first dummy variable for time, β_4 is the regression coefficient for the interaction between *therapy* and the second dummy variable for time, and ε_{it} is the “error” for subject i at time t .

Without the therapy variable, the baseline values for the two groups are assumed to be the same. Output 9.17 shows the results of a GEE analysis without the therapy variable.

The results shown in Output 9.17 indicate that the intervention effect at the first follow-up measurement is -4.059527 and at the second follow-up -2.010645 . As has been mentioned before, the analysis based on Equation 9.19 is basically the same as a longitudinal analysis of covariance, although the results in the present example are slightly different. This difference is caused by a different number of observations in both methods. If the two analyses were performed on a full dataset without any missing values, the results of the two analyses would have been exactly the same.

9.2.3 Conclusion

Although longitudinal analysis of covariance is mostly used to analyze the effect of an intervention in experimental studies with more than one follow-up measurement, one should be (very) careful with the interpretation of the results of such an

analysis. In some situations, it is better to use either an autoregressive analysis or a “combination” approach.

Output 9.17 Results of a GEE analysis with an exchangeable correlation structure to analyse the effect of the therapy intervention on systolic blood pressure at different time-points based on Equation 9.19

GEE population-averaged model		Number of obs	=	413		
Group variable:	id	Number of groups	=	152		
Link:	identity	Obs per group: min	=	1		
Family:	Gaussian	avg	=	2.7		
Correlation:	exchangeable	max	=	3		
		Wald chi2(4)	=	20.92		
Scale parameter:	205.9377	Prob > chi2	=	0.0003		
(Std. Err. adjusted for clustering on id)						

		Semirobust				
sysbp		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+						
_Itime_2		-0.1633899	1.49892	-0.11	0.913	-3.101218 2.774439
_Itime_3		-3.217221	1.343139	-2.40	0.017	-5.849725 -0.5847162
_ItimXther~2		-4.059527	1.763523	-2.30	0.021	-7.515968 -0.603085
_ItimXther~3		-2.010645	1.827062	-1.10	0.271	-5.591621 1.570331
_cons		128.2258	1.268389	101.09	0.000	125.7398 130.7118

9.3 Dichotomous outcome variables

9.3.1 Introduction

The example of an experimental study with a dichotomous outcome variable uses a dataset from a hypothetical study in which a new drug was tested on patients with a stomach ulcer. Treatment duration was 1 month, and patients are seen at three follow-up visits. The first follow-up visit was directly at the end of the intervention period (after 1 month) and the two long-term follow-up visits were scheduled 6 and 12 months after the start of the intervention. In this RCT, the intervention (i.e. the new drug) is compared to a placebo, and 60 patients were included in each of the two groups. In the follow-up period of 1 year, there was no loss to follow-up, and therefore there were no missing data. [Figure 9.8](#) shows the proportion of patients who had fully recovered at the different follow-up measurements.

Table 9.12 Results of an RCT to investigate the effect of an intervention, i.e. the number of patients recovered, the “relative risks” and 95% confidence intervals (in parentheses) for the intervention group and the corresponding *p*-values at each of the follow-up measurements

	Recovery after 1 month		Recovery after 6 months		Recovery after 12 months	
	Yes	No	Yes	No	Yes	No
Intervention	35	25	39	21	60	10
Placebo	28	32	29	31	30	30
Relative risk	1.28 (0.87–1.88)		1.48 (0.97–2.53)		3.00 (1.61–5.58)	
<i>p</i> -value	0.20		0.07		<0.01	

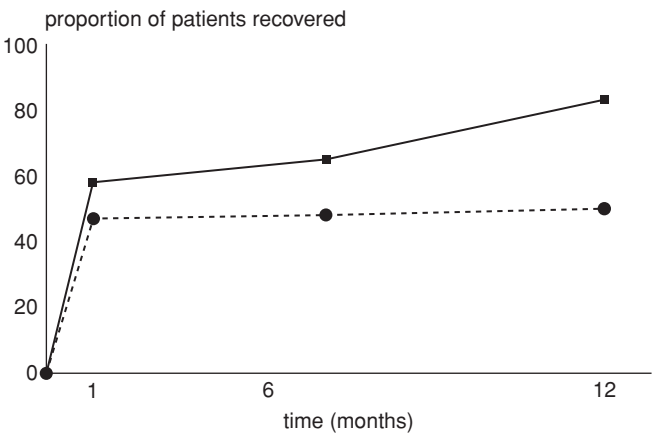


Figure 9.8 The proportion of patients recovered in an RCT to investigate the effect of an intervention (i.e. a new drug) (■ — intervention, ● - - - placebo).

9.3.2 Simple analysis

The classical way to analyze the results of such an RCT is to analyze the difference in proportion of patients experiencing full recovery between the intervention and the placebo group at each of the three follow-up measurements, by simply applying a Chi-square test. Furthermore, at each of the follow-up measurements, the effect of the intervention can be estimated by calculating the relative risk (and corresponding 95% confidence interval). The relative risk is defined as the proportion of subjects recovered in the intervention group, divided by the proportion of subjects recovered in the placebo group. [Table 9.12](#) summarizes the results of the analyses.

Table 9.13 Comparison between relative risks and odds ratios, including 95% confidence intervals (in parentheses) and *p*-values as a result of an RCT to investigate the effect of an intervention

	1 month	6 months	12 months
Relative risk	1.28 (0.87–1.88)	1.48 (0.97–2.53)	3.00 (1.61–5.58)
Odds ratio	1.60 (0.78–3.29)	1.99 (0.95–4.13)	5.00 (2.14–11.66)
<i>p</i> -value	0.20	0.07	<0.01

From the results in [Table 9.12](#) it can be seen that during the intervention period of 1 month both the intervention group and the placebo group show quite a high proportion of patients who recover, and although in the intervention group this proportion is slightly higher, the difference is not statistically significant ($p = 0.20$). After the intervention period, in both groups there is an increase in the number of patients who recovered, but in the intervention group this increase is more pronounced, which results in a significant difference between the intervention group and the placebo group after 1 year of follow-up.

9.3.3 Sophisticated analysis

It is, of course, also possible to estimate the effect of the intervention with either a logistic GEE analysis or a logistic mixed model analysis. The first thing that should be realized is that as a result of a logistic GEE analysis or a logistic mixed model analysis, odds ratios are calculated. Odds ratios can be interpreted as relative risks, but they are not the same. Owing to the mathematical background of the odds ratios and relative risks, the odds ratios are always an overestimation of the “real” relative risk. This overestimation becomes stronger as the proportion of “cases” (i.e. recovered patients) increases. To illustrate this, the odds ratios for intervention versus placebo were calculated at each of the follow-up measurements ([Table 9.13](#)).

From the results in [Table 9.13](#) it can be seen that the calculated odds ratios are bigger than the relative risks shown in [Table 9.12](#), and that the confidence intervals are wider, but that the significance levels are the same. So, when a logistic GEE analysis is carried out, one must realize that the results (i.e. odds ratios) obtained from such an analysis have to be interpreted with caution, and cannot be directly interpreted as relative risks.

Output [9.18](#) presents the results of a logistic GEE analysis (assuming an exchangeable correlation structure), in which the average effect of the intervention over time is analyzed.

Output 9.18 Results of a GEE analysis with an exchangeable correlation structure to compare an intervention with a placebo with regard to recovery (a dichotomous outcome variable) measured over a period of one year

GEE population-averaged model		Number of obs	=	360		
Group variable:	id	Number of groups	=	120		
Link:	logit	Obs per group: min	=	3		
Family:	binomial	avg	=	3.0		
Correlation:	exchangeable	max	=	3		
		Wald chi2(1)	=	8.71		
Scale parameter:	1	Prob > chi2	=	0.0032		
(Std. Err. adjusted for clustering on id)						

		Semirobust				
rec		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]

interven		0.8616212	0.2919719	2.95	0.003	0.2893669 1.433876
_cons		-0.0666914	0.1983756	-0.34	0.737	-0.4555004 3.322117

From Output 9.18 it can be seen that the intervention is highly successful over the total follow-up period. To obtain the odds ratio, $\text{EXP}[\text{regression coefficient}]$ has to be taken. In the present example, the odds ratio is $\text{EXP}[0.8616212] = 2.37$. This can be interpreted that on average over time, the odds for recovery in the intervention group is 2.37 as high as the odds for recovery in the placebo group. To obtain the 95% confidence interval around the odds ratio, $\text{EXP}[0.2893669]$ and $\text{EXP}[1.433876]$ have to be taken; the 95% confidence interval around the OR of 2.37 is therefore 1.34 to 4.19. The same analysis can also be performed with a logistic mixed model analysis (Output 9.19).

As expected, the odds ratio obtained from the logistic mixed model analysis is higher compared to the odds ratio obtained from the logistic GEE analysis. The difference between the two approaches has to do with the difference between a “population average” approach and a “subject specific” approach. In Section 6.2.5, the differences between the two approaches were extensively discussed.

It should be noted, that there is no need to adjust for the baseline values. As all patients were ill at baseline (by definition), there are no baseline differences between the groups. This is mostly the case when a dichotomous outcome variable is considered in an RCT. So the whole discussion about how and when to adjust

for baseline differences in the outcome variable between the groups is mostly not relevant for dichotomous outcome variables.

Output 9.19 Results (shown as an odds ratio) of a mixed model analysis to compare an intervention with a placebo with regard to recovery (a dichotomous outcome variable) measured over a period of one year

Mixed-effects logistic regression	Number of obs	=	360			
Group variable: id	Number of groups	=	120			
	Obs per group: min	=	3			
	avg	=	3.0			
	max	=	3			
Integration points = 7	Wald chi2(1)	=	8.38			
Log likelihood = -214.8222	Prob > chi2	=	0.0038			

rec	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
interven	4.090633	1.990225	2.90	0.004	1.576353	10.61519

Random-effects Parameters			Estimate	Std. Err.	[95% Conf. Interval]	
-----+-----						
id: Identity						
	var(_cons)		3.875641	1.325859	1.982199	7.577745

LR test vs. logistic regression: chibar2(01) =				42.88	Prob>=chibar2 = 0.0000	

The odds ratios obtained from the analysis performed before can be interpreted as “on average over time.” Although this is an interesting effect measure, but it can also be interesting to investigate the effect of the intervention at different time-points. Therefore, a logistic GEE analysis or a logistic mixed model analysis can be performed with time and an interaction with time in the model. Because in an experimental study mostly fixed time-points are used, time can be treated as a categorical variable and represented by dummy variables. Output 9.20 shows the results of the logistic GEE analysis to estimate the effect of the intervention at the different time-points.

Output 9.20 Results of a logistic GEE analysis with an exchangeable correlation structure to compare an intervention with a placebo with regard to recovery (a dichotomous outcome variable) measured over a period of one year including time and the interaction between intervention and time

GEE population-averaged model		Number of obs	=	360		
Group variable:	id	Number of groups	=	120		
Link:	logit	Obs per group: min	=	3		
Family:	binomial	avg	=	3.0		
Correlation:	exchangeable	max	=	3		
		Wald chi2(5)	=	18.94		
Scale parameter:	1	Prob > chi2	=	0.0020		
(Std. Err. adjusted for clustering on id)						

		Semirobust				
rec		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]

_Itime_2		0.06684	0.3075189	0.22	0.828	-0.5358859 0.6695659
_Itime_3		0.1335314	0.3416822	0.39	0.696	-0.5361533 0.8032161
interven		0.4700036	0.3696954	1.27	0.204	-0.254586 1.194593
_ItimXinte~2		0.215727	0.4051391	0.53	0.594	-0.578331 1.009785
_ItimXinte~3		1.139434	0.5097866	2.24	0.025	0.140271 2.138598
_cons		-0.1335314	0.2598596	-0.51	0.607	-0.6428468 0.3757841

From Output 9.20 the effects of the intervention at the different time-point can be derived. The coefficient for the intervention variable (i.e. 0.4700036) can be transformed into an odds ratio, which reflects the effect of the intervention at the first follow-up measurement. The odds ratio at the first measurement is therefore equal to 1.60. The effect of the intervention at the second follow-up measurement can be estimated by adding up the regression coefficient for the intervention variable and the regression coefficient for the interaction between the intervention and the first dummy variable for time (i.e. $0.4700036 + 0.215727 = 0.6857306$), which gives an odds ratio of 1.99. In the same way, the odds ratio at the last follow-up measurement can be estimated (i.e. $0.4700036 + 1.139434 = 1.6094376$), which gives an odds ratio of 5.00. To obtain the 95% confidence intervals around the odds ratios at the different time-points the time dummy variables should be recoded (see also Section 5.2). The odds ratios (and 95% confidence intervals) derived in this way are exactly the same as the odds ratios derived from three separate analyses; these results were shown in Table 9.13. This has to do with the fact that there are no missing values in the example RCT. When there

Of course, the same analyses could also be performed with a logistic mixed model analysis. Again, the regression coefficients (and therefore the odds ratios) were bigger than the ones obtained from a logistic GEE analysis (Output 9.21). Comparing the results of the logistic GEE analysis and the logistic mixed model analysis with the results of the analyses at the separate time-points indicates that regarding the two sophisticated methods, logistic GEE analysis has to be preferred above logistic mixed model analysis (see also Section 7.2.5).

Mixed-effects logistic regression	Number of obs	=	360			
Group variable: id	Number of groups	=	120			
	Obs per group: min	=	3			
	avg	=	3.0			
	max	=	3			
Integration points = 7	Wald chi2(5)	=	18.68			
Log likelihood = -206.55953	Prob > chi2	=	0.0022			
rec	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
_Itime_2	0.1150878	0.4799433	0.24	0.810	-0.8255838	1.055759
_Itime_3	0.2298581	0.4801381	0.48	0.632	-0.7111954	1.170912
interven	0.8269259	0.6519283	1.27	0.205	-0.45083	2.104682
_ItimXinte~2	0.3908873	0.6976773	0.56	0.575	-0.9765351	1.75831
_ItimXinte~3	1.946288	0.7732359	2.52	0.012	0.4307738	3.461803
_cons	-0.2147389	0.4515219	-0.48	0.634	-1.099706	0.6702277
Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]		
id: Identity						
	sd(_cons)	2.213432	0.3791637	1.582174	3.096549	
LR test vs. logistic regression: chibar2(01) =				49.34	Prob>=chibar2 = 0.0000	

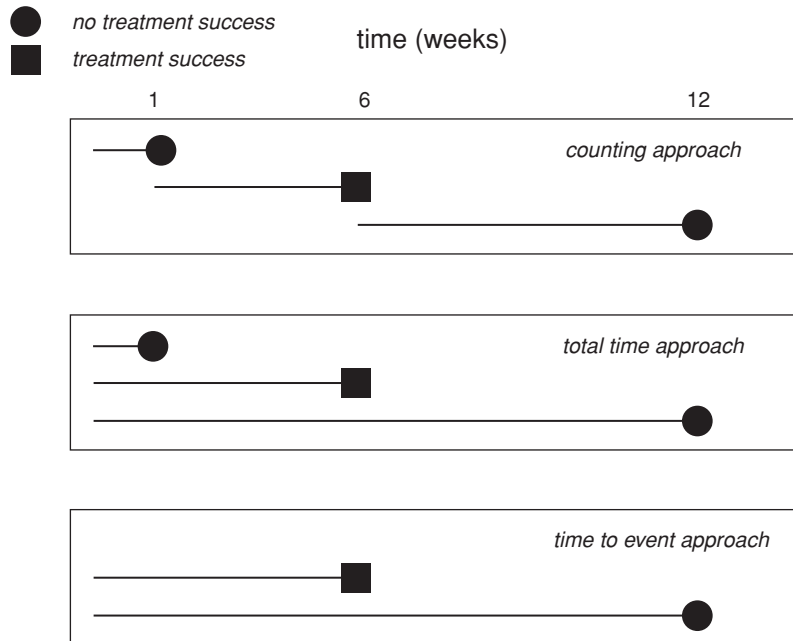


Figure 9.9 Possible definitions of the time at risk to be analyzed with Cox regression for recurrent events for a patient whose treatment was not successful at week 1, successful at week 6, and not successful at week 12.

9.3.4 Other approaches

Besides the longitudinal logistic regression approaches (i.e. logistic GEE analysis and logistic mixed model analysis), a survival approach can also be used to analyze the data from the present example. Regarding survival approaches, Cox regression for recurrent events can be performed. Although there are different estimation procedures available (Kelly and Lim, 2003), the general idea behind Cox regression for recurrent events is that the different time periods are analyzed separately and adjusted for the fact that the time periods within one subject are dependent (Glynn et al., 1993). The idea of this adjustment is that the standard error of the regression coefficient of interest is increased proportional to the correlation of the observations within the subject. One of the problems using Cox regression for recurrent events is the question of how to define the time at risk. This is especially the case in this example, because the events under study are not short-lasting events, but can be long lasting and can be considered as states, i.e. the events can continue over more than one time-point. In general, the time at risk can be defined in three different ways (Figure 9.9): (1) *The counting approach*. Each time period is analyzed separately assuming that all patients are at risk at the beginning of each period,

irrespective of the situation at the end of the foregoing period. (2) *The total time approach*. This approach is comparable to the counting approach. However, in the total time approach, the starting point for each period is the beginning of the study. (3) *The time-to-event approach*. In this approach only the transitions from no treatment success to treatment success are taken into account. So, if for a patient the treatment was successful after 3 months and stays successful at all repeated measurements, only the first time is taken into account in the analysis. When for another patient the treatment was successful after 3 months and not successful at 6 months, that particular patient is again at risk from 3 months onwards until the treatment for that patient is successful for the second time, or until the follow-up period ends.

Output 9.22 shows the results of a Cox regression for recurrent events when the time at risk is defined according to the “counting” approach.

Output 9.22 Results of a Cox regression for recurrent event when time at risk is defined according to the “counting” approach

Cox regression -- Breslow method for ties						
No. of subjects	=	120	Number of obs	=	214	
No. of failures	=	98				
Time at risk	=	385				
			Wald chi2(1)	=	3.99	
Log pseudolikelihood	=	-435.56115	Prob > chi2	=	0.0457	
(Std. Err. adjusted for 120 clusters in id)						

		Robust				
_t		Haz. Ratio	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----						
interven		1.346467	0.2004789	2.00	0.046	1.005676 1.802741

From Output 9.22 it can be seen that the hazard ratio for the intervention compared to the placebo condition is 1.35 with a 95% confidence interval that ranges between 1.01 and 1.80. The intervention effect is much lower than the one estimated with the logistic GEE analysis or logistic mixed model analysis. This is not surprising, because the hazard ratio can be interpreted as an average relative risk over time and (as has been mentioned before) a relative risk is always lower compared to an odds ratio estimated on the same data. In the example study the

difference between the two is relatively big, because the prevalence of the outcome (i.e. recovery) is quite high (around 60%). Because there are no missing data in this dataset, the results of a Cox regression analysis for recurrent events with time at risk defined according to the total time approach are exactly the same as the ones with time at risk defined according to the “counting” approach. When the time to event approach is used the hazard ratio for the intervention variable is only slightly different (i.e. 1.36 instead of 1.35) with a slightly higher standard error (i.e. 0.21 instead of 0.20).

It should be realized that Cox regression for recurrent events is especially useful for short lasting events, such as asthmatic attacks, falls, etc. In all the three ways the time at risk can be defined it is assumed that directly after an event occurs the individual is at risk again to get another event. As has been mentioned before, in the present example, this is not really the case, i.e. the events are long lasting and can be considered as “states.” Therefore, in the present example it is not recommended to use Cox regression for recurrent events.

There are also other possibilities to model recurrent events data, such as the continuous-time Markov process model for panel data (Berkhof et al., 2009) or the conditional frailty model (Box-Steffensheimer and De Boef, 2006). However, most of those alternative methods are mathematically complicated and not much used in practice. Therefore they are beyond the scope of this book.

9.4 Comments

The analyses discussed for both the continuous and dichotomous outcome variables in experimental studies were limited to “crude” analyses, in such a way that no potential confounders (apart from the value of the outcome variable at baseline in the example of a continuous outcome variable) and/or effect modifiers (apart from the interaction between therapy/intervention and time in both examples) have been discussed. Potential effect modifiers can be interesting if one wishes to investigate whether the intervention effect is different for subgroups of the population under study. The way confounders and effect modifiers are treated in longitudinal regression analysis is, however, exactly the same as in cross-sectional regression analysis. The procedures to construct prognostic/prediction models with variables measured at baseline, which is quite popular in clinical epidemiology these days, is also the same as in cross-sectional analysis.

In the examples discussed in this chapter, the first analyses performed were (simple) cross-sectional analyses. In fact, it is recommended that statistical analysis to evaluate the effect of an intervention should always start with a simple analysis. This not only provides insight into the data, but can also provide (important) information regarding the effect of the intervention. It should also be noted that,

although the simple techniques and summary statistics are somewhat limited, the interpretation of the results is often easy, and their use in clinical practice is therefore very popular. In general, to answer many research questions the simple techniques and summary statistics are quite adequate, and there is no real need to use the sophisticated techniques. Moreover, the results of the sophisticated techniques are (sometimes) difficult to interpret. However, when the number of repeated measurements differs between subjects, and/or when there are (many) missing data, it is (highly) recommended that the more sophisticated statistical analyses should be applied.

Missing data in longitudinal studies

10.1 Introduction

One of the main methodological problems in longitudinal studies is missing data, i.e. the (unpleasant) situation when not all N subjects have data on all T measurements. When subjects have missing data at the end of a longitudinal study they are often referred to as drop-outs. It is, however, also possible that subjects miss one particular measurement, and then return to the study at the next follow-up. This type of missing data is often referred to as intermittent missing data (Figure 10.1). It should be noted that, in practice, drop-outs and intermittent missing data usually occur together.

Besides the distinction regarding the missing data pattern (i.e. intermittent missing data versus drop-outs), in the statistical literature a distinction is made regarding the missing data mechanism. Three types of missing data are distinguished: (1) missing completely at random (MCAR: missing, independent of both unobserved and observed data); (2) missing at random (MAR: missing, dependent on observed data, but not on unobserved data, or, in other words, given the observed data, the unobserved data are random); and (3) missing not at random (MNAR: missing, dependent on unobserved data) (Little and Rubin, 2003). Missing at random usually occurs when data are missing by design. An illustrative example is the Longitudinal Aging Study Amsterdam (Deeg and Westendorp-de Serière, 1994). In this observational longitudinal study, a large cohort of elderly subjects was screened for the clinical existence of depression. Because the number of non-depressed subjects was much greater than the number of depressed subjects, a random sample of non-depressed subjects was selected for the follow-up, in combination with the total group of depressed subjects. So, given the fact that the subjects were not depressed, the data were missing at random (Figure 10.2).

Although the above-mentioned distinction between the three different types of missing data is important, it is rather theoretical. For a correct interpretation of

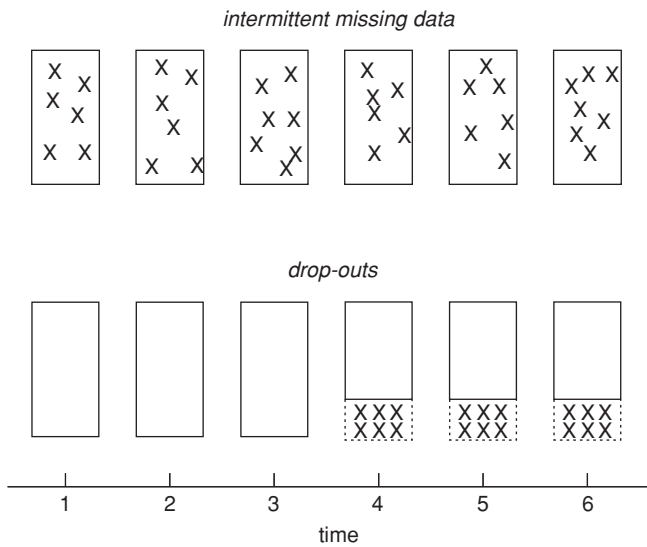


Figure 10.1 Illustration of intermittent missing data and drop-outs (X indicates a missing data point).

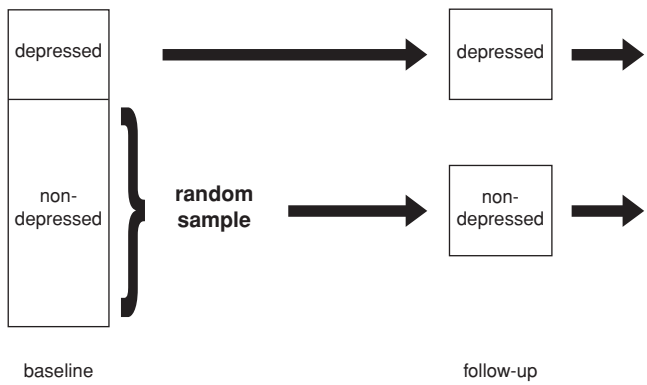


Figure 10.2 An illustration of data missing by design, i.e. missing at random.

the results of longitudinal data analysis, two issues must be considered. First of all, it is important to investigate whether or not missing data on the outcome variable Y at a certain time-point are dependent on the values of the outcome variable observed one (or more) time-point(s) earlier. In other words, it is important to investigate whether or not missing data depend on earlier observations of the outcome variable. Secondly, it is important to determine whether or not certain covariates are related to the occurrence of missing data. For example: “Are males more likely to have missing data than females?” In general, it is preferable to make a distinction between “ignorable” missing data (i.e. missing, not dependent on earlier

observations of the outcome variable and/or covariates) and “non-ignorable” or “informative” missing data (i.e. missing, dependent on earlier observations of the outcome variable and/or covariates).

10.2 Ignorable or informative missing data?

Although there is an abundance of statistical literature describing (complicated) methods that can be used to investigate whether or not one is dealing with ignorable or informative missing data in a longitudinal study (see, for instance, Diggle, 1989; Ridout, 1991; Diggle et al., 1994; Potthoff, et al., 2006; Enders, 2010), it is basically quite easy to investigate this matter. It can be done by comparing the group of subjects with data at $t = t$ with the group of subjects with missing data at $t = t$. First of all, this comparison can concern the particular outcome variable of interest measured at $t = t - 1$. Depending on the distribution of that particular variable, an independent sample t -test (for continuous variables) or a Chi-square test (for dichotomous and categorical outcome variables) can be carried out. Secondly, the influence of covariates on the occurrence of missing data can be investigated. This can be done by means of a (simple) logistic regression analysis, with missing or non-missing at each of the repeated measurements as a dichotomous outcome variable.

Up to now, a distinction has been made between missing data dependent on earlier values of the outcome variable Y and missing data dependent on values of particular covariates. Of course, this distinction is not really necessary, because in practice they both occur together and can both be investigated with logistic regression analysis, with both the earlier value of the outcome variable and the values of the covariates as possible determinants for why the data are missing. It should be noted that the distinction is made purely for educational purposes.

When there are only a few (repeated) measurements, and when the amount of missing data at each of the (repeated) measurements is rather high, the above-mentioned procedures are highly suitable to investigate whether one is dealing with ignorable or informative missing data. However, when the amount of missing data at a particular measurement is rather low, the power to detect differences between the subjects with data and the subjects without data at a particular time-point can be too low. Although the possible significance of the differences is not the most important issue in determining whether or not the pattern of missing data is ignorable or informative, it can be problematic to interpret the observed differences correctly. Therefore, the information about missing data at different time-points can be combined. This can be done in a relatively simple way in which the population is divided into two groups; i.e. the subjects without any missing

data over the longitudinal period and the subjects with missing data at one or more of the repeated measurements. The two groups are then compared to each other regarding the values of the outcome variable and/or the covariates at the first measurement. This is done because, in practice, nearly all subjects are measured at the first measurement. There are also more complicated methods available to combine the information about missing data at different time-points (Diggle, 1989; Ridout, 1991; Diggle et al., 1994). However, the statistical techniques involved are seldom used in practice.

10.3 Example

10.3.1 Generating datasets with missing data

The dataset used to illustrate the influence of missing data on the results of statistical analysis is the same example dataset which has been used throughout this book (see Section 1.4). However, for the examples used in this chapter, only the outcome variable Y , the time-dependent continuous covariate X_2 , and the time-independent dichotomous covariate X_4 are used.

From the complete dataset, four missing datasets were created. In all datasets the first measurement was complete for all subjects. The second measurement was removed for 15 subjects (approximately 10%). The third to sixth measurements were removed for 25 (17%), 35 (24%), 45 (30%), and 55 (37%) subjects, respectively. In the MCAR dataset, all missing data were randomly selected from the complete dataset. Regarding MAR, two datasets were created; one in which missing data was related to the dichotomous time-independent independent variable X_4 and one in which the missing data was related to the outcome variable Y measured at $t = t - 1$. In the first MAR dataset (MAR_1), missing data at the different time-points were randomly selected from the subpopulation for which the time-independent covariate X_4 was coded 0, while in the second MAR dataset (MAR_2), all observations were removed for subjects with the highest values for the outcome variable Y at $t = t - 1$. Finally, in the MNAR dataset, all observations were removed for subjects with the highest values for the outcome variable Y at that particular time-point. For all missing datasets, when the outcome variable was removed, the time-dependent covariate X_2 was also removed. This is comparable to missing data observed in a real life study, because when a certain subject does not attend a particular visit in a longitudinal study, all data to be collected at that measurement are generally missing. Furthermore, the missing datasets contain both intermittent missing data and drop-outs. Table 10.1 shows descriptive information regarding the outcome variable Y in the complete dataset and in the datasets with missing data.

Table 10.1 Mean value of the outcome variable Y for the different datasets with missing data at the different time-points

	Complete	MCAR	MAR_1		MAR_2	MNAR
	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	N^a
Y_{t1}	4.4 (0.7)	4.4 (0.7)	4.4 (0.7)	4.4 (0.7)	4.4 (0.7)	147
Y_{t2}	4.3 (0.7)	4.3 (0.7)	4.3 (0.7)	4.2 (0.6)	4.2 (0.5)	132
Y_{t3}	4.3 (0.7)	4.3 (0.7)	4.3 (0.7)	4.1 (0.6)	4.0 (0.5)	122
Y_{t4}	4.2 (0.7)	4.2 (0.7)	4.1 (0.6)	3.9 (0.5)	3.9 (0.5)	112
Y_{t5}	4.7 (0.8)	4.7 (0.8)	4.6 (0.8)	4.3 (0.6)	4.3 (0.5)	102
Y_{t6}	5.1 (0.9)	5.2 (0.9)	5.1 (0.9)	4.7 (0.8)	4.5 (0.5)	92
N^b	147	43	73	81	75	

^aNumber of observations in the datasets with missing data.

^bNumber of complete cases.

10.3.2 Analysis of determinants for missing data

As mentioned in the introduction of this chapter, it is important to investigate whether or not the missing data are dependent either on earlier values of the outcome variable or on the values of certain covariates. This knowledge can have important implications for the interpretation of the results of a longitudinal study with missing data.

It is quite simple to investigate whether the missing data are dependent on values of the outcome variable Y one time-point earlier. This can be done by comparing the subjects with data at $t = t$ with the subjects with missing data at $t = t$. The comparison is then performed on the value of the outcome variable at $t = t - 1$. The difference between the two groups can be tested with an independent sample t -test.

Besides the analyses at each time-point, also an analysis can be performed in which the subjects with complete data (i.e. data at all six measurements) are compared with the subjects with missing data at one of the measurements. The two groups can be compared to each other regarding the value of the outcome variable Y at $t = 1$.

To illustrate the latter, in [Table 10.2](#), the results of the independent sample t -tests comparing the complete cases with the incomplete cases are given for the four datasets with missing data.

Because the missing datasets are forced to be of a certain type, the results are as expected. In the MNAR and MAR_2 datasets, missing data was related to the outcome variable (either at the time of missing data or one time-point earlier)

Table 10.2 Results of independent sample *t*-tests to compare subjects with missing data with subjects without missing data (i.e. with complete data) regarding the value of the outcome variable *Y* at *t* = 1

		N	Mean(Y_{t1})	p -Value
MCAR				
	Missing	104	4.4	0.35
	Not missing	43	4.5	
MAR_1				
	Missing	74	4.4	0.78
	Not missing	73	4.5	
MAR_2				
	Missing	66	4.8	<0.01
	Not missing	81	4.1	
MNAR				
	Missing	72	4.8	<0.01
	Not missing	75	4.0	

and, therefore, for these two datasets the outcome variable *Y* at *t* = 1 is statistically significant different between subjects with complete data compared to subjects with missing data. For the MCAR and MAR_1 datasets no significant differences were found between subjects with complete data compared to subjects with missing data. Also, this is not very surprising because in both datasets missing data were not (forced to be) related to (earlier) values of the outcome variable *Y*.

The independent sample *t*-tests are only performed to determine whether the missing data are dependent on the outcome variable *Y* measured at *t* = 1. It is also of interest to analyze other possible determinants of the missing data. Information about these determinants can also be important for correct interpretation of the results of a longitudinal study with missing data. Logistic regression analyses were subsequently performed, with complete versus non-complete as the dichotomous outcome variable. The values of two covariates in the example dataset (*X*₂ and *X*₄) at *t* = 1 were analyzed as potential determinants for the missing data. [Table 10.3](#) summarizes the results of the logistic regression analyses.

From the results presented in [Table 10.3](#), it can be seen that it is only in the MAR_2 and MNAR datasets that subjects with the higher values of *X*₂ at *t* = 1 seem to have a higher probability of having missing data. This is not really surprising because, from earlier analyses of the example dataset, it is already known

Table 10.3 Regression coefficients and p -values of logistic regression analyses to investigate possible determinants of missing data

	X_4	X_2 at $t = 1$
MCAR	-0.49 ($p = 0.22$)	0.24 ($p = 0.12$)
MAR_1	NA ^a	0.08 ($p = 0.83$)
MAR_2	-0.50 ($p = 0.19$)	-0.61 ($p < 0.01$)
MNAR	-0.44 ($p = 0.25$)	-0.65 ($p < 0.01$)

^aNot applicable (NA), because there were no complete cases for which X_4 was coded 0.

that X_2 and Y are associated with each other. So, when missing data are found to be dependent on the value of Y at $t = 1$, it can be expected that this is also the case for X_2 .

The analyses described in this section illustrate how to investigate possible determinants of missing data. In the example datasets these analyses were not really interesting, because the datasets with missing data were forced to be of a certain type. However, in practice it is necessary to investigate both issues, because interpretation of the results highly depends on the missing data “mechanism.”

10.4 Analysis performed on datasets with missing data

In the foregoing sections it was stressed that it is important to investigate whether one is dealing with ignorable or informative missing data. First of all, it is important to invoke a correct interpretation of the results of the statistical analysis performed on the “incomplete” dataset. Secondly, it is also important because the sophisticated statistical techniques (i.e. GEE analysis and mixed model analysis) differ in the way in which they treat missing data. In fact, in the literature it is often argued that one of the most important differences between GEE analysis and mixed model analysis is found in the analysis of datasets with missing data. The difference is that with GEE analysis the missing data are assumed to be missing completely at random (MCAR), and that in mixed model analysis the missing data are assumed to be missing at random (MAR) (Little, 1995; Albert, 1999; Omar et al., 1999). When GEE analysis is performed on an incomplete dataset with informative missing data, the calculation of the working correlation structure is biased, and therefore the calculation of the regression coefficients is also assumed to be biased. However, from the literature it is not clear how important this bias really is (see also Section 10.6). It is therefore interesting to analyze the missing

Table 10.4 Table shows p -values derived from a MANOVA for repeated measurements performed on a complete dataset and on several incomplete datasets^a

	X_4	Time	Time by X_4
Complete dataset	$p = 0.01$	$p < 0.01$	$p < 0.01$
Missing data			
MCAR	$p = 0.03$	$p < 0.01$	$p < 0.01$
MAR_1	$p = 0.32$	$p < 0.01$	$p = 0.03$
MAR_2	$p = 0.53$	$p < 0.01$	$p < 0.01$
MNAR	$p = 0.67$	$p < 0.01$	$p < 0.01$

^aSee Section 10.3.1 for the description of the datasets with different types of missing data.

datasets with both GEE analysis and mixed model analysis and to compare the results.

Furthermore, as mentioned earlier, one of the advantages of both GEE analysis and mixed model analysis is that all longitudinal data are included, while with, for instance, multivariate analysis of variance (MANOVA) for repeated measurements only the subjects with complete data are included in the analysis. Therefore, in the first part of the example (see Section 10.4.1) a MANOVA for repeated measurements will be performed on the datasets with different types of missing data that were described in Section 10.3.1. In the second part of the example both a GEE analysis and a mixed model analysis will be performed.

10.4.1 Example

Table 10.4 shows the results of MANOVA for repeated measurements for the complete dataset and the datasets with missing data. With MANOVA for repeated measurements, the overall difference between two groups (indicated by X_4 , i.e. males and females), the overall development over time, and the possible difference in development over time between the two groups (indicated by the time by X_4 interaction) were investigated. For both the within-subject effects (i.e. the effects involving time) and the between-subjects effect (i.e. the overall group difference), the corresponding p -values were calculated (see Chapter 3 for an extensive explanation of MANOVA for repeated measurements). Figure 10.3 shows the corresponding graphs derived from the MANOVA for repeated measurements.

From Figure 10.3 it can be seen that the development over time in the outcome variable Y for both groups of X_4 is only slightly different for the different datasets. Furthermore, based on the p -values shown in Table 10.4, the effect of time and the

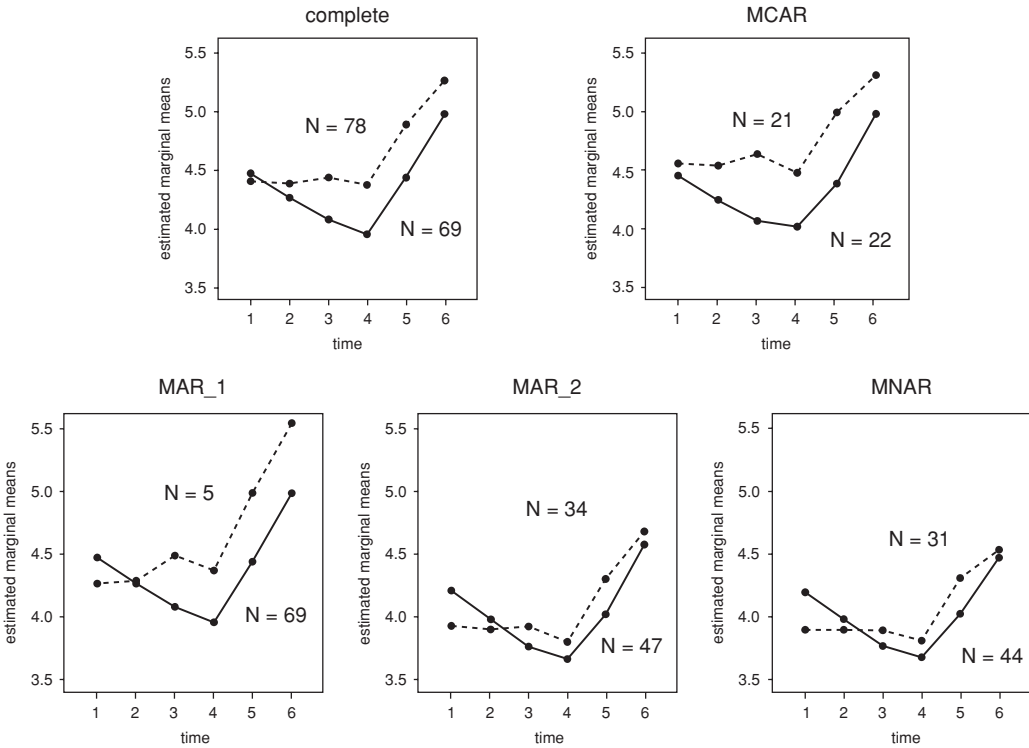


Figure 10.3 The estimated development over time for two groups indicated by X_4 , derived from a MANOVA for repeated measurements performed on a complete dataset and on several incomplete datasets.

interaction between time and X_4 seem to be hardly influenced by the missing data. This is, however, not totally true. Although the p -values are all highly significant, the test-statistics are much lower in the analysis with missing data. This is due to the fact that with MANOVA for repeated measurements only the subjects with complete data are included in the analysis. From the results it can also be seen that there seems to be no big influence of the missing data mechanism. This is not really surprising, because from Table 10.3 it was already known that the missing data were independent of the value of X_4 .

In the second part of this example both a GEE analysis and a mixed model analysis were performed on the different datasets. With both methods the longitudinal relationship between outcome variable Y and the explanatory variable X_2 was analyzed. Table 10.5 shows the results of these analyses.

From Table 10.5 it can be seen that almost the same results were obtained for a GEE analysis and a mixed model analysis, even when data are MAR or MNAR. This is remarkable because, as has been mentioned before, in the literature it is argued

Table 10.5 Regression coefficients and standard errors (in parentheses) derived from GEE analyses (with an exchangeable correlation structure) and mixed model analyses (with only a random intercept) performed on a complete dataset and several incomplete datasets^a to investigate the (longitudinal) relationship between outcome variable Y and the independent variable X_2

	GEE analysis	Mixed model analysis
Complete dataset	0.186 (0.020)	0.186 (0.018)
Missing data		
MCAR	0.176 (0.023)	0.175 (0.020)
MAR_1	0.202 (0.025)	0.202 (0.021)
MAR_2	0.154 (0.023)	0.158 (0.020)
MNAR	0.115 (0.019)	0.116 (0.018)

^aSee Section 10.3.1 for the description of the datasets with different types of missing data.

that GEE analysis is only valid when missing data are MCAR, while mixed model analysis is assumed to be valid also on MAR datasets. These results show again that the differences between GEE analysis and mixed model analysis on datasets with missing data is slightly more subtle than often suggested. In Section 10.6 this issue will be further discussed.

Because a few decades ago MANOVA for repeated measurements was the only available method for the analysis of longitudinal data, imputation techniques were developed in order to create complete datasets. In the following sections, several of the available imputation methods to replace missing data will be discussed, and the influence of different imputation methods on the results of statistical analysis will be illustrated. This will not be limited to a MANOVA for repeated measurements, but the performance of GEE analysis and mixed model analysis on imputed datasets will also be evaluated.

10.5 Imputation methods

10.5.1 Continuous outcome variables

Historically, imputation methods can be divided into cross-sectional and longitudinal imputation methods. Both can be used to replace missing data in longitudinal studies. The cross-sectional methods described here are the “mean or median of series” method, the “hot-deck” method, and the “cross-sectional linear regression” method. Longitudinal imputation methods which are discussed are the “last value

carried forward” or “last observation carried forward” method, the “linear interpolation” method, and the “longitudinal linear regression” method. Besides these historical methods, a lot of attention will be given to “multiple imputation,” which is the “state of the art” method for imputing missing data nowadays.

10.5.1.1 Cross-sectional imputation methods

All variants of the “mean or median of series” imputation method involve calculation of the mean or median of the available data for a particular variable at a particular time-point. The mean or median is imputed for the missing values. Because of its simplicity, it was by far the most frequently used imputation method in practice. A somewhat different approach is called the “hot-deck” imputation method. With this approach, the mean or median value of a subset of comparable subjects (e.g. subjects with the same gender, age, etc.) is imputed for the missing value. The minimum number of subjects in the subset can be one, and the maximum number can be the total population (which makes the “hot-deck” method the same as the “mean or median of series” method). With “cross-sectional regression” methods, a linear regression with all available (possible) explanatory variables at a certain time-point is used to provide predicted values for the outcome variable Y at that particular time-point. This predicted value is used for the imputation. It is obvious that this approach is only suitable in studies when only the outcome variable is missing and not the covariates.

10.5.1.2 Longitudinal imputation methods

The simplest longitudinal imputation method is called the “last value carried forward” (LVCF) or “last observation carried forward” (LOCF) method. In this approach the value of a variable at $t = 1$ for a particular subject is imputed for a missing value for that same subject at $t = 2$. Another longitudinal imputation method is the “linear interpolation” method. With this method, for a missing value at $t = 2$ the average of the values at $t = 1$ and $t = 3$ is imputed, assuming a linear development over time of the variables with missing data. Comparable, but somewhat more sophisticated, is the “longitudinal regression” imputation method. With this method a linear regression analysis between the outcome variable Y and time is assessed for each subject with a missing value. The predicted value for the time-point of the missing value is imputed for that missing value.

10.5.1.3 Comment

The biggest problem with imputation methods based on average values or predicted values from some sort of cross-sectional or longitudinal regression analysis is that the standard deviation of the imputed variable is artificially decreased. When a statistical analysis is performed on such an imputed dataset, the standard error

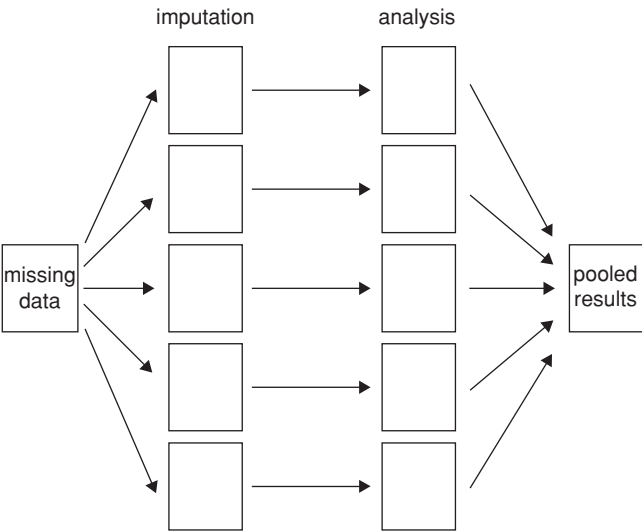


Figure 10.4 Illustration of the multiple imputation method.

of the effect estimate will be decreased as well and, therefore, the corresponding p -value will be too low. This problem can be solved by making a random draw from a distribution around the average value or around the value predicted by the cross-sectional or longitudinal regression analysis.

All imputation methods discussed before are known as single imputation methods. Nowadays, however, it is generally accepted that multiple imputation must be used instead of single imputation.

10.5.1.4 Multiple imputation

With multiple imputation, various (say M) imputation values are calculated for every missing value. With the M imputations, M complete datasets are developed, and on each dataset created in this way, statistical analyses are performed. The M complete dataset summary statistics (e.g. regression coefficients) can be combined (i.e. pooled) to form one summary statistic (Figure 10.4).

The point estimate of the summary statistic is calculated as the average of the M imputations, while the standard error of the summary statistic is usually calculated from two components (i.e. two variances). One component reflects the within-imputation variance (the average of the variances of the summary statistics of the M imputations) and the other component reflects the between-imputation variance (the difference between the summary statistic of each imputation and the

average of the summary statistics of the M imputations). Equation 10.1 shows a possible way in which the overall variance is calculated:

$$\text{var}_t = \frac{\sum_{i=1}^M \text{var}_i}{M} \quad (10.1a)$$

$$\text{var}_b = \frac{\sum_{i=1}^M (b_i - \bar{b})^2}{M - 1} \quad (10.1b)$$

$$\text{var} = \text{var}_t + \frac{M + 1}{M} \text{var}_b \quad (10.1c)$$

where var_t is the within-imputation variance, var_i is the variance of imputation i , M is the number of imputations, var_b is the between-imputations variance, b_i is the parameter of interest calculated with imputation i , and $\bar{b}$ is the average of the parameter of interest calculated with M imputations.

The major advantage of the multiple imputation method is that the combined standard error is greater than the standard error obtained from a single imputation method. This greater standard error accounts for the uncertainty introduced by estimating the missing values. In principle, the M imputations of the missing values are M repetitions from the “posterior predictive distribution” of the missing values. The “posterior predictive distribution” is related to a “model for missing data” which can (or in fact must) be based on the information derived from the simple analyses discussed in Sections 10.2 and 10.3. When the missing data are known to be dependent on earlier values of the outcome variable and on the values of several covariates, these variables can or must be used (for instance in a regression analysis) to create the “posterior predictive distribution” of the missing values (see the example in Section 10.5.3.1). It is suggested in the literature that five imputations should be enough to obtain valid results (Rubin, 1987; Schafer, 1999). However, it is questionable whether that is true (see Section 10.5.3.3). For extensive information on the multiple imputation method, reference is made to several other publications (e.g. Rubin, 1987, 1996; Schafer, 1997, 1999; Royston, 2004; Kenward and Carpenter, 2007; Royston et al., 2009; Stuart et al., 2009).

10.5.2 Dichotomous and categorical outcome variables

For dichotomous and/or categorical outcome variables a commonly used cross-sectional method is imputation of the category with the highest frequency for the subject(s) with missing data. This can either be based on the total population (“mean or median of series” method) or on a particular subset (“hot-deck” method). The most frequently used longitudinal imputation method available

Table 10.6 Table shows p -values derived from a MANOVA for repeated measurements performed on a complete dataset and on several datasets with imputed missing values.^a Imputation was performed with the LVCF method

	X_4	Time	Time by X_4
Complete dataset	$p = 0.01$	$p < 0.01$	$p < 0.01$
Imputed data			
MCAR	$p = 0.03$	$p < 0.01$	$p < 0.01$
MAR_1	$p = 0.23$	$p < 0.01$	$p = 0.03$
MAR_2	$p = 0.02$	$p < 0.01$	$p < 0.01$
MNAR	$p = 0.01$	$p < 0.01$	$p < 0.01$

^aSee Section 10.3.1 for the description of the datasets with different types of missing data.

for dichotomous and categorical missing data is the “LVCF” method. Linear interpolation can be used, but the average value of the outcome variable at the two surrounding time-points has to be rounded off. For dichotomous outcome variables, cross-sectional and longitudinal logistic regression can also be used to predict missing data. However, in these situations the predicted values also have to be rounded off, which makes the use of these techniques slightly more complicated.

10.5.3 Example

10.5.3.1 Continuous outcome variables

The influence of imputation of missing data on the results of statistical analysis will be first illustrated for MANOVA for repeated measurements. In this example only the LCVF imputation method will be illustrated. Multiple imputation is complicated because a proper summarization of the results from a MANOVA for repeated measurements performed on different imputed datasets is not possible.

Table 10.6 shows the results of MANOVA for repeated measurements for the complete datasets and several imputed datasets by the LVCF method. With this analysis, again, the development over time for the two groups indicated by X_4 were compared to each other. Figure 10.5 shows the corresponding graphs derived from the MANOVA for repeated measurements.

From both the results shown in Table 10.6 and Figure 10.5 it can be seen that the biggest difference between the LVCF imputed datasets and the complete dataset is found for the MAR dataset in which the missing data were randomly selected from one of the two groups of X_4 (i.e. the MAR_1 dataset). However, for the MAR_2 and the MNAR dataset, the LVCF method also does not seem to perform very well.

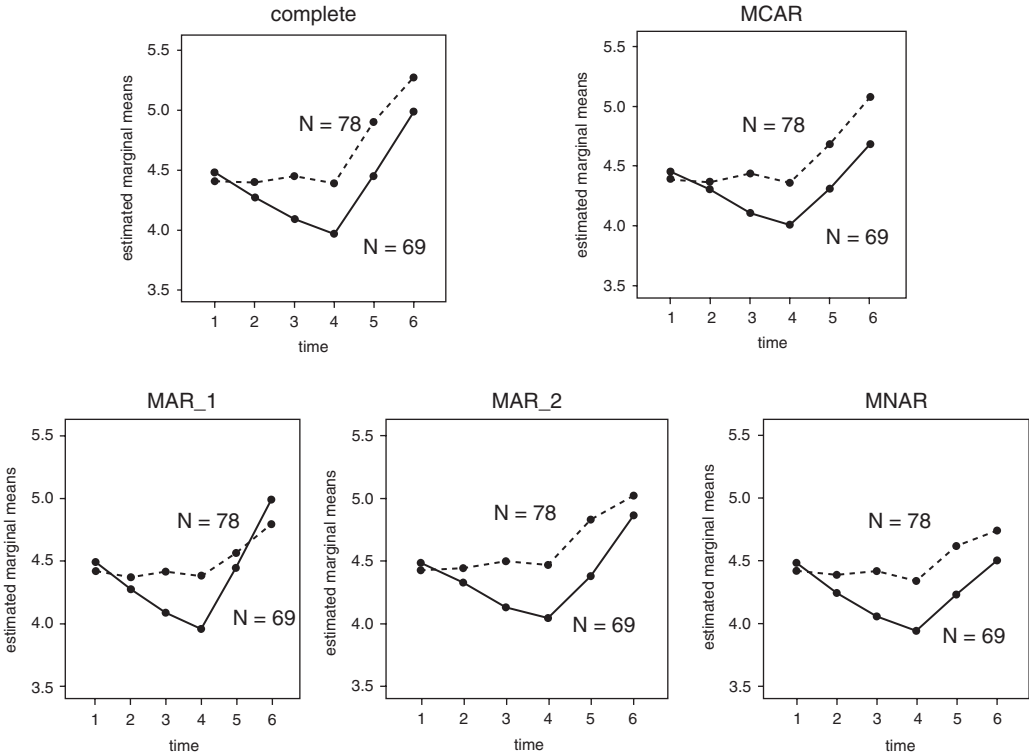


Figure 10.5 The estimated development over time for two groups indicated by X_4 derived from a MANOVA for repeated measurements performed on a complete dataset and on several imputed datasets by the LVCF method.

The second part of this example deals with the influence of the imputation of missing data on the results of a mixed model analysis. Both the use of LVCF and multiple imputation will be illustrated. It is questionable whether or not it is necessary to perform multiple imputation in combination with a mixed model analysis, because it is suggested that performing a mixed model analysis without any imputation is valid when missing data is either MCAR or MAR. However, as many researchers are not aware of the differences between a mixed model analysis on a dataset with missing data and a mixed model analysis on a multiple imputed dataset, these differences will be illustrated in this example. For the multiple imputation method used in this example, Data Augmentation (DA) was used. DA is an iterative Markov Chain Monte Carlo (MCMC) method to generate the imputed values assuming a multivariate normal distribution. DA is recognized as a state of the art method for imputing arbitrary missing data patterns (i.e. for longitudinal data with both intermittent missing data and drop-outs) (Barnard and Meng, 1999;

Table 10.7 The relationship between the outcome variable Y and the time-dependent covariate X_2 estimated with a linear mixed model analysis with only a random intercept with and without imputation on datasets with different missing data mechanisms^a

	Regression coefficient	SE
Complete dataset	0.186	0.018
MCAR		
Without imputing	0.175	0.020
LVCF	0.172	0.018
Multiple imputation ^b	0.177	0.022
MAR_1		
Without imputing	0.201	0.021
LVCF	0.181	0.018
Multiple imputation ^b	0.223	0.034
MAR_2		
Without imputing	0.158	0.020
LVCF	0.174	0.018
Multiple imputation ^b	0.155	0.032
MNAR		
Without imputing	0.116	0.018
LVCF	0.111	0.016
Multiple imputation ^b	0.104	0.025

^aSee Section 10.3.1 for the description of the datasets with different types of missing data.

^bFor multiple imputation, five imputations were used.

Fairclough et al., 2008; Stata, 2009; Stuart et al., 2009). Furthermore, for all multiple imputation models, the observed values of the outcome variable Y at the different time-points as well as the covariates used in the mixed model analyses were used to predict the missing values. For all multiple imputation, five imputations are used.

Table 10.7 shows the results of the analyses regarding the relationship between the outcome variable Y and the time-dependent covariate X_2 on the complete dataset, the datasets with missing data and the imputed datasets.

From Table 10.7 it can be seen that for the MCAR datasets, mixed model analyses without imputation only provided slightly different results compared to LVCF and

multiple imputation. The regression coefficients obtained from multiple imputation were a bit closer to the regression coefficient obtained from the complete dataset compared to the dataset without imputations and the LVCF imputed dataset. The standard errors obtained from the analyses with and without multiple imputation were, as expected, slightly higher compared to the analyses on the complete dataset. The standard error obtained from the mixed model analysis on the LVCF imputed dataset is the same as the one obtained from the complete dataset. Although this seems to be an advantage, it is not. Due to the artificial increase in the number of observations in an LVCF imputed dataset with missing data, the standard error of the regression coefficient decreases. Also, in the multiple imputation method, the number of observations is artificially increased, but the decrease in standard error due to the higher number of observations is compensated with the higher standard error based on the combination of two variances (see Equation 10.1).

For the MAR datasets, regarding the regression coefficients, the analyses led to comparable results for the mixed model analyses with and without imputation. However, they were different from the regression coefficients obtained from the complete dataset. Regarding the uncertainty of the estimates, the standard errors of the analyses with or without multiple imputation were quite different. Surprisingly, the regression coefficients obtained from a mixed model analysis on an LVCF imputed dataset were relatively close to the regression coefficients obtained from the complete dataset. The standard error, again, is underestimated.

For the MNAR datasets, the regression coefficients obtained from the different analyses are more or less the same, but they are totally different from the analyses on the complete dataset without missing data. The comparison of standard errors shows the same picture as for all other analyses.

In general, the results of the example show that the results obtained from the imputed and non-imputed datasets are different. More specifically, the standard errors of the regression coefficients are different and in general somewhat larger when they are obtained from mixed model analyses with multiple imputation than obtained from mixed model analyses without multiple imputation. Regarding the regression coefficients, the results are not straightforward in favor of one of the approaches.

In the example, the standard errors of the mixed model analysis with multiple imputation were higher than the standard errors obtained from the mixed model analysis without multiple imputation. The remaining question is whether the standard error resulting from the mixed model analysis without multiple imputation is an underestimation or whether the standard error obtained from the mixed model analysis with multiple imputation is an overestimation. In the literature, most evidence is given for the fact that the two approaches lead to similar results when the data are either MCAR or MAR, which is, however, not the case in the present

example. There are authors who suggest that a mixed model analysis without multiple imputation leads to an underestimation of the standard error (Kenward and Carpenter, 2007; Mazumdar et al., 2007), while Enders on the other hand states that the imputation phase can use an unnecessarily complex model to deal with the missing data and that this additional complexity can add a small amount of noise to the resulting estimations (Enders, 2010), which leads to an overestimation of the standard error. Furthermore, Robins and Wang suggest that the “combination rule” shown in Equation 10.1c leads to an overestimation of the standard error (Robins and Wang, 2000). In their theoretical paper they suggest an alternative way to obtain the pooled standard error. However, that alternative has never been used in applied studies.

10.5.3.2 Should multiple imputation be used in combination with a mixed model analysis?

As the comparison of regression coefficients and standard errors does not give a straight answer to the question of whether or not multiple imputation should be used in combination with a mixed model analysis, other arguments must be used to provide an answer to that question. A mixed model analysis without multiple imputation is definitely less complicated because only one dataset has to be analyzed, while with multiple imputation, multiple datasets have to be analyzed. So it is computationally more efficient to perform a mixed model analysis without multiple imputation than a mixed model analysis with multiple imputation. In addition, the results of the example show that the results obtained from a mixed model analysis with multiple imputation can be quite unstable (see Section 10.5.3.3). So, based on these arguments, it can be concluded that in general it is not necessary to perform multiple imputation before performing a mixed model analysis.

Although in the example, performing multiple imputation in combination with a mixed model analysis did not lead to more valid results, there are possible situations in which multiple imputation potentially holds an advantage (Enders, 2010). When auxiliary variables (i.e. variables that are related to the missing data mechanism but not used in the model to answer the research question) are measured, they can be included in predicting the missing data, without being included in the mixed model analysis, which probably increases the efficiency. Also in a situation when only covariates are missing, performing multiple imputation before a mixed model analysis seems to be better. However, the latter hardly exists in longitudinal studies, because in longitudinal studies mostly all data to be collected at a particular visit are missing for that particular subject.

10.5.3.3 Additional analyses

Because there is a discussion going on in the literature regarding the number of imputations needed to obtain stable results, for all multiple imputations performed

Table 10.8 The relationship between the outcome variable Y and the time-dependent covariate X_2 , estimated with a linear mixed model analysis with only a random intercept on datasets with different missing data mechanisms^a with multiple imputation with 50 imputations

	Regression coefficient	SE
MCAR	0.183	0.024
MAR_1	0.213	0.029
MAR_2	0.168	0.027
MNAR	0.113	0.021

^aSee Section 10.3.1 for the description of the datasets with different types of missing data.

in the present example, the use of 50 imputations was evaluated. Table 10.8 shows the results of these analyses.

Based on the inconsistent patterns in the magnitude of (especially) the standard errors between some of the analyses with 5 and 50 imputations (Tables 10.7 and 10.8), the (in)stability of the mixed model analyses with multiple imputation was further investigated. As an example the relationship between the outcome variable Y and the time-dependent covariate X_2 was reanalyzed. In these additional analyses, all multiple imputations were repeated 100 times. Figure 10.6 summarizes the (in)stability of the different multiple imputation for both the regression coefficients and the standard errors. Figure 10.7 shows the distribution of the standard errors of the 100 analyses performed on the MAR_2 dataset. From Figure 10.6 it is clear that using five imputations leads to very unstable results in all datasets, but especially for the MAR_2 dataset. For the latter, even the use of 50 imputations is somewhat unstable (see Figure 10.7).

The (in)stability of the multiple imputation method is in contrast with most basic multiple imputation literature (Rubin, 1987; Kenward and Carpenter, 2007), which suggests that five imputations should be sufficient to obtain valid inference. However, Graham recognized that some analyses may require much more imputations to obtain valid results (Graham, 2009). Literature with formal recommendations on how to choose the optimal number of imputations is scarce. Royston and co-workers discuss the impact of the number of imputations on the precision of the estimates and suggests ways of determining the required number of imputations by evaluating the sampling error of the multiple imputation estimates (Royston, 2004; Royston et al., 2009). In the example it is obvious that five imputations do not appear to be sufficient to obtain stable estimates.

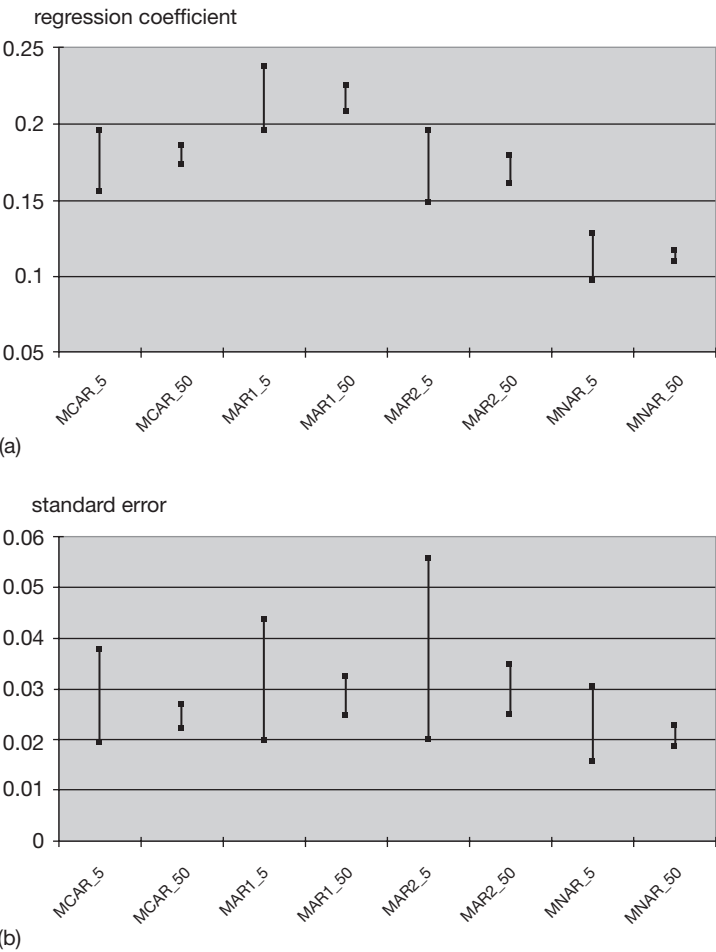


Figure 10.6 Range in regression coefficients (a) and range in standard errors (b) obtained from 100 mixed model analyses after multiple imputation for different types of missing data with either 5 or 50 imputations regarding the relationship between the outcome variable Y and the time-dependent covariate X_2 .

10.5.3.4 Dichotomous outcome variables

In the example discussed before, the analyses were performed on a continuous outcome variable Y . The reason for this is that the method used for multiple imputation in the present example assumes a multivariate normal distribution. For dichotomous outcome variables, sophisticated multiple imputation methods are only available for missing data patterns without intermittent missing data (i.e. monotone missing data). Several authors, however, suggest that for non-monotone missing data patterns (i.e. longitudinal data with both intermittent missing data

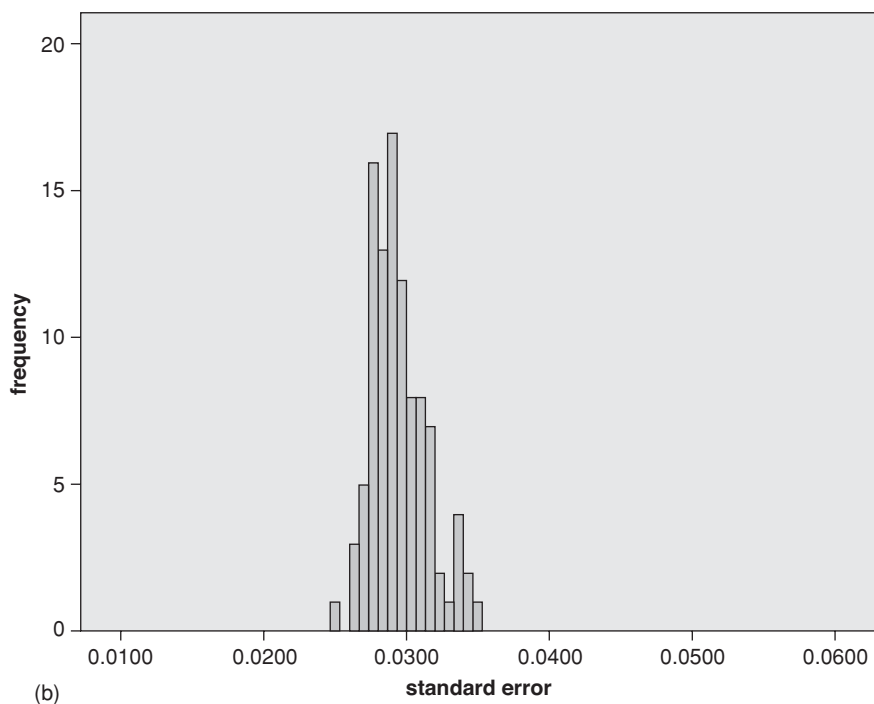
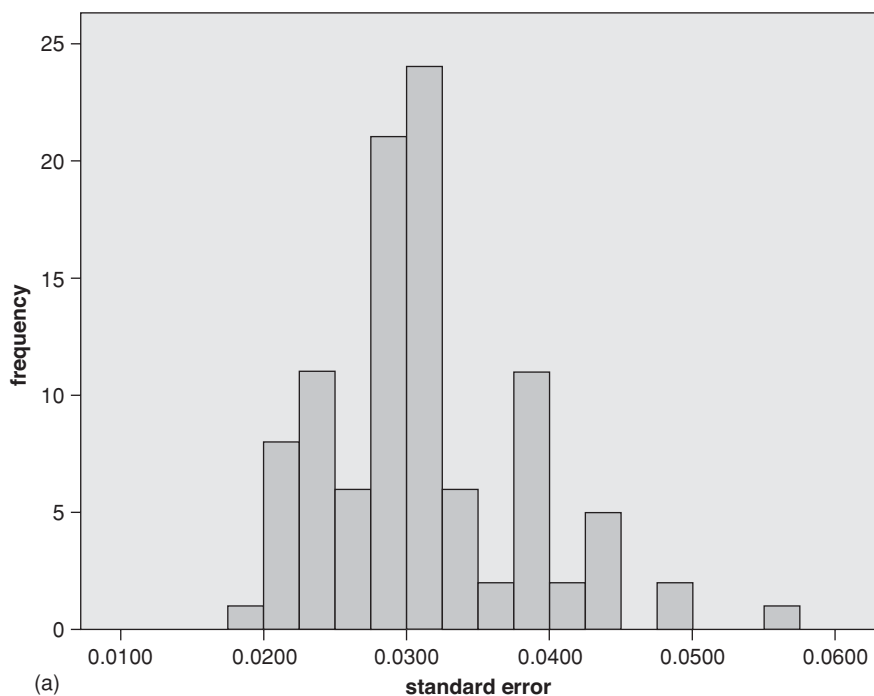


Figure 10.7 Distribution of the standard errors obtained from 100 mixed model analyses after multiple imputation regarding the relationship between the outcome variable Y and the time-dependent covariate X_2 , performed on the MAR_2 dataset with either 5 (a) or 50 (b) imputations.

Table 10.9 The relationship between the outcome variable Y_{dich} and the time-dependent covariate X_2 estimated with a logistic mixed model analysis with and without imputation on datasets with different missing data mechanisms^a

	Regression coefficient	SE
Complete dataset	0.560	0.106
MCAR		
Without imputing	0.602	0.121
LVCF	0.597	0.120
Multiple imputation ^b	0.478	0.103
MAR_1		
Without imputing	0.663	0.127
LVCF	0.695	0.129
Multiple imputation ^b	0.546	0.113
MAR_2		
Without imputing	0.442	0.122
LVCF	0.586	0.137
Multiple imputation ^b	0.367	0.132
MNAR		
Without imputing	0.114	0.122
LVCF	0.015	0.135
Multiple imputation ^b	0.160	0.124

^aSee Section 10.3.1 for the description of the datasets with different types of missing data.

^bFor multiple imputation, five imputations were used.

and drop-outs) the same procedures can also be used as for continuous outcome variables (Bernaards et al., 2007; Demirtas et al., 2008). One major problem with this approach for dichotomous data is the rounding of the imputed values which further complicates the multiple imputation method (Yucel et al., 2008). Royston (2009) suggests, on the other hand, to use multiple imputation by chained equations (MICE) to impute dichotomous and categorical missing data. Although it is not expected that the conclusions based on logistic mixed model analyses with and without multiple imputation would be different from the ones based on the linear mixed model analyses presented in the example with the continuous outcome variable Y , Table 10.9 shows the results of a logistic mixed model analysis on imputed datasets (both LVCF and multiple imputation) regarding the relationship between the dichotomous outcome variable Y_{dich} and the time-dependent covariate X_2 .

From [Table 10.9](#) it can be seen that multiple imputation only seems to work well for the MAR_1 dataset. On the other hand, for the MAR_2 dataset, LVCF seems to be the most appropriate. This is rather surprising because it is generally accepted that multiple imputation is better than any single imputation method. The fact that the standard errors of the effect estimates are not lower than the ones obtained from the logistic mixed model analysis on either the dataset with missing data or the multiple imputed datasets is rather strange. Due to the artificial increase in the number of observations, it was expected that the standard errors would be comparable to the ones obtained from the complete dataset.

It should be realized, however, that the comparison is based on only one dataset with very typical missing data mechanisms and that the results of a logistic mixed model analysis should be interpreted with caution (see [Chapter 7](#)).

10.5.4 Comments

The present example shows that mixed model analysis with or without multiple imputation does not lead to valid results when performed on a MNAR dataset. Both approaches behave equally unsatisfactorily when the results are compared with the results of the analysis on the complete dataset without missing data. This is as expected because both multiple imputation and mixed model analysis use the observed data for the calculations. For both mixed model analysis and multiple imputation, the assumption is that the methods are only valid when the incomplete dataset is either MAR or MCAR (Hogan et al., [2004](#); Kristman et al., [2005](#); Kenward and Carpenter, [2007](#)). A big problem, however, is that in real life data it is not possible to evaluate whether incomplete data is MAR or MNAR (Potthoff et al., [2006](#); Kenward and Carpenter, [2007](#); Enders, [2010](#)), which means that you will never know whether the results obtained from the mixed model analysis (either with or without multiple imputation) provides valid results. Furthermore, in real life datasets, missing data is never totally MCAR, MAR, or MNAR. Some missing observations will be totally random, while other missing observations will depend on either observed data or unobserved data. Surprisingly, even in the MCAR datasets, the results of the mixed model analyses were different from the results obtained from the dataset without missing data. This has probably to do with the fact that only one MCAR dataset was created, which makes it possible that the created MCAR dataset was not completely MCAR (Burton et al., [2006](#)).

10.5.4.1 Alternative approaches

Although in the literature most researchers use a mixed model approach with or without multiple imputation, there are some alternative methods available to deal with missing data in longitudinal studies. These alternative methods include selection models and pattern mixture models (Little, [1993](#), [1994](#); Demirtas and

Schafer, 2003; Yang, X. et al., 2008). Both are frequently used in econometrics and are supposed to provide valid results even on MNAR datasets. The general idea of a selection model is that the analysis is split into two parts. The first part is the regression equation of interest, and the second part is a regression equation that predicts the response probabilities. The basis of a pattern mixture model is more or less the same. In this approach, first, subgroups of subjects with the same missing data pattern are created. In the next step, the model coefficients are estimated within the different subgroups. In the last step, the subgroup specific coefficients are combined to get one regression coefficient that accounts for missing data being MNAR. Although both methods have some potential in adequately dealing with missing values, they also have some disadvantages (i.e. computational complexity and their reliance on knowledge about the missing data mechanism) and are therefore not much used in real life epidemiological studies.

In the literature, some alternative approaches are also suggested (e.g. Fitzmaurice et al., 1994; Greenland and Finkle, 1995; Little, 1995; Hogan and Laird, 1997; Shih and Quan, 1997; Kenward, 1998; Molenberghs et al., 1998; Haan et al., 1999; Kenward and Molenberghs, 1999; Chen et al., 2000; Verbeke and Molenberghs, 2000; Sun and Song, 2001), but unfortunately most of them are very technical and difficult to understand for non-statisticians.

10.6 GEE analysis versus mixed model analysis regarding the analysis on datasets with missing data

In the literature there is some discussion about the use of GEE analysis in datasets with missing data. It was already been mentioned that it is argued that GEE analysis only provides valid results when the missing data is MCAR. Mixed model analysis, on the other hand, also provides valid results when the missing data is MAR. However, from Table 10.5 it can be seen that the results of a GEE analysis on datasets with missing data were comparable to the results of a mixed model analysis even when the missing data are MAR or MNAR. To explore the difference between GEE analysis and mixed model analysis on datasets with missing data a very simple example dataset will be used, in which 10 subjects were measured 10 times. Half of the subjects show an increase of one unit per time interval, while the other half show an increase of two units per time interval (see Figure 10.8a). When these data are analyzed by either GEE analysis or mixed model analysis, both give the same regression coefficient of 1.5. Figure 10.8b shows a situation in which there is missing data. From the subjects with an increase of two units per time interval, half of them dropped out after the fifth measurement. When these data are analyzed with a GEE analysis, the regression coefficient will be 1.357. A mixed model analysis with only a random intercept will also give a regression coefficient

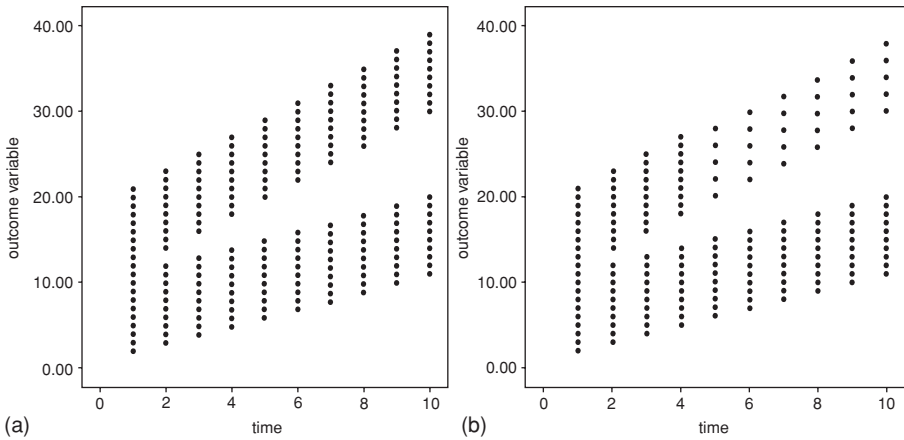


Figure 10.8 Simple longitudinal dataset used to illustrate the difference between GEE analysis and mixed model analysis regarding the analysis of a dataset with missing data; (a) complete dataset and (b) dataset with missing data.

of 1.357. However, when a random slope is added to the mixed model analysis, the regression coefficient returns to 1.5. So, in general, only if the missing data can be predicted perfectly by observed data (when the missing data are perfectly MAR), and only when a mixed model analysis is performed with a correctly specified model (i.e. including random slopes and the appropriate covariances between random intercept and random slopes), a mixed model analysis will provide valid results whereas GEE analysis will provide invalid results. However, when the missing data are not perfectly MAR, GEE analysis and mixed model analysis will both provide comparable invalid results.

10.7 Conclusions

For continuous outcome variables, the use of imputation methods is recommended when MANOVA for repeated measurements is used to analyze a longitudinal dataset with missing data. When mixed model analysis is used to analyze a longitudinal dataset with missing data, no imputations at all may be better than applying any of the imputation methods. If a decision is made to impute missing values, longitudinal methods are generally preferred above cross-sectional methods and multiple imputation is preferred above single imputation.

Beyond the question of whether or not to use imputations, it is of utmost importance to describe as well as possible the missing data mechanisms in the study dataset, because this can have important implications for the interpretation of the results of the statistical analyses performed.

Sample size calculations

11.1 Introduction

Before performing a longitudinal (experimental) study, it is “necessary” to calculate the number of subjects needed to ensure that a certain predefined effect will be significant. Sample size calculations are also a prerequisite for research grants and are used by (medical) ethical committees in their evaluation of study design protocols. Besides this, sample size calculations are part of the CONSORT statement, meaning that without a sample size calculation, a paper reporting the results of an experimental study will not be published. The importance of sample size calculations is basically a very strange phenomenon. First of all, sample size calculations are based on many assumptions which can easily be changed, in which case the number of subjects needed will be totally different. Secondly, sample size calculations are related to the importance of “significance levels” (i.e. How many subjects are needed to make a certain “effect” significant?), and that is strange because in epidemiological research the importance of significance is becoming more and more questionable. Nevertheless there is a large amount of literature discussing sample size calculations in longitudinal studies (e.g. Lui and Cumberland, 1992; Snijders and Bosker, 1993; Diggle et al., 1994; Lee and Durbin, 1994; Lipsitz and Fitzmaurice, 1994; Liu, G. and Liang, 1997; Hedeker et al., 1999).

In general, the sample size calculations used for a longitudinal experimental study are the same as for “standard” experimental studies. It should be noted that the “standard” sample size calculations are developed for experimental studies with one follow-up measurement. In fact, with the standard sample size calculations the difference in a certain outcome variable between several groups at the first follow-up measurement is used as an effect size. This assumes that the baseline values for the groups to be compared are equal, which seems to be a reasonable assumption in a randomized trial, but which is not always true (see [Chapter 9](#)). Equation 11.1 shows how the sample size can be calculated in the “standard” situation for a

continuous outcome variable.

$$N = \frac{(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2 \times \sigma^2 \times 2}{v^2} \quad (11.1)$$

where N is the sample size in either the experimental or control group, $Z_{(1-\alpha/2)}$ is the $(1 - \alpha/2)$ percentile point of the standard normal distribution, $Z_{(1-\beta)}$ is the $(1 - \beta)$ percentile point of the standard normal distribution, σ is the standard deviation of the outcome variable and v is the difference in mean value of the outcome variable between the groups.

For dichotomous outcome variables a comparable equation can be used (Equation 11.2).

$$N = \frac{(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2 \times \bar{p}(1 - \bar{p}) \times 2}{(p_1 - p_0)^2} \quad (11.2a)$$

$$\bar{p} = \frac{p_1 + (r p_0)}{1 + r} \quad (11.2b)$$

where N is the sample size in either the experimental or control group, $Z_{(1-\alpha/2)}$ is the $(1 - \alpha/2)$ percentile point of the standard normal distribution, $Z_{(1-\beta)}$ is the $(1 - \beta)$ percentile point of the standard normal distribution, $\bar{p}$ is the “weighted” average of p_0 and p_1 , p_1 is the proportion of “cases” in the intervention group, and p_0 is the proportion of “cases” in the control group.

In the standard sample size calculations, it is assumed that the number of subjects in the experimental group is equal to the number of subjects in the control group. Although equal number of subjects in both groups is the most efficient way to divide the population under study into two groups, it is also possible to have unequal numbers in both groups. When that is the case, the number 2 in the numerator of the standard sample size calculation equation must be replaced by $r + 1$, where r is the ratio in the number of subjects in the two groups. Furthermore, r should be multiplied by the effect size (either v^2 or $(p_1 - p_0)^2$) in the denominator of the standard sample size equation.

When more than one follow-up measurement is carried out, and the purpose of the study is to estimate the effect of the intervention on average over the total follow-up period Equation 11.3 can be applied. It should, however, be noted that when the purpose of the experimental study is just to compare the different groups at one single point in time, Equations 11.1 and 11.2 must be applied.

$$N = \frac{(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2 \times \sigma^2 \times 2[1 + (T - 1)\rho]}{v^2 T} \quad (11.3)$$

where N is the sample size in either the experimental or control group, $Z_{(1-\alpha/2)}$ is the $(1 - \alpha/2)$ percentile point of the standard normal distribution, $Z_{(1-\beta)}$ is the $(1 - \beta)$ percentile point of the standard normal distribution, σ is the standard

Table 11.1 Sample sizes needed to make a certain difference in a continuous outcome variable statistically significant on a 5% level with a power of 80%; studies with different within-subject correlation coefficients (ρ)

	Expected difference (in standard deviation units)			
	0.1	0.2	0.5	1
Two follow-up measurements				
$\rho = 0$	785	196	31	8
$\rho = 0.25$	981	245	39	10
$\rho = 0.5$	1178	294	47	12
Three follow-up measurements				
$\rho = 0$	523	130	21	5
$\rho = 0.25$	785	196	31	8
$\rho = 0.5$	1047	262	42	10

deviation of the outcome variable, T is the number of follow-up measurements, ρ is the correlation coefficient of the repeated measurements, and v is the difference in mean value of the outcome variable between the groups.

To illustrate how many subjects are needed to make a certain difference between groups statistically significant, Equation 11.3 is applied to several research situations in which the difference between groups is expressed in standard deviation units. Table 11.1 presents a “sample size table” for several studies in which two and three follow-up measurements are carried out.

For sample size calculations in longitudinal experimental studies with a dichotomous outcome variable, Equation 11.4 can be applied.

$$N = \frac{(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2 \times \bar{p}(1 - \bar{p}) \times 2[1 + (T - 1)\rho]}{(p_1 - p_0)^2 T} \tag{11.4}$$

where N is the sample size in either the experimental or control group, $Z_{(1-\alpha/2)}$ is the $(1 - \alpha/2)$ percentile point of the standard normal distribution, $Z_{(1-\beta)}$ is the $(1 - \beta)$ percentile point of the standard normal distribution, $\bar{p}$ is the “weighted” average of p_0 and p_1 (Equation 11.2b), T is the number of follow-up measurements, ρ is the correlation coefficient of the repeated measurements, p_1 is the proportion of “cases” in the intervention group, and p_0 is the proportion of “cases” in the control group.

Table 11.2 Sample sizes needed to make a certain difference in a dichotomous outcome variable statistically significant on a 5% level with a power of 80%; studies with different within-subject correlation coefficients (ρ)

	Expected proportion of intervention group ^a		
	0.4	0.3	0.2
Two follow-up measurements			
$\rho = 0$	194	47	20
$\rho = 0.25$	243	59	25
$\rho = 0.5$	291	71	30
Three follow-up measurements			
$\rho = 0$	130	31	13
$\rho = 0.25$	194	47	20
$\rho = 0.5$	259	59	26

^aThe expected proportion in the reference category is assumed to be 0.5.

Based on Equation 11.4, a “sample size table” for different studies with a dichotomous outcome variable can also be constructed (Table 11.2).

All sample size equations presented in this section can be used to estimate the sample size needed for a particular experimental study or to calculate the “power” of that particular study. Here again it should be noted that for the calculation of sample sizes or power, certain unknown values have to be filled in, i.e. the expected (relevant) difference between the groups, the standard deviation of the outcome variable of interest, and the within-subject correlation coefficient. Furthermore, in the equations, a specific significance level (usually 5%) is essential, and as has been mentioned before, the importance of significance levels is rather doubtful. Caution is therefore strongly advised in the use of sample size equations.

11.2 Example

Equations 11.3 and 11.4 can not only be used to calculate the sample size needed to make a certain predefined effect statistically significant, but they can also be used to estimate the power of a particular longitudinal experimental study, given a certain sample size. Besides this, they can also be used to calculate the effect that

Table 11.3 Information needed to perform a power analysis for an experimental study with a continuous outcome variable (i.e. systolic blood pressure) described in Section 9.2

N	Average sample size at the first and second follow-up measurement, $N = 69$
SD	Average standard deviation at the first and second follow-up measurement, $SD = 13.6$
ρ	Correlation between the first and second follow-up measurement, $\rho = 0.687$
T	Number of follow-up measurements, $T = 2$
$(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2$	With a significance level of 5% and a power of 80%, $(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2 = 7.85$

Table 11.4 Information needed to perform a power analysis for an experimental study with a dichotomous outcome variable (i.e. recovery from stomach ulcer) described in Section 9.3

N	Sample size per group, $N = 60$
p_0	Proportion recovered in the control group, $p_0 = 50\%$
ρ	Average correlation between the follow-up measurements $\rho = 0.15$
T	Number of follow-up measurements, $T = 3$
$(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2$	With a significance level of 5% and a power of 80%, $(Z_{(1-\alpha/2)} + Z_{(1-\beta)})^2 = 7.85$

will be statistically significant, given a certain sample size. In this section, the latter will be illustrated for the examples, which were explained in detail in [Chapter 9](#). [Table 11.3](#) shows the information that is needed to perform such an analysis for the longitudinal experimental study with a continuous outcome variable, i.e. systolic blood pressure.

With the information presented in [Table 11.3](#), the smallest difference that will be significant in this experimental study can be calculated with [Equation 11.3](#).

$$69 = \frac{7.85 \times 13.6^2 \times 2 \times [1 + (2 - 1) \times 0.687]}{v^2 \times 1 \times 2}$$
$$v = \sqrt{35.4988} = 5.96$$

So, the smallest difference in systolic blood pressure between the two groups that will be significant in this experimental study is 5.96 mmHg.

A similar calculation can be made for the example with a dichotomous outcome variable, i.e. recovery from stomach ulcer. Based on the information about the study ([Table 11.4](#)), the smallest detectable difference (i.e. the smallest difference to

be significant) in proportions can be calculated:

$$60 = \frac{7.85 \times \bar{p}(1 - \bar{p}) \times 2 \times [1 + (3 - 1) \times 0.15]}{(p_1 - 0.50)^2 \times 1 \times 3}$$

$$\bar{p} = \frac{p_1 + (1 \times 0.50)}{1 + 1}$$

$$p_1 = \pm 0.335$$

In other words, with the characteristics of the experimental study with a dichotomous outcome variable presented in [Chapter 9](#), a 16.5% (i.e. 50 – 33.5) difference in proportion of recovery between the two groups will be significant at a 5% level.

Software for longitudinal data analysis

12.1 Introduction

In the foregoing chapters many research questions have been addressed and many techniques for the analysis of longitudinal data have been discussed. In the examples, the relatively “simple” statistical techniques were performed with SPSS, while the generalized estimating equations (GEE) analyses and the mixed model analyses were performed with Stata. This chapter provides an overview of a few major software packages (i.e. SPSS, SAS, R, and MLwiN) and their ability to perform sophisticated longitudinal data analysis. In this chapter, only GEE analysis and mixed model analysis will be discussed. Multivariate analysis of variance (MANOVA) for repeated measurements can be performed with all major software packages, and can usually be found under the repeated measurements option of the generalized linear model (GLM) or as an extension of the (M)ANOVA procedure. The emphasis of this overview is on the output and syntax of the sophisticated longitudinal analysis in the different software packages, and especially is on the comparison of the results obtained with the various different packages. In this overview, only analysis with a continuous and a dichotomous outcome variable will be discussed in detail.

12.2 GEE analysis with continuous outcome variables

12.2.1 Stata

Stata was used in the examples throughout this book, so the outputs will not be repeated here. The syntax needed to perform a linear GEE analysis is very simple:

```
xtgee ycont x1 x2 x3 x4, i(id) corr(exch) robust
```

First the Stata procedure is specified (i.e. `xtgee`), directly followed by the outcome variable and the covariates. After the comma, additional information is supplied, i.e. the subject identifier (`id`) and the “working” correlation structure (`exch`). Because an exchangeable correlation structure is the default in Stata, the syntax will also run without this additional information. It should be noted that in Stata the default procedure for the estimation of the standard errors is so-called “model based.” This is rather strange, because it is generally accepted that the “robust” estimation procedure of the standard errors is preferred; for details of the robust estimation procedure of the standard errors within GEE analysis, see Section 4.5.2.

12.2.2 SAS

In SAS, the GENMOD procedure can be used to perform a GEE analysis. Output 12.1 shows a section of the output of a linear GEE analysis performed with the GENMOD procedure.

From the first part of Output 12.1 it can be seen that a linear GEE analysis has been performed (i.e. a normal distribution, and an identity link function). Furthermore, it can be seen that an exchangeable correlation structure is used and that the value of the (exchangeable) correlation is equal to 0.5145064485. At the end of the output, the parameter estimates are given, i.e. the regression coefficient, the standard error, the 95% confidence interval around the regression coefficient, the *z*-value and the corresponding *p*-value.

The syntax needed to perform a linear GEE analysis with the GENMOD procedure is slightly more complicated than the syntax for Stata:

```
proc genmod data=long.x;  
class id;  
model ycont = x1 x2 x3 x4;  
repeated subject=id/type=exch;  
run;
```

Each SAS procedure starts with the procedure specification (`proc genmod`) and ends with a `run` statement. The `class` statement in SAS is needed to indicate that the subject identifier is a categorical variable (`class id`). In the third line of the syntax the model to be analyzed is specified (`model ycont = x1 x2 x3 x4`), and in the fourth line the fact that the subjects are repeatedly measured (`repeated subject=id`), and that the correlation structure is exchangeable (`type=exch`).

Output 12.1 Results of a linear GEE analysis performed with SAS

The GENMOD Procedure						
Data Set	long.x					
Distribution	Normal					
Link Function	Identity					
Dependent Variable	YCONT	continuous outcome variable y				
GEE Model Information						
Correlation Structure	Exchangeable					
Subject Effect	ID (147 levels)					
Number of Clusters	147					
Correlation Matrix Dimension	6					
Maximum Cluster Size	6					
Minimum Cluster Size	6					
Exchangeable Working Correlation						
Correlation	0.5145064485					
Analysis Of GEE Parameter Estimates						
Empirical Standard Error Estimates						
Parameter Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z	
Intercept	3.4487	0.6646	2.1462 4.7512	5.19	<0.0001	
X1	0.1482	0.2653	-0.3717 0.6681	0.56	0.5764	
X2	0.1849	0.0225	0.1408 0.2289	8.23	<0.0001	
X3	0.0447	0.0642	-0.0812 0.1706	0.70	0.4863	
X4	0.0371	0.1282	-0.2142 0.2883	0.29	0.7724	

12.2.3 R

R is a very popular software program that can be downloaded for free from the internet (R Development Core Team, <http://www.r-project.org>). R is developed from the commercial statistical software package S-plus and it uses the same

programming environment (Venables and Ripley, 2000, 2002; Dalgaard, 2002; Fox, 2002; Maindonald and Braun, 2003). Besides the fact that it is free, it is also recognized for its flexibility and its excellent graphical features. To perform a GEE analysis in R, first the GEE library should be loaded. Output 12.2 presents a section of the output of a linear GEE analysis performed with R.

Output 12.2 Results of a linear GEE analysis performed with R

Summary of Residuals:

Min	1Q	Median	3Q	Max
-1.97781415	-0.52722867	-0.08696921	0.42706287	2.75658780

Coefficients:

	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z
(Intercept)	3.48582528	0.58226190	5.9866965	0.57277715	6.0858316
x1	0.14821326	0.26951871	0.5499183	0.26527348	0.5587187
x2	0.18486851	0.01954975	9.4563088	0.02247607	8.2251250
x3	0.04471056	0.05924418	0.7546827	0.06422056	0.6962031
x4	0.03708125	0.11888838	0.3118997	0.12818611	0.2892766

Estimated Scale Parameter: 0.5685152

Number of Iterations: 2

Working Correlation

	[0.1]	[0.2]	[0.3]	[0.4]	[0.5]	[0.6]
[1,]	1.000000	0.514503	0.514503	0.514503	0.514503	0.514503
[2,]	0.514503	1.000000	0.514503	0.514503	0.514503	0.514503
[3,]	0.514503	0.514503	1.000000	0.514503	0.514503	0.514503
[4,]	0.514503	0.514503	0.514503	1.000000	0.514503	0.514503
[5,]	0.514503	0.514503	0.514503	0.514503	1.000000	0.514503
[6,]	0.514503	0.514503	0.514503	0.514503	0.514503	1.000000

The output of R is different to what has been seen for Stata and SAS. Firstly, a “Summary of Residuals” is given. This part can be used to check whether or not the residuals are normally distributed. As has been mentioned before, for longitudinal analysis on continuous outcomes, the same assumptions hold as for cross-sectional linear regression analysis. One of the most important assumptions is a normal distribution of the residuals. From the residuals shown in Output 12.2, it can be seen that they are slightly skewed to the right. But the skewness is very

small, so it was not necessary to transform the continuous outcome variable. In the second part of the output the regression coefficients and the naïve and robust standard errors are given. With both, the *z*-values are also calculated. Below the coefficients, the scale parameter is given (0.5685152) and the magnitude of the correlation of the exchangeable working correlation matrix (0.514503).

The syntax needed to perform a linear GEE analysis in R is as follows:

```
test <- gee(formula = ycont ~ x1 + x2 + x3 + x4, id = id,  
data = data,family = gaussian, corstr = "exchangeable")
```

R is an “object-oriented” program, which means that the results of the analysis must be linked to an object. The object for the first analysis is named *test* and from the syntax it can be seen that the object *test* is linked to the results of the GEE analysis. The other part of the syntax is straightforward and is comparable to the syntax used in Stata; i.e. after the GEE procedure specification, the outcome variable and the independent variables are given. After the comma, additional information has to be specified, i.e. the subject identifier (*id = id*), the dataset used (*data = data*), the distribution of the outcome variable (*family = gaussian*), and the correlation structure (*corstr = “exchangeable”*).

12.2.4 SPSS

From version 15 onwards, in SPSS it is possible to perform GEE analyses. Output 12.3 shows part of the results of a linear GEE analysis performed in SPSS.

The output of a linear GEE analysis performed with SPSS gives the same information as the outputs from the other software packages. First, some general model information is provided, i.e. the number of individuals, the minimum and maximum number of observations within the individual, the link function, the family (here the Probability Distribution), and the correlation structure that is used. The last part of the output contains the regression coefficients, the standard errors, the 95% confidence intervals, and the results of the Wald-test for the regression coefficients. Note that in the other programs the *z*-test was used to obtain the *p*-values. However, because the Chi-square distribution with one degree of freedom (which is used in the Wald-test) is equal to the standard normal distribution squared, the results are the same.

Although one of the best parts of SPSS is the fact that the program is fully menu driven, it is also possible to use syntax to perform a GEE analysis. The following syntax was used to obtain the results presented in Output 12.3.

Table 12.1 Summary of the results (i.e. regression coefficients and standard errors (in parentheses)) of a linear GEE analysis with an exchangeable correlation structure performed with different software packages

	Stata	SPSS	SAS	R
X_1	0.15 (0.27)	0.15 (0.27)	0.15 (0.27)	0.15 (0.27)
X_2	0.18 (0.02)	0.18 (0.02)	0.18 (0.02)	0.18 (0.02)
X_3	0.04 (0.06)	0.04 (0.06)	0.04 (0.06)	0.04 (0.06)
X_4	0.04 (0.13)	0.04 (0.13)	0.04 (0.13)	0.04 (0.13)

12.2.5 Overview

Table 12.1 summarizes the results of a linear GEE analysis with an exchangeable correlation structure performed with different software packages.

From Table 12.1 it can be seen that the results of a linear GEE analysis with an exchangeable correlation structure are the same for all four software packages.

12.3 GEE analysis with dichotomous outcome variables

12.3.1 Stata

The syntax needed to perform a logistic GEE analysis in Stata is comparable to that needed to perform a linear GEE analysis, except for the part of the additional information where it is indicated that a logistic GEE analysis is performed (i.e. fam(bin) link(logit)):

```
xtgee ydich x1 x2 x3 x4 time, i(id) fam(bin) link(logit)
corr(exch)robust
```

Again, the standard errors are, by default, estimated by the “model-based” procedure. This can be changed by adding “robust” to the syntax (see Section 4.5.2).

12.3.2 SAS

Output 12.4 shows a section of the output of a logistic GEE analysis with an exchangeable correlation structure performed with the GENMOD procedure in SAS. It is obvious that the output of a logistic GEE analysis with the GENMOD procedure is comparable to the output of a linear GEE analysis. The difference is the information in the first part of the output, where it is mentioned that a “logit link” is used with a “binomial distribution,” i.e. that a logistic GEE analysis has been performed.

Output 12.4 Results of a logistic GEE analysis performed with SAS

The GENMOD Procedure						
Data Set	long.x					
Distribution	Binomial					
Link Function	Logit					
Dependent Variable	YDICH	dichotomous outcome variable y				
GEE Model Information						
Correlation Structure	Exchangeable					
Subject Effect	ID (147 levels)					
Number of Clusters	147					
Correlation Matrix Dimension	6					
Maximum Cluster Size	6					
Minimum Cluster Size	6					
Exchangeable Working						
Correlation						
Correlation	0.4904205234					
Analysis Of GEE Parameter Estimates						
Empirical Standard Error Estimates						
	Standard	95% Confidence				
Parameter Estimate	Error	Limits		Z	Pr > Z	
Intercept	-2.0664	1.9121	-5.8140	1.6812	-1.08	0.2798
X1	0.0509	0.7515	-1.4219	1.5238	0.07	0.9460
X2	0.2884	0.0581	0.1744	0.4024	4.96	<0.0001
X3	-0.2585	0.1909	-0.6327	0.1156	-1.35	0.1757
X4	0.1308	0.3747	-0.6037	0.8652	0.35	0.7271

The syntax needed to perform a logistic GEE analysis with the GENMOD procedure is comparable to that discussed for the linear GEE analysis. The only difference is found in the third line, where it has to be specified that the outcome variable is dichotomous (link=logit and dist=binomial):

```
proc genmod data=long.x;
class id;
model ydich = x1 x2 x3 x4 /link=logit dist=binomial;
repeated subject=id/type=exch;
run;
```

12.3.3 R

Output 12.5 shows the results of a logistic GEE analysis with an exchangeable correlation structure performed with R. As for Stata and SAS, the output of a logistic GEE analysis performed with R is comparable to that discussed for a linear GEE analysis.

Output 12.5 Results of a logistic GEE analysis performed with R

Summary of Residuals:						
Min	1Q	Median	3Q	Max		
-0.6549122	-0.3216171	-0.2397452	0.5534995	0.7898593		
Coefficients:						
	Estimate	Naive S.E.	Naive z	Robust S.E.	Robust z	
(Intercept)	-1.93567868	1.62431419	-1.19168982	1.62866783	-1.18850428	
x1	0.05093498	0.75026676	0.06788916	0.75147003	0.06778045	
x2	0.28839332	0.05570306	5.17733354	0.05814968	4.95949956	
x3	-0.25851584	0.17440821	-1.48224578	0.19090074	-1.35418979	
x4	0.13075120	0.33072896	0.39534245	0.37470672	0.34894276	
Estimated Scale Parameter: 0.9684559						
Number of Iterations: 5						
Working Correlation						
	[0.1]	[0.2]	[0.3]	[0.4]	[0.5]	[0.6]
[1,]	1.0000000	0.4904205	0.4904205	0.4904205	0.4904205	0.4904205
[2,]	0.4904205	1.0000000	0.4904205	0.4904205	0.4904205	0.4904205
[3,]	0.4904205	0.4904205	1.0000000	0.4904205	0.4904205	0.4904205
[4,]	0.4904205	0.4904205	0.4904205	1.0000000	0.4904205	0.4904205
[5,]	0.4904205	0.4904205	0.4904205	0.4904205	1.0000000	0.4904205
[6,]	0.4904205	0.4904205	0.4904205	0.4904205	0.4904205	1.0000000

The syntax needed to obtain a logistic GEE analysis is also fairly straightforward. The difference with the syntax needed to perform a linear GEE analysis is found in the family definition (family = binomial).

```
test <- gee(formula = ycont ~ x1 + x2 + x3 + x4, id = id,
data = data,family = binomial, corstr = "exchangeable")
```

12.3.4 SPSS

Output 12.6 shows the results of a logistic GEE analysis performed with SPSS.

Output 12.6 Results of a logistic GEE analysis performed with SPSS

Model Information	
Dependent Variable	dichotomous outcome variable y ^a
Probability Distribution	Binomial
Link Function	Logit
Subject Effect 1	identification number
Working Correlation Matrix Structure	Exchangeable

^aThe procedure models 1 as the response, treating 0 as the reference category.

Model Information	
Dependent Variable	dichotomous outcome variable y ^a
Probability Distribution	Binomial
Link Function	Logit
Subject Effect 1	identification number
Working Correlation Matrix Structure	Exchangeable

^aThe procedure models 1 as the response, treating 0 as the reference category.

Parameter Estimates							
		95% Wald Confidence Interval			Hypothesis Test		
Parameter	B	Std. Error	Lower	Upper	Wald Chi-Square	df	Sig.
(Intercept)	-2.066	1.9121	-5.814	1.681	1.168	1	0.280
x1	0.051	0.7515	-1.422	1.524	0.005	1	0.946
x2	0.288	0.0581	0.174	0.402	24.597	1	0.000
x3	-0.259	0.1909	-0.633	0.116	1.834	1	0.176
x4	0.131	0.3747	-0.604	0.865	0.122	1	0.727
(Scale)	1						

Model: (Intercept), x1, x2, x3, x4
Dependent Variable: dichotomous outcome variable y

Table 12.2 Summary of the results (i.e. regression coefficients and standard errors (in parentheses)) of a logistic GEE analysis with an exchangeable correlation structure performed with different software packages

	SPSS	Stata	SAS	R
X_1	0.05 (0.75)	0.05 (0.75)	0.05 (0.75)	0.05 (0.75)
X_2	0.29 (0.06)	0.29 (0.06)	0.29 (0.06)	0.29 (0.06)
X_3	-0.26 (0.19)	-0.26 (0.19)	-0.26 (0.19)	-0.26 (0.19)
X_4	0.13 (0.37)	0.13 (0.38)	0.13 (0.37)	0.13 (0.38)

As for all the other software packages, the output of a logistic GEE analysis is comparable to the output of a linear GEE analysis. The syntax to obtain these results is also comparable to the syntax used to perform a linear GEE analysis. The difference is found in the third line where the distribution (BINOMIAL) and the link function (LOGIT) are defined.

```
GENLIN ydich (REFERENCE=FIRST) WITH x1 x2 x3 x4
  /MODEL x1 x2 x3 x4 INTERCEPT=YES
  DISTRIBUTION=BINOMIAL LINK=LOGIT
  /CRITERIA METHOD=FISHER(1) SCALE=1 MAXITERATIONS=100
  MAXSTEPHALVING=5 PCONVERGE=1E-006 (ABSOLUTE) SINGULAR=1E-012
  ANALYSISTYPE=3 (WALD) CILEVEL=95 LIKELIHOOD=FULL
  /REPEATED SUBJECT=id SORT=YES CORRTYPE=EXCHANGEABLE
  ADJUSTCORR=YES COVB=ROBUST MAXITERATIONS=100 PCONVERGE=1e-
  006 (ABSOLUTE) UPDATECORR=1
  /MISSING CLASSMISSING=EXCLUDE
  /PRINT CPS DESCRIPTIVES MODELINFO FIT SUMMARY SOLUTION.
```

12.3.5 Overview

[Table 12.2](#) summarizes the results of the logistic GEE analysis with an exchangeable correlation structure performed with different software packages. As was the case with the linear GEE analysis, the results obtained from the logistic GEE analysis with different software packages are (almost) the same.

12.4 Mixed model analysis with continuous outcome variables

12.4.1 Stata

As for GEE analysis, Stata was used in the examples in [Chapter 4](#), and so the output will not be repeated here. The analyses discussed in that chapter were

performed with the `xtmixed` procedure. The syntax for the `xtmixed` procedure is slightly different from the syntax for the `xtgee` procedure, which was described in Section 12.2.1.

```
xtmixed ycont x1 x2 x3 x4 || id: ,var
```

The first part of the syntax is the same for all Stata procedures. First the procedure is specified and then the outcome variable and the covariates are given. Normally, after the variables a comma is included for some additional information. In the `xtmixed` procedure, however, first `|| id:` is given in order to define the “cluster variable.” In a longitudinal study, the cluster variable is the variable indicating the subject. In the example dataset this identifier is the variable `id`. After the definition of the cluster variable a comma is included after which some additional information can be provided. In the syntax above the additional information (`var`) deals with the fact that the procedure by default gives the standard deviation of the random part of the model. With the `var` statement, variances instead of standard deviations are given in the output (see Section 4.6.3). The following syntax can be used to perform a mixed model analysis with a random intercept and a random slope for X_2 .

```
xtmixed ycont x1 x2 x3 x4 || id: x2 ,var cov(unstruct)
```

The additional statement “`cov(unstruct)`” is necessary to also obtain, besides the random intercept and the random slope for X_2 , the covariance between the two (see Section 4.6.3).

12.4.2 SAS

In SAS, the `MIXED` procedure can be used to perform a linear mixed model analysis. Output 12.7 shows the results of a linear mixed model analysis with only a random intercept performed with the `MIXED` procedure in SAS. Because the length of the original output is quite large, only a section of the output is shown.

A few important messages can be derived from Output 12.7. First of all, it is shown that the method of estimation is restricted maximum likelihood (REML). In Chapter 4 it has already been mentioned that there is some debate about the use of either a maximum likelihood or restricted maximum likelihood estimation method. Both methods are available in the `MIXED` procedure in SAS, but because restricted maximum likelihood is the default procedure, these results have been reported. The next part of the output, which is of interest, is where the covariance parameter estimates are presented. The variance of the random intercept is equal to 0.2994, and the residual variance is equal to 0.2741.

Output 12.7 Results of a linear mixed model analysis with only a random intercept performed with SAS

The Mixed Procedure					
Model Information					
Data Set	long.x				
Dependent Variable	YCONT				
Covariance Structure	Variance Components				
Subject Effect	ID				
Estimation Method	REML				
Residual Variance Method	Profile				
Fixed Effects SE Method	Model-Based				
Degrees of Freedom Method	Containment				
Covariance Parameter Estimates					
	Cov Parm	Subject	Estimate		
	Intercept	ID	0.2994		
	Residual		0.2741		
Fit Statistics					
	-2 Res Log Likelihood		1671.4		
	AIC (smaller is better)		1675.4		
	AICC (smaller is better)		1675.4		
	BIC (smaller is better)		1681.4		
Solution for Fixed Effects					
	Standard				
Effect	Estimate	Error	DF	t Value	Pr > t
Intercept	3.4497	0.6648	144	5.19	<0.0001
X1	0.1479	0.2721	733	0.54	0.5868
X2	0.1847	0.01953	733	9.46	<0.0001
X3	0.04455	0.05911	733	0.75	0.4512
X4	0.03722	0.1200	733	0.31	0.7565

In addition to the variance parameters, some model fit indicators are also provided. Because REML was used, the -2 res log likelihood was estimated instead of the -2 log likelihood. The interpretation is, however, basically the same. Furthermore, Akaike's information criterion (AIC), Hurvich and Tsai's criterion (AICC), and Schwarz's Bayesian criterion (BIC) are presented. All three can be seen as "adjusted" values of the -2 (res) log likelihood, i.e. adjusted for the number of

parameters estimated by the particular model (Akaike, 1974; Schwarz, 1978; Hurvich and Tsai, 1989).

The last part of the output shows the estimates of the regression coefficients, the standard errors, the degrees of freedom, the t -values, and the corresponding p -values. It should be noted that in the MIXED procedure the t -distribution is used instead of the standard normal distribution (z -distribution), which is used in Stata. Using the t -distribution instead of the z -distribution leads to slightly higher p -values, especially when the number of observations is low.

The following syntax can be used to perform a mixed model analysis with only a random intercept in SAS.

```
proc mixed data=long.x;  
class id;  
model ycont = x1 x2 x3 x4 time/s;  
random int/ subject=id;  
run;
```

The syntax looks similar to that needed to perform a linear GEE analysis with the GENMOD procedure. The difference is firstly that the “repeated” statement is replaced by the “random” statement, which is necessary in order to identify the random part of the mixed model (i.e. intercept (int)). Secondly there is no specification of the correlation structure needed.

Output 12.8 shows a section of the output of a linear mixed model analysis with both a random intercept and a random slope for X_2 . The most important difference between Output 12.7 and Output 12.8 is the number of covariance parameters that are estimated. In the analysis with both a random intercept and a random slope with time, three covariance parameters are estimated: (1) UN(1.1) which is an estimate of the variance of the random intercept; (2) UN(2.2) which is an estimate of the variance of the random slope for X_2 ; and (3) UN(2.1) which is an estimate of the covariance between the random intercept and the random slope. The value of the -2 res log likelihood can be used to evaluate the necessity of adding a random slope for X_2 to the model. The difference between the -2 res log likelihoods of the model with only a random intercept and a model with both a random intercept and a random slope for X_2 is 2.8. This difference follows a Chi-square distribution with 2 degrees of freedom (i.e. the variance of the random slope for X_2 and the covariance between the random intercept and slope), which is not statistically significant.

The following syntax can be used to perform a linear mixed model analysis with both a random intercept and a random slope for X_2 with the MIXED procedure in SAS.

Output 12.8 Results of a linear mixed model analysis with a random intercept and a random slope for X_2 performed with SAS

The Mixed Procedure
Model Information

Data Set	long.x
Dependent Variable	YCONT
Covariance Structure	Unstructured
Subject Effect	ID
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Containment

Covariance Parameter Estimates

Cov Parm	Subject	Estimate
UN(1.1)	ID	0.4930
UN(2.1)	ID	-0.04498
UN(2.2)	ID	0.01028
Residual		0.2636

Fit Statistics

-2 Res Log Likelihood	1668.6
AIC (smaller is better)	1676.6
AICC (smaller is better)	1676.6
BIC (smaller is better)	1688.5

Null Model Likelihood Ratio Test

DF	Chi-Square	Pr > ChiSq
3	351.49	<0.0001

Solution for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	3.3645	0.6780	144	4.96	<0.0001
X1	0.1871	0.2772	587	0.68	0.4999
X2	0.1917	0.02232	146	8.59	<0.0001
X3	0.04306	0.05904	587	0.73	0.4661
X4	0.03167	0.1218	587	0.26	0.7949

```
proc mixed data=long.x;
class id;
model ycont = x1 x2 x3 x4 time/s;
random int x2/ subject=id typ3=un;
run;
```

12.4.3 R

Output 12.9 shows the result of a linear mixed model analysis with only a random intercept performed with R.

Output 12.9 Results of a linear mixed model analysis with only a random intercept performed with R

Linear mixed-effects model fit by REML

Data: data

AIC	BIC	logLik
1685.416	1718.852	-835.7082

Random effects:

Formula: ~1 | id

(Intercept)	Residual
StdDev:	0.5471496 0.5235472

Fixed effects: ycont ~ x1 + x2 + x3 + x4

	Value	Std.Error	DF	t-value	p-value
(Intercept)	3.486943	0.5875498	733	5.934719	0.0000
x1	0.147907	0.2720621	144	0.543652	0.5875
x2	0.184720	0.0195253	733	9.460520	0.0000
x3	0.044554	0.0591060	733	0.753795	0.4512
x4	0.037217	0.1199952	144	0.310155	0.7569

Correlation:

	(Intr)	x1	x2	x3
x1	-0.988			
x2	-0.204	0.110		
x3	-0.045	0.048	-0.184	
x4	-0.584	0.542	-0.167	0.064

Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-2.7788686	-0.6268489	-0.0961649	0.5184294	4.3212066

Number of Observations: 882

Number of Groups: 147

The first line of Output 12.9 shows that the parameters of this linear mixed model were estimated with restricted maximum likelihood (REML). As with the MIXED procedure in SAS, the REML estimation procedure is default, although it is also possible to perform a maximum likelihood estimation procedure. In the same block of the output, the log likelihood of the model is shown, in addition to some other model fit indicators (i.e. AIC and Schwarz's BIC; see Section 12.4.2).

The next part of the output shows the random part of the model. In a model with only a random intercept, two variance parameters are estimated (given as standard deviations in the R output): the standard deviation of the random intercept (0.5661159) and the residual (or error) standard deviation (0.491077). The following part of the output shows the estimates of the regression coefficients, the standard errors of the regression coefficients, the degrees of freedom, the t -values, and the corresponding p -values. Note that also in R the t -distribution is used instead of the standard normal distribution. However, when the number of individuals is high, the t -distribution is almost equal to the z -distribution.

The following syntax can be used to perform a linear mixed model analysis with only a random intercept in R.

```
test <- lme (ycont~x1+x2+x3+x4, data=data, random= ~1|id)
```

As for the results of the GEE analysis, the results of the mixed model analysis (the procedure lme) are linked to the object test. In the last part of the syntax, the random part of the model is specified (random= ~1|id).

Output 12.10 shows the results of a linear mixed model analysis with both a random intercept and a random slope for the independent variable X_2 performed with R. It is obvious that the most important difference between Output 12.10 and Output 12.9 is the estimation of two more parameters in the random part of the model, i.e. the random slope for X_2 and the correlation between the random intercept and random slope. Note the R provides the correlation between the random intercept and random slope instead of the covariance, which is provided by the other software programs. Although the numbers are different, the general interpretation of the two is the same.

As in all the other mixed model analyses, the necessity of the random slope for X_2 can be estimated with the likelihood ratio test. The difference between the $-2 \log$ likelihood of the analysis with only a random intercept and the $-2 \log$ likelihood of the analysis with both a random intercept and a random slope for X_2 is the same as was calculated with the MIXED procedure in SAS, i.e. 2.8.

The following syntax can be used to perform a linear mixed model analysis with both a random intercept and a random slope for X_2 in R.

```
test <- lme (ycont~x1+x2+x3+x4, data=data, random= ~ x2|id)
```

Output 12.10 Results of a mixed model analysis with a random intercept and a random slope for X_2 performed with R

```

Linear mixed-effects model fit by REML
Data: data
      AIC      BIC    logLik
1686.565 1729.554 -834.2827

Random effects:
Formula: ~x2 | id
Structure: General positive-definite, Log-Cholesky parametrization
           StdDev   Corr
(Intercept) 0.7021259 (Intr)
x2          0.1014079 -0.632
Residual    0.5134225

Fixed effects: ycont ~ x1 + x2 + x3 + x4
              Value    Std.Error   DF   t-value   p-value
(Intercept) 3.396132   0.6002742  733   5.657634   0.0000
x1          0.187152   0.2772226  144   0.675096   0.5007
x2          0.191660   0.0223166  733   8.588231   0.0000
x3          0.043061   0.0590371  733   0.729384   0.4660
x4          0.031674   0.1218082  144   0.260029   0.7952

Correlation:
      (Intr)  x1      x2      x3
x1      -0.986
x2      -0.220  0.111
x3      -0.046  0.047 -0.155
x4      -0.578  0.538 -0.163  0.062

Standardized Within-Group Residuals:
Min          Q1          Med          Q3          Max
-2.46321227  -0.62410433  -0.07425537   0.51357028   4.29624820

Number of Observations: 882
Number of Groups: 147

```

12.4.4 SPSS

Before the release of version 11, with SPSS it was not possible to perform sophisticated longitudinal data analysis. However, in version 11, linear mixed model analysis became available. This procedure, which is based on the algorithms discussed by Wolfinger et al. (1994) can be derived from the menu and is called “mixed models – linear.” Output 12.11 shows a section of the output of a linear mixed model analysis with only a random intercept performed in SPSS.

Output 12.11 Results of a linear mixed model analysis with only a random intercept performed with SPSS

Model Dimension ^a		Number of Levels	Covariance Structure	Number of Parameters	Subject Variables
Fixed Effects	Intercept	1		1	
	x1	1		1	
	x2	1		1	
	x3	1		1	
	x4	1		1	
Random Effects	Intercept ^b	1	Variance Components	1	id
Residual				1	
Total		6		7	

^aDependent Variable: continuous outcome variable y.

^bAs of version 11.5, the syntax rules for the RANDOM subcommand have changed. Your command syntax may yield results that differ from those produced by prior versions. If you are using version 11 syntax, please consult the current syntax reference guide for more information.

Information Criteria ^a	
-2 Restricted Log Likelihood	1671.416
Akaike's Information Criterion (AIC)	1675.416
Hurvich and Tsai's Criterion (AICC)	1675.430
Bozdogan's Criterion (CAIC)	1686.969
Schwarz's Bayesian Criterion (BIC)	1684.969

The information criteria are displayed in smaller-is-better forms.

^aDependent Variable: continuous outcome variable y

Estimates of Fixed Effects ^a						95% Confidence Interval	
Parameter	Estimate	Std. Error	df	t	Sig.	Lower Bound	Upper Bound
Intercept	3.449725	0.664828	151.148	5.189	0.000	2.136170	4.763281
x1	0.147907	0.272062	148.029	0.544	0.587	-0.389720	0.685534
x2	0.184720	0.019525	876.874	9.461	0.000	0.146398	0.223041
x3	0.044554	0.059106	839.889	0.754	0.451	-0.071459	0.160567
x4	0.037217	0.119995	150.941	0.310	0.757	-0.199870	0.274304

^aDependent Variable: continuous outcome variable y.

Estimates of Covariance Parameters^a

Parameter	Estimate	Std. Error
Residual	0.274102	0.014315
Intercept [subject = id] Variance	0.299373	0.040771

^aDependent Variable: continuous outcome variable y.

The first part of Output 12.11 shows some general information (Model Dimension) and the model fit indicators (Information Criteria). In the latter, first of all the -2 restricted log likelihood is provided (1671.416). This indicates that REML is used as default estimation procedure. Besides the -2 restricted log likelihood, AIC, AICC, and BIC were also provided. Again, they all can be seen as “adjusted” values of the -2 restricted log likelihood. Bozdogan’s criterion (CAIC) is slightly different but can be interpreted in more or less the same way (Bozdogan, 1987). The next part of the output shows the estimates of the fixed part of the model. In this part of the output, the regression coefficients, the standard errors, the degrees of freedom, the t -values, the corresponding p -values, and the 95% confidence interval around the regression coefficient are provided. The last part of the output shows the estimates of the covariance parameters. Because only a random intercept was added to the model, only the variance of the random intercept (0.299373) and the remaining error variance (0.274102) are provided.

As has been mentioned before, in SPSS the mixed model analysis can also be derived from the menu. The corresponding syntax looks as follows:

```
MIXED ycont WITH x1 x2 x3 x4
  /CRITERIA=CIN(95) MXITER(100) MXSTEP(5) SCORING(1)
SINGULAR(0.000000000001) HCONVERGE(0, ABSOLUTE)
LCONVERGE(0, ABSOLUTE) PCONVERGE(0.000001, ABSOLUTE)
  /FIXED=x1 x2 x3 x4 | SSTYPE(3)
  /METHOD=REML
  /PRINT=SOLUTION
  /RANDOM=INTERCEPT | SUBJECT(id) COVTYPE(VC) .
```

Output 12.12 shows a section of the output of a linear mixed model analysis with both a random intercept and a random slope for X_2 performed with SPSS. The output looks similar to the one discussed for the analysis with only a random intercept. The difference is found in the last part in which the Estimates of Covariance Parameters are given. Besides the variance of the random intercept (UN(1.1)) and

Output 12.12 Results of a linear mixed model analysis with a random intercept and a random slope for X_2 performed with SPSS

Model Dimension^a

		Number of Levels	Covariance Structure	Number of Parameters	Subject Variables
Fixed Effects	Intercept	1		1	
	x1	1		1	
	x2	1		1	
	x3	1		1	
	x4	1		1	
Random Effects	Intercept + x2 ^b	2	Unstructured	3	id
Residual				1	
Total		7		9	

^aDependent Variable: continuous outcome variable y.

Information Criteria^a

-2 Restricted Log Likelihood	1668.565
Akaike's Information Criterion (AIC)	1676.565
Hurvich and Tsai's Criterion (AICC)	1676.611
Bozdogan's Criterion (CAIC)	1699.672
Schwarz's Bayesian Criterion (BIC)	1695.672

The information criteria are displayed in smaller-is-better forms.

^aDependent Variable: continuous outcome variable y.

Estimates of Fixed Effects^a

Parameter	Estimate	Std. Error	df	t	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	3.364458	0.677978	144.479	4.962	0.000	2.024422	4.704493
x1	0.187152	0.277223	141.546	0.675	0.501	-0.360880	0.735184
x2	0.191660	0.022317	73.433	8.588	0.000	0.147187	0.236132
x3	0.043061	0.059037	828.684	0.729	0.466	-0.072819	0.158940
x4	0.031674	0.121808	144.897	0.260	0.795	-0.209077	0.272424

^aDependent Variable: continuous outcome variable y.

Estimates of Covariance Parameters^a

Parameter		Estimate	Std. Error
Residual		0.263603	0.015023
Intercept + x2 [subject = id]	UN (1.1)	0.492981	0.161907
	UN (2.1)	-0.044978	0.032438
	UN (2.2)	0.010284	0.007365

^aDependent Variable: continuous outcome variable y.

the remaining error variance, the random variance of the slope for X_2 (UN(2.2)) and the covariance between the random intercept and random slope (UN(2.1)) are also given. Similar to all other mixed model analyses, with the -2 restricted log likelihood values of the two analyses discussed, the necessity of allowing a random slope for X_2 can be evaluated.

The following syntax can be used to perform a mixed model analysis with a random intercept and a random slope for X_2 in SPSS.

```
MIXED ycont WITH x1 x2 x3 x4
  /CRITERIA=CIN(95) MXITER(100) MXSTEP(5) SCORING(1)
  SINGULAR(0.000000000001) HCONVERGE(0, ABSOLUTE)
  LCONVERGE(0, ABSOLUTE) PCONVERGE(0.000001, ABSOLUTE)
  /FIXED=x1 x2 x3 x4 | SSTYPE(3)
  /METHOD=REML
  /PRINT=SOLUTION
  /RANDOM=INTERCEPT x2 | SUBJECT(id) COVTYPE(UN) .
```

12.4.5 MLwiN

Multilevel analysis for windows (MLwiN) is specifically developed to perform mixed model analysis. [Output 12.13](#) shows the results of a linear mixed model analysis with only a random intercept. Because MLwiN is specifically developed to perform multilevel analysis, the levels of the analysis must first be defined. In the case of longitudinal data, the observations over time are nested within subjects so the repeated observations are the lowest level of analysis and the subject is the highest level of analysis. It should be noted that in the MLwiN output the lowest level (the repeated observations) is indicated by the subscript i , and the highest level (subject) is indicated by the subscript j . This is different from the regular notation, in which subjects are usually indicated by the subscript i .

Output 12.13 Results of a linear mixed model analysis with only a random intercept performed with MLwiN

$$y_{cont_{ij}} \sim N(XB, \Omega)$$

$$y_{cont_{ij}} = \beta_{0ij} \text{cons} + 0.148(0.269)x1_j + 0.185(0.019)x2_{ij} + 0.045(0.059)x3_{ij} + 0.037(0.119)x4_j$$

$$\beta_{0ij} = 3.449(0.658) + u_{0j} + e_{0ij}$$

$$[u_{0j}] \sim N(0, \Omega_u) : \Omega_u = [0.292(0.039)]$$

$$[e_{0ij}] \sim N(0, \Omega_e) : \Omega_e = [0.273(0.014)]$$

$$-2 * \log \text{likelihood}(\text{IGLS Deviance}) = 1653.664(882 \text{ of } 882 \text{ cases in use})$$

The first line of the MLwiN output shows that the outcome variable y_{cont} is assumed to be a continuous normally distributed variable. In the second line the model is given, with the intercept β_0 (“cons” stands for a row of ones, which has no meaning, but is necessary for the estimation of the intercept), and the (fixed) regression coefficients for the four independent variables (X_1 to X_4). The output provides no test statistics and/or p -values for the regression coefficients, but they can be tested for significance by dividing the coefficient by its standard error. This ratio approximately follows a standard normal distribution. In the literature, it is mentioned that the ratio of the regression coefficient and the standard error should be squared to get the Wald statistic, which follows a Chi-square distribution with one degree of freedom. However, this is just a theoretical distinction because the Chi-square distribution with one degree of freedom is equal to the standard normal distribution squared.

From the MLwiN output it can, furthermore, be seen that X_1 and X_4 are time-independent covariates (subscript j , so only varying between subjects), while X_2 and X_3 are time-dependent covariates (subscript ij , so varying between repeated observations and between subjects).

The third line of the output provides information about the random intercept β_0 . The coefficient consists of a fixed part (3.499 with standard error 0.658) and a random part which is divided into two parts: the first part μ is the random variation around the intercept and the second part ε is the remaining “error” variance. In the next two lines of the output these variances are given, with between brackets the corresponding standard errors.

The last line of the output presents the $-2 \log$ likelihood (1653.664) which can be used in comparison with the $-2 \log$ likelihood from a model with both a random intercept and a random slope for X_2 in order to evaluate the necessity of a random slope for X_2 (see Output 12.14).

Output 12.14 Results of a linear mixed model analysis with a random intercept and a random slope for X_2 performed with MLwiN

$$y_{cont_{ij}} \sim N(XB, \Omega)$$

$$y_{cont_{ij}} = \beta_{0ij} \text{cons} + 0.185(0.274)x1_j + \beta_{2j}x2_{ij} + 0.043(0.059)x3_{ij} + 0.031(0.120)x4_j$$

$$\beta_{0ij} = 3.371(0.669) + u_{0j} + e_{0ij}$$

$$\beta_{2j} = 0.191(0.022) + u_{2j}$$

$$\begin{bmatrix} u_{0j} \\ u_{2j} \end{bmatrix} \sim N(0, \Omega_u) : \Omega_u = \begin{bmatrix} 0.473(0.139) \\ -0.042(0.028) \quad 0.009(0.006) \end{bmatrix}$$

$$[e_{0ij}] \sim N(0, \Omega_e) : \Omega_e = [0.264(0.014)]$$

$$-2 * \log \text{likelihood}(\text{IGLS Deviance}) = 1651.189(882 \text{ of } 882 \text{ cases in use})$$

From Output 12.14 it can be seen that in addition to a random intercept, for the regression coefficient for X_2 (β_2) a random slope is considered (μ_{2j} in the fourth line in the output). So three variance parameters are estimated: the random variation of the intercept (0.473 with standard error 0.139), the random variation of the slope for X_2 (0.009 with standard error 0.006), and the covariance between the random intercept and random slope (-0.042 with standard error 0.028).

The likelihood ratio test can be used to decide whether or not a random slope for X_2 should be added to the model. The difference between the $-2 \log$ likelihood of the model without a random slope (i.e. 1653.664) and a model with a random slope for X_2 (i.e. 1651.189) is 2.475, which follows a Chi-square distribution with 2 degrees of freedom, and is not statistically significant. It should be noted that the log likelihood values obtained from MLwiN are slightly different to those obtained from other software packages. This is due to the fact that with MLwiN an iterative generalized least squares (IGLS) approach is used by default, which is basically the same as maximum likelihood, and that with the other packages REML is used. With MLwiN a so-called restricted iterative generalized least squares (RIGLS) estimation procedure can also be used, which is comparable to REML (Goldstein, 1986, 1989, 1995). Because with MLwiN the mixed model analysis can be directly derived from the menu, no syntax is needed.

12.4.6 Overview

Table 12.3 summarizes the results of the linear mixed model analyses with both a random intercept and a random slope for X_2 performed with different software packages. From Table 12.3 it can be seen that using a different software package does not lead to different results of the linear mixed model analysis. This is irrespective

Table 12.3 Summary of the results (i.e. regression coefficients, standard errors (in parentheses), -2 log likelihoods, and random variances) of a linear mixed model analysis with both a random intercept and a random slope for X_2 performed with different software packages

	Stata ^a	SAS ^a	MLwiN ^b	R ^a	SPSS ^a
X_1	0.19 (0.28)	0.19 (0.28)	0.19 (0.27)	0.19 (0.28)	0.19 (0.28)
X_2	0.19 (0.02)	0.19 (0.02)	0.19 (0.02)	0.19 (0.02)	0.19 (0.02)
X_3	0.04 (0.06)	0.04 (0.06)	0.04 (0.06)	0.04 (0.06)	0.04 (0.06)
X_4	0.03 (0.12)	0.03 (0.12)	0.03 (0.12)	0.03 (0.12)	0.03 (0.12)
-2 log likelihood	1668.6	1668.6	1651.2	1668.6	1668.6
Random variance					
Intercept	0.493	0.493	0.473	0.493	0.493
Slope for X_2	0.010	0.010	0.009	0.010	0.010

^aEstimated with restricted maximum likelihood.

^bEstimated with maximum likelihood.

of the estimation procedure used, because the results derived from Stata, SAS, R, or SPSS (i.e. all packages use the REML estimation procedure as default) did not differ from the results derived from MLwiN (which uses the maximum likelihood estimation procedure by default). It is often argued that the REML approach gives a “better” estimation of the variance components in the mixed model analysis. However, although there are differences in the variances, these differences are small and hardly affect the magnitude of the regression coefficients or the standard errors of the regression coefficients.

12.5 Mixed model analysis with dichotomous outcome variables

12.5.1 Introduction

In [Chapter 7](#) it has already been mentioned that a logistic mixed model analysis is quite difficult to perform. Basically there are two possible approaches. The most straightforward approach is the Gauss–Hermite technique, which is based on Gaussian quadrature points (Rabe-Hesketh and Pickles, 1999; Rabe-Hesketh et al., 2000, 2001a, 2001b). The second approach is based on so-called penalized quasi-likelihood (PQL). However, it is far beyond the scope of this book to explain these techniques in detail. For more information, reference is made to the more technical literature (e.g. Goldstein, 1991; Schall, 1991; Breslow and Clayton, 1993; Longford, 1993; Liu, Q. and Pierce, 1994; Pinheiro and Bates, 1995; Goldstein and

Rasbash, 1996; Agresti et al., 2000; Lesaffre and Spiessens, 2001) and the reference manuals of the different software packages.

12.5.2 Stata

In the examples discussed in Chapter 7, Stata was used to perform a logistic mixed model analysis. In Stata, the logistic mixed model analysis can be performed with the `xtmelogit` procedure. The syntax to perform a logistic mixed model analysis with only a random intercept in Stata is comparable to the syntax used to perform a linear mixed model analysis.

```
xtmelogit ydich x1 x2 x3 x4 || id: ,var
```

For a mixed model analysis with a random intercept and a random slope for X_2 , the following syntax can be used.

```
xtmelogit ydich x1 x2 x3 x4 || id: x2 ,var cov(unstruct)
```

In Chapter 7, it has already been mentioned that this analysis did not converged, so it was not possible to add a random slope for X_2 to the model. A random slope for the other time-dependent independent variable X_3 was, however, possible (see Section 7.2.4.2).

In the examples presented in Chapter 7, the `xtmelogit` procedure was used with the default of 7 quadrature points. In the literature it is argued that the results of a logistic mixed model analysis strongly depend on the number of quadrature points used in the estimation procedure (Lesaffre and Spiessens, 2001). In the statistical literature, it is generally accepted that 10 quadrature points are sufficient for a “valid” logistic mixed model analysis, although others suggest that 20 quadrature points are needed (e.g. Hu et al., 1998; Rodriguez and Goldman, 2001). However, in the present example, as well as in examples presented by others (e.g. Lesaffre and Spiessens, 2001), neither of these suggestions is confirmed. In the example presented here, the default option of 7 quadrature points seems to be very reasonable (Table 12.4).

The reason why the results from the `xtmelogit` procedure with a different number of quadrature points are almost the same has to do with the use of “adaptive quadrature,” which leads to a much more accurate (numeric) integration and therefore to much more valid estimations of the parameters (Rice, 1975; Pinheiro and Bates, 1995; Gandar and Gautschi, 2000).

12.5.3 SAS

In the 1990s, a SAS macro called GLIMMIX that can be used to perform logistic mixed model analysis became available (Breslow and Clayton, 1993). However, the

Table 12.4 Summary of the results (i.e. regression coefficients, standard errors (in parentheses), and random variances) of logistic mixed model analyses with a random intercept and a random slope for X_3 with a different number of quadrature points

	Number of quadrature points			
	5	7	10	15
X_1	0.18 (1.30)	0.16 (1.33)	0.16 (1.33)	0.16 (1.34)
X_2	0.58 (0.11)	0.58 (0.11)	0.58 (0.11)	0.58 (0.11)
X_3	0.02 (0.36)	0.05 (0.37)	0.05 (0.37)	0.05 (0.37)
X_4	0.33 (0.57)	0.34 (0.58)	0.34 (0.58)	0.34 (0.58)
Random variance				
Intercept	7.92	8.41	8.55	8.64
Slope for X_3	1.56	1.74	1.79	1.83

performance of this procedure was not very satisfactory (Wolfinger, 1998; Lesaffre and Spiessens, 2001). Basically, GLIMMIX is only suitable when the number of observations per subject is fairly large. Nowadays the NLMIXED procedure is available for logistic mixed model analysis. Output 12.15 shows a section of the results of a logistic mixed model analysis with only a random intercept performed with the NLMIXED procedure in SAS.

The output of the NLMIXED procedure is not very straightforward, and deserves further explanation. The first part of the output provides general information about the estimation procedure. It can be seen that a binary outcome variable is used (i.e. that a logistic mixed model analysis has been performed) and that there is one random effect (b0), which is normally distributed. Furthermore, it shows that a dual quasi-Newton algorithm and an adaptive Gaussian quadrature method are used. The dual quasi-Newton algorithm is one of the possible optimization techniques available for logistic mixed model analysis which will not be discussed any further (for details see the software manuals). From the next part of the output it can be seen that seven parameters are estimated, the starting values of which are shown below the general information. The problems with the NLMIXED procedure in SAS are that the user has to provide the starting values for the different parameters and (even more problematic) that the final results of the estimated regression coefficients and standard errors can depend on the choice of these starting values. If no starting values are provided, the NLMIXED procedure assumes a starting value of one for all parameters. The seven parameters to be estimated correspond with the intercept (beta0), the regression coefficients for the four covariates X_1 to X_4 (beta1 to beta4), the random variance of the intercept (s2b0) and the – log likelihood (NegLokLike).

Output 12.15 Results of a logistic mixed model analysis with only a random intercept performed with SAS

The NLMIXED Procedure						
Specifications						
Data Set					long.x	
Dependent Variable					YDICH	
Distribution for Dependent Variable					Binary	
Random Effects					b0	
Distribution for Random Effects					Normal	
Subject Variable					ID	
Optimization Technique					Dual Quasi-Newton	
Integration Method					Adaptive Gaussian Quadrature	
Parameters						
beta0	beta1	beta2	beta3	beta4	s2b0	NegLogLike
-4	0.5	0.1	-0.2	0.3	5	403.771624
The NLMIXED Procedure						
Fit Statistics						
-2 Log Likelihood					804.3	
AIC (smaller is better)					816.3	
AICC (smaller is better)					816.4	
BIC (smaller is better)					834.3	
Parameter Estimates						
Standard						
Parameter	Estimate	Error	DF	t Value	Pr > t	Alpha Lower Upper
beta0	-4.3723	3.5527	146	-1.23	0.2204	0.05 -11.3937 2.6491
beta1	0.1339	1.4563	146	0.09	0.9269	0.05 -2.7443 3.0121
beta2	0.5722	0.1142	146	5.01	<0.0001	0.05 0.3465 0.7978
beta3	-0.4721	0.3389	146	-1.39	0.1658	0.05 -1.1419 0.1977
beta4	0.3365	0.6346	146	0.53	0.5967	0.05 -0.9177 1.5908
s2b0	6.9656	1.5761	146	4.42	<0.0001	0.05 3.8507 10.0804

The next part of the output shows the results of the logistic mixed model analysis: firstly, the model fit indicators (i.e. the -2 log likelihood, AIC, Hurvich and Tsai’s AICC, and Schwarz’s BIC), and, secondly, both the regression coefficients and the

random part of the model, including standard errors, 95% confidence intervals, and p -values (which are based on the t -distribution).

The syntax needed to perform a logistic mixed model analysis with the NLMIXED procedure is rather complicated:

```
proc nlmixed data=long.x
parms beta0=-4 beta1=0.1 beta2=0.5 beta3 =-0.2 beta4=0.3 s2b0=5;
c1 = beta0 + b0;
eta = c1 + beta1*x1+ beta2*x2 + beta3*x3 + beta4*x4;
expeta = exp(eta);
p = expeta / (1+expeta);
model ydich ~ binary(p);
random b0~ normal([0],[s2b0]) subject=id;
run;
```

With the “parms” statement, starting values are assigned to all parameters that are to be estimated. After this, the random intercept is specified (c1). In the next lines the logistic model is specified, while in the last line of this command the random part of the model is defined.

Unfortunately, a logistic mixed model analysis with both a random intercept and a random slope for X_3 cannot be performed with the NLMIXED procedure in SAS. Nevertheless, the following syntax can be used to perform such an analysis.

```
proc nlmixed data=work.longbook;
parms beta0=-4 beta1=0.5 beta2=0.1 beta3 =-0.2 beta4=0.3
s2b0=5 s2b1=1 cb01=1;
c1 = beta0 + b0;
c2 = beta3 + b1;
eta = c1 + beta1*x2+ beta2*x1 + (c2*x3) + beta4*x4;
expeta = exp(eta);
p = expeta / (1+expeta);
model ydich ~ binary(p);
random b0 b1 ~ normal([0.0],[s2b0,cb01,s2b1]) subject=id;
run;
```

12.5.4 R

Output 12.16 shows the results of a logistic mixed model analysis performed with R.

Output 12.16 Results of a logistic mixed model analysis with only a random intercept performed with R

```
Linear mixed-effects model fit by maximum likelihood
Data: data
      AIC BIC logLik
      NA  NA   NA

Random effects:
Formula: ~1 | id
      (Intercept) Residual
StdDev:    2.197931 0.749019

Fixed effects: ydich ~ x1 + x2 + x3 + x4
              Value Std.Error DF   t-value p-value
(Intercept) -3.481032 2.5156566 733 -1.383747  0.1669
x1           0.172600 1.1637397 144  0.148315  0.8823
x2           0.493362 0.0837390 733  5.891666  0.0000
x3          -0.428718 0.2481511 733 -1.727649  0.0845
x4           0.234078 0.5074890 144  0.461247  0.6453
Correlation:
      (Intr) x1      x2      x3yes
x1      -0.987
x2      -0.232  0.134
x3      -0.043  0.047 -0.181
x4      -0.577  0.532 -0.148  0.059

Standardized Within-Group Residuals:
Min          Q1          Med          Q3          Max
-4.3107940 -0.4126610 -0.2298523  0.4666250  3.6141927

Number of Observations: 882
Number of Groups: 147
```

Output 12.16 is comparable to the output of a linear mixed model performed with R (see Output 12.9). Surprisingly at the top of the output, it is mentioned that the output shows the result of a “linear mixed-effects model fit by maximum likelihood.” This is, however, not the case. The results are the result of a logistic mixed model analysis which is fit by “penalised quasi likelihood.” This can also be seen in the first part of the output, where it is mentioned that model fit indicators, such as the log likelihood are not applicable in this analysis.

Output 12.17 Results of a logistic mixed model analysis with a random intercept and a random slope for X_3 performed with R

```
Linear mixed-effects model fit by maximum likelihood
Data: data
      AIC BIC logLik
      NA  NA    NA

Random effects:
Formula: ~x3 | id
Structure: General positive-definite, Log-Cholesky parametrization
              StdDev   Corr
(Intercept) 2.4152365 (Intr)
x3yes        1.4421442 -0.697
Residual     0.7277829

Fixed effects: ydich ~ x1 + x2 + x3 + x4
              Value Std.Error DF   t-value p-value
(Intercept) -3.521652 2.4560416 733  -1.433873  0.1520
x1           0.125596 1.1336678 144   0.110787  0.9119
x2           0.502577 0.0848114 733   5.925815  0.0000
x3          -0.178863 0.2851099 733  -0.627347  0.5306
x4           0.300268 0.4973086 144   0.603787  0.5469

Correlation:
              (Intr) x1      x2      x3
x1           -0.986
x2           -0.242  0.141
x3           -0.109  0.090 -0.165
x4           -0.566  0.521 -0.150  0.054

Standardized Within-Group Residuals:
Min          Q1          Med          Q3          Max
-4.5938692 -0.4041939 -0.2192521  0.4595710  3.5031675

Number of Observations: 882
Number of Groups: 147
```

Output 12.17 shows the results of a logistic mixed model analysis with both a random intercept and a random slope for X_3 .

In R, logistic mixed model analysis can be performed with several procedures. In this example, the glmmPQL procedure was used. GlmmPQL stands for generalized linear and mixed models Penalized Quasi Likelihood. The following syntax shows

the syntax to perform a logistic mixed model analysis with the glmmPQL procedure in R.

```
test <- glmmPQL (ydich ~ x1+x2+x3+x4, random = ~x3 | id,
family=binomial, data=data)
```

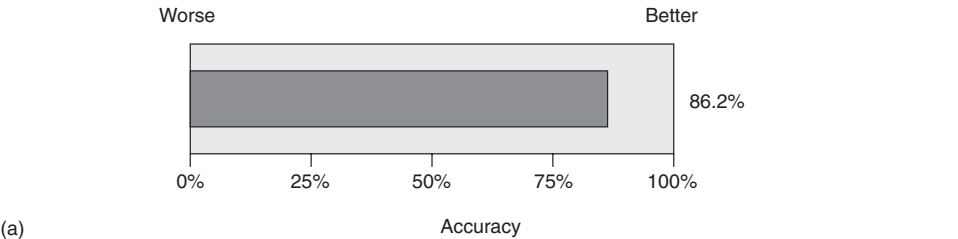
12.5.5 SPSS

From version 19 onwards, SPSS provides the possibility to perform a logistic mixed model analysis. Output 12.18 shows the output of a logistic mixed model analysis with only a random intercept performed with SPSS.

Output 12.18 Results of a logistic mixed model analysis with only a random intercept performed with SPSS

Model Summary	
Target: dichotomous outcome variable y	
Target	dichotomous outcome variable y
Probability Distribution	Binomial
Link Function	Logit
Akaike Corrected Information Criterion	4.297,114
Bayesian	4.311,416

Information criteria are based on the -2 log pseudo likelihood (4.291,087) and are used to compare models. Models with smaller information criterion values fit better. When comparing models using pseudo likelihood values, caution should be used because different data transformations may be used across the models.



Output 12.18 (cont.)

Covariance Parameters

Target: dichotomous outcome variable y

Covariance Parameters	Residual Effect	0
	Random Effects	1
Design Matrix Columns	Fixed Effects	5
	Random Effects	1 ^a
Common Subjects		147

Common subjects are based on the subject specifications for the residual and random effects and are used to chunk the data for better performance.

^a This is the number of columns per common subject.

Random Effect	Estimate	Std.Error	Z	Sig.	95% Confidence Interval	
					Lower	Upper
Var(Intercept)	3.312	0.585	5.660	0.000	2.343	4.683

Covariance Structure: Variance components
Subject Specification: id

(b)

(cont.)

The output of a logistic mixed model analysis performed with SPSS is totally different from the other outputs derived from SPSS. The first part of the output shows the general information of the analysis performed, i.e. that a logit link function is used with a binomial probability distribution. In other words, it is

Output 12.18 (cont.)

Fixed Coefficients						
Target: dichotomous outcome variable y						
Reference Category:0						
Model Term	Coefficient ▼	Std.Error	t	Sig.	95% Confidence Interval	
					Lower	Upper
Intercept	−3.519	2.533	−1.389	0.165	−8.491	1.453
x1	0.262	1.033	0.254	0.800	−1.766	2.291
x2	0.465	0.095	4.905	0.000	0.279	0.651
x3	−0.369	0.295	−1.251	0.211	−0.947	0.210
x4	0.181	0.451	0.401	0.688	−0.703	1.065

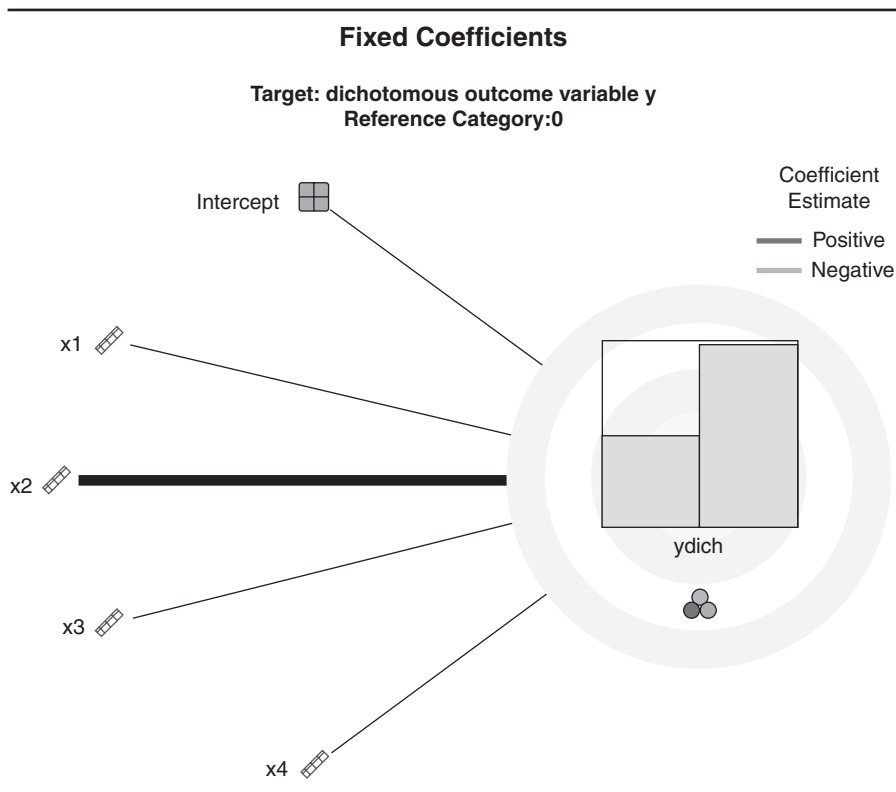
Probability distribution: Binomial
Link function: Logit

(c)

mentioned that a logistic mixed model analysis is performed. Furthermore, two model fit indicators are given (AIC and BIC) which are derived from the $-2 \log$ (pseudo) likelihood of 4291.087. The second part of the output shows the variance of the random intercept, while the third part shows the regression coefficients, the standard errors, t -values, corresponding p -values, and the 95% confidence intervals around the regression coefficients. It is also possible to obtain the results of the logistic mixed model analysis in a different way (Output 12.19). In this graphical representation, different colors indicate whether the coefficient is negative or positive and the thicker the line, the stronger the relationship (i.e. the thicker the line, the higher the regression coefficient).

The following syntax can be used to perform a logistic mixed model analysis with only a random intercept in SPSS.

Output 12.19 Graphical representation of the results of a logistic mixed model analysis with only a random intercept performed with SPSS



```

GENLINMIXED
  /DATA_STRUCTURE SUBJECTS=id
  /FIELDS TARGET=ydich TRIALS=NONE OFFSET=NONE
  /TARGET_OPTIONS DISTRIBUTION=BINOMIAL LINK=LOGIT
  /FIXED EFFECTS=x1 x2 x3 x4 USE_INTERCEPT=TRUE
  /RANDOM USE_INTERCEPT=TRUE SUBJECTS=id
COVARIANCE_TYPE=VARIANCE_COMPONENTS
  /BUILD_OPTIONS TARGET_CATEGORY_ORDER=DESCENDING
INPUTS_CATEGORY_ORDER=ASCENDING MAX_ITERATIONS=100
CONFIDENCE_LEVEL=95 DF_METHOD=RESIDUAL COVB=MODEL
  /EMMEANS_OPTIONS SCALE=ORIGINAL PADJUST=LSD.

```

Although it was possible to perform a logistic mixed model analysis with both a random intercept and a random slope for X_3 with SPSS, the program gives a warning message indicating that the results can not be trusted and therefore the

output of the analysis will not be presented. However, the following syntax can be used to perform such an analysis.

```
GENLINMIXED
  /DATA_STRUCTURE SUBJECTS=id
  /FIELDS TARGET=ydich TRIALS=NONE OFFSET=NONE
  /TARGET_OPTIONS DISTRIBUTION=BINOMIAL LINK=LOGIT
  /FIXED EFFECTS=x1 x2 x3 x4 USE_INTERCEPT=TRUE
  /RANDOM EFFECTS=x3 USE_INTERCEPT=TRUE SUBJECTS=id
COVARIANCE_TYPE=UNSTRUCTURED
  /BUILD_OPTIONS TARGET_CATEGORY_ORDER=DESCENDING
INPUTS_CATEGORY_ORDER=ASCENDING MAX_ITERATIONS=100
CONFIDENCE_LEVEL=95 DF_METHOD=RESIDUAL COVB=MODEL
  /EMMEANS_OPTIONS SCALE=ORIGINAL PADJUST=LSD.
```

12.5.6 MLwiN

Output 12.20 shows the results of a logistic mixed model analysis with only a random intercept performed with MLwiN.

Output 12.20 Results of a logistic mixed model analysis with only a random intercept performed with MLwiN

```
ydichij ~ Binomial (denomij, πij)
logit(πij) = β0jcons + 0.338(1.374)x1j + 0.538(0.110)x2ij + -0.451(0.330)x3ij + 0.267(0.592)x4j
β0j = -4.243(3.349) + u0j
```

```
[u0j] ~ N(0, Ωu) : Ωu = [5.282(0.946)]
```

```
var(ydichij|πij) = πij(1 - πij)/denomij
```

```
(882 of 882 cases in use)
```

The MLwiN output of a logistic mixed model analysis is similar to the MLwiN output of a linear mixed model analysis. The first line of the output shows that a binomial distribution is used (i.e. that a dichotomous outcome variable is analyzed). “Denom” is a specification for (generally) a row of ones, which is needed to perform the logistic mixed model analysis (for details see the software manual). The next part of the output shows that a logit link function is used, and provides the regression

coefficients and the standard errors. The regression coefficients can be tested for significance in exactly the same way as has been described for a linear mixed model analysis performed with MLwiN. Furthermore, information about the random part of the model is given: the variance of the normally distributed random intercept (5.282) and the standard error (0.946). Besides this, it can be seen that there is no error variance, which has to do with the fact that there is no error variance in a logistic regression model, i.e. the probability of having a dichotomous outcome is estimated without error. The “error” is in the comparison between the estimated probability and the observed dichotomous outcome.

Output 12.21 shows the results of a logistic mixed model analysis with both a random intercept and a random slope for X_3 performed with MLwiN.

Output 12.21 Results of a logistic mixed model analysis with a random intercept and a random slope for X_3 performed with MLwiN

```

ydichij ~ Binomial(denomij, πij)
logit(πij) = β0jcons + 0.361(1.216)x1j + 0.546(0.107)x2ij + β3jx3ij + 0.286(0.528)x4j
β0j = -4.414(2.970) + u0j
β3j = -0.074(0.304) + u3j

```

$$\begin{bmatrix} u_{0j} \\ u_{3j} \end{bmatrix} \sim N(0, \Omega_u) : \Omega_u = \begin{bmatrix} 6.540(1.234) \\ -3.039(1.075) \quad 1.108(1.186) \end{bmatrix}$$

```
var(ydichij|πij) = πij (1 - πij)/denomij
```

(882 of 882 cases in use)

In the fourth line of Output 12.21 it can be seen that a random slope for X_3 is estimated. The magnitude of the variance around the regression coefficient for X_3 is 1.108. Besides that, also the covariance between the random intercept and the random slope is estimated (i.e. -3.039).

MLwiN uses a PQL estimation procedure to obtain the regression coefficients and the variance components of the logistic mixed model analysis. It should be noted that as a result of the PQL estimation procedure, no log likelihood values are produced. It has already been mentioned that R also uses the quasi-likelihood estimation procedure and it should be noted that also in the R-output no likelihood values are given (see Outputs 12.16 and 12.17). This indicates that

Table 12.5 Summary of the results (i.e. regression coefficients, standard errors (in parentheses), and random variances) of a logistic mixed model analysis with only a random intercept performed with different software packages

	Stata	SAS	MLwiN	R	SPSS
X_1	0.14 (1.44)	0.13 (1.46)	0.34 (1.37)	0.17 (1.16)	0.26 (1.03)
X_2	0.57 (0.11)	0.57 (0.11)	0.54 (0.11)	0.49 (0.08)	0.47 (0.10)
X_3	−0.47 (0.34)	−0.47 (0.34)	−0.45 (0.33)	−0.43 (0.25)	−0.37 (0.30)
X_4	0.33 (0.63)	0.34 (0.64)	0.27 (0.59)	0.23 (0.51)	0.18 (0.45)
Random variance					
Intercept	6.82	6.97	5.28	4.83	3.31

with MLwiN and R the necessity of a random slope for X_3 cannot be estimated with the likelihood ratio test. The only possibility left is to evaluate the magnitude of the variance and the corresponding standard error. Sometimes the variance is divided by its standard error to obtain a sort of z-statistic. However, although this provides some information about the necessity of the particular random coefficient, the interpretation is not straightforward (Twisk, 2006).

Logistic mixed model analysis in MLwiN can be derived directly from the menu, so no syntax is needed.

12.5.7 Overview

Table 12.5 summarizes the results of the logistic mixed model analysis with only a random intercept performed with different software packages. From Table 12.5 it can be seen that there are (major) differences between the results obtained with the different software packages. Although the overall conclusions (if based on p -values) are the same for all analyses, the magnitude of the regression coefficients and standard errors is different.

Table 12.6 summarizes the results of a logistic mixed model analysis with both a random intercept and a random slope for X_3 performed with different software packages.

In general, a comparison between all logistic mixed model analyses illustrated in this section indicates the instability of a logistic mixed model analysis. In other words, the results obtained from a logistic mixed model analysis must be interpreted with great caution.

Table 12.6 Summary of the results (i.e. regression coefficients, standard errors (in parentheses), and random variances) of a logistic mixed model analysis with a random intercept and a random slope for X_3 performed with different software packages

	Stata	SAS ^a	MLwiN	R	SPSS ^a
X_1	0.16 (1.33)		0.36 (1.22)	0.13 (1.13)	
X_2	0.58 (0.11)		0.55 (0.11)	0.50 (0.08)	
X_3	0.04 (0.37)		−0.07 (0.30)	−0.18 (0.29)	
X_4	0.34 (0.58)		0.29 (0.53)	0.30 (0.50)	
Random variance					
Intercept	8.41		6.54	5.83	
Slope for X_3	1.74		1.11	2.07	

^aThe logistic mixed model analysis with both a random intercept and a random slope for X_3 could not be performed with either SAS or SPSS.

12.6 Categorical and “count” outcome variables

Longitudinal data analysis with a categorical outcome variable is not implemented in many software packages yet. In the example discussed in [Chapter 8](#), the GLLAMM procedure was used to analyze the longitudinal relationship between the categorical outcome variable Y_{cat} and the four covariates X_1 to X_4 . The following syntax shows the syntax needed to perform such a multinomial logistic mixed model analysis with both a random intercept and a random slope for X_3 with the GLLAMM procedure in Stata:

```
gen con=1
eq int:con
eq slope:x3
gllamm ycat x1 x2 x3 x4, i(id) fam(bin) link(mlogit)
nrf(2) eqs(int slope) adapt
```

The first line of the syntax for the GLLAMM procedure is needed to define a row of ones, which is necessary to define the random intercept (eq int:con). Furthermore, the random slope for X_3 is defined (eq slope:x3). The syntax for the GLLAMM procedure is then comparable with all other Stata syntaxes. After the procedure specification, the outcome variable is directly followed by the predictor variables, and after the comma additional information is provided, i.e. the subject identifier (i(id)), the number of random coefficients (nrf(2)), and the definition of the random coefficients (eqs(int slope)). “Adapt” stands for adaptive quadrature (see

Section 12.5.2). Detailed information can be found in the software manual (Rabe-Hesketh et al., 2001a), and in several publications (Rabe-Hesketh and Pickles, 1999; Rabe-Hesketh et al., 2000, 2001b).

All packages suitable for performing a logistic GEE analysis or a logistic mixed model analysis are also suitable for the longitudinal analysis of a “count” outcome variable. The only difference is that for the analysis of a “count” outcome variable, a “log” link function and a “Poisson” distribution have to be specified. Although the output and syntax of the different packages will not be discussed in detail, just as an example the syntax needed for a GEE analysis and a Poisson mixed model analysis in Stata is given below:

```
xtgee ycount x1 x2 x3 x4, i(id) fam(poisson)
link(log) corr(exch)
xtmepoisson ycount x1 x2 x3 x4 || id: ,var
```

12.7 The “adjustment for covariance” approach

Up to now, two sophisticated statistical techniques, developed to adjust for the dependency of observations within one subject over time, have been discussed in detail. There is, however, a widely used alternative approach for analysis of continuous outcome variables, which does not adjust for within-subject correlations, but for within-subject covariances. In Section 4.7.3 the “adjustment for covariance” approach was already introduced. The general idea behind this approach is to select a priori a certain “working” covariance structure, which is used in the estimation of the regression coefficients. It is not surprising that the possible choice of structures is comparable to the choice of “working” correlation structures in GEE analysis (see Section 4.5.2). As in GEE analysis, one possibility is the “compound symmetry” or “exchangeable” covariance structure, which assumes equal correlations (irrespective of the time interval between the repeated measurements), and equal variances of the repeated measurements. An “exchangeable” covariance structure for a longitudinal study with six repeated measurements is shown below.

$$\sigma^2 \times \begin{bmatrix} 1 & \rho & \rho & \rho & \rho & \rho \\ \rho & 1 & \rho & \rho & \rho & \rho \\ \rho & \rho & 1 & \rho & \rho & \rho \\ \rho & \rho & \rho & 1 & \rho & \rho \\ \rho & \rho & \rho & \rho & 1 & \rho \\ \rho & \rho & \rho & \rho & \rho & 1 \end{bmatrix}$$

The most extensive covariance structure is an “unstructured covariance structure”.

$$\begin{bmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \rho_{13}\sigma_1\sigma_3 & \rho_{14}\sigma_1\sigma_4 & \rho_{15}\sigma_1\sigma_5 & \rho_{16}\sigma_1\sigma_6 \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 & \rho_{23}\sigma_2\sigma_3 & \rho_{24}\sigma_2\sigma_4 & \rho_{25}\sigma_2\sigma_5 & \rho_{26}\sigma_2\sigma_6 \\ \rho_{13}\sigma_1\sigma_3 & \rho_{23}\sigma_2\sigma_3 & \sigma_3^2 & \rho_{34}\sigma_3\sigma_4 & \rho_{35}\sigma_3\sigma_5 & \rho_{36}\sigma_3\sigma_6 \\ \rho_{14}\sigma_1\sigma_4 & \rho_{24}\sigma_2\sigma_4 & \rho_{34}\sigma_3\sigma_4 & \sigma_4^2 & \rho_{45}\sigma_4\sigma_5 & \rho_{46}\sigma_4\sigma_6 \\ \rho_{15}\sigma_1\sigma_5 & \rho_{25}\sigma_2\sigma_5 & \rho_{35}\sigma_3\sigma_5 & \rho_{45}\sigma_4\sigma_5 & \sigma_5^2 & \rho_{56}\sigma_5\sigma_6 \\ \rho_{16}\sigma_1\sigma_6 & \rho_{26}\sigma_2\sigma_6 & \rho_{36}\sigma_3\sigma_6 & \rho_{46}\sigma_4\sigma_6 & \rho_{56}\sigma_5\sigma_6 & \sigma_6^2 \end{bmatrix}$$

Although the unstructured covariance structure is obviously the best choice for the “working” covariance structure, it can be seen that, when using this structure in a study with 6 repeated measurements, 21 parameters must be calculated (6 variance parameters and 15 correlation coefficients). As in GEE analysis, it is worthwhile to choose the least complicated covariance structure, which “fits” the data well. It has already been mentioned that for GEE analysis there was no indication of the fit of the longitudinal model which could be used to evaluate the different correlation structures. In the “adjustment for covariance” approach, however, the regression coefficients are estimated with maximum likelihood or REML, so models with different covariance structures can be compared with each other by using the likelihood ratio test.

12.7.1 Example

The “adjustment for covariance” approach is implemented in the MIXED procedure in SAS, which was already discussed in Section 12.4.2, and in SPSS. In SPSS, the “adjustment for covariance” approach is implemented within the linear mixed model procedure which was discussed in Section 12.4.4. Output 12.22 shows the output of a longitudinal analysis with an exchangeable covariance structure performed with the linear mixed model procedure in SPSS.

The output of the “adjustment for covariance” approach looks similar to the output of the linear mixed model analysis performed with SPSS. From Output 12.22 it can be seen that with a compound symmetry (i.e. exchangeable) covariance structure only two covariance parameters are estimated (CS diagonal offset and CS covariance). It should be noted that the two numbers are exactly the same as the variance of the random intercept and the residual variance obtained from a linear mixed model analysis with only a random intercept (see Output 12.11). In the output it can further be seen that an REML estimation procedure is used, and that as a result of that a -2 res log likelihood is estimated (i.e. 1671.4). This value can be used to compare models with different covariance structures with each other. To illustrate this, the same dataset is analyzed with an unstructured covariance structure. Output 12.23 shows the results of this analysis.

Output 12.22 Results of the “adjustment for covariance” approach with an exchangeable covariance structure performed with SPSS

Model Dimension ^a						
		Number of Levels	Covariance Structure	Number of Parameters	Subject Variables	Number of Subjects
Fixed Effects	Intercept	1		1		
	x1	1		1		
	x2	1		1		
	x3	1		1		
	x4	1		1		
Repeated Effects	time	6	Compound Symmetry	2	id	147
Total		11		7		

^aDependent Variable: continuous outcome variable y.

Information Criteria ^a	
-2 Restricted Log Likelihood	1671.416
Akaike’s Information Criterion (AIC)	1675.416
Hurvich and Tsai’s Criterion (AICC)	1675.430
Bozdogan’s Criterion (CAIC)	1686.969
Schwarz’s Bayesian Criterion (BIC)	1684.969

The information criteria are displayed in smaller-is-better forms.

^aDependent Variable: continuous outcome variable y.

Estimates of Fixed Effects ^a							
Parameter	Estimate	Std. Error	Df	t	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	3.449725	0.664828	151.148	5.189	0.000	2.136170	4.763281
x1	0.147907	0.272062	148.029	0.544	0.587	-0.389720	0.685534
x2	0.184720	0.019525	876.874	9.461	0.000	0.146398	0.223041
x3	0.044554	0.059106	839.889	0.754	0.451	-0.071459	0.160567
x4	0.037217	0.119995	150.941	0.310	0.757	-0.199870	0.274304

^aDependent Variable: continuous outcome variable y.

Estimates of Covariance Parameters ^a			
Parameter		Estimate	Std. Error
Repeated Measures	CS diagonal offset	0.274102	0.014315
	CS covariance	0.299373	0.040771

^aDependent Variable: continuous outcome variable y.

From Output 12.23 it can be seen that in this analysis 21 covariance parameters are estimated. The first parameter (UN(1.1)) is an indication of the variance of the first measurement, the second parameter (UN(2.1)) is an indication of the covariance between the first and the second measurements, and so on.

The likelihood ratio test can be used to decide which covariance structure is to be preferred in the analysis. Therefore, the difference between the –2 res log

Output 12.23 Results of the “adjustment of covariance” approach with an unstructured covariance structure performed with SPSS

Model Dimension ^a					
		Number of Covariance Levels	Structure	Number of Subject Parameters	Number of Variables Subjects
Fixed Effects	Intercept	1		1	
	x1	1		1	
	x2	1		1	
	x3	1		1	
	x4	1		1	
Repeated Effects	time	6	Unstructured	21	id 147
Total		11		26	

^aDependent Variable: continuous outcome variable y.

Information Criteria ^a	
-2 Restricted Log Likelihood	1372.222
Akaike’s Information Criterion (AIC)	1414.222
Hurvich and Tsai’s Criterion (AICC)	1415.303
Bozdogan’s Criterion (CAIC)	1535.529
Schwarz’s Bayesian Criterion (BIC)	1514.529

The information criteria are displayed in smaller-is-better forms.

^aDependent Variable: continuous outcome variable y.

Estimates of Fixed Effects^a

Parameter	Estimate	Std. Error	Df	t	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	3.869219	0.658825	149.681	5.873	0.000	2.567421	5.171017
x1	-0.017914	0.269946	147.247	-0.066	0.947	-0.551383	0.515555
x2	0.121667	0.018469	613.749	6.588	0.000	0.085397	0.157938
x3	-0.071629	0.049184	617.767	-1.456	0.146	-0.168217	0.024959
x4	0.059011	0.118853	149.238	0.497	0.620	-0.175841	0.293863

^aDependent Variable: continuous outcome variable y.

Estimates of Covariance Parameters^a

Parameter		Estimate	Std. Error
Repeated Measures	UN(1.1)	0.470608	0.060011
	UN(2.1)	0.331086	0.047857
	UN(2.2)	0.416285	0.049269
	UN(3.1)	0.301898	0.045952
	UN(3.2)	0.325920	0.045506
	UN(3.3)	0.445687	0.054345
	UN(4.1)	0.265089	0.045200
	UN(4.2)	0.320825	0.048054
	UN(4.3)	0.368600	0.054061
	UN(4.4)	0.458917	0.063225
	UN(5.1)	0.315988	0.058341
	UN(5.2)	0.286973	0.050727
	UN(5.3)	0.298496	0.050798
	UN(5.4)	0.274051	0.049093
	UN(5.5)	0.563372	0.078517
	UN(6.1)	0.409495	0.099354
	UN(6.2)	0.316314	0.087524
	UN(6.3)	0.304738	0.084764
	UN(6.4)	0.211509	0.078838
	UN(6.5)	0.577358	0.117816
	UN(6.6)	1.279897	0.191445

^aDependent Variable: continuous outcome variable y.

likelihoods of both models has to be calculated. This difference follows a Chi-square distribution with 19 degrees of freedom. Again, the number of degrees of freedom is based on the difference in the number of parameters to be calculated with each analysis. With the unstructured covariance structure 21 variance and covariance

Table 12.7 Table showing -2 restricted log likelihoods and the number of estimated variance and covariance parameters of longitudinal data analysis adjusting for different covariance structures

Covariance structure	-2 restricted log likelihood	Number of variance and covariance parameters
Exchangeable (compound symmetry)	1671.4	2
First-order autoregressive	1583.2	2
5-Dependent (Toeplitz (5))	1574.2	6
Unstructured	1372.2	21

Table 12.8 Regression coefficients and standard errors (in parentheses) obtained from the “adjustment for covariance” approach (performed in SPSS) and the ones obtained from GEE analyses using different correlation structures

	“Adjustment for covariance”		GEE analysis	
	Exchangeable	Unstructured	Exchangeable	Unstructured
X_1	0.15 (0.27)	-0.02 (0.27)	0.15 (0.27)	0.14 (0.29)
X_2	0.18 (0.02)	0.12 (0.02)	0.18 (0.02)	0.14 (0.02)
X_3	0.04 (0.06)	-0.07 (0.05)	0.04 (0.06)	0.01 (0.05)
X_4	0.04 (0.12)	0.06 (0.12)	0.04 (0.13)	0.11 (0.14)

parameters were calculated, while for the exchangeable covariance structure only 2 parameters were calculated. The difference between the -2 res log likelihoods (i.e. 299.2) is highly significant or, in other words, the model with an unstructured covariance structure is statistically “better” than the model with an exchangeable covariance structure. Because there are other possible covariance structures that can be considered with less parameters to be estimated than with the unstructured covariance structure, the data were reanalyzed with a 5-dependent (i.e. Toeplitz (5)) covariance structure, and with a first-order autoregressive covariance structure. Table 12.7 summarizes the results.

Based on the results of the -2 (restricted) log likelihoods reported in Table 12.7, an unstructured covariance structure seems to be the most appropriate in this particular situation. To illustrate the importance of a good choice of covariance structure, Table 12.8 shows the regression coefficients and standard errors of the four independent variables, calculated with an exchangeable and an unstructured covariance structure. The same table also shows the coefficients for comparable GEE analyses.

From Table 12.8 it can be seen that the results of a GEE analysis with an exchangeable correlation structure are equal to the results obtained from an analysis adjusting for an “exchangeable” covariance structure. This is not surprising, because in the two analyses either one variance parameter or one correlation coefficient is estimated. The major difference between the “adjustment for covariance” approach and GEE analysis occurs in the comparison between an unstructured covariance structure and an unstructured correlation structure. With the “adjustment for covariance” approach, different variances are estimated at the different time-points, while with GEE analysis only one variance parameter is estimated. So (probably), the greatest advantage of the “adjustment for covariance” approach, compared to GEE analysis, is the flexibility of modeling the variance over time. Furthermore, there is the possibility to evaluate the need for a certain (more complicated) covariance structure by means of the likelihood ratio test. A major disadvantage of the “adjustment for covariance” approach is that it is only suitable for continuous outcome variables.

Finally, it should again be noted that an “adjustment for covariance” with an “exchangeable” covariance structure is exactly the same as a linear mixed model analysis with only a random intercept (see Outputs 12.22 and 12.11).

Although the “adjustment for covariance” approach can be derived from the menu, the following syntax can be used to perform the analysis.

```
MIXED ycont WITH x1 x2 x3 x4
  /CRITERIA=CIN(95) MXITER(100) MXSTEP(5) SCORING(1)
SINGULAR(0.000000000001) HCONVERGE(0, ABSOLUTE)
  LCONVERGE(0, ABSOLUTE) PCONVERGE(0.000001, ABSOLUTE)
/FIXED=x1 x2 x3 x4 | SSTYPE(3)
/METHOD=REML
/PRINT=SOLUTION
/REPEATED=time | SUBJECT(id) COVTYPE(UN)
```

For illustrative purposes, Output 12.24 shows the results of the “adjustment for covariance” approach with an exchangeable covariance structure performed with the MIXED procedure in SAS, while output 12.25 shows the results of the same analysis with an unstructured covariance structure. Not surprisingly, the results of the analyses are comparable to the results obtained from SPSS.

The syntax needed to perform the “adjustment for covariance” approach with an unstructured covariance structure in the MIXED procedure in SAS is given below.

```
proc mixed data=long.x;
class id time;
model ycont = x1 x2 x3 x4 /s;
repeated time/ subject=id type=un;
run;
```


Output 12.24 Results of the “adjustment for covariance” approach with an exchangeable covariance structure performed with SAS

The Mixed Procedure					
Model Information					
Data Set	long.x				
Dependent Variable	YCONT				
Covariance Structure	Variance Components				
Subject Effect	ID				
Estimation Method	REML				
Residual Variance Method	Parameter				
Fixed Effects SE Method	Model-Based				
Degrees of Freedom Method	Between-Within				
Covariance Parameter Estimates					
Cov Parm	Subject	Estimate			
TIME	ID	0.5678			
Fit Statistics					
-2 Res Log Likelihood		2020.1			
AIC (smaller is better)		2022.1			
AICC (smaller is better)		2022.1			
BIC (smaller is better)		2025.0			
Null Model Likelihood Ratio Test					
DF	Chi-Square	Pr > ChiSq			
0	0.00	1.0000			
Solution for Fixed Effects					
	Standard				
Effect	Estimate	Error	DF	t Value	Pr > t
Intercept	3.3209	0.3629	144	9.15	<0.0001
X1	0.1885	0.1461	144	1.29	0.1991
X2	0.2026	0.01938	733	10.45	<0.0001
X3	0.07509	0.06824	733	1.10	0.2716
X4	0.02171	0.06517	144	0.33	0.7396

Output 12.25 Results of the “adjustment for covariance” approach with an unstructured covariance structure performed with SAS

The Mixed Procedure		
Model Information		
Data Set	long.x	
Dependent Variable	YCONT	
Covariance Structure	Unstructured	
Subject Effect	ID	
Estimation Method	REML	
Residual Variance Method	None	
Fixed Effects SE Method	Model-Based	
Degrees of Freedom Method	Between-Within	
Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1.1)	ID	0.4706
UN(2.1)	ID	0.3311
UN(2.2)	ID	0.4163
UN(3.1)	ID	0.3019
UN(3.2)	ID	0.3259
UN(3.3)	ID	0.4457
UN(4.1)	ID	0.2651
UN(4.2)	ID	0.3208
UN(4.3)	ID	0.3686
UN(4.4)	ID	0.4589
UN(5.1)	ID	0.3160
UN(5.2)	ID	0.2870
UN(5.3)	ID	0.2985
UN(5.4)	ID	0.2741
UN(5.5)	ID	0.5634
UN(6.1)	ID	0.4095
UN(6.2)	ID	0.3163
UN(6.3)	ID	0.3047
UN(6.4)	ID	0.2115
UN(6.5)	ID	0.5774
UN(6.6)	ID	1.2799
Fit Statistics		
-2 Res Log Likelihood		1372.2
AIC (smaller is better)		1414.2
AICC (smaller is better)		1415.3
BIC (smaller is better)		1477.0

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
20	647.83	<0.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	3.8692	0.6588	144	5.87	<0.0001
X1	-0.01791	0.2699	144	-0.07	0.9472
X2	0.1217	0.01847	144	6.59	<0.0001
X3	-0.07163	0.04918	144	-1.46	0.1475
X4	0.05901	0.1189	144	0.50	0.6203

The syntax is comparable to what has been discussed earlier for the MIXED procedure used to perform a linear mixed model analysis. The differences are observed in the last line of the syntax, where the “random” statement is replaced by the “repeated” statement, and in the additional specification of the covariance structure (type=un). A detailed description of the (almost infinite) possibilities within the SAS MIXED procedure can be found in the software manual (SAS Institute, Inc., [1997](#)).

One step further

13.1 Introduction

In the foregoing chapters, different techniques for longitudinal data analysis were described and illustrated. In these chapters a distinction was made between the analysis of continuous, dichotomous, and categorical outcome variables. In this chapter, some “new” features for longitudinal data analysis will be introduced. In Section 13.2 longitudinal data analysis with outcome variables with upper or lower censoring will be discussed, while in Section 13.3 a short introduction will be given of the analysis aiming to classify individuals with the same developmental trajectories. Because it basically goes beyond the scope of this book, both explanations will be relatively short and basic, but many references will be given in which more detailed and/or mathematical information can be found.

13.2 Outcome variables with upper or lower censoring

13.2.1 Introduction

In Section 9.2.1 several possibilities were discussed to define the change in a particular continuous outcome variable over time. It was suggested that for a longitudinal study in which the outcome variable reaches either a ceiling or a floor a different definition of change should be used. When the development over time is analyzed in a particular continuous outcome variable, it is quite common that the variable reaches either a ceiling or a floor. For instance, in rehabilitation research, most of the patients will recover after a certain amount of time. On the instrument to measure the rehabilitation process, these patients cannot score any higher than the maximum. It is also possible that so-called floor effects occur. For instance, when pain medication is investigated and the outcome variable pain is measured on a visual analog scale, some patients will report “no pain” after a certain amount of time. They cannot score lower than the “no pain” level. Also in studies when

there is some detection limit (e.g. for blood parameters or environmental factors, such as pesticides), these floor effects are present. In fact these problems always arise when a measurement instrument is used that has upper and lower limits and when some of the subjects in the study reach these upper or lower limits.

In most longitudinal epidemiological studies, this upper or lower censoring is ignored. The outcome variables are analyzed as if they were normally distributed over the whole period of time. This is not the case, because when patients reach the floor or ceiling, the outcome variable is not normally distributed anymore. In cross-sectional studies (especially in econometrics) the problem of upper and lower censoring is solved by using so-called tobit models, after Tobin's (1958) classical example on household expenditures. Within epidemiology, only a few examples are available in which cross-sectional tobit regression analysis is used (Twisk and Rijmen, 2009). However, for longitudinal epidemiological studies tobit analysis is almost never used, while it has some nice theoretical advantages above the "classical" longitudinal data analysis. Tobit regression analysis is an example of a so-called two-part model. The general idea behind two-part models (which are also known as mixed response or mixed distribution models) is that the outcome variable has a mixed distribution, i.e. a binomial distribution for either reaching the floor or ceiling, and another distribution for the part between the floor and ceiling. Within tobit regression analysis it is assumed that the distribution of the outcome variable between the floor and ceiling is approximately normal. However, in theory it can be another distribution as well. One two-part model that is used quite often within epidemiology is zero-inflated Poisson regression. A zero-inflated Poisson regression is a two-part model that is used when a Poisson outcome variable has an excess of zeros. In this situation, the binomial distribution of being zero or not is mixed with a Poisson distribution for the non-zero part of the distribution. Within the two-part models a distinction must be made between standard two-part regression models and two-part joint regression models. In the standard two-part regression models the two processes are split and two sets of regression coefficients are obtained, which means that different sets of covariates can be included for the two processes. Zero-inflated Poisson regression is an example of a standard two-part model. With zero-inflated Poisson regression, one set of covariates is used for the binomial process (zero versus non-zero) and one set for the Poisson process. For some research questions (e.g. investigating the determinants of smoking behavior) this is a nice feature. However, in many situations one regression coefficient for each covariate would be preferable (e.g. the analysis of hypoglycemic events). Models that provide one set of regression coefficients are known as two-part joint regression models and tobit regression is an example of a two-part joint regression

model. Longitudinal tobit regression analysis is implemented in Stata in the `xttobit` procedure. This procedure is a mixed model approach on a mixed distribution of the outcome variable (i.e. a binomial distribution for either reaching the floor or ceiling or not and a normal distribution for the part of the distribution between the floor and ceiling). The problem with the `xttobit` procedure is that it only allows a random intercept. For more flexible two-part models the GLLAMM procedure can be used (Rabe-Hesketh and Skondral, 2001; Rabe-Hesketh et al., 2001a). The GLLAMM procedure was already mentioned in Section 8.1.5.4 in which it was used to perform a longitudinal mixed model analysis with a categorical outcome variable, i.e. a longitudinal multinomial logistic regression analysis. The GLLAMM procedure is flexible because it allows random slopes for the time-dependent covariates and because the binomial distribution can be mixed with several other distributions besides a normal distribution, i.e. a Poisson distribution, a gamma distribution, etc.

13.2.2 Example

The first example is taken from a longitudinal rehabilitation study among stroke patients (Kwakkel et al., 1999). The main purpose of the study was to analyze the development of the Barthel index. An outcome variable represents a patient's ability to carry out 10 everyday tasks (i.e. bladder and bowel control, toilet use, dressing, feeding, walking, personal toilet, transfer activities, bathing, and stair climbing). The lowest score for the Barthel index is 0 and the highest possible score is 20. The Barthel index was assessed weekly during the first 10 weeks after stroke onset, then every 2 weeks until week 20, and finally the Barthel index was assessed at week 26, week 38, and week 52. The study population consisted of 101 patients with on average 16.3 measurements (range 2–18) per patient. Forty-seven patients have a full dataset, while 33 patients only missed the first measurement. Furthermore, there are 7 patients who dropped-out (varying from the second measurement to the measurement after 26 weeks); there are 6 patients with intermittent missing data with more than 3 missing observations; and 8 patients with intermittent missing data with only one or two missing observations. The research question to be answered was related to the development over time. Output 13.1 shows the result of a longitudinal tobit analysis to analyze the development over time in the Barthel index in which a quadratic development over time was modeled.

The output of the longitudinal tobit analysis looks somewhat similar to the outputs of the other mixed model analyses performed throughout this book. The first part of the output shows some general information. It is mentioned that a random effects tobit regression is performed, that the data are clustered within the

Output 13.1 Results of a longitudinal tobit analysis to analyze the development over time in the Barthel index in which a quadratic development over time was modeled

Random-effects tobit regression	Number of obs	=	1644			
Group variable: id	Number of groups	=	101			
Random effects u_i ~ Gaussian	Obs per group: min	=	2			
	avg	=	16.3			
	max	=	18			
	Wald chi2(2)	=	2859.15			
Log likelihood = -3527.6622	Prob > chi2	=	0.0000			

barthel		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----						
time		0.7588106	0.016988	44.67	0.000	0.7255148 0.7921065
time2		-0.0111579	0.000333	-33.51	0.000	-0.0118105 -0.0105052
_cons		6.008142	0.5195729	11.56	0.000	4.989797 7.026486
-----+-----						
/sigma_u		5.041176	0.3622171	13.92	0.000	4.331243 5.751108
/sigma_e		2.422371	0.0477902	50.69	0.000	2.328704 2.516038
-----+-----						
rho		0.8124159	0.0225832			0.7649858 0.8534247

Observation summary:						
			0 left-censored observations			
			1388 uncensored observations			
			256 right-censored observations			

individual (Group variable: id) and that the analysis assumes that the distribution of the intercepts is normal (Random effects $u_i \sim \text{Gaussian}$). Besides that, also the log likelihood of the model is given. From the right column it can be seen that there are 1644 observations within 101 individuals used in the analysis. On average there are 16.3 observations for each individual, and the minimum number of observations was 2 and the maximum number of observations was 18. Furthermore, the result of the Chi-square test relating to the fixed effects (i.e. the variables in the model) is given.

The second part of the output contains the regression coefficients, standard errors, z-values, corresponding p -values, and the 95% confidence intervals around the regression coefficients. The last part shows the random part of the model,

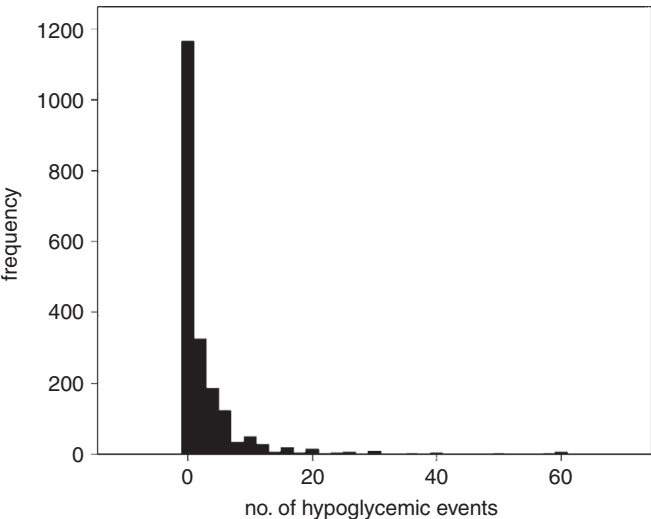


Figure 13.1 Overall distribution of the number of hypoglycemic events in the Spirit dataset.

From Output 13.2 it can be seen that the regression coefficients were only slightly different between the two methods. However, the log likelihood of the longitudinal tobit analysis is much lower than the one obtained from the mixed model analysis assuming a normal distribution of the outcome variable. Although it is questionable whether the likelihood ratio test can be used to choose between the longitudinal tobit analysis and the mixed model analysis, it gives an indication that the longitudinal tobit analysis provides a better fit. This conclusion was also confirmed by the Schwarz’s Bayesian criterion (BIC) values for both models (i.e. the BIC was 7891.347 for the mixed model analysis and 7092.349 for the longitudinal tobit analysis) and by the comparison between the observed and predicted values (for details see Twisk and Rijmen, 2009).

The second example is taken from the SPIRIT study (Hajos et al., 2011). This study aimed to examine the effect of initiation of insulin glargine (a long acting insulin analog) on general emotional well-being, diabetes symptom distress, and worries about hypoglycemia in Dutch type 2 diabetes patients who previously used oral anti-hyperglycemic medication. Type 2 diabetes patients who used oral anti-hyperglycemic agents were recruited from 363 Dutch primary care practices, which were spread across the country. This resulted in a total sample of 889 patients. At baseline, patients initiated insulin glargine either combined with oral medications, a rapid-acting insulin analog, or both at the discretion of their treating physician. Measurements were conducted at baseline, after three and after six months and, in this example, the development over time in hypoglycemic events for diabetic patients was assessed. Figure 13.1 shows the overall distribution of the number of

hypoglycemic events in the dataset used in this example. From [Figure 13.1](#) it can be seen that the distribution is highly zero-inflated, or in other words, the distribution has a strong floor effect.

Output [13.3](#) shows the result of a two-part joint regression analysis in which a binomial distribution was used for the zero versus non-zero part of the distribution and a Poisson distribution for the non-zero part. In the longitudinal two-part joint regression analysis, only a random intercept is considered. Because there are only two fixed time-points in this study, time was treated as a categorical variable and represented by two dummy variables.

Output 13.3 Results of a longitudinal two-part joint regression model with only a random intercept to analyze the development over time in hypoglycemic events performed with the GLLAMM procedure

log likelihood = -3312.5361

Robust standard errors

hyp_al	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
_Itime_2	-0.1682755	0.1210002	-1.39	0.164	-0.4054316	0.0688806
_Itime_3	-0.2791713	0.1050495	-2.66	0.008	-0.4850646	-0.073278
_cons	0.0132694	0.1223562	0.11	0.914	-0.2265443	0.2530832

Variances and covariances of random effects

***level 2 (id)

var(1): 3.5673155 (.30504197)

Output [13.3](#) is very simple and very straightforward. First the log likelihood of the model is given (i.e. -3312.5361), and after that the regression coefficients, the z -values, the corresponding p -values, and the 95% confidence intervals around the regression coefficients. From this part of the output it can be seen that the development of hypoglycemic events between baseline and 3 months is not statically significant (p -value equals 0.164), while the development between baseline and

6 months is statistically significant (p -value equals 0.008). The last part of Output 13.3 shows the variance of the random intercept (i.e. 3.5673155).

To compare the results of the longitudinal two-part joint regression model with only a random intercept with a comparable Poisson mixed model analysis, Output 13.4 shows the results of the Poisson mixed model analysis with only a random intercept.

Output 13.4 Results of a longitudinal Poisson mixed model analysis with only a random intercept to analyze the development over time in hypoglycemic events

Mixed-effects Poisson regression				Number of obs	=	1984
Group variable: id				Number of groups	=	889
				Obs per group: min	=	1
				avg	=	2.2
				max	=	3
Integration points = 7				Wald chi2(2)	=	38.51
Log likelihood = -3976.4112				Prob > chi2	=	0.0000

hyp_a1		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----						
_Itime_2		-0.1055194	0.0376571	-2.80	0.005	-0.1793259 -0.0317129
_Itime_3		-0.2646151	0.042647	-6.20	0.000	-0.3482016 -0.1810286
_cons		-0.5001234	0.0891871	-5.61	0.000	-0.6749269 -0.32532

-----+-----						
Random-effects Parameters			Estimate	Std. Err.	[95% Conf. Interval]	
-----+-----						
id: Identity						
var(_cons)			3.946494	0.3103373	3.382799	4.604121

LR test vs. Poisson regression:				chibar2(01) =	7885.91	Prob>=chibar2 = 0.0000

When the results reported in Output 13.3 are compared with the results reported in Output 13.4, it can be seen that the results are slightly different, especially regarding the development of hypoglycemic events between the first measurement and the measurement after 3 months. This difference is statistically significant in the longitudinal Poisson mixed model analysis (p -value equals 0.005), but it was not significant in the longitudinal two-part model. Again, although it is doubtful whether the log likelihood values can be compared to each other, the comparison indicates that the longitudinal two-part model provides a better fit compared to the

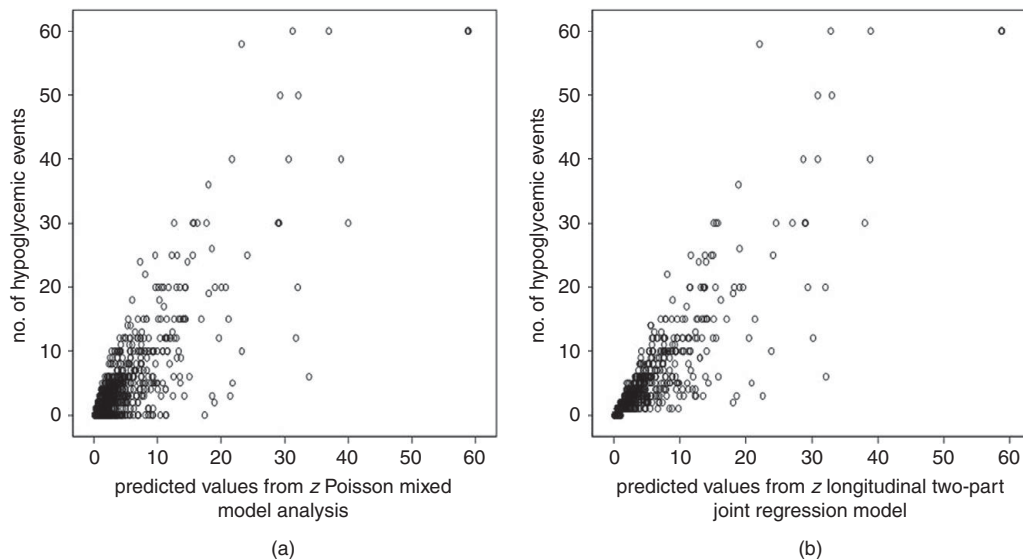


Figure 13.2 Observed versus predicted values derived from a Poisson mixed model analysis (a) and a longitudinal two-part joint regression model (b).

Poisson mixed model. Also in this example, this was confirmed by the BIC values for both models, which were 6295.563 for the two-part model and 7983.194 for the Poisson mixed model, respectively, and by the comparison between the predicted and observed values (Figure 13.2).

For detailed information regarding both examples one is referred to Twisk and Rijmen (2009) and Spriensma et al. (2012).

13.2.3 Remarks

In both examples it was chosen to analyze the longitudinal data with lower or upper censoring as a two-part joint regression model. An important reason why the two-part joint regression model was chosen is that the outcome variable in the examples (i.e. either the Barthel index or the number of hypoglycemic events) should be seen as one process that cannot be split into two processes with separate regression coefficients for the development over time. Sometimes it is better to analyze the data with a standard two-part regression model, leading to separate regression coefficients for both parts of the process. An example could be the analysis of determinants of smoking behavior, which can be different for the logistic part of the analysis (i.e. the probability of smoking versus non-smoking) and the Poisson part (i.e. the number of cigarettes smoked).

The biggest problem in using longitudinal two-part joint regression models is the interpretation of the regression coefficients. The general idea is that the regression

coefficient combines two interpretations: (1) The difference in the outcome variable if above the limit, weighted by the probability of being above the limit; and (2) the difference in the probability of being above the limit, weighted by the expected value of the outcome variable if above the limit. Theoretically this makes sense; however, in practice if, for instance, one is interested in the magnitude of the effect of a certain intervention over time, it is difficult to get a proper interpretation of the regression coefficient.

Maybe because of the difficult interpretation of the regression coefficients, the two-part approach is almost never used in epidemiological longitudinal studies. Mostly, the floor or ceiling effects are ignored in the analysis. Other ways to analyze these kind of data is to reduce the information in the data to either a dichotomous outcome variable (mostly comparing zero to non-zero) or a categorical outcome variable (mostly comparing zero with two groups of non-zero outcomes in which the groups are divided according to the median of the non-zero part). Sometimes, researchers try to transform (with a logarithmic transformation) a distribution with many zeros into a normally distributed variable. However, due to the bunch of zeros at the beginning of the distribution, this transformation will not work because it will never result in a normal distribution.

13.3 Classification of subjects with different developmental trajectories

In [Chapter 5](#) a detailed discussion was given regarding the modeling of time and the analysis of the development of a particular outcome variable over time. Although the analysis of the development over time is of great interest, researchers are often interested in dividing the cohort under study into groups of subjects with comparable developmental trajectories: firstly, as a tool to describe the population under study; and, secondly, as a first step to study either the determinants of different trajectories or the consequences of different trajectories. Although the division into subgroups can be done in many different ways, most approaches are based on structural equation modeling (Duncan et al, [1999](#)), a fairly complex statistical technique particularly popular in psychology and social science, but not so much in medical science and epidemiology. The general idea behind structural equation modeling is that the development over time in a particular outcome variable is captured by so-called latent or unobserved variables. In a longitudinal study these latent variables are the growth curve parameters. When a linear development over time is modeled, there are two growth curve parameters, i.e. the intercept and the slope. [Figure 13.3](#) is a typical illustration of a structural equation model, in which, in this case, a linear growth curve example for a longitudinal study with six repeated measurements is shown. The six repeated measurements of the outcome variable

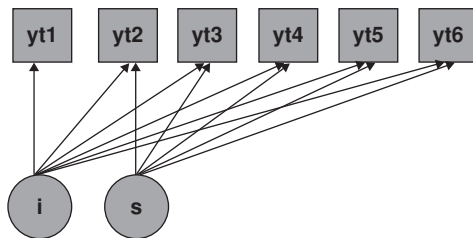


Figure 13.3 Schematic representation of the latent variable modeling framework. The six repeated measurements are represented by squares and the latent growth curve parameters (i and s) are represented by circles.

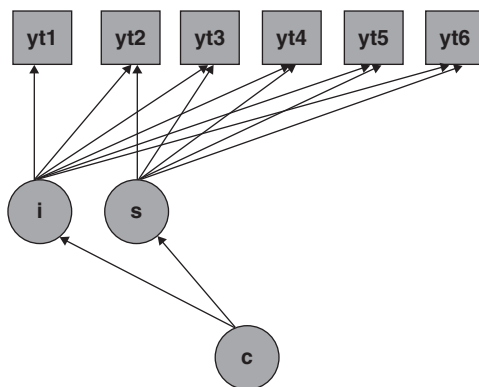


Figure 13.4 Schematic representation of the latent variable modeling framework. The six repeated measurements are represented by squares and the latent growth curve parameters (i and s) and the latent class variable (c) are represented by circles.

are the observed variables and they are summarized into two latent growth curve parameters; the intercept and the slope.

It should be noted that in a situation when one is interested in the development over time for a particular continuous outcome variable, the results of a structural equation model are exactly the same as the results from a linear mixed model analysis. This not only holds for the regression coefficients of the growth parameters, but also for the variances and covariances of the random intercept and random slope.

When one is interested in the classification of subjects with different developmental trajectories, the structural equation model shown in [Figure 13.3](#) is extended with another latent variable, i.e. a categorical variable defining the different classes of developmental trajectories. This classification is based on the intercepts and slopes for the individual growth curve parameters ([Figure 13.4](#)).

Within this structural equation framework, two approaches are often used to classify subjects with the same developmental trajectories, i.e. latent class growth analysis (LCGA) and latent class growth mixture modeling (LCGMM). LCGMM can be seen as an extension of LCGA. The difference between LCGA and LCGMM has to do with the assumptions regarding the individual trajectories within a certain class. Looking at the development from a LCGA perspective, the cohort under study consists of a number of classes, each of which has his own growth trajectory. The classes differ in trajectory shape, but within the classes the individuals are assumed to have a similar growth trajectory (there is no within class variation). Looking at the development from a LCGMM perspective, the interpretation of the classes is similar, but within the classes, individuals are allowed to differ in growth trajectory (so there can be within class variation). This indicates that subjects with slightly different growth parameters are easier classified to the same class within LCGMM compared to LCGA.

There is some discussion about the use of LCGA or LCGMM (Connell and Frye, 2006a; 2006b; Hoeksma and Kelderman, 2006; Muthén, 2006). The LCGA methodology is developed by Nagin and colleagues (Nagin, 1999; Nagin and Tremblay, 2001) and implemented in the SAS procedure Traj (Jones et al., 2001), whereas the LCGMM methodology is developed by Muthén and colleagues (Muthén and Shedden, 1999; Muthén and Muthén, 2000; Muthén, 2004) and implemented in the Mplus software program (Mplus, Version 7, 1998–2012). As been mentioned before, LCGMM can be seen as an extension of LCGA. The difference between the two methods has to do with the variation within a certain class. With LCGA this variation is set to zero, while for LCGMM this is not the case. This implies that the (optimal) number of classes derived from LCGA is always bigger than the (optimal) number of classes derived from LCGMM. Within LCGA, subjects with slightly different growth parameters are sooner defined to a different class compared to LCGMM in which the growth parameters within a class are allowed to differ. To find the optimal number of classes, various approaches are available. Probably the best way is a “forward” classifying approach which starts with a one class solution (i.e. there are no subgroups; all individuals follow the same trajectory over time), then adding extra classes one at a time to investigate whether or not model fit becomes better due to the additional class. This procedure ends the moment the model fit does not improve any more. However, with this procedure, one has to be very careful. It could be that the solution that statistically optimally describes the data is a solution with one (or more) clinically uninterpretable classes, or with (a) class(es) with very few subjects. Both issues should be kept in mind when analysing the data with these classification methods.

Within epidemiology, creating groups with the same developmental trajectory is increasingly popular these days (Twisk and Hoekstra, 2012). It is mostly used

as a descriptive tool, but the trajectories are also used either as a determinant for future (health) outcomes or as an outcome variable in order to investigate potential predictors of these trajectories. However, it should be realized that both additional questions can be answered with different methods as well. Regarding the investigation of potential predictors for the development over time in a certain outcome, it is not necessary to classify the population under study into several groups. The outcome variable itself can be analyzed with a mixed model analysis. Regarding the relationship between different developments over time and future (health) outcomes, it is also not necessary to classify the population under study into several groups. Instead of using the categorical group variable as a determinant for future (health) outcomes, it is also possible to use the individual growth parameters as determinants. In general, it should be realized that classifying developmental trajectories is mostly not the only solution to answer certain research questions.

For detailed insight into the (complicated) mathematical background of the approaches used to classify subjects with different developmental trajectories one is referred to several more fundamental papers (Muthén and Shedden, 1999; Nagin, 1999; Muthén and Muthén, 2000; Muthén, 2004; Jung and Wickrama, 2008; Muthén and Asparouhov, 2008; Feldman et al., 2009).

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